

Unconditional all-orders asymptotics for two OEIS permutation sequences

Stable path forests, Poisson expansions, and index inversion

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Abstract

The OEIS entries A189281 and A110128 count permutations avoiding a prescribed signed or absolute difference between entries two positions apart. Their displayed higher asymptotic expansions are associated with conjectured recurrences. We give a direct, recurrence-free proof of those expansions, together with a finite exact formula for every coefficient. More generally, for any fixed positive offsets r, s , the complete probability generating function of the number of violations has an all-orders expansion consisting of $e^{\theta u}$ times polynomial corrections, where $\theta = 1$ in the oriented case and $\theta = 2$ in the absolute case. The remainder is uniform on compact complex u -sets.

The structural ingredient is an exact stabilization theorem for tilings of long path forests: low factorial moments depend on the numbers of paths, not their individual lengths. We also prove a quantitative total-variation universality bound, a factorial denominator bound for all asymptotic coefficients, and an all-orders Lambert- W index inversion. Exact rational computations recover the OEIS coefficients through order ten and extend both expansions through order sixteen. Independent permutation enumeration and Bonferroni certificates accompany the proofs. The specific conjectured recurrences and the stronger all-orders integrality suggested by A189281 are not assumed or claimed proved. A concrete rational-function conjecture is proposed as a route to the latter question.

Status. This is a proof-focused research draft, not a claim of peer review or a Lean-checked formalization. The theorems below have mathematical proofs; the computations are reproducible exact checks. Historical priority beyond the sources reviewed here remains to be assessed. The results establish the algebraic asymptotic sector, not a complete exponentially improved transseries.

Editorial note (1 October 2026, batch 73O2). This report is manuscript 60 of batch 73 of ProveIt’s incoming-reports intake, placed in the research-report collection in commit [6e193dd4f](#). It is AI-assisted, unrefereed and not formalized. The text is the delivered manuscript; the write added only the label prefix `spf:`, the bracketed notes marked “[Added 1 October 2026, batch 73O2]”, Remarks [7.2](#) and [9.1](#), three bibliography entries, and [section B](#), which records the provenance and every editorial change. The rational-collapse statement of [section 11](#) remains a conjecture.

Editorial note (1 October 2026, batch 74). Since batch 74 the report has two Parts. Part I is manuscript 60, as described above. Part II ([section 14](#)) adds manuscript 03 of batch 74, an independent second derivation of Part I’s main theorems, and prints only what it adds: uniformity over forest shapes, comparison bounds, closed forms of $c_3(r, s)$, and proved integrality for the oriented one-path family. Its labels carry the prefix `fga:`.

Editorial note (3 October 2026, batch 85). Since batch 85 the report has three Parts. Part III ([section 23](#)) adds manuscript 03 of batch 85, which proves the rational collapse, Conjecture 11.1, for every h ([theorem 26.1](#)), extends it to every pair of directed offsets, and proves that every oriented correction polynomial $B_f^{(1)}(u; r, s)$ has integer coefficients, with an exact degree law ([theorem 28.1](#)). Its labels carry the prefix `spf:rc:`. The abstract, the batch-7302 note above, [section 11](#) and the conclusion of Part I describe manuscript 60 as delivered, in which the collapse is a conjecture and integrality is not claimed; dated notes at Conjecture 11.1, Proposition 11.2, questions 1 and 4 of [section 12](#), in Part II and in [section B](#) record what has changed. The conjectured recurrences of A189281 and A110128 are still not proved, and the absolute case is not covered by Part III.

Editorial note (4 October 2026, batch 97). Since batch 97 the report has five Parts. Part IV ([section 36](#)) adds manuscript 05 of batch 97, which proves the all-orders expansion uniformly over every pair of path forests, with no hypothesis on the path lengths, in normalized subpath densities; it gives the first shape-dependent term of a fixed short path and the corrections of moving and proportional offsets. Part V ([section 54](#)) adds manuscript 06 of batch 97, which sums the oriented $(2, 2)$ algebraic sector along the positive Borel ray, determines the growth of its coefficients, and proves that the counts of A189281 differ from that sum by a nonzero flat term. Their labels carry the prefixes `spf:mg:` and `spf:bc:`. Dated notes in the questions of Parts I–III, in Part II and in [section B](#) record what has changed. The conjectured recurrences of A189281 and A110128 are still not proved.

Editorial note (5 October 2026, batch 101). Since batch 101 the report has seven Parts. Part VI ([section 71](#)) adds Report 83 of the session bundle, a third derivation of Part I’s main theorem, written on the same day as Parts I and II and independently of both; on Part II’s precedent it prints only what Report 83 adds: exact Bonferroni enclosures of the A189281 probability at $n = 100, 200$ and 1000 , the threshold bracket stated as a theorem, an inverse in the exact digamma coordinate, and four literature items. Part VII ([section 77](#)) adds Report 232, which proves the joint all-orders expansion for two reciprocal forbidden gaps, with selected components that may contain cycles, and the second corrections of the camel and knight counts A189358 and A137774. Their labels carry the prefixes `spf:fd:` and `spf:cg:`. Dated notes in [sections 9](#) and [10](#), in Remark 7.4, in the questions of Parts I–IV, in Part IV’s front matter and in [section B](#) record what has changed. The conjectured recurrences of A189281 and A110128 are still not proved.

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Part I

Unconditional all-orders asymptotics for two OEIS permutation sequences

1 The target and the precise contribution

For a uniformly random $\pi \in S_n$, define, for fixed $r, s \geq 1$,

$$X_{r,s,n}^{(1)} = \sum_{i=1}^{n-r} \mathbf{1}\{\pi(i+r) - \pi(i) = s\}, \quad (1.1)$$

$$X_{r,s,n}^{(2)} = \sum_{i=1}^{n-r} \mathbf{1}\{|\pi(i+r) - \pi(i)| = s\}. \quad (1.2)$$

An empty sum is zero. Write

$$A_{r,s}^{(\theta)}(n) = n! \mathbb{P}(X_{r,s,n}^{(\theta)} = 0), \quad \theta \in \{1, 2\}. \quad (1.3)$$

In this notation,

$$A189281(n) = A_{2,2}^{(1)}(n), \quad A110128(n) = A_{2,2}^{(2)}(n). \quad (1.4)$$

These identifications and the sequence definitions are recorded in the OEIS entries [1, 2].

1.1 What the sources do, and do not, already establish

The A189281 entry lists an order-eight, degree-eleven recurrence as conjectural, and describes its displayed terms through n^{-10} as consequences of that recurrence. It also records independent dynamic-programming verification of sequence values through $n = 300$ [1]. The recurrence-guessing work of Kauers and Koutschan is an important source for this problem [5]. For A110128, the entry lists a guessed order-24, degree-64 operator and attributes its higher asymptotic terms to that operator [2].

The common leading limits are not new. Tauraso proved, in the diagonal absolute case, the first correction

$$\frac{A_{r,r}^{(2)}(n)}{n!} = e^{-2} \left(1 + \frac{4(r-1)}{n} + O(n^{-2}) \right) \quad (1.5)$$

and gave exact counting formulas [4]. Spahn and Zeilberger developed an exact matching-of-tilings formula for both offset families, explained its computational use, and discussed Matsuo's alternative approach [3]. We use their combinatorial framework, with a self-contained proof, rather than claiming the underlying inclusion-exclusion formula as new.

There is also a relevant recent development. Padhi's August 2026 preprint reports holonomicity for every fixed r, s , while explicitly recording only a partial result toward the specific A189281 operator [6]. Our arguments neither assume that preprint's holonomicity theorem nor prove a particular recurrence. A general existence theorem for a recurrence and a proof of a specific guessed operator are distinct questions. [Added 1 October 2026, batch 73O2: the intake confirmed that the arXiv record of [6] exists under the title given in the bibliography, but did not check its content. Nothing below depends on it.]

1.2 The gap addressed here

We supply an unconditional route from the counting problem to every coefficient and a controlled remainder. The important point is not merely to fit more terms. One must justify replacing

an n -dependent inclusion-exclusion sum by an infinite coefficient calculation, including the contributions of arbitrarily many isolated selected edges. The exact stabilization theorem and a uniform factorial-moment bound supply that justification.

The resulting contribution has four connected parts. First, the coefficients attached to the two conjectural recurrences become theorems about the original permutation counts. Second, the same proof gives every fixed pair of offsets and the full violation distribution. Third, it explains why congruence classes of n do not enter any algebraic-order coefficient. Fourth, it gives an inverse expansion with a precise interpretation for a discrete sequence.

The ProveIt repository was consulted as the project context. Its Fabius asymptotic audit carefully separates a printed approximation, a necessary correction, and an all-orders theorem [9]. Here the analogous separation is between finite data, formal coefficient extraction, a uniform asymptotic theorem, and an unproved stronger closed form. No declaration from the repository is a mathematical dependency of the present proof, and no new repository build is reported.

1.3 A preview of the main theorem

Put

$$G_{r,s,n}^{(\theta)}(u) = \mathbb{E}(1+u)^{X_{r,s,n}^{(\theta)}}. \quad (1.6)$$

The use of $1+u$, rather than u , makes the coefficients factorial moments.

Theorem 1.1 (All-orders polynomial correction theorem). *For fixed positive integers r, s , $\theta \in \{1, 2\}$, every integer $M \geq 0$, and every $0 \leq U < \infty$, there are polynomials $B_J^{(\theta)}(u; r, s) \in \mathbb{Q}[u]$ such that, uniformly for $|u| \leq U$,*

$$G_{r,s,n}^{(\theta)}(u) = e^{\theta u} \sum_{J=0}^M \frac{B_J^{(\theta)}(u; r, s)}{n^J} + O_{M,r,s,U}(n^{-M-1}). \quad (1.7)$$

They are given by the finite formula (5.11). They satisfy

$$B_0 = 1, \quad \deg B_J \leq 2J, \quad B_J(0) = 0 \quad (J \geq 1), \quad (1.8)$$

$$B_J(u; r, s) = B_J(u; s, r), \quad (2J)!B_J(u; r, s) \in \mathbb{Z}[u], \quad (1.9)$$

$$B_1^{(\theta)}(u; r, s) = \theta(1-r-s)u + \theta(1-\theta)u^2. \quad (1.10)$$

In particular, for $c_J^{(\theta)}(r, s) = B_J^{(\theta)}(-1; r, s)$,

$$\frac{A_{r,s}^{(\theta)}(n)}{n!} = e^{-\theta} \left(\sum_{J=0}^M \frac{c_J^{(\theta)}(r, s)}{n^J} + O(n^{-M-1}) \right), \quad c_0 = 1, \quad c_1 = \theta(r+s-\theta). \quad (1.11)$$

The proof occupies sections 2 to 6. The arithmetic assertions are proved in section 7.

2 Selected violations and pairs of path tilings

2.1 Why paths are the right objects

The directed graph on $\{1, \dots, n\}$ with edges $i \rightarrow i+r$ is a disjoint union of r directed paths, when $n \geq r$. Their vertex counts are $\lfloor n/r \rfloor$ and $\lceil n/r \rceil$. The value graph with edges $j \rightarrow j+s$ is similarly a union of s paths.

More generally, consider any directed path forest on the position vertices and any directed path forest on the value vertices, both having n vertices. The oriented statistic counts position edges mapped to directed value edges by a uniform bijection. The absolute statistic counts position edges mapped to value edges in either direction. The proofs below apply to this more general model as well.

Selecting a subset of the position edges gives a collection of nontrivial path components and singleton vertices. For the oriented statistic, each selected component must map in its prescribed order to a value-path interval. For the absolute statistic there are exactly two choices of direction per nontrivial component. A reversal of direction *within* one component is impossible: at the first change of sign it would repeat the preceding value. Thus the orientation weight is 2^c , not 2^k , where c is the number of components and k the number of selected edges.

Definition 2.1 (Component profile). A profile is a finitely supported sequence $\mathbf{b} = (b_2, b_3, \dots)$ of nonnegative integers. Define

$$\begin{aligned} c &= \sum_{\ell \geq 2} b_\ell, & k &= \sum_{\ell \geq 2} (\ell - 1)b_\ell, \\ v &= \sum_{\ell \geq 2} \ell b_\ell = k + c, & d &= k - c = \sum_{\ell \geq 2} (\ell - 2)b_\ell, \end{aligned} \quad (2.1)$$

and $\mathbf{b}! = \prod_{\ell \geq 2} b_\ell!$. The number d is the *defect*: dimers have defect zero, and each extra edge in a component adds one unit of defect.

For a position forest $\mathcal{P}$, let $C_{\mathcal{P}}(\mathbf{b})$ be its number of tilings with b_ℓ components of size $\ell \geq 2$, and $n - v$ singletons. A tiling is simply a partition into consecutive intervals along each path; unused edges may lie between neighboring intervals.

Proposition 2.2 (Exact profile identity). *For position and value forests $\mathcal{P}, \mathcal{V}$ on n vertices,*

$$\mathbb{E}(1 + u)^X = \sum_{\mathbf{b}: v \leq n} \theta^c u^k C_{\mathcal{P}}(\mathbf{b}) C_{\mathcal{V}}(\mathbf{b}) \frac{(n - v)! \mathbf{b}!}{n!}. \quad (2.2)$$

Infeasible profiles have zero tiling count.

Proof. For each permutation, expand $(1 + u)^X$ as a sum over its selected violation sets. Such a set determines its input tiling and its image tiling, with the same component profile. Match the b_ℓ input components of each length to the b_ℓ output components in $b_\ell!$ ways. Choose a direction for each nontrivial component when $\theta = 2$, and match the $n - v$ singletons arbitrarily. This gives exactly $\theta^c (n - v)! \mathbf{b}!$ permutations for each pair of tilings, counted with the selected set distinguished. Division by $n!$ proves the identity.

Additional violations outside the selected set are allowed. This is essential: the identity counts marked subsets of violations, not permutations having exactly that selected set. $\square$

At $u = -1$, (2.2) is the matching-of-tilings inclusion-exclusion formula of Spahn and Zeilberger [3]. Its extension to general u makes the later distributional statements possible.

3 Exact stabilization on long path forests

Write $x^{\underline{j}} = x(x - 1) \cdots (x - j + 1)$ and $x^{\bar{j}} = x(x + 1) \cdots (x + j - 1)$, with both empty products equal to one. For a profile $\mathbf{b}$, put

$$E_h(\mathbf{b}) = [z^h] \prod_{\ell \geq 2} (1 + (\ell - 1)z)^{b_\ell}. \quad (3.1)$$

Define an integer polynomial

$$F_r(n; \mathbf{b}) = \sum_{h=0}^c (-1)^h (r - 1)^{\bar{h}} E_h(\mathbf{b}) (n - k - h)^{\underline{c-h}}. \quad (3.2)$$

Its degree in n is c and its leading coefficient is one. For $r = 1$, it reduces to $(n - k)^{\underline{c}}$.

Theorem 3.1 (Stable tiling polynomial). *Let $\mathcal{P}$ be a union of r directed paths with vertex counts $L_1, \dots, L_r$ and total n . If each $L_i \geq k$, then*

$$C_{\mathcal{P}}(\mathbf{b}) = \frac{F_r(n; \mathbf{b})}{\mathbf{b}!}. \quad (3.3)$$

Thus, in this range, the answer depends on $n, r, \mathbf{b}$, but not on the individual path lengths.

3.1 A formal coefficient identity

The following residue calculation is included to make the stabilization mechanism explicit. Only finitely many variables are needed for any specified coefficient.

Let $z_2, z_3, \dots$ be formal variables and let W , with zero constant term, solve

$$W = \sum_{\ell \geq 2} z_{\ell} (1 + W)^{-(\ell-1)}. \quad (3.4)$$

Define

$$\mathcal{D} = 1 + \sum_{\ell \geq 2} (\ell - 1) z_{\ell} (1 + W)^{-\ell}. \quad (3.5)$$

For every integer N and every profile,

$$[\mathbf{z}^{\mathbf{b}}] \frac{(1 + W)^N}{\mathcal{D}} = \frac{(N - k)^c}{\mathbf{b}!}. \quad (3.6)$$

To prove this, replace every z_{ℓ} by tz_{ℓ} and put $\phi(w) = \sum z_{\ell} (1 + w)^{-(\ell-1)}$. Then $W = t\phi(W)$ and $\mathcal{D} = 1 - t\phi'(W)$. Work first over the fraction field in finitely many z_{ℓ} , so that $\phi(0)$ is invertible; the resulting identity is polynomial coefficientwise. The change of formal residue variable $t = w/\phi(w)$ gives

$$[t^c] \frac{f(W)}{1 - t\phi'(W)} = [w^c] f(w) \phi(w)^c. \quad (3.7)$$

Indeed, the Jacobian factor cancels $1 - t\phi'(W)$. The coefficient of $\mathbf{z}^{\mathbf{b}}$ in $\phi(w)^c$ is $c!(1 + w)^{-k}/\mathbf{b}!$. Taking $f(w) = (1 + w)^N$ proves (3.6).

3.2 Proof of the stabilization theorem

For a single path of length L , a profile with $v \leq L$ has $L - k$ tiles in all. Ordering those tiles gives

$$C_L(\mathbf{b}) = \frac{(L - k)!}{(L - v)! \mathbf{b}!} = \frac{(L - k)^c}{\mathbf{b}!}. \quad (3.8)$$

If $k \leq L < v$, the falling factorial is zero, again giving the correct count. Consequently, through weighted edge degree $k \leq L$, the actual tiling series agrees with the formal surrogate $(1 + W)^L/\mathcal{D}$.

For r paths, all allocations of a total profile of edge degree k to the paths have local edge degree at most k . Under $L_i \geq k$, multiplying the surrogates is therefore legitimate at the required coefficient. Their product is

$$\frac{(1 + W)^n}{\mathcal{D}^r}. \quad (3.9)$$

Expand the additional factor $\mathcal{D}^{-(r-1)}$. A selected subprofile $\mathbf{a} \leq \mathbf{b}$, with $|\mathbf{a}| = h$, contributes

$$\frac{(-1)^h (r - 1)^{\bar{h}}}{\mathbf{a}!} \prod_{\ell \geq 2} (\ell - 1)^{a_{\ell}} \mathbf{z}^{\mathbf{a}} \frac{(1 + W)^{n - \sum \ell a_{\ell}}}{\mathcal{D}}. \quad (3.10)$$

Applying (3.6) to the remaining profile gives

$$\frac{(-1)^h (r-1)^{\bar{h}} \prod (\ell-1)^{a_\ell}}{\mathbf{a}!(\mathbf{b}-\mathbf{a})!} (n-k-h)^{\underline{c-h}}. \quad (3.11)$$

Summing over subprofiles of size h replaces the product of binomial coefficients by $E_h(\mathbf{b})/\mathbf{b}!$. This is exactly (3.2)–(3.3), proving theorem 3.1. $\square$

Corollary 3.2 (Stable factorial moments). *If the position forest has r paths, the value forest has s paths, every path has at least k vertices, and $2k \leq n$, then*

$$\mu_k := \mathbb{E} \binom{X}{k} = \sum_{\mathbf{b}: \sum (\ell-1)b_\ell = k} \frac{\theta^c F_r(n; \mathbf{b}) F_s(n; \mathbf{b})}{\mathbf{b}! n^v}. \quad (3.12)$$

For the ordinary offset model, this is exact whenever

$$k \leq \min \left(\left\lfloor \frac{n}{r} \right\rfloor, \left\lfloor \frac{n}{s} \right\rfloor, \left\lfloor \frac{n}{2} \right\rfloor \right). \quad (3.13)$$

Proof. Use theorem 3.1 on both tiling counts in (2.2) and extract u^k . The condition $2k \leq n$ ensures $v = k + c \leq n$ for every profile in the displayed sum. $\square$

As a first check, a single dimer has $F_r = n - r$, so, for $n \geq \max(r, s, 2)$,

$$\mathbb{E} X = \frac{\theta(n-r)(n-s)}{n(n-1)}. \quad (3.14)$$

The same formula follows directly by counting the available input and output edges.

4 Uniform bounds and beyond-all-orders universality

An exact formula for fixed moments is not, by itself, enough to justify an asymptotic expansion of a probability. A bound uniform in the moment order is needed.

Lemma 4.1 (Uniform factorial-moment bound). *For either statistic and arbitrary path forests on $n \geq 1$ vertices,*

$$0 \leq \mu_k \leq \frac{(\theta e^2)^k}{k!} \quad (k \geq 0). \quad (4.1)$$

Consequently, for $|u| \leq U$ and $a = \theta e^2 U$,

$$\left| G(u) - \sum_{k=0}^K \mu_k u^k \right| \leq e^a \frac{a^{K+1}}{(K+1)!}. \quad (4.2)$$

Proof. Fix a selected set of k position edges, with c components and $v = k + c$ vertices. There are at most $\theta^c n^c$ ways to assign starts and directions to its components in the value forest, even before excluding overlaps. Hence the probability that all the selected edges are violations is at most $\theta^c n^c / n^v$.

For $0 \leq v \leq n$, the geometric mean of the largest v factors in $n!$ is at least the geometric mean of all n factors. Also $n! \geq (n/e)^n$. Thus $n^v \geq (n/e)^v$, and the last probability is at most $\theta^k e^{2k} n^{-k}$, since $c \leq k$ and $v \leq 2k$. There are at most $n^k/k!$ possible selected edge sets. Summing proves (4.1). Summing its exponential-series tail proves (4.2). For $k = 0$ the assertion is immediate. $\square$

Theorem 4.2 (Quantitative independence of long path lengths). *Consider two position-value forest pairs, each on n vertices, with the same numbers r and s of paths. Suppose every path in both pairs has at least K vertices and $2K \leq n$. Let X and $\tilde{X}$ be the corresponding statistics for the same θ . For every $R > 1$, with $a = \theta e^2(R + 1)$,*

$$d_{\text{TV}}(\mathcal{L}(X), \mathcal{L}(\tilde{X})) \leq \frac{R}{R-1} e^a \frac{a^{K+1}}{(K+1)!}. \quad (4.3)$$

If all path lengths are at least ηn for a fixed $\eta > 0$, the right-hand side can be chosen to be $\exp(-\eta' n \log n + O(n))$ for every fixed $0 < \eta' < \min(\eta, 1/2)$.

Proof. By Corollary 3.2, the two sets of factorial moments agree through order K . On the circle $|t| = R$, use $u = t - 1$ in (4.2); the two ordinary probability generating functions differ by at most $2e^a a^{K+1}/(K+1)!$. Cauchy's coefficient inequality bounds the difference of the probabilities at j by this quantity times R^{-j} . Sum over $j \geq 0$ and divide by two to obtain (4.3). Finally, take $K = \lfloor \eta' n \rfloor$ and use $\log(K!) = K \log K - K + O(\log K)$. $\square$

This statement is stronger than agreement of a few asymptotic coefficients. It concerns the entire probability law. It also clarifies a limitation: the bound does not identify the leading exponentially small difference between two particular forest geometries. That finer sector is left open.

5 A finite formula for every asymptotic coefficient

5.1 Normalization and the defect filtration

Put $x = 1/n$ and define

$$f_{r,\mathbf{b}}(x) = \sum_{h=0}^c (-1)^h (r-1)^{\bar{h}} E_h(\mathbf{b}) x^h \prod_{a=0}^{c-h-1} (1 - (k+h+a)x), \quad (5.1)$$

$$D_v(x) = \prod_{a=0}^{v-1} (1 - ax), \quad (5.2)$$

$$Q_{\mathbf{b}}(x) = \frac{f_{r,\mathbf{b}}(x) f_{s,\mathbf{b}}(x)}{D_v(x)}. \quad (5.3)$$

Then $f_{r,\mathbf{b}}(1/n) = n^{-c} F_r(n; \mathbf{b})$, and a stabilized profile contributes

$$\frac{\theta^c u^k}{\mathbf{b}!} n^{-d} Q_{\mathbf{b}}(1/n) \quad (5.4)$$

to the generating function. This displays why defect $d > J$ cannot contribute to the coefficient of n^{-J} .

Separate the dimers from the other components:

$$m = b_2, \quad \beta = (b_3, b_4, \dots). \quad (5.5)$$

For the latter profile write

$$C = \sum_{\ell \geq 3} \beta_\ell, \quad K = \sum_{\ell \geq 3} (\ell - 1) \beta_\ell, \quad d = K - C, \quad \beta! = \prod_{\ell \geq 3} \beta_\ell!. \quad (5.6)$$

Thus $c = m + C$, $k = m + K$, $v = 2m + K + C$. The profiles β of a given defect d are in bijection with ordinary integer partitions of d : a component of length ℓ contributes a part $\ell - 2$.

Lemma 5.1 (Polynomial dependence on the number of dimers). *For fixed β, j, r, s ,*

$$P_{\beta,j}(m) := [x^j]Q_{(m,\beta)}(x) \quad (5.7)$$

is a polynomial in m over $\mathbb{Q}$ of degree at most $2j$. For positive integer r, s , it takes integer values at every $m \geq 0$.

Proof. The coefficient $E_h(m, \beta)$ is a polynomial in m of degree at most h , since the dimer contribution is $(1+z)^m$. For a product of consecutive linear factors

$$\prod_{a=0}^{m+C-h-1} (1 - (m + K + h + a)x), \quad (5.8)$$

the coefficient of x^q is polynomial in m of degree at most $2q$. One way to see the bound is to take its formal logarithm. The coefficient of x^i is a constant multiple of a sum of i th powers over a consecutive interval with affine endpoints, and therefore has degree at most $i + 1$. Exponentiating shows degree at most $2q$, because a product of power sums of total x -degree q has degree at most $q + q$. The same argument applies to $D_v(x)^{-1}$.

In a term carrying the additional factor $E_h x^h$, the degree at coefficient x^j is at most $h + 2(j - h) \leq 2j$. Products preserve the total bound. For the few small nonnegative m for which $c < h$, $E_h = 0$; thus the polynomial extension gives the same zero contribution. Equivalently one can define the consecutive power sums by their polynomial extensions before multiplying by E_h .

Finally, for nonnegative m , the numerator polynomials in (5.3) have integer coefficients and $D_v^{-1} \in \mathbb{Z}[[x]]$. Hence every value in (5.7) is an integer. $\square$

5.2 Poisson summation by finite differences

Let $\Delta P(m) = P(m+1) - P(m)$. For a polynomial of degree at most q ,

$$P(m) = \sum_{t=0}^q \frac{\Delta^t P(0)}{t!} m^t. \quad (5.9)$$

Therefore the elementary identity

$$\sum_{m \geq 0} \frac{z^m}{m!} P(m) = e^z \sum_{t=0}^q \frac{\Delta^t P(0)}{t!} z^t \quad (5.10)$$

turns an infinite dimer sum into a finite polynomial. It follows either by summing each falling factorial, or by differentiating e^z exactly t times.

Definition 5.2 (Coefficient formula). For $J \geq 0$, define

$$B_J^{(\theta)}(u; r, s) = \sum_{d=0}^J \sum_{\substack{\beta \geq 0 \\ \sum_{\ell \geq 3} (\ell-2)\beta_\ell = d}} \frac{\theta^C u^K}{\beta!} \sum_{t=0}^{2(J-d)} \frac{\Delta^t P_{\beta, J-d}(0)}{t!} (\theta u)^t. \quad (5.11)$$

All sums on the right are finite.

Only evaluations at $m = 0, 1, \dots, 2(J-d)$ are required. Their sufficiency is a proved degree bound, not a polynomial guess. No term $A(n)$ and no recurrence for $A(n)$ is needed to compute B_J .

6 Proof of the uniform all-orders expansion

We now justify the asymptotic interpretation of (5.11). This is the analytic step that a purely formal expansion of the profile formula would leave unresolved.

Lemma 6.1 (Uniform analytic bound for a profile). *For every profile of edge degree k , put $\rho_k = 1/[8(k+1)^2]$. There is a constant $C_{r,s}$, independent of the profile and k , such that*

$$\sup_{|x| \leq \rho_k} |Q_{\mathbf{b}}(x)| \leq C_{r,s}. \quad (6.1)$$

One may take $C_{r,s} = e^{2^{r+s}}$. In particular,

$$|[x^j]Q_{\mathbf{b}}(x)| \leq C_{r,s}[8(k+1)^2]^j, \quad (6.2)$$

and, if $n \geq 16(k+1)^2$,

$$\left| Q_{\mathbf{b}}(1/n) - \sum_{j=0}^q [x^j]Q_{\mathbf{b}}(x)n^{-j} \right| \leq 2C_{r,s} \left(\frac{8(k+1)^2}{n} \right)^{q+1}. \quad (6.3)$$

Proof. The elementary symmetric function bound $E_h(\mathbf{b}) \leq k^h/h!$ follows by expanding the h th power of the sum of the component edge lengths. Every integer appearing in a numerator factor is at most $2k$, and there are at most k such factors. Thus, on the stated disk,

$$|f_{r,\mathbf{b}}(x)| \leq e^{2k^2\rho_k} \sum_{h \geq 0} \frac{(r-1)^{\bar{h}}}{h!} (k\rho_k)^h = e^{2k^2\rho_k} (1 - k\rho_k)^{-(r-1)}. \quad (6.4)$$

For $r = 1$, the last sum is simply one. Also $v \leq 2k$ and $a\rho_k < 1/2$ for $0 \leq a < v$, so

$$|D_v(x)^{-1}| \leq \exp \left(2\rho_k \sum_{a=0}^{v-1} a \right) \leq e^{4k^2\rho_k}. \quad (6.5)$$

Combining these bounds gives (6.1) with the stated generous constant. The denominator has no zero on the disk. Cauchy's coefficient bound then proves (6.2), and its geometric tail at $1/n \leq \rho_k/2$ proves (6.3). $\square$

Proof of theorem 1.1: analytic assertion. Fix M, r, s, U , and for sufficiently large n choose

$$T_n = \left\lfloor \frac{(2M+4) \log n}{\log \log n} \right\rfloor. \quad (6.6)$$

The factorial-moment tail (4.2) with $K = T_n$ is $n^{-(2M+4)+o(1)}$, hence is $O(n^{-M-1})$. For fixed offsets, every position and value path has more than T_n vertices eventually. Also $2T_n \leq n$ and $16(T_n+1)^2 \leq n$. Thus all retained moments are given exactly by (3.12), and Lemma 6.1 applies to every retained profile.

High defect. The sum of absolute profile weights, marked by their defect, has the convergent generating function

$$\sum_{\mathbf{b}} \frac{\theta^c U^k}{\mathbf{b}!} z^d = \exp \left(\sum_{q \geq 1} \theta U^q z^{q-1} \right) = \exp \left(\frac{\theta U}{1 - Uz} \right). \quad (6.7)$$

For $U = 0$ the assertion is immediate; otherwise use a disk strictly inside $|z| < 1/U$. Its Taylor tail of degree M at $z = 1/n$ is $O(n^{-M-1})$. Together with (6.1), this bounds all retained profiles of defect $d \geq M+1$ by $O(n^{-M-1})$.

Low defect. For each $d \leq M$, there are only finitely many β . Expand $Q_{(m,\beta)}$ through order $M - d$. By (6.3), after multiplying by n^{-d} the error is bounded by n^{-M-1} times a fixed polynomial in m times $(\theta U)^m/m!$, with a constant depending only on the fixed parameters and β . The sum over m is finite uniformly in n . This is why no extra power of $\log n$ appears in the remainder.

We may now extend the retained dimer sums to $m = \infty$. Indeed, for each retained coefficient, (6.2) bounds the omitted summands by a polynomial in m times $(\theta U)^m/m!$. Since $k = m + K > T_n$ implies $m > T_n - K$, their tails are again $n^{-(2M+4)+o(1)}$. Summing each polynomial in m by (5.10) gives exactly (5.11). The total error is $O(n^{-M-1})$, uniformly on $|u| \leq U$. This proves (1.7) and, on setting $u = -1$, (1.11). $\square$

The proof also works for any fixed numbers r, s of paths whose shortest length is bounded below by a positive multiple of n . It does not assert uniformity for arbitrarily growing offsets or for forests with a bounded short path. [Added 1 October 2026, batch 74: Part II proves the expansion uniformly over forest shapes, on the disk $|1 + u| \leq 1$, whenever the shortest path has at least $K_n \asymp \log n / \log \log n$ vertices (theorem 16.6 and corollary 16.7); remark 16.8 notes that the proof above already gives this on every disk $|u| \leq U$ with T_n in place of K_n .]

7 Arithmetic, first corrections, and the full distribution

7.1 Degree and denominator control

Proposition 7.1 (Arithmetic of the correction polynomials). *The structural assertions (1.9) hold.*

Proof. For a defect- d profile, $C \leq d$ and $K \leq 2d$. A term in (5.11) has u -degree at most $K + 2(J - d) \leq 2J$. The formula is symmetric in r, s because $Q_{\mathbf{b}}$ is. For $J = 0$ it gives $B_0 = 1$. When $J > 0$, any nonempty β contributes positive u -degree. For the empty profile the constant term is $P_{\emptyset, J}(0) = 0$, since the completely empty $\mathbf{b}$ has $Q_{\mathbf{b}} = 1$. Thus $B_J(0) = 0$.

By Lemma 5.1, every $\Delta^t P(0)$ is an integer. A summand's only possible denominator is $t!\beta!$. This divides $(t + C)!$, by the multinomial coefficient, and

$$t + C \leq 2(J - d) + d \leq 2J. \quad (7.1)$$

Hence $t!\beta!$ divides $(2J)!$. Multiplying the finite sum by $(2J)!$ therefore gives a polynomial over $\mathbb{Z}$. $\square$

This denominator bound is unconditional. It is deliberately weaker than claiming all the oriented coefficients themselves are integers; that stronger phenomenon is discussed in section 11.

7.2 The first two corrections

For the dimer-only profile,

$$P_{\emptyset, 1}(m) = -m^2 + (2 - r - s)m = -m^2 + (1 - r - s)m. \quad (7.2)$$

The only defect-one profile is a single component of length three, with $C = 1$, $K = 2$, and $P_{\beta, 0} = 1$. Applying (5.10) immediately gives (1.10). For $\theta = 1$, a calculation of the four profile types of defect at most two gives

$$B_2^{(1)}(u; r, s) = (r - 1)(s - 1)u + \binom{r + s - 1}{2}u^2. \quad (7.3)$$

For completeness, the required dimer-only polynomial is

$$P_{\emptyset,2}(m) = \frac{1}{2}m^4 + (r + s - 3)m^3 + \frac{r^2 + 2rs + s^2 - 5r - 5s + 7}{2}m^2 + \frac{-r^2 + r - s^2 + s}{2}m. \quad (7.4)$$

For one length-three component, $P_{\beta,1}(m) = -m^2 - (r + s)m + 3 - 2r - 2s$. The defect-two profiles are one length-four component or two length-three components; both have $P_{\beta,0} = 1$. These data and (5.11) verify (7.3) directly.

In the absolute case a useful general expression is

$$B_2^{(2)}(u; r, s) = 2(r - 1)(s - 1)u + 2((r + s)^2 - r - s - 1)u^2 + 2(2r + 2s - 1)u^3 + 2u^4. \quad (7.5)$$

Thus (1.5) is recovered as a special case of the first correction, while the theorem supplies every later order.

Remark 7.2 (Offsets (1,1): two classical sequences). [Added 1 October 2026, batch 73O2.] The manuscript does not discuss the smallest offsets, which give two classical sequences. The intake ran the shipped coefficient generator `code/path_forests.py` on a copy with `-r 1 -s 1 -order 10`.

- *Oriented case, no successions.* Here $A_{1,1}^{(1)}(n)$ counts permutations of $[n]$ with no $\pi(i + 1) = \pi(i) + 1$, which is $A000255(n - 1) = D_n + D_{n-1}$ with D_n the derangement numbers [10]. The formula gives $c_J^{(1)}(1, 1) = 1, 1, 0, 0, \dots, 0$ for $J = 0, \dots, 10$. This is consistent with $(D_n + D_{n-1})/n! = e^{-1}(1 + 1/n) + O(1/n!)$, so every later coefficient is zero as well.
- *Absolute case, Hertzprung's problem.* Here $A_{1,1}^{(2)}(n)$ is A002464, the number of permutations with no rising or falling succession [11]. The formula gives

$$c_J^{(2)}(1, 1) = 1, 0, -2, -\frac{10}{3}, -6, -\frac{154}{15}, -\frac{88}{9}, \frac{5336}{105}, \frac{1612}{3}, \frac{2098234}{567}, \frac{36500686}{1575} \quad (J = 0, \dots, 10).$$

These are, term for term, the Abramson–Moser expansion displayed in the OEIS entry (computed by Kotesovec, 2011, extended 2020), through n^{-10} .

Both cases agree with $c_1 = \theta(r + s - \theta)$ of (1.11). As an independent check, the intake computed both sequences exactly to $n = 800$, from the recurrences printed in the two OEIS entries (themselves checked against brute-force enumeration for $n \leq 8$). It found the scaled residuals $n^{M+1}(e^\theta A(n)/n! - \sum_{j \leq M} c_j n^{-j})$ for $M = 4, 8$ approaching c_{M+1} at $n = 200, 400, 800$. These finite checks are evidence; the proof is [theorem 1.1](#). The theorem covers the pair (1,1), and the manuscript's brute-force tests already include it (it is one of the six offset pairs of [section 10](#)). Only the comparison with the two OEIS entries is new here.

7.3 Poisson corrections for all probabilities

The zero-violation event is only one coefficient of the probability generating function. Write $B_J(u) = \sum_h b_{J,h} u^h$ and define

$$H_h(q; \theta) = \sum_{i=0}^{\min(h,q)} \binom{h}{i} (-1)^{h-i} \frac{q^i}{\theta^i}. \quad (7.6)$$

These are a normalization of the Charlier correction polynomials. Direct coefficient extraction gives

$$[t^q](t - 1)^h e^{\theta(t-1)} = e^{-\theta} \frac{\theta^q}{q!} H_h(q; \theta). \quad (7.7)$$

Consequently define the signed approximation

$$p_M(q) = e^{-\theta} \frac{\theta^q}{q!} \sum_{J=0}^M \frac{1}{n^J} \sum_h b_{J,h} H_h(q; \theta). \quad (7.8)$$

Corollary 7.3 (An all-orders approximation of the entire law). *For fixed M, r, s, θ ,*

$$\sum_{q \geq 0} \left| \mathbb{P}(X_{r,s,n}^{(\theta)} = q) - p_M(q) \right| = O(n^{-M-1}). \quad (7.9)$$

In particular the law converges in total variation to $\text{Poisson}(\theta)$, with error $O(n^{-1})$.

Proof. Apply [theorem 1.1](#) on $|t| = R > 1$, using $u = t - 1$. Cauchy's inequality gives an error $O(n^{-M-1}R^{-q})$ in the coefficient of t^q . Summing the geometric series proves [\(7.9\)](#). $\square$

For $M \geq 1$, p_M need not be nonnegative at every q , so it is a signed asymptotic approximation, not a new probability model. It nevertheless has total mass one, because $B_0(0) = 1$ and $B_J(0) = 0$ for $J > 0$.

Remark 7.4 (A clique-model analogue). [Added 1 October 2026, batch 73O2, reciprocal note.] The companion report [a330266-balanced-smirnov-poisson](#) of this collection (batch 73, manuscript 57) studies the number $X_{n,k}$ of equal-rank adjacencies in a uniform shuffle of kn cards, k in each of n ranks. There the value relation is n cliques K_k , not a fixed number of long paths, so the hypotheses of this report do not apply, and the limit is $\text{Poisson}(\lambda)$ with $\lambda = k - 1$. Its first-order local law ([Corollary 8.3](#) there, label `bsw:cor:local`),

$$\mathbb{P}(X_{n,k} = j) = e^{-\lambda} \frac{\lambda^j}{j!} \left[1 - \frac{(\lambda - j)^2 - j}{2kn} + O_{k,j}(n^{-2}) \right],$$

is the analogue of the case $M = 1$ of [\(7.8\)–\(7.9\)](#). That report proves it at first order only, and the analytic step behind its expansion is sketched; its [Remark 6.3](#) names a uniform factorial-moment bound of the type of [Lemma 4.1](#) as the device that would make it rigorous. [Added 5 October 2026, batch 101: both statements of the last sentence are out of date, since 2 October 2026. That report's Part II proves the analytic step for fixed k , with the bound $\mathbb{E}X_{n,k}^M \leq (k - 1)^M$ for its own model ([Theorem II.3.1](#) there, label `bsw:rp:thm:bound`; see its [Remark II.6.3](#), `bsw:rp:rem:settles`), and its Part III (batch 101) proves the step a second time and gives the local law to every fixed order, $\mathbb{P}(X_{n,k} = j) = \sum_{s \leq J} N^{-s} [u^j] (e^{\lambda(u-1)} \mathcal{P}_s(u - 1)) + O_{k,j,J}(N^{-J-1})$ with $N = kn$, $\lambda = k - 1$ and that report's polynomials $\mathcal{P}_s$ ([Remark III.7.5](#) there, `bsw:ut:rem:local`). It proves its bound directly; neither report uses the other's lemma.]

8 The two OEIS expansions, unconditionally

8.1 A189281: oriented distance-two avoidance

The first correction polynomials for $r = s = 2$, $\theta = 1$ are

$$B_0 = 1, \quad B_1 = -3u, \quad B_2 = u + 3u^2, \quad (8.1)$$

$$B_3 = u + u^2 - u^3, \quad B_4 = u - u^2 - 2u^3, \quad B_5 = u + u^2 - 3u^3, \quad (8.2)$$

$$B_6 = u + 23u^2 - 4u^3, \quad B_7 = u + 121u^2 + 19u^3. \quad (8.3)$$

Further examples are

$$B_8 = u + 479u^2 + 378u^3 + 24u^4, \quad (8.4)$$

$$B_9 = u + 1681u^2 + 3737u^3 + 672u^4 + 24u^5, \quad (8.5)$$

$$B_{10} = u + 5543u^2 + 28936u^3 + 11088u^4 + 984u^5 + 24u^6. \quad (8.6)$$

Setting $u = -1$ in these polynomials proves

$$\frac{A189281(n)}{n!} = e^{-1} \left(1 + \frac{3}{n} + \frac{2}{n^2} + \frac{1}{n^3} + \frac{3}{n^5} + \frac{26}{n^6} + \frac{101}{n^7} + \frac{124}{n^8} - \frac{1409}{n^9} - \frac{13266}{n^{10}} + O(n^{-11}) \right). \quad (8.7)$$

The coefficient of n^{-4} is exactly zero. The terms through n^{-10} agree with the recurrence-dependent list in the OEIS [1]; the proof above does not assume that recurrence.

8.2 A110128: absolute distance-two avoidance

For $r = s = 2$, $\theta = 2$, the first polynomials are

$$B_1 = -6u - 2u^2, \quad (8.8)$$

$$B_2 = 2u + 22u^2 + 14u^3 + 2u^4, \quad (8.9)$$

$$B_3 = 2u + 6u^2 - 60u^3 - 56u^4 - 16u^5 - \frac{4}{3}u^6, \quad (8.10)$$

$$B_4 = 2u - 18u^2 - 122u^3 + 74u^4 + 154u^5 + 70u^6 + 12u^7 + \frac{2}{3}u^8. \quad (8.11)$$

It follows that

$$\begin{aligned} \frac{A110128(n)}{n!} = e^{-2} & \left(1 + \frac{4}{n} + \frac{8}{n^2} + \frac{68}{3n^3} + \frac{242}{3n^4} + \frac{1692}{5n^5} \right. \\ & + \frac{72802}{45n^6} + \frac{2725708}{315n^7} + \frac{16083826}{315n^8} \\ & \left. + \frac{186091480}{567n^9} + \frac{32213578294}{14175n^{10}} + O(n^{-11}) \right). \end{aligned} \quad (8.12)$$

This recovers the higher list attributed to the guessed operator in the OEIS [2], again by a direct argument from the permutation model. The fractions here also show why an integer-coefficient assertion cannot be transferred indiscriminately from the oriented case to the absolute case.

8.3 Additional coefficients and how to use them

J	$c_J^{(1)}(2, 2)$	$c_J^{(2)}(2, 2)$
11	-59103	2620164054016/155925
12	-9448	61699558213516/467775
13	2525459	6658719787519636/6081075
14	26632378	136086447719175784/14189175
15	152270749	5133067332126139204/58046625
16	106221948	546416517735316663856/638512875

Table 1: Exact evaluations of the finite coefficient formula beyond the order-ten lists. The accompanying JSON includes every polynomial B_J through order sixteen, not only its value at $u = -1$.

For any fixed M , adding terms through $c_M n^{-M}$ gives a remainder $O(n^{-M-1})$ inside the parentheses. This is a Poincaré statement with M fixed as $n \rightarrow \infty$. It does not establish convergence of the infinite series, justify setting M proportional to n , or identify an optimal truncation rule. The exact finite coefficients in table 1 are supplied with their generating code rather than inferred from a numerical fit.

9 Lambert- W inversion with a discrete interpretation

Remark 9.1 (Relation to the repository’s inversion apparatus). [Added 1 October 2026, batch 7302.] This section re-derives, for its own forward map, machinery that the canonical volume *Transseries and Inversion* of the ProveIt repository already contains [12], and the manuscript does not cite it. The forward map $e^{-\theta}\Gamma(n+1)C(1/n)$ is the gamma carrier of that volume’s chapter on the increasing inverse of Gamma (theorem “All-orders expansion of Γ_+^{-1} ”, label `p6:thm:gamma`), perturbed by a power series in $1/n$, which is the situation of its theorem “Perturbed inversion” (`p0:thm:perturbed-inversion`). The volume’s chapter “The subfactorial” (`p8:sec:top`) treats the closest analogue, the derangement numbers, by citing that carrier. Normalizations differ: here $z = L/W_0(L/e)$ with $L = \log y$, and the constant $\frac{1}{2}\log(2\pi)$ sits in κ_θ ; the volume uses $R = \log y - \frac{1}{2}\log(2\pi)$ and $T = R/W_0(R/e)$. The discussion of integer thresholds at the end of the section is an instance of the volume’s staircase theorem (`p0:thm:staircase`). [Theorems 9.2](#) and [9.3](#) are therefore printed as a self-contained second route for this family of sequences. No novelty is claimed for the inversion method; what is specific to this report is the forward input, the coefficients c_j of [theorem 1.1](#).

9.1 Logarithmic coefficients

For either sequence, or any fixed r, s, θ , suppress the parameters and write

$$\frac{A(n)}{n!} \sim e^{-\theta}C(1/n), \quad C(x) = 1 + \sum_{j \geq 1} c_j x^j. \quad (9.1)$$

The right side is a formal asymptotic series. Define $\log C(x) = \sum_{j \geq 1} \ell_j x^j$. An explicit coefficient recursion is

$$\ell_j = c_j - \frac{1}{j} \sum_{k=1}^{j-1} k \ell_k c_{j-k}. \quad (9.2)$$

Combining the proved probability expansion with Stirling’s expansion, in its usual remainder-controlled form [7], yields

$$\log A(n) = n \log n - n + \frac{1}{2} \log n + \kappa_\theta + \sum_{j=1}^M a_j n^{-j} + O(n^{-M-1}), \quad (9.3)$$

where

$$\kappa_\theta = \frac{1}{2} \log(2\pi) - \theta, \quad a_j = \ell_j + \frac{B_{j+1}}{j(j+1)}. \quad (9.4)$$

Here B_m are Bernoulli numbers; the additional term vanishes for even j . For A189281,

$$\ell_1 = 3, \quad \ell_2 = -\frac{5}{2}, \quad \ell_3 = 4, \quad \ell_4 = -\frac{29}{4}, \quad a_1 = \frac{37}{12}, \quad a_2 = -\frac{5}{2}, \quad a_3 = \frac{1439}{360}. \quad (9.5)$$

For A110128, $a_1 = 49/12$.

9.2 The first explicit inverse terms

Given $y = A(n)$, let $L = \log y$ and

$$z = \frac{L}{W_0(L/e)}, \quad w = \log z. \quad (9.6)$$

The principal Lambert function is characterized by $W_0(x)e^{W_0(x)} = x$ for $x > 0$ [8]. Thus $z(\log z - 1) = L$ exactly.

Theorem 9.2 (First inverse corrections). *Along the sequence values $y = A(n)$ as $n \rightarrow \infty$,*

$$n = z - \frac{1}{2} - \frac{\kappa_\theta}{w} - \frac{1}{z} \left(\frac{c_1 - 1/24}{w} + \frac{\kappa_\theta^2}{2w^3} \right) + O\left(\frac{1}{z^2 w}\right). \quad (9.7)$$

For A189281 the coefficient $c_1 - 1/24$ is $71/24$; for A110128 it is $95/24$.

Proof. Set $n = z + D$. The derivative of $x \log x - x$ at z is w , and its second derivative is $1/z$. The constant-order terms of (9.3), after subtracting L , give $wD + \frac{1}{2}w + \kappa_\theta = 0$, hence $D = d_0 = -1/2 - \kappa_\theta/w$ at that order. The coefficient of z^{-1} is

$$wd_1 + \frac{d_0^2 + d_0}{2} + a_1. \quad (9.8)$$

Since $a_1 = c_1 + 1/12$, this gives

$$d_1 = -\frac{c_1 - 1/24}{w} - \frac{\kappa_\theta^2}{2w^3}. \quad (9.9)$$

The residual in the logarithmic equation is $O(z^{-2})$; division by a derivative asymptotic to w gives the asserted index error. A precise all-orders version of this argument follows next. $\square$

9.3 An all-orders inverse recursion

Put $t = 1/z$ and seek $D(t, w) = \sum_{j \geq 0} d_j(w)t^j$. Define

$$\begin{aligned} \mathcal{H}(t, D) = & \frac{(1 + tD) \log(1 + tD) - tD}{t} + \frac{1}{2} \log(1 + tD) \\ & + \sum_{j \geq 1} a_j t^j (1 + tD)^{-j}. \end{aligned} \quad (9.10)$$

The quotient by t is a formal series divisible by t , so no singular constant term occurs. The inverse equation is

$$wD + \frac{1}{2}w + \kappa_\theta + \mathcal{H}(t, D) = 0. \quad (9.11)$$

It gives the triangular recursion

$$d_0(w) = -\frac{1}{2} - \kappa_\theta/w, \quad (9.12)$$

$$d_j(w) = -\frac{1}{w} [t^j] \mathcal{H} \left(t, \sum_{q=0}^{j-1} d_q(w)t^q \right) \quad (j \geq 1). \quad (9.13)$$

Only $a_1, \dots, a_j$ enter the j th step.

Theorem 9.3 (All-orders index inversion). *For every fixed $J \geq 0$, the functions in (9.13) are rational in w , with coefficients in $\mathbb{Q}[\kappa_\theta]$ for the rational forward coefficients considered here, and*

$$n = z + \sum_{j=0}^J \frac{d_j(w)}{z^j} + O\left(\frac{1}{z^{J+1}w}\right), \quad y = A(n). \quad (9.14)$$

Moreover, $d_0 = O(1)$ and $d_j = O(1/w)$ for $j \geq 1$ as $w \rightarrow \infty$.

Proof. The recursion first proves the algebraic form of the coefficients by induction. Since $\mathcal{H}$ has no t^0 term, its coefficient of t^j depends only on the earlier d_q , and stays bounded when those earlier coefficients are bounded. Dividing by w proves the stated size bounds.

Fix J and retain sufficiently many terms of (9.3). Substitution of the truncated D cancels every term through z^{-J} , leaving a residual $O(z^{-J-1})$. All relevant derivatives of the finite logarithmic model are controlled on a bounded neighborhood of $z + d_0$, and its first derivative is $\log z + O(1/z)$. The actual index is in such a neighborhood: the leading term first gives $n/z \rightarrow 1$, and then the mean value theorem applied to $n \log n - n - z \log z + z = O(\log n)$ gives $n - z = O(1)$. A further application of the mean value theorem divides the residual by a quantity asymptotic to w , proving (9.14). $\square$

9.4 What is being inverted

A discrete sequence has no distinguished analytic interpolation. The preceding theorems make no such choice: their statements are evaluated at the actual values $y = A(n)$. Alternatively, define an explicit smooth approximation

$$\mathcal{A}_M(x) = e^{-\theta} \Gamma(x+1) \sum_{j=0}^M c_j x^{-j}. \quad (9.15)$$

It is positive and strictly increasing for all sufficiently large x , because the finite correction tends to one and the logarithmic derivative is $\log x + O(1/x)$. Its inverse satisfies

$$\mathcal{A}_M^{-1}(A(n)) - n = O\left(\frac{n^{-M-1}}{\log n}\right). \quad (9.16)$$

This is a well-defined model inverse, with its discrepancy from the sequence controlled.

For an integer threshold $N(y) = \min\{n : A(n) \geq y\}$, an asymptotic estimate alone does not justify unconditional nearest-integer rounding near a jump. Certified bounds for $A(n)$ can resolve the remaining adjacent candidates. The Bonferroni method in the next section provides one such tool. [Added 1 October 2026, batch 73O2: the general form of this remark, exact rounding $N(y) = \lceil \nu(y) \rceil$ for an increasing interpolation ν and the separation condition under which an approximate inverse decides it, is the staircase theorem `p0:thm:staircase` of [12]. Its separation step is formalized in Lean as `Fabius.staircase_separation` in `Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`. That formalization is context only; it does not verify any statement of this report.] [Added 5 October 2026, batch 101: Part VI states this bracket for $\mathcal{A}_M$ as a theorem, with a complete proof (theorem 73.1), and expands the model inverse in the exact digamma coordinate rather than in z, w (proposition 73.2); Part VII proves the same bracket, with explicit envelope roots, for the camel and knight counts (theorem 84.1). Neither makes the constant or the onset explicit.]

10 Reproducible computations and exact certificates

10.1 Coefficient generation

The core implementation, `code/path_forests.py`, uses only the Python standard library. It forms the numerator factors and reciprocal denominator in (5.3) as truncated integer polynomials; evaluates $P_{\beta,j}(m)$ at the proved sufficient set of integers; takes forward differences; and assembles (5.11) using exact rational arithmetic. The work depends on the requested asymptotic order, not on a large sequence index n .

At order J , profiles are indexed by partitions of $d \leq J$, and each profile uses at most $2(J-d)+1$ dimer evaluations. No integer-sequence terms are used as fitting data. The archive contains all B_J and c_J through order sixteen in `data/coefficients_order16.json`.

10.2 Independent finite checks

The included validation program completed the following checks.

Check	Exact equalities/cases
Full tiling enumeration against both OEIS lists, $n = 0, \dots, 21$	44
Brute-force full distributions for $n = 1, \dots, 8$, six offset pairs, and both statistics	96
Stable factorial moments against those brute-force distributions	246
Stable tiling polynomial on specified unequal path-length forests	86
Additional finite-difference tests beyond the interpolation degree	375
Factorial denominator checks on polynomial coefficients through order 16	396

The six offset pairs were $(1, 1), (1, 2), (2, 2), (2, 3), (3, 2), (3, 3)$. Full tiling enumeration is independent of the stable-polynomial implementation, and brute permutation enumeration is independent of both. These checks detect finite implementation and bookkeeping errors; none replaces the proofs for arbitrary n or arbitrary expansion order.

10.3 Bonferroni probability intervals

When (3.13) holds through $2q + 1$, exact factorial moments give

$$S_{2q+1} \leq \mathbb{P}(X = 0) \leq S_{2q}, \quad S_j = \sum_{k=0}^j (-1)^k \mu_k. \quad (10.1)$$

For completeness, this follows pointwise from $\sum_{k=0}^j (-1)^k \binom{x}{k} = (-1)^j \binom{x-1}{j}$ when $x \geq 1$, while both partial sums equal one for $x = 0$. Thus the lower and upper bounds are rigorous rational numbers, even though the full count has not been computed.

The certificate program uses S_{31} and S_{30} at $n = 80, 160, 320$. It brackets e^{-1} between the odd and even Taylor sums of degrees 101 and 102, and squares that interval for e^{-2} . All arithmetic in the resulting error intervals is rational. The quantity displayed below is

$$E_M(n) = n^{M+1} \left(e^\theta \frac{A_{2,2}^{(\theta)}(n)}{n!} - \sum_{j=0}^M c_j n^{-j} \right). \quad (10.2)$$

By the theorem its limit is c_{M+1} .

θ	n	Certified interval for $E_4(n)$	Certified interval for $E_8(n)$
1	80	[3.3409847, 3.3409848]	[-1584.0080, -1584.0079]
1	160	[3.1664733, 3.1664734]	[-1494.2195, -1494.2194]
1	320	[3.0822399, 3.0822400]	[-1451.0335, -1451.0334]
2	80	[360.08331, 360.08332]	[359523.99, 359524.00]
2	160	[348.86238, 348.86239]	[343097.56, 343097.57]
2	320	[343.54178, 343.54179]	[335473.66, 335473.68]

Table 2: Outward-rounded exact intervals. The limits are $3, -1409$ for $\theta = 1$, and $1692/5, 186091480/567$ for $\theta = 2$. The intervals certify these finite residuals; their convergence is a consequence of the proof, not an inference from three samples.

The archive includes the unrounded rational endpoints and additional order-ten residual certificates. With the same moment cutoff, those order-ten intervals become wider at larger n , especially for $\theta = 2$: the factor n^{11} magnifies the fixed Bonferroni width. This is an honest

precision limit of that finite computation; increasing the cutoff addresses it without using a guessed recurrence. [Added 5 October 2026, batch 101: Part VI adds exact enclosures of the oriented avoidance probability at $n = 100, 200$ and 1000 , with the cutoffs S_{23} , S_{27} and S_{29} , from Report 83's program (table 27); at $n = 1000$ the order-ten residual interval contains $c_{11} + c_{12}/n$ computed from table 1.]

10.4 Inverse diagnostics and reproducibility

Using exact enumerated sequence values at $n = 20$, the Lambert backbone z has index errors about 0.52023 for A189281 and 0.20749 for A110128. Including both displayed correction levels in (9.7) reduces these to approximately -0.00066084 and 0.00089419 , respectively. These are numerical diagnostics, not certified rounding guarantees.

The programs were run with Python 3.13.5. The core coefficient, enumeration, and rational-certificate programs need no third-party package. The optional rational-identity exploration uses SymPy 1.14.0; the inverse diagnostic uses mpmath 1.3.0. The commands and expected outputs are recorded in the archive's README. No Lean code was executed or supplied as an asserted formal proof. [Added 1 October 2026, batch 73O2: the delivered README is not shipped. The report's README.md replaces it and gives rerun commands for the shipped layout; see section B.]

11 A sharper conjecture suggested by the exact computations

There is a striking simplification in the oriented $(2, 2)$ case that is not needed by any preceding theorem. Define the universal rational moments

$$\mu_k^*(n) = \sum_{\mathbf{b}: \sum (\ell-1)b_\ell = k} \frac{F_2(n; \mathbf{b})^2}{\mathbf{b}! n^{\mathbf{v}}}, \quad (11.1)$$

as rational functions in an indeterminate n . They agree with the actual moments whenever the stability conditions hold. Remove the leading Poisson factor by setting

$$R_h(n) = \sum_{k=0}^h \frac{(-1)^{h-k}}{(h-k)!} \mu_k^*(n). \quad (11.2)$$

Exact finite algebra gives

$$R_0 = 1, \quad (11.3)$$

$$R_1 = -\frac{3n-4}{n^2}, \quad (11.4)$$

$$R_2 = \frac{3n^2 - 17n + 26}{n^4}, \quad (11.5)$$

$$R_3 = -\frac{(n-6)(n^2 - 7n + 16)}{n^6}. \quad (11.6)$$

Conjecture 11.1 (Rational collapse). For every $h \geq 4$,

$$R_h(n) = \frac{24(n-h+1)^{h-4}}{n^{2h}} = \frac{24}{n^{h-1}(n-2h+5)^5}. \quad (11.7)$$

This has been checked as an exact identity of rational functions for each $h = 4, \dots, 16$. The check clears the common denominator n^{2h} and compares polynomials; it is stronger than testing finitely many numerical values of n , but it is still only a finite check in h . No induction or creative-telescoping certificate for all h has been obtained here. [Added 3 October 2026, batch

85: the conjecture is now proved for every $h \geq 4$, together with the ratio recurrence (11.8), in Part III (theorem 26.1). It is the case $r = s = 2$ of a closed form of R_h for every pair of directed offsets (theorem 25.1), proved by a single-sum formula for the stable moments and an alternating Vandermonde identity rather than by telescoping. The heading “Conjecture” and the label are kept so that references to this statement stay valid.]

The proposed formula would imply the recurrence

$$\frac{R_{h+1}(n)}{R_h(n)} = \frac{(n-2h+5)(n-2h+4)}{(n-h+1)(n-2h)(n-2h-1)} \quad (h \geq 4). \quad (11.8)$$

Proving (11.8) directly from (11.2) is therefore a sharply specified next target.

Proposition 11.2 (What the rational collapse would imply). *If Conjecture 11.1 holds, then $B_J^{(1)}(u; 2, 2) \in \mathbb{Z}[u]$ for every J , and therefore all $c_J^{(1)}(2, 2)$ are integers. For $J \geq 4$ it would also give*

$$\deg B_J^{(1)}(u; 2, 2) \leq \max(3, J-4). \quad (11.9)$$

Proof. The first four rational functions $R_h(1/x)$ have integer Taylor coefficients. Under the conjecture, each later one is $24x^{h+4}$ times a product of factors $(1-ax)^{-1}$ with nonnegative integers a , after cancellation. It too has integer coefficients and begins at degree $h+4$. At any fixed power x^J , only finitely many of these R_h can contribute.

By (11.2), the coefficient of u^h in the Poisson-normalized stable moment series is $R_h(n)$. Comparing with the already proved coefficient formula gives $B_J^{(1)}(u; 2, 2) = \sum_h [x^J] R_h(1/x) u^h$. The integrality and degree claims follow. $\square$

This conditional proposition must not be confused with the unconditional factorial denominator theorem. The conjecture packages the observed integrality into a more structured rational-function identity, making it more approachable than a bare statement about an infinite list of integers. [Added 3 October 2026, batch 85: since theorem 26.1 proves the hypothesis, the proposition now holds unconditionally (corollary 26.3). Part III also proves, by a second route, that $B_J^{(1)}(u; r, s) \in \mathbb{Z}[u]$ for every pair of directed offsets, and that the degree bound is attained: $\deg B_J^{(1)}(u; 2, 2) = J-4$, with leading coefficient 24, for every $J \geq 8$ (theorem 28.1). For the absolute case $\theta = 2$, which Part III does not cover, Proposition 7.1 remains the denominator statement.]

12 Further research questions and proposed next steps

1. Prove or refute the rational collapse. The most concrete next step is an identity in two symbolic parameters rather than a fit to sequence values. Clear denominators in (11.2), organize profiles by their component count, and seek a telescoping proof of (11.8). A combinatorial cancellation proof would be particularly valuable: it should explain both the factor 24 and the loss of four powers of n beyond the moment index. A failure at some later h would also be informative, because the first discrepancy would identify a missing component interaction. The all-orders integrality issue for the oriented family remains separate until such a proof exists. [Added 1 October 2026, batch 74: for the oriented one-path subfamily, $r = 1$ or $s = 1$, all-orders integrality is proved in Part II (corollary 18.2); the case $(2, 2)$ of A189281 remains open.] [Added 3 October 2026, batch 85: the rational collapse is proved in Part III (theorem 26.1), by a single-sum formula for the stable moments (theorem 24.1) and Vandermonde convolutions, not by telescoping. The factor 24 is $4!$, from a component-index shift of four, and the four lost powers of n are the vanishing binomial coefficients with $j < r + s$ (section 25). All-orders integrality now holds for every oriented pair (r, s) , A189281 included (theorem 28.1). A combinatorial cancellation proof, also asked for here, is still open (section 34, question 9).]

2. Connect the asymptotic theorem to the specific guessed operators. A proved asymptotic expansion does not establish the A189281 or A110128 recurrence. Conversely, a rigorous operator certificate could supply a second, independent route to large-order coefficient behavior. One useful project is to translate the stable moment framework into a finite-dimensional holonomic representation with boundary terms retained, then compare its annihilator with the listed operators. All boundary terms must be proved, not discarded because their numerical influence is small. [Added 4 October 2026, batch 97: Part V proves that the order-13 operator listed in the OEIS, in Mark van Hoeij’s factorization VUT , annihilates the analytic completion $A_*(n) = \Gamma(n+1)C_*(n)/e$ of the oriented $(2,2)$ expansion, and that A_* satisfies a fourth-order equation (theorem 63.1). It does not prove the recurrence for the counts, which differ from A_* by a nonzero flat term (theorem 64.1); Part V’s question 2 restates what remains.]

3. Identify the first geometry-dependent exponential sector. Theorem 4.2 shows that long path-length differences are invisible to every algebraic order. It does not give their leading amplitude, sign, or possible dependence on n modulo the offsets. The first profiles that can see a short boundary have edge degree comparable to the shortest path. A saddle-point analysis of that moving profile boundary could reveal the first exponentially small sector, and would turn the present algebraic expansion into part of a genuine transseries description. [Added 4 October 2026, batch 97: for A189281 Part V proves that a beyond-all-orders correction is present: the normalized count $e A189281(n)/n!$ differs from its positive-ray Borel completion $C_*(n)$ by a flat term δ_n that is nonzero at least once in every four consecutive indices (theorem 64.1). Its amplitude and sign, and the geometry dependence asked for here, remain open (Part V, question 1; question 50.3). For a short path of fixed length, Part IV shows that the shape already enters at an algebraic order (theorem 45.1).]

4. Determine large-order growth and justified optimal truncation. The coefficients through order sixteen already show that sign and magnitude can change markedly. The present proof allows constants to depend on M . It would be useful to prove explicit growth bounds in M , decide whether a Gevrey-type estimate is sharp, and derive a valid truncation rule for a given n . The rational-collapse conjecture, if proved, would make the oriented case a tractable testing ground, but the absolute case needs its own analysis. [Added 3 October 2026, batch 85: for the oriented case Part III proves that the Borel transform of the coefficients is entire (theorem 30.1), so that $|c_J| \leq \gamma_A A^{J-1}(J-1)!$ for every $A > 0$. Sharp growth, a summation method and an optimal truncation rule remain open (section 34, questions 3 and 4).] [Added 4 October 2026, batch 97: for the oriented case at $(2,2)$ Part V determines the growth, $\limsup_m (\log m) |c_{m+1}/m!|^{1/m} = 2$, so that no Gevrey bound of order $s < 1$ holds, and shows that (c_J) is not P-recursive (theorems 60.2 and 60.4). It names a summation method, the positive-ray Borel–Laplace sum C_* , valid exactly for $\Re n > 1$ and not sectorial (theorem 58.2 and corollary 59.4), and proves a truncation rule with error $\exp(-\frac{1}{2}n \log n + O(n))$ for C_* , not for the counts (theorem 61.1). Optimal truncation, a lower bound, the counts themselves, other offsets and the absolute case remain open (Part V, questions 4, 5, 7 and 8).]

5. Allow offsets or path counts to grow with n . The current proof uses fixed r, s , both in the shortest-path condition and in the analytic bound on Q_b . When r or s grows, the shortest path may meet the moment cutoff, and the number of path boundaries changes the first corrections. Determine the largest growth range in which a useful uniform expansion persists, and the transition law when path lengths become comparable to $\log n / \log \log n$. This question requires new uniform estimates; it is not an immediate corollary of the fixed-offset theorem. [Added 1 October 2026, batch 74: for fixed r, s , Part II shows that the expansion survives whenever the shortest path is long compared with $\log n / \log \log n$ (corollary 16.7); the

transition at that scale remains open (section 22, question 2).] [Added 4 October 2026, batch 97: the first half is answered in Part IV: for every pair of path forests, with no hypothesis on the path lengths, and in particular for all offsets $1 \leq r, s \leq n$, the expansion holds uniformly in the actual subpath densities, the order n^{-M} depending only on subpaths of at most $M + 1$ edges (theorem 38.1 and corollary 46.1); new corrections appear at reciprocal-integer ratios (theorem 46.2). The transition law at path lengths comparable to $\log n / \log \log n$ remains open (question 50.3).]

6. Extend from paths to several forbidden offsets and to cycles. For several allowed differences, selected components can branch or contain cycles. Their embeddings are no longer determined by one starting value and a single orientation. A useful extension would classify component types by an appropriate defect and prove an analogue of the uniform moment bound and finite coefficient formula. Cyclic boundaries are another natural test: short cycles can contribute at a finite algebraic order, whereas long cycles may only change a beyond-all-orders sector. [Added 5 October 2026, batch 101: answered for reciprocal pairs of gaps by Part VII. For the absolute gaps (r, s) and (s, r) , $r < s$, marked separately, the selected components are two-coloured graphs that may branch and contain cycles; the compatible ones are triangle-free and 2-degenerate, are classified by their deficit $v - 2c$ (not the defect of section 5), and have at most twice as many core edges as their deficit (lemma 80.1). A uniform moment bound (lemma 80.3) and a finite coefficient construction (theorem 81.1, (82.7)) give the joint all-orders expansion (theorem 78.1). Short cycles do contribute at a finite algebraic order: the alternating square at n^{-2} (remark 80.2). Other gap sets, the directed model, more colours and cyclic boundaries remain open (section 87).]

7. Formalize the proof and build a certified coefficient library. The finite profile identity, the stable tiling polynomial, and the integer coefficient arithmetic are natural first formalization targets. The analytic layer then needs the uniform factorial-moment tail, Cauchy bounds, and finite asymptotic truncation. A robust implementation should expose the exact stability hypotheses instead of silently applying F_r outside its range. The rational Bonferroni certificates are suitable artifacts for an independent checker. This is a proposed formalization route, not a report of completed Lean verification.

8. Combine asymptotic inversion with exact threshold certification. For very large y , the Lambert backbone and a few correction terms should reduce the inverse problem to a small number of candidate indices. Use exact Bonferroni intervals to compare each candidate count with y , adaptively increasing the moment cutoff only when necessary. The interesting algorithmic question is a proved cost bound that accounts for near-threshold inputs, where asymptotic rounding alone cannot decide the answer.

13 Conclusion

The main result is a direct all-orders theorem for a natural pair of permutation sequences whose higher displayed OEIS expansions were tied to guessed recurrences. Its proof does not solve those recurrence problems by assumption: it bypasses them. The stable path-tiling identity gives exact low moments, the factorial tail controls the remaining moments, and finite differences sum the unlimited dimer contributions. Together they turn a formal asymptotic calculation into a uniform theorem.

The same structure yields more than two lists of numbers: a full-law Poisson correction expansion, a quantitative explanation of boundary-length universality, an arithmetic denominator theorem, and a rigorous inverse construction. The stronger rational-collapse pattern is separated

from these proved results and posed as a specific next problem. That distinction is central to the usefulness of the report: the established theorems can be used now, while the remaining conjectures have explicit, testable targets.

Part II

Fixed-gap asymptotics: a second derivation and its own results

14 Provenance, scope and notation of Part II

[Added 1 October 2026, batch 74.] Part II was added in the write of batch 74. It is built from manuscript 03 of that batch and prints only what that manuscript adds to Part I. Where the two manuscripts prove the same statement, the statement is printed once, in Part I, and manuscript 03’s route is recorded in [section 15](#). Everything in Part II is written in Part I’s notation ([table 4](#)).

14.1 Provenance

Manuscript 03 of batch 74 is the archive `ProveIt_Fixed_Gap_Asymptotics.zip`, which arrived in commit `c664fc2f0` and was placed in this report by commit `b669cff87`. Its article is titled *All-Orders Asymptotics and Boundary Universality for Fixed-Gap Permutations* (25 pages, dated October 1, 2026); its title page reads “Prepared for Vladimir Reshetnikov” and its PDF metadata names “ChatGPT, research report prepared for Vladimir Reshetnikov” as author. Its pinned repository commit is `291fbe6bb4477e0cb2131f1e041b12a9abe363fd` (2026-10-01, 18:10; [\[14\]](#)).

It was written without sight of manuscript 60, the source of Part I, and proves Part I’s main theorems in independent text: its structure, notation, constants and several proof devices differ throughout, and the two texts share about 1.1 % of their word 8-grams, almost all of them bibliography entries and the digits of the coefficient list. Every number the two manuscripts share agrees exactly: the coefficients of A189281 and A110128 through order twelve, the general second correction in r, s for both orientations, the polynomials B_1, B_2, B_3 at $(r, s) = (2, 2)$, the logarithmic coefficients and the first inverse coefficients.

[Added 1 October 2026, batch 74: manuscript 03 states that “a repository code search for A189281 returned no match at the time of the audit”, and its delivered `02-fixed-gap-sources.md` repeats it. The pinned commit `291fbe6bb` already held the archive of manuscript 60, `docs/incoming/oeis_path_forests.zip` (arrival commit `aa43cc555`), which code search cannot read; that archive was placed an hour later as Part I of this report (commit `6e193dd4f`). Part II is an independent derivation of the same theorems plus the results listed below.]

Scope and non-claims of manuscript 03, kept. Its contribution is “the explicit stable-profile and finite-defect framework, the uniform error analysis, and the quantitative comparison of whole laws”. The leading Poisson behaviour, Tauraso’s first correction and the matching-of-tilings inclusion–exclusion of Spahn and Zeilberger are credited as prior work. The proofs are ordinary mathematics, not a Lean-certified development, and no publication priority is claimed. “All orders” means a remainder theorem at each fixed order, not a convergent or resurgent completion; no Borel summability, optimal truncation, Stokes constant or complete family of exponentially small sectors is asserted. The conjectured OEIS recurrences of A189281 and A110128 are not proved, and neither is all-orders integrality when both component counts

exceed one. Its numerical tables are not interval certificates, and its values of A189281 at $n = 80, 120$ are b-file inputs, not regenerated. It made no OEIS submission and modified no repository file.

14.2 What Part II adds, and how it compares with Part I

Topic	Part I (manuscript 60)	Part II (manuscript 03)
All-orders remainder	fixed offsets r, s ; uniform on every disk $ u \leq U$ (theorem 1.1)	only on $ 1 + u \leq 1$, i.e. $ u \leq 2$, but uniform over all forest shapes whose shortest path has at least $K_n \asymp \log n / \log \log n$ vertices (theorem 16.6 and corollary 16.7)
Coefficients	B_J through order 16, $(2J)!B_J \in \mathbb{Z}[u]$ (Proposition 7.1)	same values through order 12 (not new); B_J polynomial in r, s of total degree at most J (proposition 17.1); closed forms of $c_3^{(\theta)}(r, s)$ (proposition 17.2)
Integrality	for $(2, 2)$ oriented only conditional on Conjecture 11.1	<i>proved</i> for the oriented case with $r = 1$ or $s = 1$ (corollary 18.2); $(2, 2)$ untouched
Universality	total variation bound (theorem 4.2)	bounds for the whole generating function and the avoidance probability, a second total-variation bound, and an $n = 12$ example (theorem 16.3 and remark 16.4)
Whole law	ℓ^1 approximation of the law (Corollary 7.3)	explicit first-order local laws at $(2, 2)$ (19.2)–(19.3)
Inverse	all orders in the coordinates z, w (theorem 9.3)	explicit terms through N^{-3} in a coordinate absorbing κ_θ ; a_2, a_3 for A110128 (proposition 20.1)
Certificates	certified Bonferroni intervals (table 2)	uncertified decimals only (section 21)

Table 3: Part II against Part I. The remainder theorem of Part II is weaker in u and stronger in geometry; see also [remark 16.8](#). [Added 4 October 2026, batch 97: Part IV removes the length hypothesis, on every disk ([theorem 38.1](#)).]

14.3 Notation

Manuscript 03 and Part I use several letters for different objects, and manuscript 03 overloads D, w, T, A and B internally. [Table 4](#) fixes the convention: Part II uses Part I’s names for every shared object and new names $(K_n, D_{n,k}, \Lambda, N, \Delta, \omega_j, \varepsilon_M, \alpha)$ for the rest. No normalization is changed.

15 Manuscript 03’s second routes to Part I’s theorems

[Table 5](#) records, for each result of Part I that manuscript 03 proves again, the device its proof uses. None of these statements is reprinted. Where a device is also needed for a new result of Part II, it is stated there.

Notion	Part I and Part II	Manuscript 03
orientation weight (1 or 2)	θ	η (Part I's η is a length proportion)
length proportion	α (Part II); η in theorem 4.2	α
generating function $\mathbb{E}\binom{X}{k}$	$G(1+u) = \mathbb{E}(1+u)^X$ μ_k	$G(w)$, $w = 1+z$, $z = u$ S_k (Part I's S_j are Bonferroni sums)
profile, multiplicity factorial	$\mathbf{b} = (b_\ell)$, $\mathbf{b}!$	$\mathbf{m} = (m_j)$, $\mathbf{m}!$
tiling count	$C_{\mathcal{P}}(\mathbf{b})$	$T_{\mathcal{F}}(\mathbf{m})$
stable tiling polynomial	$F_r(n; \mathbf{b})$	$\mathbf{b}!^{-1} F_r$ is its $T_r(n; \mathbf{m})$
formal root	$1+W$, $\mathcal{D}^{-1}$	λ , A
dimers; long profile	m ; β , C , K	h ; $\mathbf{m}$, c_0 , k_0
normalized profile kernel	$Q_{\mathbf{b}}(x)$	$H_h(t)$ (<i>not</i> Part I's H_h)
Charlier polynomials	$H_h(q; \theta)$	$Q_m(j; \eta)$ (<i>not</i> Part I's $Q_{\mathbf{b}}$)
correction polynomials	$B_J^{(\theta)}(u; r, s)$	$C_J(r, s; \eta, z)$
avoidance coefficients	$c_J^{(\theta)}(r, s)$	$c_J^{(\eta)}(r, s)$ (same)
moment cutoff $(1 - 2k/n)^{-2k}$	T_n (Part I), K_n (Part II) $D_{n,k}$	K (Part I's K is an edge degree) $D_{n,k}$ (its D is also $\log N$)
logarithmic coefficients	a_j	d_j (Part I's d_j are inverse coefficients)
$\frac{1}{2} \log(2\pi) - \theta$	κ_θ	γ
inverse coordinates	Λ , N , $\Delta = \log N$ (Part II)	L , N , D
inverse coefficients	$\omega_j(\Delta)$	$u_j(D)$
unscaled truncation error	$\varepsilon_M(n)$	$E_M(n)$ (Part I's E_M is scaled by n^{M+1})

Table 4: Notation of manuscript 03 against Part I. Part II uses the middle column throughout.

Part I	Manuscript 03's route
Proposition 2.2	the same matching-of-tilings argument, stated for binomial moments μ_k , with the same orientation lemma (weight θ^c , not θ^k); adds the symmetry $A_{r,s} = A_{s,r}$ via $\pi \mapsto \pi^{-1}$
(3.6)	a formal residue identity for the root $\lambda = 1 + \sum_{\ell} z_{\ell} \lambda^{1-\ell}$, with $A = (1 + \sum_{\ell} (\ell-1) z_{\ell} \lambda^{-\ell})^{-1}$; here $\lambda = 1 + W$ and $A = \mathcal{D}^{-1}$
theorem 3.1	the identical polynomial, written $T_r(n; \mathbf{m}) = F_r(n; \mathbf{b})/\mathbf{b}!$
Lemma 4.1	$\mu_k \leq \theta^k D_{n,k}/k!$ for $1 \leq k < n/2$ (lemma 16.1): sharper for $k \ll n$, Part I's holds for every k
(4.2)	a Taylor remainder on the unit disk $ 1+u \leq 1$ (lemma 16.2), not an exponential-series tail
theorem 4.2	total variation through Parseval and Cauchy-Schwarz (theorem 16.3); the $\exp(-\delta n \log n + O(n))$ form is Part I's last assertion with η' renamed δ
Definition 5.2	the same finite formula; the dimer sum is evaluated by Touchard polynomials, $e^{-x} \sum_m x^m m^q / m! = T_q(x)$ with $T_{q+1} = x(T_q + T'_q)$, instead of finite differences, and the denominator by exponentials of power sums
Proposition 7.1	$B_0 = 1$, $\deg_u B_J \leq 2J$, $B_J(0) = 0$ and symmetry; adds the degree in r, s (proposition 17.1); Part I's denominator theorem is not in manuscript 03
(1.10), (7.3), (7.5)	B_1 by the same two-profile hand calculation; general B_2 and B_3 in symbolic r, s in <code>data/02-fixed-gap-general_pgf_first_three.txt</code> ; B_2 agrees with Part I identically in r, s, u
theorem 1.1	theorem 16.6: a defect-sensitive moment bound and a kernel bound on $ x \leq 1/(D(1+k)^2)$ in place of Lemma 6.1
(8.7), (8.12), table 1	the coefficients of both sequences through order 12, equal to Part I's (see below)
residue classes (section 1)	the same explanation by theorem 3.1
(3.14)	the same count; adds $\mathbb{E}X = \theta + \theta(1-r-s)/n + \theta(r-1)(s-1) \sum_{q \geq 2} n^{-q}$
(9.3), theorems 9.2 and 9.3	the same logarithmic series; the inverse in a coordinate absorbing κ_{θ} (proposition 20.1)
section 9, last subsection	the same smooth carrier $\mathcal{A}_M$ and threshold bracket

Table 5: Results of Part I proved again by manuscript 03, and the device of each second route.

Order twelve is not new. Manuscript 03 presents its coefficients through order twelve as an extension of the OEIS lists (“extend them through order twelve”) and proposes appending

$$-\frac{59103}{n^{11}} - \frac{9448}{n^{12}} \text{ to A189281, } \frac{2620164054016}{155925 n^{11}} + \frac{61699558213516}{467775 n^{12}} \text{ to A110128.}$$

[Added 1 October 2026, batch 74: Part I’s [table 1](#) already gives orders 11 to 16 of both sequences. Manuscript 03’s values at orders 0 to 12, computed independently by its own engine, agree with Part I’s exactly; they confirm Part I’s table and are not new.]

The inverse is not new as a method. As for Part I (Remark 9.1), manuscript 03’s inverse is an instance of the transseries volume’s apparatus [12]: its coordinates are those of `p6:thm:gamma` with $\log Y$ shifted by θ , its inverse expansion is a perturbed inversion around that gamma carrier (`p0:thm:perturbed-inversion`), and its threshold bracket is `p0:thm:staircase`. No novelty is claimed for the method.

16 Uniformity over forest shapes

Throughout this section $\mathcal{P}$ and $\mathcal{V}$ are arbitrary directed path forests on n vertices with r and s paths, as in [section 2](#), and

$$D_{n,k} = \left(1 - \frac{2k}{n}\right)^{-2k} \quad (1 \leq k < n/2). \quad (16.1)$$

Lemma 16.1 (A sharper binomial-moment bound; manuscript 03). *For either statistic and arbitrary path forests, $\mu_k \leq \theta^k D_{n,k}/k!$ for $1 \leq k < n/2$.*

Proof (manuscript 03). As in Lemma 4.1, a selected set of k position edges with c components has probability at most $\theta^c n^c / n^{k+c}$. Since $c \leq k$ and $k + c \leq 2k$, this is at most $\theta^k n^{-k} D_{n,k}$; there are at most $n^k/k!$ such sets. $\square$

Lemma 16.2 (Taylor remainder on the unit disk; manuscript 03). *For $|1 + u| \leq 1$ and every integer $K \geq 1$,*

$$\left| G(1 + u) - \sum_{k=0}^{K-1} \mu_k u^k \right| \leq |u|^K \mu_K. \quad (16.2)$$

Proof (manuscript 03). For an integer $x \geq K$, Taylor’s integral remainder for w^x about $w = 1$ along the segment from 1 to $w = 1 + u$, which stays in the closed unit disk, has modulus at most $\binom{x}{K} |u|^K$; for $x < K$ it vanishes. Take expectations. $\square$

Theorem 16.3 (Quantitative boundary universality; manuscript 03). *Let $(\mathcal{P}, \mathcal{V})$ and $(\tilde{\mathcal{P}}, \tilde{\mathcal{V}})$ be two pairs of forests on n vertices with r position and s value paths, every path of all four having at least K vertices, $1 \leq K < n/2$, and let $X, \tilde{X}$ be their statistics for the same θ . Then $\mu_k = \tilde{\mu}_k$ for $0 \leq k \leq K$, and*

$$\sup_{|w| \leq 1} |G(w) - \tilde{G}(w)| \leq \frac{2(2\theta)^K}{K!} D_{n,K}, \quad (16.3)$$

$$|\mathbb{P}(X = 0) - \mathbb{P}(\tilde{X} = 0)| \leq \frac{2\theta^K}{K!} D_{n,K}, \quad (16.4)$$

$$d_{\text{TV}}(\mathcal{L}(X), \mathcal{L}(\tilde{X})) \leq \sqrt{n+1} \frac{(2\theta)^K}{K!} D_{n,K}. \quad (16.5)$$

If every path has at least αn vertices, all three are at most $\exp\{-\delta n \log n + O_{\alpha, \delta, \theta}(n)\}$ for every fixed $0 < \delta < \min(\alpha, 1/2)$.

Proof (manuscript 03). Moments of order $k \leq K$ are stable in all four forests and agree by Corollary 3.2. Subtract the common Taylor polynomial of degree $K - 1$ and apply lemmas 16.1 and 16.2 to both remainders, with $|u| = |w - 1| \leq 2$ on the disk and $|u| = 1$ at $w = 0$. For the total variation, write $G - \tilde{G} = \sum_{j \leq n} q_j w^j$; Parseval on the unit circle gives $\sum |q_j|^2 \leq \sup_{|w|=1} |G - \tilde{G}|^2$, and Cauchy–Schwarz gives $\frac{1}{2} \sum |q_j| \leq \frac{1}{2} \sqrt{n+1} \sup |G - \tilde{G}|$. For the last assertion take $K = \lfloor \delta n \rfloor$: $\log D_{n,K} = O(n)$ and $\log K! = \delta n \log n + O(n)$. $\square$

The bounds (16.3)–(16.4) on the whole generating function and on the avoidance probability are new here; (16.5) and the $\exp(-\delta n \log n)$ form are a second route to theorem 4.2. Boundary placement can change the exact count, but, when all components are long, its normalized influence is smaller than every power of $1/n$; the theorem compares shape-dependent corrections and does not identify an exponentially small expansion for a single sequence.

Remark 16.4 (Agreement of moments is not equality of laws; manuscript 03). At $n = 12$ compare the position and value path lengths $(6, 6)/(6, 6)$ with $(5, 7)/(4, 8)$. Their moments agree through $k = 4$ and first differ at $k = 5$. The oriented avoidance counts are 222822578 and 222822556, the absolute ones 91310696 and 91284560, by manuscript 03’s independent exact tiling routine. “All algebraic orders agree” must not be read as “the finite distributions are identical”.

Lemma 16.5 (A defect-sensitive moment bound; manuscript 03). *Let $\mu_{k,c}$ be the part of μ_k from selected sets with exactly c components, so $d = k - c$. For $1 \leq c \leq k < n/2$,*

$$\mu_{k,c} \leq \frac{\theta^c}{c!} \binom{k-1}{c-1} n^{-d} D_{n,k} \leq \frac{\theta^k}{k!} \left(\frac{k^2}{n}\right)^d D_{n,k}. \quad (16.6)$$

Proof (manuscript 03). There are at most $\frac{n^c}{c!} \binom{k-1}{c-1}$ selected sets with c components and k edges: choose the c initial vertices, order them, and give them a composition of k into c positive edge counts (overlaps are overcounted). Each has probability at most $\theta^c n^c / n^{k+c} \leq \theta^c n^{-k} D_{n,k}$, because $n^{k+c} \geq n^{k+c} (1 - 2k/n)^{2k}$ as in lemma 16.1. Multiplying, and using $n^c \cdot n^{-k} = n^{-d}$, gives the first bound. For the second, $\binom{k-1}{c-1} = \binom{k-1}{d} \leq k^d$, $k!/c! \leq k^d$ and $\theta \geq 1$. $\square$

Theorem 16.6 (All orders, uniformly over forest shapes; manuscript 03). *Fix $r, s \geq 1$, $\theta \in \{1, 2\}$ and $M \geq 0$, and for large n put*

$$K_n = \left\lfloor \frac{(M+4) \log n}{\log \log n} \right\rfloor. \quad (16.7)$$

For every pair of path forests on n vertices with r and s paths, each path having at least K_n vertices,

$$G(1+u) = e^{\theta u} \sum_{J=0}^M \frac{B_J^{(\theta)}(u; r, s)}{n^J} + O_{M,r,s}(n^{-M-1}), \quad |1+u| \leq 1, \quad (16.8)$$

with B_J the polynomials of (5.11). The implied constant does not depend on the individual path lengths.

Proof (manuscript 03, condensed). Tail. By lemmas 16.1 and 16.2 with $|u| \leq 2$, truncating at $k < K_n$ costs at most $(2\theta)^{K_n} D_{n,K_n} / K_n!$. Since $K_n^2/n \rightarrow 0$, $D_{n,K_n} = \exp(O(K_n^2/n))$, and $\log K_n! = (M+4 + o(1)) \log n$; so the cost is $o(n^{-M-1})$. Every retained profile has $k < K_n$ edges and is stable in both forests (Corollary 3.2).

High defect. By lemma 16.5, with $k^2/n < 1/2$ for $k < K_n$, the retained profiles of defect $d \geq M+1$ contribute at most $O(n^{-M-1}) \sum_{k \geq 1} (2\theta)^k k^{2M+2} / k! = O_M(n^{-M-1})$, uniformly in u and in the path lengths.

Low defect. For each of the finitely many long profiles β of defect $d \leq M$, expand the normalized kernel $Q_{(m,\beta)}(1/n)$ of (5.3) through order $M-d$. Manuscript 03 proves that the

kernel is analytic and bounded by a constant depending only on r, s on $|x| \leq 1/(D_0(1+k)^2)$ for a suitable D_0 , which is the role of Lemma 6.1; Cauchy's estimate gives a remainder $O((1+k)^{2(M-d)+2}n^{-(M-d)-1})$, summable against the weights $(\theta|u|)^m/m!$ of the dimers.

Completion. Extending each retained dimer sum from $m + K < K_n$ to all $m \geq 0$ adds tails of order $\exp(-K_n \log K_n + O(K_n)) = o(n^{-M-1})$. The completed sums are the Poisson transforms (5.10) of the polynomials (5.7), and collecting the powers of $1/n$ gives (5.11). No step uses the path lengths beyond the stability of the retained profiles. $\square$

Corollary 16.7 (How long is long enough; manuscript 03). *For fixed r, s , the expansion (16.8) holds to every fixed order for every family of forests whose shortest path length $L_{\min}(n)$ satisfies*

$$\frac{L_{\min}(n) \log \log n}{\log n} \rightarrow \infty. \quad (16.9)$$

In particular the paths need not have lengths proportional to n .

Proof (manuscript 03). For each fixed M , (16.9) eventually gives $L_{\min}(n) \geq K_n$. $\square$

Remark 16.8 (Part I's proof gives the same uniformity on every disk). [Added 1 October 2026, batch 74; this observation is the write's, not either manuscript's.] The remark after the proof of theorem 1.1 claims the theorem for forests whose shortest path is a positive multiple of n . Its proof uses the path lengths only to know that every retained profile, of edge degree at most $T_n = \lfloor (2M+4) \log n / \log \log n \rfloor$, is stable: the moment tail (4.2), the kernel bound Lemma 6.1 and the defect majorant (6.7) hold for arbitrary forests. Read so, it gives theorem 1.1 uniformly on $|u| \leq U$ for all forests whose shortest path has at least T_n vertices, which also covers the condition (16.9). Manuscript 03 states and proves the shape-uniform form, on the unit disk only. [Added 4 October 2026, batch 97: superseded in generality by Part IV, whose theorem 38.1 holds on every disk for every pair of forests, with no length hypothesis, in local-density coordinates. At fixed path counts, Remark 45.4 shows that a shortest path of M vertices already suffices for Part I's coefficients at order M .]

The theorem gives a remainder at each fixed order; it does not assert that the series in $1/n$ converges, and its proof provides no small optimized numerical constant for a given finite n .

17 The correction polynomials as polynomials in the offsets

Proposition 17.1 (Degree in the offsets; manuscript 03). *For every J and θ there is a polynomial in $\mathbb{Q}[r, s, u]$ of total degree at most J in (r, s) whose values at positive integers r, s are $B_J^{(\theta)}(u; r, s)$; it is symmetric in r, s and has degree at most $2J$ in u .*

Proof (manuscript 03, in Part I's notation). In (5.3), the offset r enters $f_{r,b}(x)$ only through the rising factorials $(r-1)^{\overline{h}}$, of degree h in r , and each is accompanied by x^h ; so $[x^j]f_{r,b}$ has degree at most j in r , and likewise for s . Hence $P_{\beta,j} = [x^j]Q_{(m,\beta)}$ has total degree at most j in (r, s) , and (5.11) uses $P_{\beta,J-d}$ with $d \geq 0$. The other assertions are Proposition 7.1. $\square$

Proposition 17.2 (The third avoidance coefficient; manuscript 03). *For all positive integers r, s ,*

$$\begin{aligned} c_1^{(1)} &= r + s - 1, & c_1^{(2)} &= 2(r + s - 2), \\ c_2^{(1)} &= \frac{1}{2}(r^2 - r + s^2 - s), & c_2^{(2)} &= 2(r^2 + rs - 2r + s^2 - 2s), \\ c_3^{(1)} &= \frac{1}{6}(r + s - 1)(r^2 - 4rs + 4r + s^2 + 4s - 6), & & \\ c_3^{(2)} &= \frac{2}{3}(2r^3 + 3r^2 + 9rs - 17r + 2s^3 + 3s^2 - 17s + 10). & & \end{aligned} \quad (17.1)$$

The first four are Part I's (1.11), (7.3) and (7.5) at $u = -1$; the closed forms of $c_3^{(\theta)}(r, s)$ are new. Manuscript 03 obtains them by evaluating its finite formula symbolically in r, s (the full B_3 for both orientations is in `data/02-fixed-gap-general_pgf_first_three.txt`); this write checked both against Part I's generator at $(r, s) = (1, 1), (1, 3), (2, 3), (3, 3), (2, 5)$. At $r = s = d$, $c_1^{(2)} = 4(d - 1)$ is Tauraso's correction (1.5).

18 The oriented one-path family: proved integrality

Proposition 18.1 (Rational collapse for one path; manuscript 03). *Let the position forest be a single path ($r = 1$), the value forest have s paths, and $\theta = 1$. Then*

$$n! \mu_k = \binom{n-s}{k} (n-k)! \quad (0 \leq k \leq n-s), \quad (18.1)$$

so that $G(1+u) = \frac{1}{n^s} \sum_{k=0}^{n-s} \frac{u^k}{k!} (n-k)^s$, and, for $|u| \leq 2$ and a suitable constant C_s ,

$$G(1+u) = \frac{e^u}{n^s} \sum_{t \geq 0} \frac{\Delta^t p_n(0)}{t!} u^t + O(\exp\{-n \log n + C_s n\}), \quad p_n(m) = (n-m)^s. \quad (18.2)$$

For $s = 1$ the rational factor is $1 - u/n$; for $s = 2$ it is $1 - 2u/n + u^2/(n(n-1))$.

Proof (manuscript 03). Work on the value side. Any k of its $n-s$ edges contract the value set to $n-k$ blocks, and these blocks can be laid along the position path in $(n-k)!$ ways, with no further restriction; this is (18.1). Then use $(n-s)^k/n^k = (n-k)^s/n^s$. Completing the sum to all $k \geq 0$ adds only zero terms through $k = n$ and a factorial tail bounded by a degree- s polynomial times $2^k/k!$, divided by n^s ; the completed sum is e^u times the finite-difference Poisson transform (5.10) of p_n , a polynomial of degree s in m . $\square$

Corollary 18.2 (An integrality subfamily; manuscript 03). *In the oriented case, if $r = 1$ or $s = 1$, then $B_J^{(1)}(u; r, s) \in \mathbb{Z}[u]$ for every J , and every avoidance coefficient $c_J^{(1)}(r, s)$ is an integer.*

Proof (manuscript 03). The finite sum in (18.2) is $\sum_q a_q(n) T_q(u)$, where $p_n(m) = \sum_q a_q(n) m^q$ has integer coefficients and $T_q(u) = e^{-u} \sum_{m \geq 0} u^m m^q / m!$ are the Touchard polynomials, which have integer coefficients; so it is an integer polynomial in n and u of degree s in n . Dividing by $n^s = n^s \prod_{i < s} (1 - i/n)$ and expanding $\prod_{i < s} (1 - i/n)^{-1}$, whose power-series coefficients are integers, gives an expansion in $1/n$ with coefficients in $\mathbb{Z}[u]$. By uniqueness of asymptotic coefficients it is $\sum_J B_J^{(1)}(u; 1, s) n^{-J}$. The case $s = 1$ follows by the symmetry $B_J(u; r, s) = B_J(u; s, r)$. $\square$

This proves integrality only for the oriented one-path subfamily. It says nothing about the case (2, 2) of A189281, whose all-orders integrality remains conditional on Conjecture 11.1 (Proposition 11.2); manuscript 03 likewise warns that the subfamily does not justify extrapolating convergence, termination or integrality to two paths on both sides. At $(r, s) = (1, 1)$ the factor $1 - u/n$ gives $c_J^{(1)}(1, 1) = 1, 1, 0, 0, \dots$, the values found in Remark 7.2. [Added 3 October 2026, batch 85: the case (2, 2), and every other pair of directed offsets, is now proved in Part III (theorem 28.1), of which corollary 18.2 is the case $r = 1$ or $s = 1$; the conditional proposition cited above holds unconditionally (corollary 26.3). Proposition 18.1 is likewise the case $r = 1$ of theorems 24.1 and 25.1, which manuscript 03 of batch 85 derives without citing it (remark 25.2).]

19 Fixed local probabilities

For fixed j and M , coefficient extraction from (1.7) on a fixed circle gives

$$\mathbb{P}(X = j) = e^{-\theta} \frac{\theta^j}{j!} \left\{ \sum_{J=0}^M \frac{1}{n^J} \sum_h b_{J,h} H_h(j; \theta) + O_j(n^{-M-1}) \right\}, \quad (19.1)$$

with $B_J = \sum_h b_{J,h} u^h$ and the Charlier polynomials (7.6). This is the pointwise form of Corollary 7.3, which is the stronger statement (it controls the summed ℓ^1 error); it asserts no relative-error expansion uniformly for j growing with n . Manuscript 03 makes the first order explicit for the two sequences: from $B_1^{(1)} = -3u$ and $B_1^{(2)} = -6u - 2u^2$ at $(r, s) = (2, 2)$,

$$\mathbb{P}(X_{2,2,n}^{(1)} = j) = \frac{e^{-1}}{j!} \left\{ 1 + \frac{3(1-j)}{n} + O_j(n^{-2}) \right\}, \quad (19.2)$$

$$\mathbb{P}(X_{2,2,n}^{(2)} = j) = e^{-2} \frac{2^j}{j!} \left\{ 1 + \frac{4-j(j+1)/2}{n} + O_j(n^{-2}) \right\}. \quad (19.3)$$

At $j = 0$ these are the first avoidance corrections; at $j = 1, 2$ they show how probability mass is redistributed, which an avoidance count alone does not.

20 The inverse in a coordinate absorbing the constant

For A189281 and A110128 the logarithmic coefficients (9.4) are

Sequence	a_1	a_2	a_3
A189281	37/12	-5/2	1439/360
A110128	49/12	0	4319/360

The first row and a_1 for A110128 are in section 9; a_2 and a_3 for A110128 are added by manuscript 03.

Manuscript 03 inverts (9.3) in coordinates that absorb κ_θ . For large Y put

$$\Lambda = \log Y - \kappa_\theta, \quad N = \frac{\Lambda}{W_0(\Lambda/e)} = \exp(W_0(\Lambda/e) + 1), \quad \Delta = \log N, \quad (20.1)$$

so that $N(\log N - 1) = \Lambda$ exactly. Part I's coordinates (9.6) are the same construction with $\log Y$ in place of Λ ; so $N(Y)$ is Part I's z evaluated at $e^{-\kappa_\theta} Y$.

Proposition 20.1 (First inverse corrections; manuscript 03). *The formal inverse $n_*(Y)$ of (9.3) satisfies*

$$n_*(Y) = N - \frac{1}{2} + \frac{1/8 - a_1}{N\Delta} + \frac{1/24 - a_1/2 - a_2}{N^2\Delta} + O\left(\frac{1}{N^3\Delta}\right), \quad (20.2)$$

and to every fixed order $n_*(Y) = N - \frac{1}{2} + \sum_{j \leq J} \omega_j(\Delta) N^{-j} + O(N^{-J-1} \Delta^{-1})$, with each ω_j rational in Δ and computable from $a_1, \dots, a_j$; for instance

$$\omega_3(\Delta) = -\frac{1}{\Delta} \left\{ \frac{(1/8 - a_1)^2}{\Delta} + \frac{(1/8 - a_1)^2}{2\Delta^2} + \frac{a_1}{4} + a_2 + a_3 - \frac{1}{64} \right\}. \quad (20.3)$$

For the two sequences,

$$n_*(Y) = N - \frac{1}{2} - \frac{71}{24N\Delta} + \frac{1}{N^2\Delta} + \dots, \quad n_*(Y) = N - \frac{1}{2} - \frac{95}{24N\Delta} - \frac{2}{N^2\Delta} + \dots, \quad (20.4)$$

with κ_1 and κ_2 respectively in (20.1).

Proof (manuscript 03). Put $n = N + \delta$. The dominant part $x \log x - x$ has derivative Δ and second derivative $1/N$ at N ; after subtracting Λ the equation becomes $0 = \Delta\delta + \frac{\delta^2}{2N} - \frac{\delta^3}{6N^2} + \dots + \frac{\Delta}{2} + \frac{1}{2} \log(1 + \delta/N) + \sum_j a_j(N + \delta)^{-j}$. The constant term gives $\delta = -\frac{1}{2} + O(1/(N\Delta))$; with $\delta = -\frac{1}{2} + \sum_j \omega_j N^{-j}$ the next two orders read $\Delta\omega_1 + a_1 - \frac{1}{8} = 0$ and $\Delta\omega_2 + \frac{1}{2}a_1 + a_2 - \frac{1}{24} = 0$. At each later order the new coefficient enters only as $\Delta\omega_j$. For a fixed truncation the uncanceled residual is $O(N^{-J-1})$ and the derivative of the smooth carrier is $\Delta + O(1/N)$, so the mean value theorem divides it by a quantity asymptotic to Δ . $\square$

The first coefficient agrees with Part I: $1/8 - a_1 = 1/24 - c_1$ is the coefficient $-(c_1 - 1/24)$ of [theorem 9.2](#), which in Part I's coordinates also carries the terms in κ_θ that are absorbed here. Manuscript 03 interprets n_* as the inverse of the same smooth carrier as Part I, with $X_M(A(n)) = n + O(n^{-M-1}/\log n)$, and brackets the integer threshold $\nu(Y) = \min\{n : A(n) \geq Y\}$ between $\lceil X_M(Y) \mp \epsilon_Y \rceil$, $\epsilon_Y = O(N^{-M-1}/\log N)$, with an existence constant only; this bracket is `p0:thm:staircase` of [\[12\]](#). It notes that expanding W_0 in logarithms would give the familiar nested logarithmic transseries in Y , and that Λ depends on θ , so one Λ must not be reused for both sequences.

21 Manuscript 03's computations

Manuscript 03's `verify.py` (shipped as `code/02-fixed-gap-verify.py`) checks, with assertions: 90 avoidance counts and 90 complete histograms against brute-force permutations for $n = 0, \dots, 8$ and the offset pairs $(1, 1), (1, 2), (2, 2), (2, 3), (3, 3)$, both orientations; its exact tiling enumerator against A189281 for $n = 0, \dots, 26$ and $n = 40$; 132 stable-profile identities and a nonrealizable zero case; the moment agreement and disagreement of [remark 16.4](#); 50 one-path identities ([18.1](#)); all posted coefficients of both sequences through order ten, and orders 11 and 12; the general first correction and the mean through order three; and that the residual of ([20.4](#)) vanishes through N^{-3} . Its exact enumerator and its coefficient engine are independent of each other and of Part I's programs. At order twelve the engine enumerates the 77 long profiles of defect twelve, 272 of defect at most twelve, with the number of dimers summed exactly.

n	$\varepsilon_2(n)$	$\varepsilon_5(n)$	$\varepsilon_{10}(n)$	$\varepsilon_{12}(n)$
20	$1.2642 \cdot 10^{-4}$	$4.8572 \cdot 10^{-7}$	$-2.3683 \cdot 10^{-10}$	$5.4066 \cdot 10^{-11}$
40	$1.5661 \cdot 10^{-5}$	$6.9763 \cdot 10^{-9}$	$-1.3658 \cdot 10^{-13}$	$4.8917 \cdot 10^{-15}$
80	$1.9541 \cdot 10^{-6}$	$1.0406 \cdot 10^{-10}$	$-6.8418 \cdot 10^{-17}$	$5.2425 \cdot 10^{-19}$
120	$5.7883 \cdot 10^{-7}$	$8.9918 \cdot 10^{-12}$	$-7.9394 \cdot 10^{-19}$	$2.5777 \cdot 10^{-21}$

Table 6: Manuscript 03: the unscaled errors $\varepsilon_M(n) = e \text{A189281}(n)/n! - \sum_{j \leq M} c_j^{(1)}(2, 2)n^{-j}$, at 110-digit working precision, rounded. The values A189281(80) and A189281(120) are b-file inputs [\[13\]](#), not regenerated. These are uncertified decimals; Part I's [table 2](#) gives certified intervals for the scaled errors at $n = 80, 160, 320$.

[Added 1 October 2026, batch 74: scaled by n^{M+1} , the last two columns are consistent with Part I's higher coefficients: $n^{11}\varepsilon_{10}(n) = -58771$ and -58990 at $n = 80, 120$, against $c_{11}^{(1)}(2, 2) = -59103$; $n^{13}\varepsilon_{12}(n) = 2.882 \cdot 10^6$ and $2.758 \cdot 10^6$, approaching $c_{13}^{(1)}(2, 2) = 2525459$ from above as the next term $c_{14}^{(1)}/n$ with $c_{14}^{(1)} = 26632378$ predicts $(2525459 + 26632378/80 \approx 2.858 \cdot 10^6)$.] The four-term inverse ([20.4](#)) at $Y = \text{A189281}(n)$ misses n by about $2.0566 \cdot 10^{-4}$, $1.9555 \cdot 10^{-5}$, $1.9282 \cdot 10^{-6}$ and $5.0462 \cdot 10^{-7}$ at $n = 20, 40, 80, 120$; the signs are not predicted by the theorem. Manuscript 03 records its run as “Completed in 22.44 seconds” and its README speaks of “tens of seconds”. On this repository's machine the order-twelve run took more than 170 seconds at the intake and 104 seconds at the write, and an order-nine run 72 seconds. The run time depends on the machine and its load.

22 Further questions raised by manuscript 03

Manuscript 03 poses twelve research questions, as proposals only. Six overlap [section 12](#): a certificate for the particular recurrence (Part I’s question 2); integrality beyond the one-path case, which asks whether $B_J^{(1)}(u; r, s) \in \mathbb{Z}[u]$ for all r, s and so whether all $c_J^{(1)}(2, 2)$ are integers (question 1, and [corollary 18.2](#) for what is proved; [added 3 October 2026, batch 85:] both now hold, by [theorem 28.1](#)); large-order growth (question 4); growing gaps and component counts (question 5); cycles and other sparse constraint graphs (question 6); and a small formal package, starting from the profile identity, the stable polynomial, a verified coefficient checker and the universality bound (question 7). The other six are new here.

1. *A sharper oriented degree bound.* Does $\deg_u B_J^{(1)}(u; r, s) \leq J$ hold at every order, sharpening $2J$? The explicit polynomials through order three show the cancellation; such a bound would constrain the factorial cumulants of X . [Added 3 October 2026, batch 85: yes, for every r, s . Part III proves $\deg_u B_J^{(1)}(u; r, s) \leq \min\{J, \max(r + s - 1, J - r - s)\}$, with equality $J - r - s$ for $J \geq 2(r + s)$ when $r, s \geq 2$ ([theorem 28.1](#)).]
2. *Sharp short-path thresholds.* Condition [\(16.9\)](#) is sufficient, not shown optimal. What is the first shape-dependent term when a shortest path has fixed length, or grows like $c \log n / \log \log n$? [Added 4 October 2026, batch 97: for fixed length, answered in the absolute model by Part IV: one path of L vertices changes the avoidance probability by $-2e^{-2}n^{-L-1} + O(n^{-L-2})$ ([theorem 45.1](#)). At each fixed order a shortest path of fixed length suffices ([Remark 45.4](#)), so [\(16.9\)](#) is far from optimal at fixed order. In the directed model the corresponding order cancels and the first nonzero term is open ([question 50.1](#)); the regime $c \log n / \log \log n$ is open ([question 50.3](#)).]
3. *Optimal quantitative universality.* Find the correct constant in the $n \log n$ exponent of [theorem 16.3](#) for given proportions, and the path balances that maximize the difference.
4. *Several gaps simultaneously.* The joint law of violations for several position/value gap pairs, with interacting edge colours and components containing cycles; a first target is two pairs sharing one gap. [Added 5 October 2026, batch 101: answered by Part VII for the reciprocal pair (r, s) , (s, r) of absolute gaps: joint law, interacting colours and components with cycles ([theorem 78.1](#) and [remark 80.2](#)). Two pairs sharing one gap, the first target named here, remains open.]
5. *Growing local counts and moderate deviations.* Extend [\(19.1\)](#) to a quantified range $j = j(n) \rightarrow \infty$.
6. *Faster generation and certified errors.* Reduce the cost of many orders through profile symmetries, falling-factorial bases or direct Touchard operations, and extract explicit constants from [theorem 16.6](#) to obtain a certified finite-input threshold algorithm (compare Part I’s question 8).

OEIS and formalization, as manuscript 03 proposes them. It suggests an OEIS addition stating that the expansions through n^{-10} have a recurrence-independent proof, appending orders 11 and 12, and keeping the recurrences’ conjectural status; it submitted nothing, and Part I already has orders 11 to 16. It proposes a formalization in three layers (finite combinatorics; algebra of the stable polynomial and coefficient generation; analysis), with a coefficient checker that verifies equality with the finite formula rather than with a guessed recurrence. It supplies no Lean source, and nothing in Part II is formalized.

The shipped files of manuscript 03. The ten files of manuscript 03 carry the prefix 02-fixed-gap-: its two scripts in `code/`, its six recorded outputs and `requirements.txt` in

`data/`, and `sources.md` at the report root, all byte-identical to the delivery. `verify.py` imports its sibling by the delivered name (`from fixed_gap import ...`) and writes unprefixes files into `data/` beside its own directory; run from `code/` it would fail to import, and if made to run it would write beside Part I's files under unprefixes names. The report README gives a copy-based recipe.

Part III

Rational collapse and universal integrality for directed offsets

23 Provenance, scope and notation of Part III

[Added 3 October 2026, batch 85.] Part III was added in the write of batch 85. It is built from manuscript 03 of that batch, which proves Conjecture 11.1 of Part I for every h , extends the identity behind it to every pair of directed offsets (r, s) , and derives from it that every oriented correction polynomial $B_j^{(1)}(u; r, s)$ has integer coefficients, with an exact degree law. Where manuscript 03 proves a statement already printed in Part I or Part II, the statement is not reprinted: its route is recorded in table 9, and its special cases that repeat Part II's propositions 17.2 and 18.1 are printed as notes crediting Part II (remarks 25.2 and 28.2). Everything in Part III is written in Part I's notation (table 8).

23.1 Provenance

Manuscript 03 of batch 85 is the archive `OEIS_Rational_Collapse_and_Integrality.zip` (16 files, one of them the checksum ledger `MANIFEST.sha256`, which lists the other 15), which arrived in commit 317c1ce2e and was placed in this report by commit 713149ded. Its article is titled *Rational collapse and universal integrality for directed permutation avoidance: Resolving the A189281 path-forest conjecture and extending it to every fixed pair of offsets* (23 pages, dated 3 October 2026); its title page reads “Prepared for Vladimir Reshetnikov, AI-assisted mathematical research report”, and its PDF metadata names “AI-assisted research report prepared for Vladimir Reshetnikov” as author. Its pinned repository commit is d68b65ea064084fbb121df9bb431b21d086f7be81 (2026-10-03, 17:08; [19]). It read this report's `article.tex` as the Git blob 5b60a1ad447afe7dce58534fba447b05e401f172, which is also the blob of this file before the batch-85 write: manuscript 03 saw Parts I and II exactly as they stood.

Two different manuscripts numbered 03 have contributed to this report. In Part III, “manuscript 03” means manuscript 03 of batch 85; in Part II it means manuscript 03 of batch 74, the source of Part II, and Part III always names that one with its batch.

Manuscript 03 states correctly what Parts I and II contain: the stable tiling theorem and the all-orders expansion, Conjecture 11.1 checked for $h = 4, \dots, 16$, its conditional integrality consequence (Proposition 11.2), and Part II's integrality for the oriented one-path family. It does not cite Part II's proposition 18.1 or proposition 17.2, which it re-derives as special cases; this write credits them.

Scope and non-claims of manuscript 03, kept. It is an unrefereed, proof-focused, AI-assisted research draft, not a Lean-checked formalization, and it makes no claim of exhaustive historical priority; its firm novelty comparison is with the explicitly open conjecture of this report. It does *not* prove, or assume, the order-eight, degree-eleven recurrence listed in the OEIS for the finite sequence A189281: the recurrences it proves are in the moment order h ,

(11.8), and for auxiliary inverse-factorial coefficients, (29.8), not in n . Its integrality theorem is for directed (oriented) differences only; the absolute model ($\theta = 2$, A110128) is not covered, and integrality must not be transplanted to it. Stable rational moments equal actual moments only in the stability range (3.13). Its hypergeometric and inverse-factorial series are formal and are not asserted to converge; its entire Borel transform does not establish Laplace summability or convergence of $\sum_J c_J n^{-J}$, and no beyond-all-orders term is identified, since forests with equal stable moments can differ. No sign rule for the c_J is claimed. Its index reconstruction numerics are uncertified, and no explicit onset for rounding is given. Its certificate limits follow from the theorem, not from extrapolation. Missing proofs on an OEIS page are not evidence that no proof exists, and its OEIS note is an unsubmitted draft, not an accepted contribution. Finite computations support its proofs but are not their logical basis, and no Lean kernel run is asserted.

23.2 What Part III adds

Topic	Parts I and II	Part III (manuscript 03 of batch 85)
Rational collapse at $(2, 2)$	Conjecture 11.1, checked for $h = 4, \dots, 16$	<i>proved</i> for every $h \geq 4$, with (11.8) (theorem 26.1)
Stable moments	a sum over all profiles of edge degree k (3.12)	a single sum of $k + 1$ terms, every (r, s) (theorem 24.1)
Normalized moments R_h	$(2, 2)$ only, explicit for $h \leq 3$ (section 11)	a closed form for every (r, s) and h (theorem 25.1)
Integrality, $\theta = 1$	$(2J)!B_J \in \mathbb{Z}[u]$ (Proposition 7.1); $B_J \in \mathbb{Z}[u]$ if $r = 1$ or $s = 1$ (corollary 18.2); $(2, 2)$ conditional (Proposition 11.2)	$B_J \in \mathbb{Z}[u]$ for every (r, s) , by an explicit Stirling-number formula (theorem 28.1); Proposition 11.2 unconditional (corollary 26.3)
Degree in u , $\theta = 1$	$\leq 2J$; $\leq \max(3, J - 4)$ at $(2, 2)$ under the conjecture; Part II asks whether $\leq J$ (section 22, question 1)	$\leq \min\{J, \max(\sigma - 1, J - \sigma)\}$, $\sigma = r + s$; equal to $J - \sigma$, with leading coefficient w_σ , for $J \geq 2\sigma$ when $r, s \geq 2$
Coefficient growth	none proved (Part I, question 4)	entire Borel transform; $ c_J \leq \gamma_A A^{J-1} (J-1)!$ for every $A > 0$ (theorem 30.1)
Coefficients	$c_J(2, 2)$ for $J \leq 16$ (table 1); $c_3^{(\theta)}(r, s)$ (proposition 17.2)	$c_J(2, 2)$ for $J \leq 100$ (data), printed to $J = 24$; A189282–A189284 through c_6 (table 10)
Inverse	all orders (theorem 9.3); two terms in Part II's coordinates (proposition 20.1)	the same two terms, the second in closed form $(r-1)(s-1)$ for every pair (proposition 32.1)
Certificates	$n = 80, 160, 320$, $\theta = 1, 2$ (table 2)	$n = 64, 128, 256$, $\theta = 1$, from single-sum moments (table 11)

Table 7: Part III against Parts I and II. Part III treats the oriented case $\theta = 1$ only.

Source and novelty ledger of manuscript 03, restated. The OEIS sequence definitions and their displayed initial asymptotics are external, and no authorship is claimed for them; the A189281 entry, inspected on 3 October 2026, displays coefficients through n^{-10} and notes their integrality, an observation that is not by itself a theorem about every order. The paired path tilings are credited to Spahn and Zeilberger [3]. The stable tiling polynomial and the earlier all-orders expansion are credited to Part I; manuscript 03 reconstructs the stable identity with proof and gives a new proof of the analytic transfer through the collapse. Manuscript 03 claims as its own: the single-sum moment formula, the universal normalized identity and the A189281

product, integrality at every order with the sharp degree law, the inverse-factorial kernel and its auxiliary recurrence, the entire Borel transform, and a simplified parameter-dependent index reconstruction. Two of these overlap Part II as special cases, and the reconstruction is an instance of the inversion apparatus Part I already cites (Remark 9.1); table 9 says where.

23.3 Notation

Throughout Part III the orientation weight is $\theta = 1$, and we drop the superscript: $X = X_{r,s,n}^{(1)}$, $G_n(u) = G_{r,s,n}^{(1)}(u) = \mathbb{E}(1+u)^X$, $A(n) = A_{r,s}^{(1)}(n)$, $B_J = B_J^{(1)}(u; r, s)$, $c_J = c_J^{(1)}(r, s)$. The moments $\mu_k = \mathbb{E}(\binom{X}{k})$ are Part I's. The stable rational moments $\mu_k^*(n) = \mu_k^*(n; r, s)$ are the right side of (3.12) at $\theta = 1$, read as rational functions of an indeterminate n ; for $r = s = 2$ they are the μ_k^* of section 11. Likewise $R_h^{r,s}(n)$ is defined by (11.2) from $\mu_k^*(n; r, s)$, and $R_h^{2,2} = R_h$. Manuscript 03 overloads several letters that Parts I and II already use; table 8 fixes the convention. No normalization is changed.

Notion	Part III	Manuscript 03
count, statistic, generating function	$A(n), X, G_n(u)$	$a_{r,s}(n), X_{r,s,n}, G_n(u)$
offsets less one	$r-1, s-1$ (written out)	p, q (Part II's p_n is a polynomial)
sum of the offsets	$\sigma = r + s$	D (Part I's $\mathcal{D}$, D_v , D ; Part II's $D_{n,k}$)
weights $(r-1)^{\bar{j}}(s-1)^{\bar{j}}/j!$	w_j	w_j (Part I's w is $\log z$)
profile; component of ℓ vertices	$\mathbf{b} = (b_\ell)_{\ell \geq 2}$	$(b_d)_{d \geq 1}$, $d = \ell - 1$ edges (Part I's d is the defect)
reciprocal-factorial functional	$\mathcal{L}_n^{(a)}, \bar{n} = n - k$	$\mathcal{L}_m^{(p)}$, $m = n - k$ (Part I's $\mathcal{L}$ is a law, m a dimer count)
profile series	$\mathcal{T}(z; y_1, y_2) = \tau_0 + \tau_1 y_1$	$T(z; y, w) = A + By$
auxiliary variables	$\zeta = z/(1-z); t, \xi$	ω (Part II's ω_j); t, v
moment cutoff; constants	$h_n = \lfloor \log n \rfloor; \gamma_{r,s}, \gamma_A$	$K; C_{p,q}, C_{r,s}, K_A$
inverse-factorial polynomials	$\Phi_i(u), \Phi(z, u)$	$P_m(u), \mathcal{P}(z, u)$ (Part I's $P_{\beta,j}$, forest $\mathcal{P}$)
their values at $u = -1$	$\psi_i, \Psi(z)$	$d_m, \mathcal{H}(z)$ (Part I's $d_j(w)$, $\mathcal{H}(t, D)$)
exponential generating function	$\mathcal{E}(\xi) = \sum \psi_i \xi^i / i!$	$\mathcal{E}(y)$
Borel transform of $C(x)$	$\widehat{C}(t)$	$\widehat{C}(t)$ (same)
inverse coordinates	$\Lambda, N, \Delta = \log N$ (Part II, $\theta = 1$)	T, z, ℓ
logarithmic coefficients	a_1, a_2	α, β
inverse corrections	ω_1, ω_2 (Part II)	A, B

Table 8: Notation of manuscript 03 against Part III. Other symbols ($R_h, \mu_k^*, B_J, c_J, F_r(n; \mathbf{b}), E_a(\mathbf{b}), \mathbf{b}!, E_M(n), S_j$) mean the same in both. In Part III, z is a formal variable, not Part I's Lambert backbone, which Part III calls N as Part II does.

Two of these renamings carry a tempting false reading. The weight w_j is not the logarithm $w = \log z$ of section 9, and $\psi_i = \Phi_i(-1)$ are not Part I's inverse coefficients $d_j(w)$: they are the coefficients of an inverse-factorial re-expansion of the avoidance probability (section 29).

24 A single sum for the stable moments

Put

$$\sigma = r + s, \quad w_j = \frac{(r-1)^{\bar{j}}(s-1)^{\bar{j}}}{j!} \quad (j \geq 0). \quad (24.1)$$

Parts I and II	Manuscript 03's route
theorem 3.1 , Corollary 3.2	the same proof (the identity (3.6) and the expansion of $\mathcal{D}^{-(r-1)}$ over subprofiles), credited to Part I and to Spahn–Zeilberger, with components indexed by their edge counts
(3.14)	the case $k = 1$ of the single-sum formula (remark 24.2)
Lemma 4.1 , $\theta = 1$	the same argument; adds a bound on the normalized moments (lemma 27.1)
theorem 1.1 , $\theta = 1$	a second proof through the collapse (theorem 27.2); its shape clause is Part I's, weaker than Part II's corollary 16.7
Corollary 7.3 , $\theta = 1$	the same Cauchy argument on $ t = R$
(1.10), (7.3); proposition 17.2 , $\theta = 1$	B_1, B_2, c_1, c_2 and c_3 for general (r, s) from the Stirling formula (remark 28.2)
proposition 18.1 , corollary 18.2	the cases $r = 1$ or $s = 1$ of theorems 24.1, 25.1 and 28.1 (remark 25.2)
(8.7), section 8 , table 1	the polynomials $B_0, \dots, B_8$ and the coefficients through order 16 at $(2, 2)$, equal to Part I's (section 31)
theorem 9.2 , proposition 20.1 , $\theta = 1$	the first two corrections in Part II's coordinates (proposition 32.1)
table 2	the same Bonferroni method at new sizes (table 11)

Table 9: Results of Parts I and II proved again by manuscript 03, and where Part III records each route.

Generalized binomial coefficients mean $\binom{a}{b} = a^b/b!$ for $b \geq 0$, also for negative integers a , and $\binom{a}{b} = 0$ for $b < 0$. The weights are nonnegative integers: if $r, s \geq 2$, then

$$w_j = j! \binom{r+j-2}{j} \binom{s+j-2}{j}, \quad (24.2)$$

and if $r = 1$ or $s = 1$, then $w_0 = 1$ and $w_j = 0$ for $j > 0$. For $r = s = 2$, $w_j = j!$.

Theorem 24.1 (Single-sum stable moments; manuscript 03). *For all $r, s \geq 1$ and $k \geq 0$, as an identity in $\mathbb{Q}(n)$,*

$$\mu_k^*(n) = \sum_{j=0}^k \frac{w_j \binom{n-\sigma+1-j}{k-j}}{n^{k+j}}. \quad (24.3)$$

Equivalently, for integers $n \geq 2k$,

$$\mu_k^*(n) = \frac{1}{n!} \sum_{j=0}^k w_j (n-k-j)! \binom{n-\sigma+1-j}{k-j}. \quad (24.4)$$

Equality with the actual moments μ_k additionally requires the stability conditions of [Corollary 3.2](#).

The profile formula (3.12) has one term for each partition of k ; (24.3) has $k+1$.

24.1 A reciprocal-factorial linear functional

Put $\bar{n} = n - k$ and take n so large that $\bar{n} \geq k$. For an integer $a \geq 0$ and polynomials of degree at most $\bar{n}$ in a variable y , define

$$\mathfrak{L}_{\bar{n}}^{(a)}(y^i) = \frac{(-1)^i a^{\bar{i}}}{\bar{n}^i} \quad (0 \leq i \leq \bar{n}). \quad (24.5)$$

For $a = 0$ this is evaluation at $y = 0$. Since $(n - k - i)^{\overline{c-i}} = \bar{n}^c / \bar{n}^i$, the stable polynomial (3.2) becomes

$$F_r(n; \mathbf{b}) = \bar{n}^c \mathfrak{L}_{\bar{n}}^{(r-1)} \prod_{\ell \geq 2} (1 + (\ell - 1)y)^{b_\ell}. \quad (24.6)$$

Introduce the profile series

$$\begin{aligned} \mathcal{T}(z; y_1, y_2) &= 1 + \sum_{\ell \geq 2} z^{\ell-1} (1 + (\ell - 1)y_1)(1 + (\ell - 1)y_2) \\ &= \frac{1}{1 - z} + \frac{z(y_1 + y_2)}{(1 - z)^2} + \frac{z(1 + z)y_1 y_2}{(1 - z)^3}. \end{aligned} \quad (24.7)$$

In the multinomial expansion of $\mathcal{T}^{\bar{n}}$, a profile $\mathbf{b}$ of edge degree k has coefficient $\bar{n}^c / \mathbf{b}!$, and $(\bar{n}^c)^2 / n^{k+c} = \bar{n}! \bar{n}^c / n!$. With (24.6) for both forests, (3.12) at $\theta = 1$ becomes

$$\mu_k^*(n) = \frac{\bar{n}!}{n!} [z^k] \mathfrak{L}_{\bar{n}, y_1}^{(r-1)} \mathfrak{L}_{\bar{n}, y_2}^{(s-1)} \mathcal{T}(z; y_1, y_2)^{\bar{n}}. \quad (24.8)$$

24.2 The first functional

Write $\mathcal{T} = \tau_0 + \tau_1 y_1$. Since τ_1 has the factor z , the finite binomial theorem and (24.5) give, through degree $\bar{n}$ in z ,

$$\mathfrak{L}_{\bar{n}}^{(r-1)} (\tau_0 + \tau_1 y_1)^{\bar{n}} \equiv \tau_0^{\bar{n}} \sum_{i=0}^{\bar{n}} \frac{(-1)^i (r-1)^{\bar{i}}}{i!} \left(\frac{\tau_1}{\tau_0} \right)^i \equiv \frac{\tau_0^{\bar{n}+r-1}}{(\tau_0 + \tau_1)^{r-1}}. \quad (24.9)$$

The omitted terms start at $z^{\bar{n}+1}$, so they cannot affect $[z^k]$. With $\zeta = z/(1 - z)$ one has $\tau_0 = (1 + \zeta y_2)/(1 - z)$ and $\tau_0 + \tau_1 = (1 + 2\zeta y_2)/(1 - z)^2$, so the last expression is

$$(1 - z)^{r-1-\bar{n}} \frac{(1 + \zeta y_2)^{\bar{n}+r-1}}{(1 + 2\zeta y_2)^{r-1}}. \quad (24.10)$$

All its z -coefficients through degree $\bar{n}$ are polynomials in y_2 of degree at most $\bar{n}$, so the second functional is defined on them.

24.3 The second functional and the endpoint

For $s \geq 2$, a beta integral represents the second functional exactly:

$$\mathfrak{L}_{\bar{n}}^{(s-1)} f(y) = \frac{(\bar{n} + s - 1)!}{\bar{n}! (s - 2)!} \int_0^1 t^{s-2} (1 - t)^{\bar{n}} f\left(-\frac{t}{1 - t}\right) dt; \quad (24.11)$$

on y^i this follows from $\int_0^1 t^{s-2+i} (1 - t)^{\bar{n}-i} dt = (s - 2 + i)! (\bar{n} - i)! / (\bar{n} + s - 1)!$.

Assume first $r, s \geq 2$. Substitute (24.10) into (24.11) coefficientwise and then put $t = (1 - z)\xi$. For real $z < 0$ close to zero the result is

$$\frac{(\bar{n} + s - 1)!}{\bar{n}! (s - 2)!} (1 - z)^{\sigma-2-\bar{n}} \int_0^{1/(1-z)} \frac{\xi^{s-2} (1 - \xi)^{\bar{n}+r-1}}{[1 - (1 + z)\xi]^{r-1}} d\xi. \quad (24.12)$$

Only its Taylor coefficients through degree $\bar{n}$ are needed, not an equality of convergent series, and replacing the upper limit by 1 does not change them. To see the size of the change, put $a = -z > 0$: on the added interval $1 - \xi = O(a)$ and $1 - (1 - a)\xi \geq a/(1 + a)$, so the integral over it is $O(a^{\bar{n}+1})$. There is also a differentiation argument, not only a size estimate. For the integral with upper limit 1, differentiating $\ell' \leq \bar{n}$ times in z leaves an integrand bounded near $\xi = 1$ by a constant times $(1 - \xi)^{\bar{n}-\ell'}$, so dominated convergence computes its Taylor coefficients through degree $\bar{n}$. The integral before the substitution has the same property: with $a = -z$ and

$v = 1 - t$, its nonregular factor is $(v + a)^{\bar{n}+r-1}/(v + a(1+t))^{r-1}$, whose derivatives of order at most $\bar{n}$ are bounded by constant multiples of $(v + a)^{\bar{n}-\ell'}$. Its Taylor coefficients are therefore exactly the coefficientwise values of (24.11), and the change of endpoint is legitimate at every degree used.

Expand the denominator of the integrand at each fixed $0 \leq \xi < 1$:

$$\frac{(1 - \xi)^{\bar{n}+r-1}}{[1 - (1 + z)\xi]^{r-1}} = \sum_{j \geq 0} \frac{(r-1)^{\bar{j}}}{j!} z^j \xi^j (1 - \xi)^{\bar{n}-j}. \quad (24.13)$$

Only the terms with $j \leq \bar{n}$ are integrated, by the differentiation argument; they give $B(s-1+j, \bar{n}-j+1) = (s-2+j)! (\bar{n}-j)! / (\bar{n}+s-1)!$. Hence, as an equality of Taylor coefficients through degree $\bar{n}$,

$$\mathfrak{L}_{\bar{n}, y_1}^{(r-1)} \mathfrak{L}_{\bar{n}, y_2}^{(s-1)} \mathcal{T}^{\bar{n}} \equiv \frac{(1-z)^{\sigma-2-\bar{n}}}{\bar{n}!} \sum_{j=0}^{\bar{n}} w_j (\bar{n}-j)! z^j. \quad (24.14)$$

Nothing is claimed about coefficients beyond that degree.

If $r = 1$, evaluate at $y_1 = 0$ first. The remaining functional acts on $(1-z)^{-\bar{n}}(1+\zeta y_2)^{\bar{n}}$, and the same finite binomial calculation gives $(1-z)^{s-1-\bar{n}}$ through degree $\bar{n}$, which is (24.14) with $w_j = 0$ for $j > 0$. The case $s = 1$ follows by symmetry.

24.4 Completion of the proof

Proof of theorem 24.1 (manuscript 03). Insert (24.14) into (24.8) and use $\bar{n} = n - k$. The coefficient of z^{k-j} in $(1-z)^{\sigma-2-\bar{n}}$ is $\binom{\bar{n}-\sigma+2+k-j-1}{k-j} = \binom{n-\sigma+1-j}{k-j}$. This proves (24.4) for all large integers n . Both sides are rational functions of n , so the identity holds in $\mathbb{Q}(n)$ and gives (24.3). $\square$

Remark 24.2 (A first check; manuscript 03). For $k = 1$,

$$\mu_1^*(n) = \frac{n - \sigma + 1}{n} + \frac{(r-1)(s-1)}{n(n-1)} = \frac{(n-r)(n-s)}{n(n-1)},$$

which is Part I's exact mean (3.14) at $\theta = 1$. It confirms both the direction convention and the normalization.

25 Universal rational collapse

Theorem 25.1 (Universal rational-collapse identity; manuscript 03). *For all $r, s \geq 1$ and $h \geq 0$,*

$$R_h^{r,s}(n) = \sum_{j=0}^h \frac{w_j \binom{h-\sigma}{h-j}}{n^{\underline{h+j}}}. \quad (25.1)$$

In particular $R_0^{r,s} = 1$. For $h \geq \sigma$ the formula reduces to

$$R_h^{r,s}(n) = \sum_{j=\sigma}^h \frac{w_j \binom{h-\sigma}{j-\sigma}}{n^{\underline{h+j}}}, \quad (25.2)$$

and, when $r, s \geq 2$, its first term at infinity is

$$R_h^{r,s}(n) = w_\sigma n^{-h-\sigma} + O(n^{-h-\sigma-1}) \quad (h \geq \sigma). \quad (25.3)$$

When $r = 1$ or $s = 1$ it terminates:

$$R_h^{r,s}(n) = \begin{cases} (-1)^h \binom{\sigma-1}{h} / n^{\underline{h}}, & 0 \leq h \leq \sigma-1, \\ 0, & h \geq \sigma. \end{cases} \quad (25.4)$$

Proof (manuscript 03). Substitute (24.4) into (11.2), put $k = j + i$ and $H = h - j$, and interchange the two finite sums. For fixed j write $N' = n - 2j$ and $A' = n - \sigma + 1 - j$. The inner sum, before division by $n!$, is

$$\sum_{i=0}^H \frac{(-1)^{H-i}}{(H-i)!} (N' - i)! \binom{A'}{i} = (N' - H)! \sum_{i=0}^H (-1)^{H-i} \binom{A'}{i} \binom{N' - i}{H - i}. \quad (25.5)$$

The alternating Vandermonde identity

$$\sum_{i=0}^H (-1)^{H-i} \binom{A'}{i} \binom{N' - i}{H - i} = \binom{A' - N' + H - 1}{H} \quad (25.6)$$

holds because its left side is the coefficient of t^H in $\sum_i \binom{A'}{i} t^i (1-t)^{N'-i} = (1-t)^{N'} (1+t/(1-t))^{A'} = (1-t)^{N'-A'}$; generalized binomial series justify it coefficientwise. Here $A' - N' + H - 1 = h - \sigma$ and $N' - H = n - h - j$, so (25.5) equals $(n - h - j)! \binom{h - \sigma}{h - j}$. This proves (25.1) for large n , and then as an identity of rational functions.

For $h \geq \sigma$ the binomial coefficient has a nonnegative upper argument, so it vanishes unless $j \geq \sigma$; the $j = \sigma$ term has coefficient w_σ and denominator $n^{h+\sigma}$, and later terms are of higher order at infinity. If $r = 1$ or $s = 1$, only $j = 0$ survives, and $\binom{h-\sigma}{h} = (-1)^h \binom{\sigma-1}{h}$, which is zero for $h \geq \sigma$. $\square$

What the cancellation explains. Before normalization each fixed moment has the nonzero Poisson limit $1/k!$. After multiplication by e^{-u} , low powers of $1/n$ still occur for $h < \sigma$, but every normalized coefficient with $h \geq \sigma$ loses the first σ possible powers of $1/n$. The loss is exact and uniform in h : it is encoded by the vanishing binomial coefficients with $j < \sigma$. This explains the late appearance of high powers of u in the correction polynomials. The formula is symmetric in r, s , which also follows combinatorially: $\pi \mapsto \pi^{-1}$ exchanges a directed position gap r and a directed value gap s (table 5 records the same observation from Part II's source).

Remark 25.2 (The one-path case is Part II's Proposition 18.1). [Note of the batch-85 write.] For $r = 1$, (24.3) reduces to $\mu_k^*(n) = \binom{n-s}{k}/n^k$, which is (18.1), and (25.4) gives the finite inverse-factorial sector

$$e^{-u} G_n(u) \sim \sum_{h=0}^s \frac{(-1)^h \binom{s}{h} u^h}{n^h}, \quad (25.7)$$

which is the rational factor of (18.2) (for $s = 2$, $1 - 2u/n + u^2/(n(n-1))$). Manuscript 03 derives both as special cases of its general theorems and credits the one-path inclusion–exclusion to Spahn and Zeilberger [3], but does not cite Part II's proposition, which proves the same statements directly; the credit belongs to Part II (manuscript 03 of batch 74). At $u = -1$ the sector is $\sum_{h=0}^s \binom{s}{h}/n^h$; for $r = s = 1$ it is $1 + 1/n$, the values of Remark 7.2. For $(r, s) = (1, 3)$ the ordinary inverse-power expansion is infinite, $c_J = 1, 3, 3, 4, 6, 10, 18, 34, \dots$, although the inverse-factorial representation has four terms.

26 The A189281 conjecture is a special case

For $r = s = 2$, $\sigma = 4$ and $w_j = j!$. At $h \leq 3$, (25.1) gives the four rational functions $R_0, \dots, R_3$ displayed before Conjecture 11.1, and every later order collapses further.

Theorem 26.1 (Resolution of the rational-collapse conjecture; manuscript 03). *For every integer $h \geq 4$,*

$$R_h(n) = \frac{24(n-h+1)^{h-4}}{n^{2h}} = \frac{24}{n^{h-1}(n-2h+5)^{\frac{5}{2}}}. \quad (26.1)$$

That is, Conjecture 11.1 holds, and so does the ratio recurrence (11.8).

Proof (manuscript 03). For large integers n , (25.2) gives $n!R_h(n) = \sum_{j=4}^h j! \binom{h-4}{j-4} (n-h-j)!$. Put $H = h - 4$, $N' = n - h - 4$ and $j = i + 4$, so that the sum is $\sum_{i=0}^H \binom{H}{i} (i+4)! (N' - i)!$. With $M' = N' - H \geq 0$ this is

$$24H!M'! \sum_{i=0}^H \binom{i+4}{4} \binom{M' + H - i}{M'},$$

and the convolution equals $\binom{M'+H+5}{H}$, the coefficient of x^H in $(1-x)^{-5}(1-x)^{-M'-1}$. Hence

$$n!R_h(n) = \frac{24(N' - H)!(N' + 5)!}{(N' - H + 5)!} = \frac{24(n - 2h)!(n - h + 1)!}{(n - 2h + 5)!}.$$

Dividing by $n!$ gives (26.1); rational continuation removes the restriction to large n , and dividing consecutive instances gives (11.8). $\square$

This proves the conjecture for every h ; it does not merely extend the finite range $h = 4, \dots, 16$ checked in Part I. It also explains the constant: the Vandermonde convolution starts with a component-index shift of four, which produces $4! = 24$. Part I's question 1 asked for both a proof and an explanation of the factor 24 and of the loss of four powers of n ; the loss is the case $\sigma = 4$ of the cancellation above. It asked for a telescoping or combinatorial proof; the proof given is a single-sum identity followed by Vandermonde convolutions, and a bijective explanation is still open (section 34, question 9).

Remark 26.2 (A recurrence that is not proved here; manuscript 03). Equation (11.8) is a recurrence in the moment order h for rational functions of n . It is not the OEIS recurrence in the permutation size n for A189281(n). Confusing the two would turn a correct identity into an unsupported claim about the sequence. The order-eight, degree-eleven recurrence of A189281 remains unproved.

Corollary 26.3 (Proposition 11.2 is unconditional). $B_J^{(1)}(u; 2, 2) \in \mathbb{Z}[u]$ for every J , every $c_J^{(1)}(2, 2)$ is an integer, and $\deg B_J^{(1)}(u; 2, 2) \leq \max(3, J - 4)$ for $J \geq 4$.

The write's statement. The proof of Proposition 11.2 used only Conjecture 11.1, which theorem 26.1 proves. Manuscript 03 proves more, by a second route: theorem 28.1 at $(r, s) = (2, 2)$, which also shows that the degree bound is attained for $J \geq 8$. $\square$

27 From the collapse to the asymptotic expansion

Manuscript 03 proves the all-orders theorem again, directly from the collapse and without assuming Part I's theorem. The route is printed here because it is genuinely different from Part I's: instead of the defect filtration and finite differences in the number of dimers, it bounds the normalized moments R_h uniformly and expands finitely many rational functions.

Lemma 27.1 (Moment bounds; manuscript 03). *For arbitrary directed position and value path forests on n vertices, $0 \leq \mu_k \leq e^{2k}/k!$, and the actual normalized moments $\tilde{R}_h(n) = [u^h]e^{-u}G_n(u)$ satisfy*

$$|\tilde{R}_h(n)| \leq \frac{(1 + e^2)^h}{h!}. \quad (27.1)$$

Proof (manuscript 03). The first bound is Lemma 4.1 at $\theta = 1$, with the same proof. Then $|\tilde{R}_h| \leq \sum_{k=0}^h e^{2k}/(k!(h-k)!) = (1 + e^2)^h/h!$. $\square$

Theorem 27.2 (Uniform expansion through the collapse; manuscript 03). *For fixed $r, s \geq 1$, every $M \geq 0$ and every finite U , uniformly on $|u| \leq U$,*

$$G_n(u) = e^u \left(\sum_{J=0}^M \frac{B_J(u)}{n^J} + O_{M,r,s,U}(n^{-M-1}) \right), \quad (27.2)$$

where

$$B_0 = 1, \quad B_J(u) = \sum_{h \geq 1} [x^J] R_h^{r,s}(1/x) u^h \quad (J \geq 1), \quad (27.3)$$

each sum being finite. These are Part I's $B_J^{(1)}(u; r, s)$, by uniqueness of asymptotic coefficients; in particular $B_J(0) = 0$ for $J > 0$. The same polynomials apply to forests whose shortest path is at least a fixed positive fraction of n .

Second proof of [theorem 1.1](#) at $\theta = 1$ (manuscript 03). Let $h_n = \lfloor \log n \rfloor$. For large n the stable range contains every $h \leq h_n$, so by (27.1), with $\tilde{a} = (1 + e^2)U$,

$$e^{-u} G_n(u) = \sum_{h=0}^{h_n} R_h^{r,s}(n) u^h + O\left(e^{\tilde{a}} \frac{\tilde{a}^{h_n+1}}{(h_n+1)!}\right), \quad (27.4)$$

and the error is smaller than every power of $1/n$, its logarithm being $-\log n \log \log n + O(\log n)$.

Suppose $r, s \geq 2$. For $\sigma \leq h \leq h_n$ and $n > 2h_n$ all terms of (25.2) are nonnegative. Uniformly in $j \leq h \leq h_n$, $1/n^{h+j} \leq 2n^{-h-j}$ for large n , because the product of the factors $1 - i/n$ with $i < 2h_n$ is $1 + O(h_n^2/n)$. Moreover $w_{j+1}/w_j = (r-1+j)(s-1+j)/(j+1) \leq \gamma_{r,s}(h_n+1)$ for $\sigma \leq j < h_n$, so that $\sum_{j=\sigma}^{h_n} w_j n^{-j} \leq 2w_\sigma n^{-\sigma}$ for large n . Bounding the binomial coefficients by 2^h gives

$$|R_h^{r,s}(n)| \leq \gamma_{r,s} 2^h n^{-h-\sigma} \quad (\sigma \leq h \leq h_n). \quad (27.5)$$

Put $h_* = \max(\sigma - 1, M - \sigma)$. Summing (27.5) against $|u|^h$ over $h_* < h \leq h_n$ gives $O(n^{-M-1})$. Only the finite family $R_0, \dots, R_{h_*}$ remains; each is rational in n , regular at $n = \infty$, with an ordinary Taylor remainder there. Expanding them proves (27.2) and (27.3). If $r = 1$ or $s = 1$, (25.4) makes the sum in (27.4) terminate at $h = \sigma - 1$, and the conclusion is immediate. For other sufficiently long forests the stability theorem and the moment bound are unchanged. Uniqueness follows by multiplying the difference of two expansions by successive powers of n and taking limits. $\square$

The identity (27.3) is the one Part I's proof of Proposition 11.2 uses at (2, 2). The shape clause of [theorem 27.2](#) is the one stated after Part I's proof of [theorem 1.1](#); it is weaker than Part II's [theorem 16.6](#) and [corollary 16.7](#) and [Remark 16.8](#), which need only $L_{\min}(n) \log \log n / \log n \rightarrow \infty$, and it is not an improvement on them. Evaluating at $u = -1$ gives (1.11) with $c_J = B_J(-1)$, and the Cauchy argument on $|t| = R > 1$ gives the whole-law statement of [Corollary 7.3](#) at $\theta = 1$; manuscript 03 says that this mechanism was already in Part I and that what the collapse adds is the arithmetic structure of the polynomials.

28 Integer coefficients and an exact degree law

The inverse falling factorial has the expansion

$$\frac{1}{(1/x)^{\overline{m}}} = \frac{x^m}{\prod_{i=1}^{m-1} (1 - ix)} = \sum_{J \geq m} \left\{ \begin{matrix} J-1 \\ m-1 \end{matrix} \right\} x^J \quad (m \geq 1), \quad (28.1)$$

where $\left\{ \begin{smallmatrix} a \\ b \end{smallmatrix} \right\}$ is the Stirling number of the second kind, zero outside $0 \leq b \leq a$, with $\left\{ \begin{smallmatrix} 0 \\ 0 \end{smallmatrix} \right\} = 1$.

It follows by induction from $\left\{ \begin{smallmatrix} a \\ b \end{smallmatrix} \right\} = \left\{ \begin{smallmatrix} a-1 \\ b-1 \end{smallmatrix} \right\} + b \left\{ \begin{smallmatrix} a-1 \\ b \end{smallmatrix} \right\}$, or by multiplying the generating functions of the columns.

Theorem 28.1 (Universal integrality and exact degrees; manuscript 03). *For all $r, s \geq 1$ and $J \geq 1$,*

$$B_J(u) = \sum_{h=1}^J u^h \sum_{j=0}^h w_j \binom{h-\sigma}{h-j} \left\{ \begin{smallmatrix} J-1 \\ h+j-1 \end{smallmatrix} \right\}. \quad (28.2)$$

In particular,

$$B_J^{(1)}(u; r, s) \in \mathbb{Z}[u], \quad c_J^{(1)}(r, s) \in \mathbb{Z} \quad \text{for every } J \geq 0. \quad (28.3)$$

The degree satisfies

$$\deg_u B_J \leq \min\{J, \max(\sigma-1, J-\sigma)\}, \quad (28.4)$$

and if $r, s \geq 2$ and $J \geq 2\sigma$ it is exact:

$$\deg_u B_J = J - \sigma, \quad [u^{J-\sigma}]B_J = w_\sigma. \quad (28.5)$$

All coefficients of u^h with $h \geq \sigma$ are nonnegative. When $r = 1$ or $s = 1$, $\deg B_J \leq \sigma - 1$ for all J .

Proof (manuscript 03). Substitute (25.1) and (28.1) into (27.3); this is (28.2). Every factor of every summand is an integer, the generalized binomial coefficient with a possibly negative upper argument included, so integrality follows without a denominator-clearing bound. For $h < \sigma$, $R_h(1/x)$ begins no earlier than x^h ; for $h \geq \sigma$, (25.2) begins no earlier than $x^{h+\sigma}$. This proves (28.4). If $J \geq 2\sigma$, the largest possible h is $J - \sigma \geq \sigma$, where only $j = \sigma$ contributes, with coefficient $w_\sigma \binom{J-2\sigma}{J-2\sigma} \left\{ \begin{smallmatrix} J-1 \\ J-1 \end{smallmatrix} \right\} = w_\sigma > 0$. For $h \geq \sigma$ every summand of (28.2) is nonnegative. The one-offset assertion follows from (25.4). $\square$

Only the summation over $j \leq \min(h, J-h)$ contributes, because $\left\{ \begin{smallmatrix} J-1 \\ h+j-1 \end{smallmatrix} \right\} = 0$ for $h+j > J$. For A189281, (28.5) says that for every $J \geq 8$

$$\deg B_J^{(1)}(u; 2, 2) = J - 4, \quad [u^{J-4}]B_J^{(1)}(u; 2, 2) = 24, \quad (28.6)$$

so Part I's conditional bound $\max(3, J-4)$ is attained. Moreover every coefficient of u^h with $h \geq 4$ is divisible by 24, because only the weights $j!$ with $j \geq 4$ contribute. No sign rule for $c_J = B_J(-1)$ is inferred: evaluation at $u = -1$ reintroduces cancellation.

Two earlier questions are answered here. Part I's question 1 and Part II's overlapping question asked whether all $c_J^{(1)}(2, 2)$ are integers, and whether $B_J^{(1)}(u; r, s) \in \mathbb{Z}[u]$ for all r, s : both hold. Part II's question 1 asked whether $\deg_u B_J^{(1)}(u; r, s) \leq J$ at every order: it holds, by (28.4). The absolute case $\theta = 2$, whose coefficients are not integers by (8.12), is not covered, and Proposition 7.1 remains its denominator statement.

28.1 The first corrections for arbitrary offsets

Formula (28.2) gives

$$\begin{aligned} B_1 &= -(\sigma - 1)u, & B_2 &= (r - 1)(s - 1)u + \binom{\sigma - 1}{2}u^2, \\ B_3 &= (r - 1)(s - 1)u + \left(\binom{\sigma - 1}{2} - (\sigma - 2)(r - 1)(s - 1) \right)u^2 - \binom{\sigma - 1}{3}u^3, \end{aligned} \quad (28.7)$$

and consequently

$$c_0 = 1, \quad c_1 = \sigma - 1, \quad c_2 = \binom{\sigma - 1}{2} - (r - 1)(s - 1), \quad c_3 = \binom{\sigma}{3} - (\sigma - 1)(r - 1)(s - 1). \quad (28.8)$$

For example $c_3 = 0$ at $(3, 3)$, but $c_3 = -7$ and -24 at $(4, 4)$ and $(5, 5)$; such cancellations are outputs of the formula, not extrapolations from data.

Remark 28.2 (These are Parts I and II's first corrections). [Note of the batch-85 write.] B_1 and B_2 are Part I's (1.10) and (7.3) at $\theta = 1$. The values c_1, c_2, c_3 are those of Part II's [proposition 17.2](#) at $\theta = 1$, which manuscript 03 does not cite: $\binom{\sigma - 1}{2} - (r - 1)(s - 1) = \frac{1}{2}(r^2 - r + s^2 - s)$, and

$$\binom{\sigma}{3} - (\sigma - 1)(r - 1)(s - 1) = \frac{1}{6}(r + s - 1)(r^2 - 4rs + 4r + s^2 + 4s - 6),$$

which is (17.1); the intake checked the identity algebraically and the write checked both values for $1 \leq r, s \leq 5$ against (28.2). The closed form of $c_3^{(1)}(r, s)$ is therefore Part II's result (manuscript 03 of batch 74), here re-derived as a special case; the absolute $c_3^{(2)}$ of [proposition 17.2](#) has no counterpart in Part III.

29 An inverse-factorial kernel and an auxiliary recurrence

Define integer polynomials

$$\Phi_i(u) = \sum_{j=0}^{\lfloor i/2 \rfloor} w_j \binom{i - j - \sigma}{i - 2j} u^{i-j} \quad (i \geq 1), \quad \Phi_0(u) = 1. \quad (29.1)$$

Then the algebraic asymptotic sector of $e^{-u}G_n(u)$ is

$$\sum_{i \geq 0} \frac{\Phi_i(u)}{n^i}, \quad (29.2)$$

in the sense of equality after re-expansion in $1/n$, coefficient by coefficient. The displayed sum is not an ordinary convergent series at an integer n ; its denominators vanish for $i > n$.

Proposition 29.1 (Formal hypergeometric kernel; manuscript 03). *As a formal power series in z with polynomial coefficients in u ,*

$$\Phi(z, u) := \sum_{i \geq 0} \Phi_i(u) z^i = (1 - uz)^{\sigma-1} \sum_{j \geq 0} w_j \left(\frac{uz^2}{1 - uz} \right)^j = (1 - uz)^{\sigma-1} {}_2F_0\left(r - 1, s - 1; ; \frac{uz^2}{1 - uz}\right), \quad (29.3)$$

where ${}_2F_0(a, b; ; v) = \sum_{j \geq 0} a^{\bar{j}} b^{\bar{j}} v^j / j!$ is a formal series.

Proof (manuscript 03). In (25.1) write $h = j + i'$ and $i = h + j$. For fixed j the remaining sum is $\sum_{i' \geq 0} \binom{j + i' - \sigma}{i'} (uz)^{i'} = (1 - uz)^{\sigma-j-1}$. Multiply by $w_j u^j z^{2j}$ and sum over j . Each coefficient of z uses finitely many terms, so the rearrangements are formal and well defined. $\square$

Avoidance coefficients as a Stirling transform. Put $\psi_i = \Phi_i(-1)$ and

$$\Psi(z) = \sum_{i \geq 0} \psi_i z^i = (1+z)^{\sigma-1} \sum_{j \geq 0} w_j \left(-\frac{z^2}{1+z}\right)^j. \quad (29.4)$$

Then

$$c_0 = \psi_0 = 1, \quad c_J = \sum_{i=1}^J \psi_i \left\{ \begin{matrix} J-1 \\ i-1 \end{matrix} \right\} \quad (J \geq 1). \quad (29.5)$$

The inverse-power coefficients are thus a shifted Stirling transform of an integer sequence with a compact formal generating function, an alternative to constructing every B_J . For $r, s \geq 2$ the ${}_2F_0$ series has radius of convergence zero in its argument, because $w_{j+1}/w_j \sim j$. That does not invalidate (29.3), since formal substitution into a series of positive valuation is legitimate; it does forbid treating the formula as an analytic generating function without a stated summation procedure.

A short recurrence at (2, 2). For $r = s = 2$ write $\mathcal{F}(t) = \sum_{j \geq 0} j! t^j$ (a formal series in a local variable t). Comparing coefficients proves

$$t^2 \mathcal{F}'(t) + (t-1)\mathcal{F}(t) = -1. \quad (29.6)$$

Substituting $t = -z^2/(1+z)$ into $\Psi(z) = (1+z)^3 \mathcal{F}(t)$ and using the chain rule gives

$$z^3(1+z)\Psi'(z) + (2+3z+3z^2-2z^3)\Psi(z) = (z+2)(z+1)^4, \quad (29.7)$$

and hence

$$\boxed{2\psi_i + 3\psi_{i-1} + (i+1)\psi_{i-2} + (i-5)\psi_{i-3} = 0 \quad (i \geq 6),} \quad (29.8)$$

with $(\psi_0, \dots, \psi_5) = (1, 3, 2, -1, 1, 2)$; the next values are $-6, 0, 24, -24, -96, 216, 384, -1704, -1296, 13656$. This is an exact recurrence for the *auxiliary* inverse-factorial coefficients ψ_i , not for $A189281(n)$. It generates $\psi_0, \dots, \psi_M$ in $O(M)$ integer operations, and the Stirling transform then takes $O(M^2)$; these are counts of arithmetic operations, not bit-complexity bounds. For general fixed offsets, (29.1) gives $\psi_0, \dots, \psi_M$ in $O(M^2)$ operations, and (28.2) gives the whole array B_J in $O(M^3)$, with no enumeration of partitions and no recurrence guessing. [Added in the batch-85 write: the write recomputed Ψ from (29.4) to order 39 in exact arithmetic and found (29.8) satisfied for $6 \leq i \leq 39$, with the values above.]

30 An entire Borel transform of the algebraic sector

Define the exponential generating function

$$\mathcal{E}(\xi) = \sum_{i \geq 0} \frac{\psi_i}{i!} \xi^i. \quad (30.1)$$

Unlike Ψ , it is a genuine entire function. Recall Part I's formal series $C(x) = 1 + \sum_{j \geq 1} c_j x^j$ (section 9).

Theorem 30.1 (Entire Borel transform; manuscript 03). *For all fixed $r, s \geq 1$ the series (30.1) is entire, and the order-one Borel transform of the nonconstant part of C is the entire function*

$$\boxed{\hat{C}(t) := \sum_{J \geq 1} \frac{c_J}{(J-1)!} t^{J-1} = \mathcal{E}'(e^t - 1).} \quad (30.2)$$

In particular, for every $A > 0$ there is a constant γ_A such that

$$|c_J| \leq \gamma_A A^{J-1} (J-1)! \quad (J \geq 1). \quad (30.3)$$

Proof (manuscript 03). Suppose $r, s \geq 2$. By (24.2), $w_j \leq \gamma_{r,s}(j+1)^{\sigma-4}j!$. For large i and $j \leq i/2$, the binomial coefficient in (29.1) has a nonnegative upper argument and modulus at most 2^i ; the finitely many remaining i are absorbed in the constant. Hence

$$|\psi_i| \leq \gamma_{r,s} 2^i (i+1)^{\sigma-3} [i/2]!. \quad (30.4)$$

The i th root of $[i/2]!/i!$ tends to zero, so (30.1) converges for every complex ξ . If $r = 1$ or $s = 1$, $\Psi(z) = (1+z)^{\sigma-1}$ is a polynomial and the conclusion is immediate. The exponential generating function of the Stirling numbers, $\sum_{J \geq i} \left\{ \begin{smallmatrix} J-1 \\ i-1 \end{smallmatrix} \right\} t^{J-1}/(J-1)! = (e^t - 1)^{i-1}/(i-1)!$, combined with (29.5), gives (30.2). The interchange is justified locally by the absolute convergence of the entire series for $\mathcal{E}'$, and the composition is entire. Cauchy's estimate on $|t| = 1/4$ gives (30.3). $\square$

What this does not imply. An entire Borel transform need not grow slowly enough for a Laplace integral along the positive axis, and composition with $e^t - 1$ can create very rapid growth. Theorem 30.1 is therefore neither a proof of Borel–Laplace summability nor a proof that $\sum c_J n^{-J}$ converges. Nor can the algebraic sector determine contributions smaller than every power of $1/n$: different long path forests can share their stable moments to increasing order, hence the whole array B_J , while their exact counts differ (Remark 16.4). The theorem describes the algebraic sector and its Borel transform, not the beyond-all-orders corrections; this is the boundary between the results proved here and a full transseries theory. It bears on Part I's question 4, which asks for growth bounds in M and a justified truncation rule; those remain open (section 34, questions 3 and 4). [Added 4 October 2026, batch 97: at (2, 2) Part V now proves positive-ray summation, with a sum C_* that is not sectorially summable, the sharp growth of $\widehat{C}$ and of the c_J , a truncation rule for C_* , and that a beyond-all-orders correction is present: the counts differ from C_* by a flat term that is nonzero in every block of four indices (theorems 58.2, 60.2, 61.1 and 64.1). Signs of the c_J , an optimal truncation rule, the amplitude of the flat term and every other pair of offsets remain open.]

31 Explicit OEIS consequences

A189281. The polynomials $B_0, \dots, B_8$ that (28.2) gives at (2, 2) are Part I's (section 8); the term u^4 first appears at order eight, in $B_8 = u + 479u^2 + 378u^3 + 24u^4$, with coefficient 24, as (28.6) requires. The avoidance expansion through n^{-10} is Part I's (8.7), with $c_4 = 0$; manuscript 03 says expressly that these displayed terms were already proved in Part I and that its new content is the all-orders identity and the integrality theorem, not these finite terms. Its coefficients through order 16 equal table 1. Beyond them, (28.2) gives

J	$c_J^{(1)}(2, 2)$	J	$c_J^{(1)}(2, 2)$
17	−9526718817	21	57992423942395
18	−136484458282	22	1360005443948778
19	−1110549722719	23	16301210950936317
20	−3629525434664	24	113926603325586836

and `data/03-rational-collapse-coefficients.json` holds every B_J , c_J and ψ_J through order 100. The runs of alternating signs neither undermine integrality nor indicate a proved periodic sign pattern; large-order sign and growth behaviour is a separate problem.

Three further diagonal sequences. The theorem applies directly to the OEIS entries for the directed gaps (3, 3), (4, 4) and (5, 5), A189282, A189283 and A189284 [15, 16, 17], whose pages display expansions through n^{-2} . For every fixed truncation order the rows of [table 10](#) continue by the same finite formula, with a rigorous next-order remainder.

OEIS	r	s	c_1	c_2	c_3	c_4	c_5	c_6
A189281	2	2	3	2	1	0	3	26
A189282	3	3	5	6	0	-13	-3	235
A189283	4	4	7	12	-7	-73	-88	1029
A189284	5	5	9	20	-24	-234	-390	3374

Table 10: Manuscript 03: $A_{r,r}^{(1)}(n)/n! = e^{-1}(1 + \sum_{J \geq 1} c_J n^{-J})$ in the Poincaré sense; not a claim that the series converges. The first row is Part I's (8.7).

For the nondiagonal pair $(r, s) = (2, 3)$ the coefficients through order eight are 1, 4, 4, 2, -1, 6, 81, 388, 775, so the result does not depend on equal offsets. The data file also holds examples with one offset equal to one, as degenerate checks ([remark 25.2](#)).

32 Index reconstruction with Lambert W

Manuscript 03 specializes the repository's inversion apparatus, which it cites through this report, to a simple parameter-dependent formula along exact sequence values, so that no interpolation of the integer sequence is chosen. In Part II's coordinates (20.1) at $\theta = 1$, for large y put

$$\Lambda = \log y - \kappa_1 = \log y + 1 - \frac{1}{2} \log(2\pi), \quad N = \frac{\Lambda}{W_0(\Lambda/e)}, \quad \Delta = \log N, \quad (32.1)$$

so that $N(\log N - 1) = \Lambda$.

Proposition 32.1 (Index reconstruction; manuscript 03). *For $y = A_{r,s}^{(1)}(n)$ and fixed offsets, as $n \rightarrow \infty$,*

$$n = N - \frac{1}{2} + \frac{1/24 - (\sigma - 1)}{N\Delta} + \frac{(r-1)(s-1)}{N^2\Delta} + O_{r,s}\left(\frac{1}{N^3\Delta}\right). \quad (32.2)$$

Every further fixed-order correction is computable from (28.2) and the Stirling expansion of $\log \Gamma$.

Proof (manuscript 03). The expansion (1.11) and Stirling's expansion [7] give, as in (9.3), $\Lambda = n(\log n - 1) + \frac{1}{2} \log n + a_1/n + a_2/n^2 + O(n^{-3})$ with $a_1 = c_1 + \frac{1}{12}$ and $a_2 = c_2 - \frac{1}{2}c_1^2$. Write $n = N + \delta$ and expand about N . The constant order gives $\delta = -\frac{1}{2}$; with $\delta = -\frac{1}{2} + \omega_1/N + \omega_2/N^2$ the next two coefficients are $\omega_1\Delta + a_1 - \frac{1}{8}$ and $\omega_2\Delta + a_2 + \frac{1}{2}a_1 - \frac{1}{24}$. They vanish for $\omega_1 = (1/24 - c_1)/\Delta$ and $\omega_2 = (c_1(c_1 - 1)/2 - c_2)/\Delta = (r-1)(s-1)/\Delta$, by (28.8). The residual is $O(N^{-3})$ and the smooth main part has derivative $\log N + O(1/N)$, so the mean value theorem turns it into an index error $O(N^{-3}/\log N)$. The initial bound $n - N = O(1)$ follows from the same equation and the monotonicity of $x(\log x - 1)$ for large x . $\square$

[Note of the batch-85 write.] This is Part II's [proposition 20.1](#) at $\theta = 1$, with the same coordinates and the same two coefficients ($1/8 - a_1 = 1/24 - c_1$, and $1/24 - a_1/2 - a_2 = (r-1)(s-1)$); what is new is that the second coefficient is written in closed form for every pair of offsets. At (2, 2) it is the $1/(N^2\Delta)$ of (20.4). In Part I's coordinates it is [theorem 9.2](#), and the method is the inversion apparatus of [Remark 9.1](#), an instance of `p0:thm:perturbed-inversion` of [12]; no novelty is claimed for the method.

For the exact A189281 values at $n = 20$ and $n = 30$ the displayed approximation misses n by about $2.05663 \cdot 10^{-4}$ and $5.16805 \cdot 10^{-5}$ (the first agrees with Part II’s [section 21](#), which evaluates the same formula). These are numerical diagnostics, not certified rounding thresholds. The proposition implies eventual nearest-integer recovery along exact sequence values, but gives no explicit onset without an additional effective remainder estimate.

33 Manuscript 03’s exact checks and certificates

Manuscript 03 separates proof, exact finite algebra and numerical diagnostics; its proofs of arbitrary order are those of [sections 24](#) and [25](#), and its checks exercise independent implementations to detect transcription or indexing errors. Its `verify.py` (shipped as `code/03-rational-collapse-verify.py`, standard library only) builds F_r directly from (3.2) and sums over integer partitions, and separately implements (25.1); each comparison clears the denominator by multiplying by $h!n^{2h}$ and compares integer polynomials in n . A separate finite tiling routine enumerates ordered tile-size multisets on each actual path, without the stable polynomial or the single-sum theorem, and a third routine enumerates permutations and records the whole violation histogram. The recorded run passed:

Exact check	Number
Polynomial identities between profile sums and (25.1), $h = 0, \dots, 16$, nine offset pairs	153
Complete violation histograms against brute permutations, $n = 0, \dots, 8$, $r, s \in \{1, 2, 3\}$	81
Single-sum moments against brute histograms in the stable range	190
A189281 values, $n = 0, \dots, 30$, from independent finite tilings	31
The recurrence (29.8), $i = 6, \dots, 100$	95
(29.5) against the polynomials B_J , $J = 1, \dots, 100$	100
Sharp-degree and leading-coefficient checks across offset pairs	178

The nine offset pairs were $(1, 1), (1, 2), (1, 4), (2, 2), (2, 3), (2, 4), (3, 2), (3, 3), (4, 4)$. It also compared (26.1) with the general expression for $h = 4, \dots, 16$, and the first seventeen $c_J(2, 2)$ with Part I’s values. The A189281 values come from a supplied excerpt of the OEIS b-file [18]; they are validation targets only, never inputs to coefficient generation or to any proof. The program computes all B_J , c_J and ψ_J at $(2, 2)$ through order 100 in exact integers. [Added in the batch-85 write: if the b-file excerpt is missing under its delivered name, the program skips those 31 checks without failing and prints no line for them; the report README gives a rerun recipe that restores the delivered names.]

Rational Bonferroni certificates. The single-sum formula gives exact low moments without computing the full count. With the inequalities (10.1) and the scaled residual (10.2) at $\theta = 1$, manuscript 03’s `certify.py` uses S_{31} and S_{30} at $n = 64, 128, 256$, where these moments are in the stable range, and brackets e^{-1} by its alternating Taylor sums through degrees 101 and 102. Dividing the probability bounds by that positive rational interval gives exact rational intervals, rounded outward in [table 11](#); the unrounded fractions are in `data/03-rational-collapse-bonferroni.json`.

Manuscript 03’s optional inverse diagnostic, `inverse_demo.py`, needs `mpmath`; neither the verifier nor the mathematical results depend on it. Repeating the finite tests to a larger order adds implementation checks, not a substitute for the proofs, and no Lean kernel run is asserted. Its delivered rerun commands use the delivered layout; the report README gives the commands for the shipped names.

n	Certified interval for $E_4(n)$	Certified interval for $E_8(n)$
64	[3.43128403, 3.43128404]	[−1630.56920464, −1630.56920463]
128	[3.20934303, 3.20934304]	[−1516.24227317, −1516.24227316]
256	[3.10311068, 3.10311069]	[−1461.72210309, −1461.72210308]

Table 11: Manuscript 03: exact finite certificates for A189281 ($\theta = 1$), generated without a recurrence for A189281(n). The limits 3 and -1409 follow from the asymptotic theorem, not from extrapolating three rows. They interleave with Part I’s intervals at $n = 80, 160, 320$ (table 2).

34 Further questions raised by manuscript 03

Manuscript 03 poses ten research projects, as proposals only, ordered to separate exact finite enumeration, the algebraic sector and beyond-all-orders information. Several continue questions of Parts I and II, as noted.

1. *Prove the original A189281 recurrence* (Part I’s question 2). The order-eight, degree-eleven recurrence remains a separate target. One route writes the exact finite tiling sum as its stable contribution plus explicit boundary corrections and seeks a telescoping certificate for the whole; agreement of its formal asymptotic solution with (26.1) is a consistency check, not a proof. The first deliverable should be a finite identity whose boundary terms are fully specified. [Added 4 October 2026, batch 97: Part V proves that the OEIS operator annihilates the analytic completion of the expansion, not the counts (theorem 63.1), and reduces the problem to a recurrence for the flat discrepancy (Part V, question 2).]
2. *The first beyond-all-orders term* (Part I’s question 3, Part II’s question 2). Find the leading difference between two forest shapes with the same r, s, n and long paths, whose low moments agree until a component can meet a whole short path: is it a single spanning component or a family of competing profiles? A first case changes two path lengths by $+1$ and -1 . [Added 4 October 2026, batch 97: still open. For A189281 itself Part V proves that the count differs from its positive-ray Borel completion by a flat term that is nonzero in every block of four indices (theorem 64.1); for a short path of fixed length Part IV finds the shape at an algebraic order (theorem 45.1).]
3. *Large-order growth of c_J* (Part I’s question 4). Use (30.3) and the composition $\mathcal{E}'(e^t - 1)$ to find sharp asymptotics, oscillation and signs, first at $(2, 2)$ and then for all fixed offsets; (29.8) may be a first step. [Added 4 October 2026, batch 97: answered in part at $(2, 2)$ by Part V: $\limsup_m (\log m) |c_{m+1}/m!|^{1/m} = 2$, the series $\sum c_J z^J$ has radius zero, no Gevrey bound of order $s < 1$ holds, and (c_J) is not P-recursive (theorems 60.2 and 60.4). Sharp asymptotics of individual c_J , oscillation and signs, and the other offsets remain open (Part V, questions 3 and 7).]
4. *Summation and optimal truncation* (Part I’s question 4). Determine the growth of $\hat{C}$ in sectors and whether an accelerated or nonstandard Laplace transform recovers a natural analytic object; a complete answer names the function reconstructed, the directions, the optimal truncation order in n , and the relation to the finite counts. [Added 4 October 2026, batch 97: answered at $(2, 2)$ by Part V, except for optimality. The function reconstructed is the positive-ray sum $C_*(n) = 1 + \int_0^\infty e^{-nt} \hat{C}(t) dt$, convergent exactly for $\Re n > 1$ and meromorphic in n (theorem 58.2 and corollary 58.3); the admissible geometry is a horizontal strip of half-width $\pi/4$, and no open angular sector about the positive axis carries an exponential bound (theorem 59.2 and corollary 59.4); the truncation order $M = \lfloor n(\frac{1}{2} \log n - 1) \rfloor$ gives the error $\exp(-\frac{1}{2}n \log n + O(n))$ (theorem 61.1), with no proof

of optimality; and the finite counts differ from C_* by a flat term that is nonzero in every block of four indices ([theorem 64.1](#)). The optimal truncation and its prefactor remain open (Part V, question 4).]

5. *A finite-algebraic proof of the single-sum lemma.* The beta integral is used only to evaluate a finite Taylor jet; a telescoping proof of ([24.14](#)), a polynomial identity for the reciprocal-factorial functionals, would simplify formalization and may show a direct combinatorial cancellation.
6. *Formalize the finite core* (Part I's question 7, Part II's formal package). A Lean development can first define falling products, generalized integer binomials and ([24.3](#)), ([25.1](#)); after clearing denominators the collapse is a finite rearrangement and a Vandermonde convolution. Stable tilings and the analytic remainder are further layers, and a formal proof of the algebraic identity alone must not be labelled a formal proof of the asymptotic statement.
7. *Absolute differences.* For $|\pi(i+r) - \pi(i)| \neq s$ each component has two orientations (weight 2^c , not 2^k), and the cancellation behind ([25.1](#)) changes. Find an equally explicit kernel, optimal denominator bounds, and which arithmetic survives; the directed integrality theorem must not be transplanted without proof.
8. *Growing offsets* (Part I's question 5). If σ grows, the sharp-degree threshold, the weight w_σ and the stability range move. Find ranges such as $r+s = o(\log n)$ with a uniform theorem, and the crossovers where Poisson(1) normalization fails; the exact mean $(n-r)(n-s)/(n(n-1))$ constrains any scaling limit. [Added 4 October 2026, batch 97: answered in local-density coordinates by Part IV: the expansion holds uniformly for all $1 \leq r, s \leq n$ with the exact parameter $\lambda_n = \theta(1-r/n)(1-s/n)$ in place of a Poisson(1) normalization, the law tends to Poisson($\theta(1-\alpha)(1-\beta)$) when $r/n \rightarrow \alpha$ and $s/n \rightarrow \beta$ ([corollary 46.1](#)), and the corrections have kinks at $r/n = 1/q$ ([theorem 46.2](#)). Part IV's coefficients are polynomials in local densities, not Part III's integer polynomials; how the degree law and the weight w_σ behave for growing σ remains open.]
9. *Positivity and divisibility, combinatorially.* For $h \geq \sigma$ the coefficients of u^h in B_J are nonnegative sums of integer weights and Stirling numbers; find a labelled-object interpretation. At (2, 2) the factor 24 suggests a four-element ordering or symmetry, but the proof gives a convolution identity, not a bijection (compare Part I's question 1).
10. *Several forbidden gaps* (Part I's question 6, Part II's question 4). The graphs are then not unions of paths, and components may branch or cycle. Determine which sparse families still admit stable component polynomials and a finite hypergeometric normalization; a first project is a two-gap directed model of bounded component width. [Added 5 October 2026, batch 101: answered in part by Part VII, for the absolute model with the reciprocal gaps (r, s) and (s, r) : components branch and cycle, but compatible ones are triangle-free and 2-degenerate ([lemma 80.1](#)), and finite packing dictionaries with eventual embedding polynomials take the place of stable component polynomials ([theorem 81.1](#) and [proposition 79.2](#)). The directed two-gap model remains open.]

Manuscript 03 suggests starting with questions 5 and 6 and the asymptotics of ([29.8](#)); questions 1 and 2 are the closest to the outstanding OEIS problem, and the Borel-growth questions 3 and 4 and the growing offsets of question 8 are broader analytic projects.

Conclusion of manuscript 03, restated. The conjecture of Part I is resolved by [theorem 26.1](#), one specialization of [theorem 25.1](#), valid for every pair of directed offsets. The integer formula ([28.2](#)) proves the arithmetic phenomenon at every order, explains the delayed

high-degree terms and gives efficient coefficient generation; the Borel transform and the index reconstruction are further consequences. The remaining distinction is between the stable asymptotic information, proved here with controlled remainders, and exact finite enumeration: the original guessed recurrence and a complete beyond-all-orders expansion are not claimed.

OEIS note, as manuscript 03 proposes it. Its `data/03-rational-collapse-OEIS_proposed_note.md` drafts an OEIS remark that every c_J of A189281 is an integer, with the finite formula (28.2) at $u = -1$, that the rational-collapse conjecture is proved, that the A189281 recurrence is not, and that the same holds for every directed pair including A189282–A189284. It is marked “not submitted”; it asks for a stable public citation and independent review first. Nothing was submitted.

35 Formula sheet

With $\sigma = r + s$ and $w_j = (r - 1)^{\bar{j}}(s - 1)^{\bar{j}}/j!$ (manuscript 03, Appendix A):

$$\begin{aligned}
\mu_k^*(n) &= \sum_{j=0}^k \frac{w_j \binom{n-\sigma+1-j}{k-j}}{n^{\underline{k+j}}}, \\
R_h^{r,s}(n) &= \sum_{j=0}^h \frac{w_j \binom{h-\sigma}{h-j}}{n^{\underline{h+j}}}, \\
B_J(u) &= \sum_{h=1}^J u^h \sum_{j=0}^{\min(h, J-h)} w_j \binom{h-\sigma}{h-j} \left\{ \begin{matrix} J-1 \\ h+j-1 \end{matrix} \right\} \quad (J \geq 1), \\
\psi_i &= \sum_{j=0}^{\lfloor i/2 \rfloor} (-1)^{i-j} w_j \binom{i-j-\sigma}{i-2j}, \\
c_J &= \sum_{i=1}^J \psi_i \left\{ \begin{matrix} J-1 \\ i-1 \end{matrix} \right\} \quad (J \geq 1), \\
\Psi(z) &= (1+z)^{\sigma-1} {}_2F_0\left(r-1, s-1; ; -\frac{z^2}{1+z}\right), \\
\widehat{C}(t) &= \mathcal{E}'(e^t - 1), \quad \mathcal{E}(\xi) = \sum_{i \geq 0} \psi_i \xi^i / i!.
\end{aligned} \tag{35.1}$$

The conventions for $J = 0$ are $B_0 = c_0 = \psi_0 = 1$. The formulas for μ_k^* and R_h are rational identities; actual moments require (3.13). Ψ is formal, while $\mathcal{E}$ and $\widehat{C}$ are entire.

The shipped files of manuscript 03. The prefix `03-rational-collapse-` marks its twelve shipped files. Its three scripts and its `Makefile` are in `code/`; its five recorded outputs, the b-file excerpt and its OEIS note are in `data/`; its `SOURCES.md` is at the report root. All are byte-identical to the delivery. `certify.py` imports its sibling by the delivered name (`from verify import ...`); `verify.py` and `inverse_demo.py` read the excerpt by its delivered name, `b189281_0_30.txt`; all three write unprefixed files into the `data/` directory beside their own; and the `Makefile` calls the scripts by their delivered names. In the shipped layout the scripts would fail to import, skip the OEIS-value checks or fail to find their input, and write beside the files of Parts I and II. The report `README` gives a copy-based recipe.

Part IV

Moving gaps: uniform local expansions, short paths and finite-size boundaries

36 Provenance, scope and notation of Part IV

[Added 4 October 2026, batch 97.] Part IV was added in the write of batch 97. It is built from manuscript 05 of that batch, which drops every hypothesis on the lengths of the paths: it proves the all-orders expansion of the whole violation law uniformly over *all* pairs of path forests on n vertices, with coefficients that are universal polynomials in normalized subpath counts, and derives from it the first shape-dependent term of a fixed short path, the behaviour of moving and proportional offsets, and an index reconstruction along rational offset rays. Where manuscript 05 proves a statement already printed in Parts I–III, its proof is kept as a marked second route and the earlier statement is named (table 13). Manuscript 05’s symbols that clash with Parts I–III are renamed or declared in table 14.

36.1 Provenance

Manuscript 05 of batch 97 is the archive `ProveIt_Moving_Gap_Permutations.zip` (21 files, one of them the checksum ledger `SHA256SUMS.txt`, which lists the other 20), which arrived in commit `e88ed8bf6` and was placed in this report by commit `5a69916e8`. Its article is titled *Moving-Gap Permutation Avoidance: Uniform local expansions, sharp short-path effects, and a hierarchy of finite-size boundaries* (26 pages, dated 4 October 2026); its title page reads “Prepared for Vladimir Reshetnikov” and “AI-assisted mathematical research report”, and its PDF metadata names “AI-assisted research report prepared for Vladimir Reshetnikov” as author. Its pinned repository commit is `d1680cdfa40c44abf0825416b6f36a3e9ec2662c` (2026-10-04, 18:28; [20]). It read this report’s `article.tex` as the Git blob `73a53d9784260c2f0d083f2baa62a5535f6d6105`, which is also the blob of this file immediately before the batch-97 write: manuscript 05 saw Parts I–III exactly as they stood.

Manuscript 05 states correctly what Parts I–III contain: the fixed-offset expansions (Parts I and II), the directed rational collapse and integrality (Part III), the exact profile identity (Proposition 2.2), the moment bound (Lemma 4.1), the one-path mechanism of Part II (proposition 18.1) and the Lambert- W inversion (theorem 9.3), and it claims none of them anew. Its “Section 22” is Part II’s section 22, and the “growing gaps” question it answers is Part I’s question 5 (section 12), listed there among the overlaps. A second batch-97 manuscript, the source of Part V, continues this report independently; the two share no theorem.

Scope and non-claims of manuscript 05, kept. It is an AI-assisted, unrefereed research manuscript with conventional proofs and reproducible checks, not a Lean or Rocq formalization. Its infinite results rest on its written proofs; its finite checks support but do not replace them. Historical priority is not established: its source audit was targeted, not exhaustive, and “failure to find a prior result in this audit is not evidence that none exists”. Its expansion is an all-orders algebraic sector, not a resurgent or exponentially complete transseries; its index reconstruction asserts neither Borel summability nor control of exponentially small sectors, and no numerical value of the eventual rounding threshold. It proves no guessed OEIS recurrence, submitted no OEIS edit and changed no repository file. Its uniform theorem concerns one source forest and one target forest, not unions of differently coloured gap constraints. [Added 5 October 2026, batch 101: Part VII treats one such union, two reciprocal fixed gaps marked separately

(theorem 78.1).] The first shape-dependent term for a shortest path that grows with n is left open; its short-path theorem is for fixed L , with constants not controlled as $L \rightarrow \infty$, so it licenses no substitution $L = L(n)$; in the directed model it proves cancellation only at the displayed order. The correction boundaries of section 46 are kinks of correction terms, not macroscopic phase transitions, and the cancellation at $\sqrt{2} - 1$ does not imply cancellation at later orders. Its inverse decimals are not interval certificates; its rational test vectors need not be realizable by forests, and the probability statements are asserted for actual forests only; finitely many Bonferroni intervals do not prove the remainder. The exact profile identity, the moment bound, connected-graph expansions, Stirling's formula and Lambert- W inversion are not advertised as new, and Spahn–Zeilberger and Tauraso are credited.

36.2 What Part IV adds

Topic	Parts I–III	Part IV (manuscript 05 of batch 97)
All-orders law	fixed offsets r, s (theorem 1.1); every forest whose shortest path has at least $K_n \asymp \log n / \log \log n$ vertices (theorem 16.6 and corollary 16.7, Remark 16.8)	<i>every</i> pair of path forests, no length hypothesis, uniformly on every disk $ u \leq U$, with coefficients $B_j^{(\theta)}(u; \mathbf{a}, \mathbf{b})$ in normalized subpath counts (theorem 38.1)
Locality	stable moments depend on path counts only (theorem 3.1)	order n^{-M} uses only subpaths of at most $M + 1$ edges, and this cutoff is sharp among actual forests (theorem 41.1 and proposition 46.3)
Short paths	the first shape-dependent term asked for (section 22, question 2)	one fixed short path of L vertices changes the absolute-model avoidance probability by $-2e^{-2}n^{-L-1} + O(n^{-L-2})$; cancellation in the directed model (theorem 45.1)
Growing offsets	open (Part I, question 5; Part III, question 8)	all $1 \leq r, s \leq n$, limit $\text{Poisson}(\theta(1 - \alpha)(1 - \beta))$, correction boundaries at reciprocal integers (corollary 46.1 and theorem 46.2)
Explicit terms	B_1, B_2 at fixed offsets (1.10), (7.3)	B_1, B_2 in six local densities (proposition 43.1)
Inverse	fixed offsets (theorem 9.3, proposition 20.1, proposition 32.1)	rational offset rays $r = \alpha n, s = \beta n$ (theorem 49.1)
Certificates	fixed offsets (tables 2 and 11)	the ray $(1/3, 1/2)$ and the short-path difference (tables 18 and 19)

Table 12: Part IV against Parts I–III.

Source and novelty ledger of manuscript 05, restated. The matching-of-tilings formula is credited to Spahn and Zeilberger [3], the fixed-gap dinner-table asymptotics to Tauraso [4], and the recurrence-independent fixed-gap expansions to Parts I and II of this report; the OEIS sequence definitions are external. Manuscript 05 claims as its own: arbitrary-forest uniformity, including linearly many bounded components and vanishing leading density; the finite locality cutoff, the finite coefficient formula and the sharpness of the cutoff among actual forests; the highest-path sensitivity, the reciprocal-integer boundaries and the cancellation at $\sqrt{2} - 1$; the leading absolute-model term of one fixed short path, its directed cancellation and the one-target-path shape independence in its stated form; and the moving-ray coefficients,

Parts I–III	Manuscript 05’s route
Proposition 2.2	the same identity for arbitrary forests, proved again with rods indexed by edges (proposition 39.2); credited to Spahn–Zeilberger at $u = -1$ and to this report for general u
Lemma 4.1	the same bound with the same proof (lemma 39.3)
proposition 18.1	the same moment formula with the two forests exchanged (proposition 45.5)
Corollary 7.3	the same Cauchy argument, now for every forest pair (corollary 38.2)
(1.11), $c_1 = \theta(r + s - \theta)$	recovered from proposition 43.1 at fixed offsets (43.12), as a consistency check
(10.1)	the same inequalities (48.8)
theorem 9.3 , proposition 32.1	the same Lambert- W apparatus along rational rays, with new coefficient input (theorem 49.1)

Table 13: Results of Parts I–III proved again by manuscript 05, and where Part IV records each route.

index reconstruction and certificates. The inequality behind its tree bound is classical and is credited in this write ([Remark 40.4](#)).

36.3 Notation

Part IV keeps manuscript 05’s notation except where it collides with Parts I–III; [table 14](#) fixes the convention. Its local-density objects are written in sans-serif type: the correction polynomials B_j , the subpath counts S_q and the sensitivity polynomials H_q . No normalization is changed.

Notion	Part IV	Manuscript 05 (and the clash)
correction polynomials in local densities	$B_j^{(\theta)}(u; \mathbf{a}, \mathbf{b})$	$B_j^{(\theta)}(u; \mathbf{a}, \mathbf{b})$ (Part I’s $B_j^{(\theta)}(u; r, s)$ at fixed offsets)
subpath counts with q edges	$S_q(F)$	$S_q(F)$ (Part I’s Bonferroni sums S_j of (10.1))
sensitivity polynomials	$H_q^{(\theta)}(\mathbf{b})$	$H_q^{(\theta)}(\mathbf{b})$ (Part I’s Charlier $H_h(q; \theta)$ of (7.6))
density vectors	$\mathbf{a} = (a_q), \mathbf{b} = (b_q)$	$\mathbf{a}, \mathbf{b}$ (Part I’s profile $\mathbf{b} = (b_\ell)_{\ell \geq 2}$; Part I’s logarithmic a_j)
rod profile, rods of ℓ edges	$\mathbf{m} = (m_\ell)_{\ell \geq 1}$	same (Part I’s $b_{\ell+1}$, components of $\ell + 1$ vertices)
falling factorial	$x^{\underline{v}}$	$(x)_v$
logarithmic coefficients on a ray	χ_j	ψ_j (Parts III and V’s ψ_i)
inverse coordinates	$\Lambda, N, \Delta = \log N$; input y ; $\hat{n} = n + \frac{1}{2}$	$\mathcal{L}, Y, \ell; T; y$
inverse coefficients	$d_j(\Delta)$	$d_j(\ell)$ (Part I’s $d_j(w)$)

Table 14: Notation of manuscript 05 against Part IV. The inverse coordinates are Part II’s [\(20.1\)](#) and Part III’s [\(32.1\)](#) with the constant λ in place of θ : at fixed offsets they coincide. All other symbols are manuscript 05’s, listed below with their clashes.

The following symbols keep manuscript 05’s meaning inside Part IV and mean something else elsewhere in the report: $G(u) = \mathbb{E}(1 + u)^X$, as in Part I (Part II writes $G(w) = \mathbb{E}w^X$);

$\lambda = \theta a_1 b_1$, an n -dependent leading Poisson parameter that is not the exact mean (38.2); $\alpha = r/n$ and $\beta = s/n$ (Part II's α is a length proportion, Part I's ratio is η), and the coefficients $\beta_{j,h}$ of (38.7) (Part I's $b_{J,h}$); the ray coefficients $c_j = B_j(-1; \mathbf{a}(\alpha), \mathbf{b}(\beta))$ (Part I's $c_j^{(\theta)}(r, s)$ are at fixed offsets); $K_F, \kappa_{\ell_B, q}, F_{\ell, h}, F_\ell, R_{J, k}, D_v, d_j(v), T_N, L_q, C, D, E$ and $\mathcal{A}, \mathcal{B}, \mathcal{R}, \mathcal{S}$ (Part I's $K, \kappa_\theta, F_r, R_h$ of (11.2), $\mathcal{D}, D$, its defect d , the series $C(x)$, and Part III's $\mathcal{T}$); the moment cutoff $K_n = \lfloor (M+3) \log n / \log \log n \rfloor$ (Part II's K_n has $M+4$, Part I's cutoff is T_n); $\mathcal{B}(t) = \sum_\ell b_\ell t^\ell$. Manuscript 05 itself overloads d (the defect, the density b_2 in section 43, $d_j(v)$ and $d_j(\Delta)$), c (the rod count, the density a_2 , the ray coefficients c_j), e (the density a_3 in section 43, next to Euler's number in (43.5)) and L (a path's number of vertices, and L_q); the section determines the meaning.

Two of these carry a tempting false reading. The polynomial $B_j^{(\theta)}(u; \mathbf{a}, \mathbf{b})$ evaluated at the densities $a_q = 1 - qr/n$, $b_q = 1 - qs/n$ of fixed offsets is *not* Part I's $B_j^{(\theta)}(u; r, s)$: the densities and λ depend on n , and Part I's polynomials are obtained only after re-expanding $e^{(\lambda-\theta)u}$ and the densities in powers of $1/n$ (section 43 does this for the first order). And H_q is a polynomial in the target densities, not a Charlier polynomial.

37 Manuscript 05: the question, the sources, and the contribution boundary

The ProveIt report *Unconditional all-orders asymptotics for two OEIS permutation sequences* develops fixed-gap avoidance using stable path forests [20]. Its Parts I and II prove all-orders asymptotics when the position and value offsets are fixed. Part III proves a directed rational-collapse and integrality statement. Those results are not claimed again here. The present starting points are two questions that remain in the inspected report's Section 22: growing gaps and component counts, and the first shape-dependent term caused by a short path. [Note of the batch-97 write: the inspected Section 22 is Part II's section 22; the growing gaps are Part I's question 5 (section 12), listed there among the overlaps, and the short-path term is its question 2.]

We work at repository revision

d1680cdfa40c44abf0825416b6f36a3e9ec2662c.

The inspected article has blob identifier

73a53d9784260c2f0d083f2baa62a5535f6d6105.

Pinning matters here: earlier versions explicitly conjectured statements that Part III subsequently proved.

The public combinatorial setting predates that report. Spahn and Zeilberger give the matching-of-tilings inclusion–exclusion formula for fixed position and value gaps [3]. Tauraso studies the rectangular dinner-table problem and derives fixed-gap asymptotics [4]. OEIS A189281 counts permutations avoiding $\pi(i+2) - \pi(i) = 2$; A110128 counts permutations avoiding $|\pi(i+2) - \pi(i)| = 2$ [1, 2]. The OEIS entries associate higher expansions with guessed recurrence operators. The earlier ProveIt report already supplies recurrence-independent fixed-gap expansions; *we do not present that fact as a new result of this manuscript*.

37.1 What changes when gaps move

A gap r on $[n]$ decomposes the labels into r paths. With fixed r , the shortest path tends to infinity. With r proportional to n , most paths have bounded length, and the stable long-path

polynomial is no longer an adequate uniform description. The natural replacement is a finite list of local statistics:

$$S_q(F) = \sum_{\text{paths of } F} \max(L - q, 0), \quad q \geq 1, \quad (37.1)$$

where L is the number of vertices in a path. Thus S_q counts consecutive subpaths of q edges, not q vertices.

The key observation is a filtration by the cost of connected overlap. A cluster spanning q edges cannot first affect an asymptotic coefficient before order $q - 1$. This leads both to a uniform theorem and to a sharp formula for where new geometry becomes visible.

Topic	Prior input or established context	Addition proved here
Exact counting	Matching of path tilings [3]; fixed-gap report [20]	Reproved as the finite foundation, not claimed new
All-orders law	Fixed path counts and long paths [20]	Uniformity over arbitrary path forests, including linearly many short paths
Local information	Long-path stabilization [20]	Order M needs only $S_1/n, \dots, S_{M+1}/n$ in each forest; the bound is sharp
Short paths	First shape-dependent term posed as a question [20]	Explicit first absolute-model term for a fixed short path; directed cancellation
Moving ratios	Growing gaps posed as a question [20]	Reciprocal-integer correction boundaries and an exact cancellation locus
Inversion	Lambert- W carriers and fixed-gap inversion [20, 8]	Specialization to rational moving-gap rays, with actual-sequence diagnostics

Table 15: Contribution boundary relative to the sources actually inspected.

37.2 Limits of the claims

The literature review is targeted, not exhaustive. The statements proved below are new additions to the inspected report, but the review does not establish that every theorem is historically new. No particular guessed OEIS recurrence is proved, no OEIS edit is submitted, and no repository file is changed. The uniform theorem concerns one source forest and one target forest; it does not cover an arbitrary union of differently colored gap constraints. The first exponentially small shape-dependent term for a shortest path that itself grows with n is also left open.

38 The model and the main uniform theorem

Let $\mathcal{P}$ and $\mathcal{V}$ be directed path forests on n vertices. Every component is a path with a consistent orientation, including paths with one vertex. Choose a uniformly random bijection π from the vertices of $\mathcal{P}$ to those of $\mathcal{V}$. In the directed model an edge of $\mathcal{P}$ is counted when its image is a directed edge of $\mathcal{V}$. In the absolute model it is counted when its image is an edge in either direction. Write $\theta = 1$ or 2 , respectively, and let X be the resulting count.

Define

$$G_{\mathcal{P}, \mathcal{V}}^{(\theta)}(u) = \mathbb{E}(1 + u)^X, \quad a_q = \frac{S_q(\mathcal{P})}{n}, \quad b_q = \frac{S_q(\mathcal{V})}{n}, \quad \lambda = \theta a_1 b_1. \quad (38.1)$$

All densities lie in $[0, 1]$. They may depend on n in any way compatible with actual forests. In particular λ is an n -dependent leading Poisson parameter, not the exact mean:

$$\mathbb{E}X = \frac{\theta S_1(\mathcal{P})S_1(\mathcal{V})}{n(n-1)} = \frac{\lambda}{1-1/n}, \quad n \geq 2. \quad (38.2)$$

Theorem 38.1 (Uniform all-orders finite-locality theorem; manuscript 05). *For $\theta \in \{1, 2\}$ there are universal polynomials*

$$B_j^{(\theta)} \in \mathbb{Q}[u, a_1, \dots, a_{j+1}, b_1, \dots, b_{j+1}] \quad (j \geq 0), \quad (38.3)$$

with $B_0 = 1$, symmetric under exchanging $\mathbf{a}$ and $\mathbf{b}$, such that for any fixed $M \geq 0$ and $U < \infty$,

$$G_{\mathcal{P}, \mathcal{V}}^{(\theta)}(u) = e^{\lambda u} \sum_{j=0}^M \frac{B_j^{(\theta)}(u; \mathbf{a}, \mathbf{b})}{n^j} + O_{M,U}(n^{-M-1}), \quad |u| \leq U. \quad (38.4)$$

The error constant is independent of both forests. Moreover,

$$\deg_u B_j \leq 2j, \quad B_j(0; \mathbf{a}, \mathbf{b}) = 0 \quad (j \geq 1). \quad (38.5)$$

No lower bound on a path length or on $a_1 b_1$ is required. A finite coefficient formula is given in [equation \(41.7\)](#).

The theorem is not a statement that fixed-gap coefficient polynomials can simply be evaluated at growing offsets. Its parameters are the *actual local densities*, and its proof remains valid after the long-path stabilization range disappears.

Corollary 38.2 (Avoidance and the complete law; manuscript 05). *Uniformly over the same forests,*

$$\mathbb{P}(X = 0) = e^{-\lambda} \sum_{j=0}^M n^{-j} B_j^{(\theta)}(-1; \mathbf{a}, \mathbf{b}) + O_M(n^{-M-1}). \quad (38.6)$$

Writing $B_j(u) = \sum_{h=0}^{2j} \beta_{j,h} u^h$, define

$$p_k^{[M]} = \sum_{j=0}^M n^{-j} \sum_{h=0}^{2j} \beta_{j,h} \frac{\partial^h}{\partial \lambda^h} \left(e^{-\lambda} \frac{\lambda^k}{k!} \right). \quad (38.7)$$

Then

$$\sum_{k \geq 0} |\mathbb{P}(X = k) - p_k^{[M]}| = O_M(n^{-M-1}). \quad (38.8)$$

In particular $d_{TV}(\mathcal{L}(X), \text{Poisson}(\lambda)) = O(n^{-1})$, uniformly. The same conclusion holds with the exact mean in place of λ .

Proof (manuscript 05). Set $u = -1$ for avoidance. For the complete law, use the identity $\partial_\lambda^h e^{\lambda(z-1)} = (z-1)^h e^{\lambda(z-1)}$. On a fixed circle $|z| = R > 1$, [theorem 38.1](#) bounds the remainder by $C_M n^{-M-1}$ because $|z-1| \leq R+1$. Cauchy's estimate bounds its k th coefficient by $C_M n^{-M-1} R^{-k}$; summing proves (38.8). The derivative form in (38.7) is valid also at $\lambda = 0$ and introduces no division by the parameter. Finally, Poisson variables with parameters differing by δ can be coupled with mismatch probability at most $|\delta|$, and (38.2) gives $\delta = O(1/n)$. $\square$

39 Exact profiles and a moment bound independent of geometry

We use falling factorials $x^{\underline{v}} = x(x-1)\cdots(x-v+1)$, with $x^0 = 1$. A *rod* of length ℓ is an interval of ℓ edges and $\ell+1$ vertices in a path.

Definition 39.1 (Rod profile; manuscript 05). A profile $\mathbf{m} = (m_1, m_2, \dots)$ has finite support and nonnegative integer entries. Put

$$c = \sum_{\ell \geq 1} m_\ell, \quad k = \sum_{\ell \geq 1} \ell m_\ell, \quad v = k + c, \quad d = k - c = \sum_{\ell \geq 1} (\ell - 1) m_\ell, \quad \mathbf{m}! = \prod_{\ell \geq 1} m_\ell!. \quad (39.1)$$

The quantity d is the intrinsic defect. Let $C_F(\mathbf{m})$ count collections of vertex-disjoint rods in F with this profile.

Proposition 39.2 (Exact profile identity; manuscript 05). *For arbitrary path forests on n vertices,*

$$G(u) = \sum_{\mathbf{m}: v \leq n} \theta^c u^k C_P(\mathbf{m}) C_V(\mathbf{m}) \frac{\mathbf{m}!}{n^v}. \quad (39.2)$$

Second proof of Proposition 2.2 (manuscript 05). Expand $(1+u)^X$ by marking a subset of the violating edges. The selected source edges decompose into vertex-disjoint rods. Their images are rods in the target forest with the same profile. Match rods of equal length in $\mathbf{m}!$ ways, choose one of θ orientations for each rod, and map the remaining $n-v$ isolated vertices arbitrarily. Each pair of configurations therefore yields $\theta^c \mathbf{m}!(n-v)!$ bijections, with the selected subset distinguished. Divide by $n!$ and sum. Unselected violations are allowed; excluding them at this stage would invalidate the identity. $\square$

At $u = -1$ this is the matching-of-tilings inclusion-exclusion formula of [3], expressed for arbitrary path forests. Its general- u form is also used in [20]. We include the proof to keep the present argument independent of any guessed recurrence. [Note of the batch-97 write: this is Part I's Proposition 2.2, which already holds for arbitrary position and value forests. Manuscript 05 indexes rods by their numbers of edges: its m_ℓ is Part I's $b_{\ell+1}$, and $\mathbf{m}!/n^v = (n-v)! \mathbf{b}!/n!$.]

Let $\mu_k = \mathbb{E} \binom{X}{k} = [u^k] G(u)$.

Lemma 39.3 (Uniform factorial-moment bound; manuscript 05). *For all forests, all $n \geq 1$, and all $k \geq 0$,*

$$0 \leq \mu_k \leq \frac{(\theta e^2)^k}{k!}. \quad (39.3)$$

Consequently, for $A = \theta e^2 U$,

$$\sup_{|u| \leq U} \left| G(u) - \sum_{k=0}^K \mu_k u^k \right| \leq e^A \frac{A^{K+1}}{(K+1)!}. \quad (39.4)$$

Second proof of Lemma 4.1 (manuscript 05). Fix k selected source edges, decomposing into c rods and $v = k + c$ vertices. Their images can be assigned starts and directions in at most $\theta^c n^c$ ways, even before requiring disjointness. For $v \leq n$,

$$n^v \geq (n/e)^v. \quad (39.5)$$

Indeed the geometric mean of the largest v factors in $n!$ is at least that of all n factors, while $\log(n!) \geq \int_1^n \log t \, dt \geq n \log n - n$. Thus the probability that every selected edge is a violation is at most $\theta^c n^c / n^v \leq \theta^k e^{2k} n^{-k}$. There are at most $n^k/k!$ selected edge sets. Summing proves the bound; if $v > n$ no such selected set occurs. The exponential-series tail gives (39.4). $\square$

This bound is available in the fixed-gap report, and is reproved here. [Note of the batch-97 write: it is Part I's Lemma 4.1, with the same proof.] It is crucial that its constant does not deteriorate when the forest has many short components.

40 Connected overlaps and exact local cluster polynomials

The next construction replaces long-path stabilization by an exact local identity. Temporarily label all c rods individually, with lengths $\ell = (\ell_1, \dots, \ell_c)$. For an arbitrary placement of these rods in F , its incompatibility graph joins two labels when their rods share a vertex. Rods with identical occupied intervals are also incompatible.

40.1 The connected graph weight

For a graph H on a nonempty label set define

$$\phi(H) = \sum_{\substack{J \subseteq H \\ J \text{ spanning and connected}}} (-1)^{|E(J)|}. \quad (40.1)$$

For a singleton graph the sum equals 1; for a disconnected graph with more than one vertex it is zero. For a nonempty set of rod labels B , let

$$K_F(\ell_B) = \sum_{\text{placements of the rods in } B} \phi(H). \quad (40.2)$$

Lemma 40.1 (Locality of a connected cluster; manuscript 05). *There are integers $\kappa_{\ell_B, q}$, independent of F , such that*

$$K_F(\ell_B) = \sum_{q=1}^{k_B} S_q(F) \kappa_{\ell_B, q}, \quad k_B = \sum_{i \in B} \ell_i. \quad (40.3)$$

In particular $K_F(\ell_B) = n \kappa_{\ell_B}(\mathbf{a})$, where $\kappa_{\ell_B}(\mathbf{a}) = \sum_{q \leq k_B} a_q \kappa_{\ell_B, q}$. For a single rod of length ℓ , $\kappa_{(\ell)}(\mathbf{a}) = a_\ell$.

Proof (manuscript 05). Only connected incompatibility graphs contribute. In a connected placement the union of the rods is a single interval of some edge length $q \leq k_B$. Translate its left endpoint to zero. There are only finitely many normalized placements of the given labeled rods with this union; sum their integer weights (40.1) to define $\kappa_{\ell_B, q}$. Each normalized pattern has exactly $S_q(F)$ translations into F . This proves the identity. $\square$

For $h \geq 0$ define the merger polynomial

$$F_{\ell, h}(\mathbf{a}) = \sum_{\substack{\rho \text{ a set partition of } [c] \\ c - |\rho| = h}} \prod_{B \in \rho} \kappa_{\ell_B}(\mathbf{a}), \quad F_{\ell}(\mathbf{a}, x) = \sum_{h \geq 0} x^h F_{\ell, h}(\mathbf{a}). \quad (40.4)$$

Here a block of m rods costs $m - 1$ mergers. The empty list has $F_{\emptyset} = 1$. The leading term is $F_{\ell, 0}(\mathbf{a}) = \prod_{i=1}^c a_{\ell_i}$.

Proposition 40.2 (Exact labeled placement polynomial; manuscript 05). *For every profile $\mathbf{m}$ and any labeling ℓ of its rods,*

$$C_F(\mathbf{m}) = \frac{n^c}{\mathbf{m}!} F_{\ell}(\mathbf{a}, 1/n). \quad (40.5)$$

This identity holds even if the profile cannot fit in the forest, in which case its right side vanishes.

Proof (manuscript 05). For each arbitrary placement, the indicator of disjointness is the product over all rod pairs of $1 - \mathbf{1}\{\text{overlap}\}$. Expand the product as a sum over graphs on the labels. Group each graph by its connected components. The sum belonging to a block is exactly $K_F(\ell_B)$; different blocks place their rods independently because there is no imposed cross-block edge in that graph term. Inserting (40.3) gives $n^c F_{\ell}(\mathbf{a}, 1/n)$ labeled disjoint placements. Dividing by $\mathbf{m}!$ removes the labels on rods of equal length. $\square$

40.2 A coefficient bound with controlled dependence on the moment order

Lemma 40.3 (Connected-weight bound; manuscript 05). *If $\tau(H)$ is the number of spanning trees of a finite graph H , then $|\phi(H)| \leq \tau(H)$, with both sides interpreted as 1 for a singleton. Consequently, for realizable densities, a rod list of total edge length $k \geq 1$ satisfies*

$$|F_{\ell,h}(\mathbf{a})| \leq \frac{k^{3h}}{h!}. \quad (40.6)$$

Proof (manuscript 05). For completeness, the first inequality follows by deletion and contraction. Let $w(H) = (-1)^{|V(H)|-1}\phi(H)$ for connected graphs and $w(H) = 0$ for disconnected graphs with at least two vertices. Separating connected spanning subgraphs according as they contain an edge e gives $w(H) = w(H - e) + w(H/e)$. Parallel edges after contraction may be simplified: the alternating sum over a nonempty subset of parallel edges is -1 , the same as for one edge. Induction gives $w(H) \geq 0$. The spanning-tree recurrence for the contraction with multiplicities, and the fact that simplification cannot increase its tree count, then give $w(H) \leq \tau(H)$. Trees and singleton graphs start the induction.

For the second inequality, bound the connected weight for each block by a sum over spanning trees. The resulting collection across blocks is a forest with h edges on the c rod labels. Root each tree. Its root rod has at most n placements. Given a parent's placement, a child has at most $\ell_i + \ell_j + 1 \leq 2k$ placements overlapping it. Hence a specified forest has at most $n^{c-h}(2k)^h$ compatible placements. There are at most $\binom{c}{h}$ such forests. Divide by n^{c-h} , as in (40.4), to obtain

$$|F_{\ell,h}(\mathbf{a})| \leq (2k)^h \binom{c}{h} \leq \frac{k^{3h}}{h!},$$

since $c \leq k$. The argument bounds sums of absolute weights, so cancellation presents no problem. $\square$

Remark 40.4 (Penrose's tree-graph inequality). [Added 4 October 2026, batch 97; a note of the write.] The first inequality of lemma 40.3, $|\phi(H)| \leq \tau(H)$ for the alternating sum (40.1) over connected spanning subgraphs, is Penrose's tree-graph inequality [21], a classical tool of cluster expansions with hard-core interactions. Manuscript 05 proves it by deletion and contraction but does not cite it, and its source ledger does not list it. Only the second inequality, the bound (40.6) on the merger polynomials, is specific to Part IV.

41 A finite formula for every correction polynomial

Put

$$D_v(x) = \prod_{i=0}^{v-1} (1 - ix), \quad d_j(v) = [x^j] D_v(x)^{-1}. \quad (41.1)$$

For fixed j , $d_j(v)$ is a rational polynomial in v of degree at most $2j$. This follows by expanding $\log D_v(x)^{-1} = \sum_{m \geq 1} x^m \sum_{i=0}^{v-1} i^m / m$; each power sum is polynomial of degree $m + 1$. Also, for $v \leq 2k$,

$$0 \leq d_j(v) \leq (2k^2)^j, \quad (41.2)$$

because a complete homogeneous polynomial is bounded by the corresponding power of the sum of its nonnegative arguments.

Substitution of (40.5) into (39.2) shows that a profile contributes

$$\frac{\theta^c u^k}{\mathbf{m}!} x^d \frac{F_{\ell}(\mathbf{a}, x) F_{\ell}(\mathbf{b}, x)}{D_{k+c}(x)}, \quad x = 1/n. \quad (41.3)$$

The total cost of a term is therefore

$$J = d + h + h' + j : \quad \text{intrinsic defect, source mergers, target mergers, denominator order.} \quad (41.4)$$

For any fixed $J, k \geq 0$ define the finite polynomial

$$R_{J,k}(\mathbf{a}, \mathbf{b}) = \sum_{\mathbf{m}: \sum \ell m_\ell = k} \frac{\theta^c}{\mathbf{m}!} \sum_{d+h+h'+j=J} F_{\ell,h}(\mathbf{a}) F_{\ell,h'}(\mathbf{b}) d_j(k+c). \quad (41.5)$$

The empty profile supplies $R_{0,0} = 1$ and $R_{J,0} = 0$ for $J > 0$. The expression is independent of the chosen rod labels.

Theorem 41.1 (Finite coefficient extraction and locality; manuscript 05). *The formal series*

$$e^{-\lambda u} \sum_{k \geq 0} R_{J,k}(\mathbf{a}, \mathbf{b}) u^k \quad (41.6)$$

is a polynomial of degree at most $2J$. It is B_J in [theorem 38.1](#), and it has the finite formula

$$B_J(u; \mathbf{a}, \mathbf{b}) = \sum_{q=0}^{2J} u^q \sum_{k=0}^q \frac{(-\lambda)^{q-k}}{(q-k)!} R_{J,k}(\mathbf{a}, \mathbf{b}). \quad (41.7)$$

Only the densities a_q, b_q with $q \leq J+1$ occur.

Proof (manuscript 05). Fix a term in (41.5) with cost (41.4). Call rods with $\ell \geq 2$ long rods. There are at most d long rods, and their total edge length is at most $2d$, because $\ell \leq 2(\ell-1)$ for $\ell \geq 2$. A partition with h mergers has at most $2h$ labels in nonsingleton blocks: each block of size $m \geq 2$ satisfies $m \leq 2(m-1)$. The source and target partitions thus involve at most $2(h+h')$ rod labels in their nonsingleton blocks.

Mark all long rods and all unit rods appearing in a nonsingleton block of either partition. Once the finite marked pattern has been fixed, every unmarked unit rod is a singleton in both partitions and contributes $\theta a_1 b_1 u = \lambda u$. Summing an arbitrary number t of these unmarked rods with the factorial normalization gives $e^{\lambda u}$. The only remaining dependence on t is polynomial: the denominator coefficient $d_j(k+c)$ has degree at most $2j$ in t . Summing $t^m (\lambda u)^t / t!$ gives $(\lambda u)^m e^{\lambda u}$, so this dependence contributes at most $2j$ additional powers of u .

More explicitly, summing over choices of marked labels converts their multiplicities to falling factorials; division by the original rod factorials leaves the factorial of the unmarked count. Taking the union of source-marked and target-marked unit rods avoids any division by a_1 or b_1 . The argument is therefore a polynomial identity, including when either leading edge density vanishes. The remaining u -degree is at most $2d+2h+2h'+2j = 2J$. This proves the polynomial assertion and then the finite extraction formula.

For locality, consider a connected source block with m rods, intrinsic defect d_B , and total edge length $k_B = d_B + m$. Its local polynomial uses a_q only for

$$q \leq k_B = d_B + (m-1) + 1 \leq d + h + 1 \leq J + 1.$$

The same argument applies to target blocks. The denominator introduces no new densities. Finally, swapping forests leaves (41.5) unchanged, and the u^0 coefficient vanishes for $J > 0$. $\square$

Remark 41.2 (What the finite formula does and does not say; manuscript 05). The intermediate moments in (41.7) go as far as $k = 2J$, and a straightforward computation may temporarily use local symbols through that weight. The theorem proves cancellation of symbols beyond $J+1$ in the final answer. It is not necessary to assume this cancellation when running the coefficient engine.

42 Uniform remainder estimates: proof of the main theorem

Formal extraction alone would not justify avoidance at $u = -1$. We now control both the Taylor remainder for low moments and the tail of all higher moments.

Lemma 42.1 (Low-moment Taylor estimate; manuscript 05). *For every fixed $M \geq 0$ there is a constant C_M such that, for $k \geq 1$ and $4k^3 \leq n$,*

$$\left| \mu_k - \sum_{J=0}^M n^{-J} R_{J,k}(\mathbf{a}, \mathbf{b}) \right| \leq C_M \frac{\theta^k (1+k)^{3M+3}}{k!} n^{-M-1}. \quad (42.1)$$

For every fixed J ,

$$|R_{J,k}(\mathbf{a}, \mathbf{b})| \leq C_J \frac{\theta^k (1+k)^{3J}}{k!} \quad (42.2)$$

on realizable densities, with constants independent of the forests.

Proof (manuscript 05). We give the estimates explicitly enough to identify the uniformity. By lemma 40.3 and (41.2), the absolute coefficient series of the kernel in (41.3), after removing $x^d \theta^c / \mathbf{m}!$, is bounded coefficientwise by

$$\frac{\exp(2k^3 x)}{1 - 2k^2 x}. \quad (42.3)$$

For $0 \leq x \leq 1/(4k^3)$ its tail above degree m is at most $C_m (k^3 x)^{m+1}$. For example, expand the exponential and geometric series; at total degree p their coefficient is at most $C(2k^3)^p$ times a fixed harmless constant, and the remaining geometric ratio is at most $1/2$. Equivalently one may bound the two Taylor tails separately. Constants here may depend on the fixed cutoff m , not on k or the forest.

The profile multiplicity sum at given k and $d = k - c$ is

$$\sum_{\substack{\mathbf{m}: \sum \ell m_\ell = k \\ \sum m_\ell = k-d}} \frac{1}{\mathbf{m}!} = \frac{\binom{k-1}{d}}{(k-d)!} \leq \frac{k^{2d}}{d! k!}. \quad (42.4)$$

The equality counts ordered compositions of k into $k-d$ positive parts and divides by $(k-d)!$. The inequality follows from $\binom{k-1}{d} \leq k^d/d!$ and $k!/(k-d)! \leq k^d$.

For $d \leq M$, use the kernel remainder after order $M-d$, multiply by x^d , and sum profiles using (42.4) and $\theta^c \leq \theta^k$. The result is at most

$$\frac{C_M \theta^k}{k!} x^{M+1} \sum_{d=0}^M \frac{k^{2d+3(M+1-d)}}{d!} \leq \frac{C'_M \theta^k (1+k)^{3M+3}}{k!} x^{M+1}.$$

For $d > M$, bound the whole kernel by (42.3); it is bounded by an absolute constant under $4k^3 x \leq 1$. Then the exponential tail in $k^2 x$ from (42.4) bounds these profiles by

$$\frac{C \theta^k}{k!} e^{k^2 x} \frac{(k^2 x)^{M+1}}{(M+1)!},$$

which is smaller than the required bound. For these low moments every profile has $v \leq 2k \leq n$, so no inadmissible denominator was introduced. This proves (42.1). The same argument at one fixed coefficient, rather than at a tail, gives (42.2). $\square$

Proof of theorem 38.1 (manuscript 05). For sufficiently large n , choose

$$K_n = \left\lfloor \frac{(M+3) \log n}{\log \log n} \right\rfloor. \quad (42.5)$$

Then $4K_n^3 \leq n$. Apply lemma 42.1, multiply by $|u|^k$, and sum over $k \leq K_n$. The series $\sum_k (\theta U)^k (1+k)^{3M+3}/k!$ converges, so the accumulated error is $O_{M,U}(n^{-M-1})$, with a forest-independent constant.

The actual moment tail is bounded by (39.4). Since $K_n \log K_n = (M+3 + o(1)) \log n$, this tail is $n^{-M-3+o(1)}$. The omitted tails of the finitely many coefficient series $\sum_k R_{J,k} u^k$ have the same factorial decay by (42.2). They can therefore be reinstated within the error $O_{M,U}(n^{-M-1})$. Finally, theorem 41.1 identifies each complete coefficient series as $e^{\lambda u} \mathbf{B}_J(u)$. All constants used above depend only on M, U , and the two possible values of θ . $\square$

42.1 A useful comparison consequence

Corollary 42.2 (Finite-locality comparison; manuscript 05). *If two forest pairs on n vertices have the same a_q, b_q for $1 \leq q \leq M+1$, then their probability generating functions and complete laws differ by $O_{M,U}(n^{-M-1})$ and $O_M(n^{-M-1})$ in the respective norms of (38.4) and (38.8). More generally, if the common leading parameter is fixed and every relevant density differs by at most ε , the coefficient differences are $O_M(\varepsilon)$.*

Proof (manuscript 05). Apply the theorem to both pairs. For the last assertion use bounded first derivatives of finitely many polynomials on the density cube. If the leading parameter itself differs, its exponential factor must also be included; it is not suppressed by a power of n . $\square$

43 Explicit first and second corrections

The first two orders depend on only six local densities. To keep the formulas legible, write

$$a = a_1, \quad c = a_2, \quad e = a_3, \quad b = b_1, \quad d = b_2, \quad f = b_3, \quad \lambda = \theta ab, \quad (43.1)$$

and introduce

$$\mathcal{A} = a + 2c, \quad \mathcal{B} = b + 2d, \quad \mathcal{R} = a + 6c + 3e, \quad \mathcal{S} = b + 6d + 3f. \quad (43.2)$$

The letter d in this section is the target density b_2 , not the profile defect of (39.1).

Define three symmetric polynomials

$$C = 2\lambda^2 - \frac{\theta^2}{2}(a^2\mathcal{B} + b^2\mathcal{A}) + \theta cd, \quad (43.3)$$

$$D = \frac{\theta^2}{2} \left[22a^2b^2 - 6(a^2\mathcal{B} + b^2\mathcal{A}) + \mathcal{A}\mathcal{B} \right] + 3\theta cd, \quad (43.4)$$

$$E = \theta^3 \left[\frac{28}{3}a^3b^3 - 4ab(a^2\mathcal{B} + b^2\mathcal{A}) + ab\mathcal{A}\mathcal{B} + \frac{a^3\mathcal{S} + b^3\mathcal{R}}{3} \right] + \theta^2 [6abcd - 2ac(d+f) - 2bd(c+e)] + \theta ef. \quad (43.5)$$

Proposition 43.1 (Two explicit orders; manuscript 05). *For both orientations,*

$$\mathbf{B}_1(u) = \lambda u + Cu^2, \quad (43.6)$$

$$\mathbf{B}_2(u) = \lambda u + (D + \lambda^2/2)u^2 + (E + \lambda C)u^3 + (C^2/2)u^4. \quad (43.7)$$

Equivalently, choosing the logarithm continuous from $u = 0$, on every fixed compact u -disk and for sufficiently large n ,

$$\log G(u) = \lambda u + \frac{\lambda u + Cu^2}{n} + \frac{\lambda u + Du^2 + Eu^3}{n^2} + O_U(n^{-3}). \quad (43.8)$$

Proof (manuscript 05). We give small-profile formulas that permit a direct check, rather than requiring trust in a symbolic output. In a forest with densities a, c, e , the counts of one, two, and three disjoint unit rods, normalized by n, n^2, n^3 , are respectively

$$a, \quad \frac{a^2}{2} - \frac{x\mathcal{A}}{2}, \quad \frac{a^3}{6} - \frac{xa\mathcal{A}}{2} + \frac{x^2\mathcal{R}}{3}, \quad x = 1/n. \quad (43.9)$$

For two unit rods the identity is $C_F(2, 0, \dots) = \binom{na}{2} - nc$. For three, inclusion–exclusion on their overlapping pairs gives the last expression. Alternatively it follows from the path-polynomial recurrence in [section 48.1](#). The normalized count of one unit rod and one two-edge rod is

$$ac - 2x(c + e), \quad (43.10)$$

because the two rods can overlap in a two-edge interval in two positions, or meet at an endpoint in a three-edge interval in two positions. A single three-edge rod has normalized count e .

For instance the exact second factorial moment is

$$\mu_2 = \frac{\theta^2(n^2a^2 - n\mathcal{A})(n^2b^2 - n\mathcal{B})}{2n^4} + \frac{\theta n^2cd}{n^3}. \quad (43.11)$$

The three terms for μ_3 correspond to three unit rods, a unit rod and a two-edge rod, and a single three-edge rod. Substitute (43.9)–(43.10) in (39.2), expand the denominators through order two, and remove $e^{\lambda u}$ as in (41.7). This gives (43.3)–(43.7). At order two the four-unit-rod contribution supplies the coefficient $C^2/2$; it follows either by the same finite extraction or by separating its two cost-one marked patterns. The shipped symbolic check independently performs the full extraction through the needed fourth moment, verifies every coefficient, and checks this quartic identity. Finally expand the logarithm of $1 + B_1/n + B_2/n^2$ to obtain (43.8). The uniform approximation to $e^{\lambda u}$ guarantees a nonzero generating function on each fixed disk once n is sufficiently large, so this logarithm exists there. $\square$

Remark 43.2 (What in [proposition 43.1](#) is computer-verified). [Added 4 October 2026, batch 97; a note of the write.] The printed proof supplies the small tiling counts (43.9), (43.10) and the exact second moment (43.11). From them $B_1 = \lambda u + Cu^2$ follows by hand; the intake recomputed it, and at the fixed-offset densities it reproduces (43.12) and Part I’s (1.10). The second-order polynomials D and E of (43.4)–(43.5) and the quartic coefficient $C^2/2$ of (43.7) are *not* derived in print: they rest on manuscript 05’s program `symbolic_checks.py` (shipped as `code/04-moving-gaps-symbolic_checks.py`). That program runs the finite extraction (41.7) with symbolic densities a, c, e, b, d, f and symbolic θ over every rod profile of defect at most two through the fourth moment, with the profile counts of the coefficient engine `local_expansions.py`, and asserts in exact SymPy arithmetic: the vanishing of the coefficients of B_1 and B_2 above degrees 2 and 4; the coefficients λ and C of B_1 ; the quartic coefficient $C^2/2$; $\partial B_2 / \partial a_3 = u^3 H_3^{(\theta)}(\mathbf{b})$, with no a_3 in the other coefficients; the compact forms of D and E ; and every coefficient of B_2 . It sets $a_4 = b_4 = 0$, which is no restriction because B_2 involves no density beyond a_3, b_3 ([theorem 41.1](#)). The engine’s profile counts are checked separately against independent finite path products (1618 assertions of [table 20](#)), and the standard-library verifier checks the first correction and the quartic identity at sixteen rational density vectors each. Thus (43.7), and the terms n^{-2} of (43.8), are exact computer-algebra identities under the proved extraction formula, not hand derivations. The batch-97 write reran the program on a copy (all assertions passed; see the report README); an independent hand derivation of D and E is not available.

43.1 Consistency with fixed gaps

For fixed position and value gaps r, s and fixed q , eventually $a_q = 1 - qr/n$ and $b_q = 1 - qs/n$. Inserting these n -dependent densities into the formula gives

$$\mathbb{P}(X = 0) = e^{-\theta} \left(1 + \frac{\theta(r + s - \theta)}{n} + O(n^{-2}) \right). \quad (43.12)$$

Indeed $\lambda = \theta - \theta(r + s)/n + O(n^{-2})$, whereas $B_1(-1)$ at $a = b = c = d = 1$ is $-\theta^2$. For $\theta = 2$ and $r = s$, (43.12) is the familiar $4(r - 1)/n$ correction in the fixed-gap setting [4, 20]. This is a consistency check, not the main new regime.

43.2 The matching regime

When $r, s \geq n/2$, both offset forests consist of isolated vertices and single edges. Let $m_P = n - r$ and $m_V = n - s$. Only unit rods are possible, so the whole finite law has factorial moments

$$\mu_k = \frac{\theta^k m_P^k m_V^k}{k! n^{2k}}, \quad 0 \leq k \leq \min(m_P, m_V), \quad (43.13)$$

and $\mu_k = 0$ thereafter. Thus

$$\mathbb{P}(X = 0) = \sum_{k=0}^{\min(m_P, m_V)} (-1)^k \mu_k. \quad (43.14)$$

This finite formula is also an independent way to test the moving-gap expansion at the boundary $r = s = n/2$.

44 The longest visible path: an exact sensitivity formula

The locality upper bound becomes more informative when one can determine whether the last permissible density actually occurs. For a formal target density sequence define

$$\mathcal{B}(t) = \sum_{\ell \geq 1} b_\ell t^\ell, \quad H_q^{(\theta)}(\mathbf{b}) = [t^q] \frac{\theta \mathcal{B}(t)}{1 + \theta \mathcal{B}(t)}. \quad (44.1)$$

Only $b_1, \dots, b_q$ matter. Equivalently,

$$H_q^{(\theta)}(\mathbf{b}) = \sum_{m=1}^q (-1)^{m-1} \theta^m \sum_{\substack{\ell_1 + \dots + \ell_m = q \\ \ell_i \geq 1}} b_{\ell_1} \cdots b_{\ell_m}. \quad (44.2)$$

The first examples are

$$\begin{aligned} H_2 &= \theta b_2 - \theta^2 b_1^2, \\ H_3 &= \theta b_3 - 2\theta^2 b_1 b_2 + \theta^3 b_1^3, \\ H_4 &= \theta b_4 - \theta^2 (2b_1 b_3 + b_2^2) + 3\theta^3 b_1^2 b_2 - \theta^4 b_1^4. \end{aligned} \quad (44.3)$$

Theorem 44.1 (Highest-path sensitivity; manuscript 05). *For every $q \geq 2$,*

$$\boxed{\frac{\partial B_{q-1}^{(\theta)}}{\partial a_q} = u^q H_q^{(\theta)}(\mathbf{b}), \quad \frac{\partial B_{q-1}^{(\theta)}}{\partial b_q} = u^q H_q^{(\theta)}(\mathbf{a}).} \quad (44.4)$$

In particular B_{q-1} is separately linear in a_q and b_q , and its mixed derivative with respect to them is θu^q .

Proof (manuscript 05). A source block whose cluster uses a_q must span a q -edge interval. By the locality proof its cost $d_B + (|B| - 1)$ is at least $q - 1$. At total order $q - 1$ this consumes the entire budget. Therefore every other rod is a unit rod isolated in both partitions; there are no target mergers and no positive-order denominator term. The distinguished source block has total rod edge length exactly q .

A connected collection of intervals whose union has edge length equal to the sum of their edge lengths has no repeated occupied edge. Its intervals meet only at endpoints and form a chain covering the union. If it has m rods, their left-to-right lengths form an ordered composition $\ell_1 + \dots + \ell_m = q$. The incompatibility graph is a path, hence its connected weight is $(-1)^{m-1}$. The target partition has singleton blocks, contributing $b_{\ell_1} \dots b_{\ell_m}$, and orientation contributes θ^m .

The labeling factorials convert the chain placements into precisely the ordered compositions in (44.2). Their u -weight is u^q . All remaining unit rods sum to $e^{\lambda u}$, which cancels in the definition of B_{q-1} . Because $q \geq 2$, λ is independent of a_q . This proves the first identity; symmetry proves the second. The only term of $H_q(\mathbf{b})$ containing b_q is θb_q , proving the mixed derivative assertion. $\square$

One can record the entire dependence on the last two density variables as

$$u^q [a_q H_q(\mathbf{b}) + b_q H_q(\mathbf{a}) - \theta a_q b_q], \quad (44.5)$$

plus a polynomial involving only densities with indices below q . This identity is particularly useful at a geometric boundary.

45 A fixed short path: the first shape-dependent term

We now answer the fixed-length part of the short-path question posed in [20]. The comparison keeps the total number of vertices and the numbers of paths fixed; it changes only their lengths.

Theorem 45.1 (Sharp absolute-model effect of one fixed short path; manuscript 05). *Fix integers $r \geq 2$, $s \geq 1$, and $L \geq 1$. For each n , let $\mathcal{P}_n^{\text{short}}$ have r paths, one with exactly L vertices and all the others with lengths tending to infinity. Let $\mathcal{P}_n^{\text{long}}$ have r paths, all with lengths tending to infinity. Both forests have n vertices. Let $\mathcal{V}_n$ have s paths, all with lengths tending to infinity, also on n vertices.*

In the absolute model,

$$G_{\text{short}}^{(2)}(u) - G_{\text{long}}^{(2)}(u) = \frac{2(-1)^L e^{2u} u^{L+1}}{n^{L+1}} + O_U(n^{-L-2}), \quad |u| \leq U. \quad (45.1)$$

In particular,

$$\mathbb{P}(X_{\text{short}}^{(2)} = 0) - \mathbb{P}(X_{\text{long}}^{(2)} = 0) = -\frac{2e^{-2}}{n^{L+1}} + O(n^{-L-2}). \quad (45.2)$$

In the directed model the difference of generating functions is instead $O_U(n^{-L-2})$; the displayed leading absolute-model order cancels.

Proof (manuscript 05). For each fixed q , once all growing paths exceed q vertices,

$$a_q^{\text{short}} - a_q^{\text{long}} = \frac{(q-L)_+}{n}, \quad a_q^{\text{long}} = 1 - \frac{rq}{n}, \quad b_q = 1 - \frac{sq}{n}. \quad (45.3)$$

For $q \leq L$ the two source densities coincide. Their first difference is $1/n$ at $q = L + 1$. Their leading parameters λ are equal exactly because the number of edges is the same.

Set $q = L + 1$. The terms B_j for $j < q - 1$ agree by locality. At $j = q - 1$, [theorem 44.1](#) gives the exact difference $u^q H_q^{(\theta)}(\mathbf{b})/n$. At any higher fixed order the polynomial coefficient difference is $O(1/n)$ by [\(45.3\)](#). Using the main theorem through order q gives

$$G_{\text{short}}^{(\theta)}(u) - G_{\text{long}}^{(\theta)}(u) = n^{-q} e^{\lambda u} u^q H_q^{(\theta)}(\mathbf{b}) + O_U(n^{-q-1}). \quad (45.4)$$

All finite target densities are $1 + O(1/n)$. At the constant density sequence $\mathbf{b} = (1, 1, \dots)$,

$$\frac{\theta \mathcal{B}(t)}{1 + \theta \mathcal{B}(t)} = \frac{\theta t}{1 + (\theta - 1)t}, \quad H_q^{(\theta)}(1, 1, \dots) = \theta(1 - \theta)^{q-1}. \quad (45.5)$$

Also $\lambda = \theta + O(1/n)$. For $\theta = 2$ this proves [\(45.1\)](#); substitution $u = -1$ gives the negative coefficient in [\(45.2\)](#). For $\theta = 1$ and $q \geq 2$, [\(45.5\)](#) vanishes, leaving the stated smaller error. $\square$

Remark 45.2 (Multiplicity and scope; manuscript 05). If exactly h short paths have the same minimal fixed vertex length L , while all other paths are longer than L and the remaining finite local densities have the corresponding fixed-count behavior, their contribution at the first order is h times [\(45.1\)](#). Paths of larger fixed length first enter later. The theorem is proved for fixed L ; its error constants are not controlled as $L \rightarrow \infty$. It therefore does not settle the logarithmically growing short-path question by substituting $L = L(n)$.

Remark 45.3 (The preceding remark, made precise). [Added 4 October 2026, batch 97; the precise hypothesis and the argument are the write's.] The preceding remark leaves its hypothesis on “the remaining finite local densities” informal. A precise version: fix integers $s \geq 1$, $L \geq 1$, a finite set I_0 of integers $\geq L$ containing L , and multiplicities $h_\ell \geq 1$ ($\ell \in I_0$) with $r > \sum_\ell h_\ell$. Let $\mathcal{P}_n^{\text{short}}$ have r paths on n vertices, exactly h_ℓ of them with ℓ vertices for each $\ell \in I_0$ and the others with lengths tending to infinity, and let $\mathcal{P}_n^{\text{long}}$ and $\mathcal{V}_n$ be as in [theorem 45.1](#). Then for every fixed q , eventually

$$a_q^{\text{short}} - a_q^{\text{long}} = \frac{1}{n} \sum_{\ell \in I_0} h_\ell (q - \ell)_+,$$

so the densities agree for $q \leq L$, differ by exactly h_L/n at $q = L + 1$, and differ by $O(1/n)$ at every fixed q ; both forests have $n - r$ edges, so λ is the same. The proof of [theorem 45.1](#) then applies verbatim: B_j agree for $j < L$, the affine dependence of B_L on a_{L+1} ([theorem 44.1](#)) gives the difference $(h_L/n)u^{L+1}H_{L+1}^{(\theta)}(\mathbf{b})$, and every later B_j , a polynomial with bounded gradient on the density cube, changes by $O(1/n)$. Hence

$$G_{\text{short}}^{(\theta)}(u) - G_{\text{long}}^{(\theta)}(u) = h_L n^{-L-1} e^{\lambda u} u^{L+1} H_{L+1}^{(\theta)}(\mathbf{b}) + O_U(n^{-L-2}),$$

which in the absolute model is h_L times [\(45.1\)](#), with avoidance difference $-2h_L e^{-2} n^{-L-1} + O(n^{-L-2})$, and in the directed model is $O_U(n^{-L-2})$. Paths of the larger lengths in I_0 do not enter at this order.

Remark 45.4 (Fixed-offset coefficients survive down to paths of fixed length). [Added 4 October 2026, batch 97; this consequence of [theorems 38.1](#) and [45.1](#) is the write's, not manuscript 05's.] Fix θ , r , s and an order $M \geq 1$. Let $\mathcal{P}_n$ have r paths and $\mathcal{V}_n$ have s paths on n vertices, every path having at least M vertices. Then, uniformly on $|u| \leq U$,

$$G_{\mathcal{P}_n, \mathcal{V}_n}^{(\theta)}(u) = e^{\theta u} \sum_{J=0}^M \frac{B_J^{(\theta)}(u; r, s)}{n^J} + O_{M,U,r,s}(n^{-M-1}),$$

with Part I's fixed-offset polynomials. Indeed, compare with the offset forests of gaps r and s on $[n]$, for which this is [theorem 1.1](#). For $q \leq M$ every path of either pair has at least q vertices, so $S_q = n - qr$ (respectively $n - qs$) for both pairs; at $q = M + 1$ the subpath counts differ by

at most r (respectively s); the numbers of edges, hence λ , coincide. By [theorem 38.1](#) with this M , the two generating functions differ by $e^{\lambda u} n^{-M} (\mathbf{B}_M(u; \mathbf{a}, \mathbf{b}) - \mathbf{B}_M(u; \mathbf{a}', \mathbf{b}')) + O(n^{-M-1})$, and the bracket is $O(1/n)$. The length M cannot be lowered to $M - 1$ in the absolute model when $M \geq 2$: by [theorem 45.1](#) with $L = M - 1$, one path of $M - 1$ vertices changes the generating function by $2(-1)^{M-1} e^{2u} u^M n^{-M} + O(n^{-M-1})$. So Part II's sufficient condition (16.9), a shortest path long compared with $\log n / \log \log n$, is far from necessary at each fixed order: a fixed length M suffices at order M , on every disk. The growing regime of Part II's question 2 is a different problem, since a fixed statement for every M at once needs constants uniform in M ([question 50.3](#)).

45.1 Exact directed shape blindness with one target path

Proposition 45.5 (A one-path identity; manuscript 05). *If $\mathcal{V}$ is one directed path and $\mathcal{P}$ has r paths, then in the directed model*

$$\mu_k = \binom{n-r}{k} \frac{(n-k)!}{n!}. \quad (45.6)$$

Thus the complete law depends on the source forest only through its edge count $n - r$, and two such forests with the same r have exactly equal laws, not merely equal asymptotic expansions.

Proof (manuscript 05). Fix any k selected source edges, with c components and $v = k + c$ vertices. The number of labeled disjoint placements of their rods in one directed target path is $n - k^c$: contract each rod to a tile and order the labeled rods among the other tiles. After mapping the unused vertices, the number of bijections is $n - k^c(n - v)! = (n - k)!$, independent of the component profile. There are $\binom{n-r}{k}$ selected source edge sets. Divide by $n!$. $\square$

The underlying one-path inclusion–exclusion mechanism is part of the classical fixed-gap framework and appears in the repository's one-path family. The formulation for an arbitrary source shape makes clear why some proposed “universal short-path effects” must fail in the directed model.

For reproducible absolute-model comparisons one can similarly use

$$\mu_k = \frac{(n-k)!}{n!} [t^k] \prod_{L \in \mathcal{P}} R_{L-1}(t; \theta), \quad (45.7)$$

where R_m enumerates subsets of a row of m edges, with a factor θ for each nonempty run. A two-state recurrence, distinguishing whether the last edge is selected, gives

$$R_0 = 1, \quad R_1 = 1 + \theta t, \quad R_m = (1 + t)R_{m-1} + (\theta - 1)tR_{m-2} \quad (m \geq 2). \quad (45.8)$$

Equation (45.7) follows from the preceding proof with the additional factor θ^c for a selected set having c runs. It supplies the independent rational short-path certificates in [section 48](#).

46 Moving gaps and reciprocal-integer correction boundaries

For offsets $1 \leq r, s \leq n$ define

$$\begin{aligned} X_{r,s,n}^{(1)} &= \sum_{i=1}^{n-r} \mathbf{1}\{\pi(i+r) - \pi(i) = s\}, \\ X_{r,s,n}^{(2)} &= \sum_{i=1}^{n-r} \mathbf{1}\{|\pi(i+r) - \pi(i)| = s\}. \end{aligned} \quad (46.1)$$

The subpath counts are particularly simple:

$$S_q(\mathcal{P}) = \max(n - qr, 0), \quad S_q(\mathcal{V}) = \max(n - qs, 0). \quad (46.2)$$

Indeed a q -edge source interval starts at an integer i with $i + qr \leq n$.

Corollary 46.1 (Arbitrary moving offsets; manuscript 05). *The all-orders expansion (38.4) holds uniformly for all $1 \leq r, s \leq n$, after the exact substitutions*

$$a_q = \left(1 - q\frac{r}{n}\right)_+, \quad b_q = \left(1 - q\frac{s}{n}\right)_+, \quad \lambda_n = \theta \left(1 - \frac{r}{n}\right) \left(1 - \frac{s}{n}\right). \quad (46.3)$$

If $r/n \rightarrow \alpha$ and $s/n \rightarrow \beta$, then the law converges to $\text{Poisson}(\theta(1 - \alpha)(1 - \beta))$. The error relative to the Poisson law with parameter λ_n is uniformly $O(1/n)$, regardless of the rates of convergence.

This includes fixed offsets, offsets growing slower than n , proportional offsets, and offsets tending to n . The limiting parameter alone does not encode the correction terms.

For real $\alpha, \beta \in [0, 1]$, regard $a_q(\alpha) = (1 - q\alpha)_+$ and $b_q(\beta) = (1 - q\beta)_+$ as macroscopic density coordinates. At order j , $B_j(u; \alpha, \beta)$ is piecewise polynomial and can have new boundaries only at $\alpha = 1/q$ or $\beta = 1/q$, with $2 \leq q \leq j + 1$.

Theorem 46.2 (First appearance and jump at a boundary; manuscript 05). *Fix $q \geq 2$ and a target ratio β . No coefficient of order below $n^{-(q-1)}$ has a boundary at $\alpha = 1/q$. The right derivative minus the left derivative of the coefficient B_{q-1} there is*

$$qu^q H_q^{(\theta)}(\mathbf{b}(\beta)). \quad (46.4)$$

For the corresponding coefficient of the avoidance probability, including its $e^{-\lambda}$ factor, the derivative jump is

$$q(-1)^q e^{-\lambda} H_q^{(\theta)}(\mathbf{b}(\beta)), \quad \lambda = \theta(1 - 1/q)(1 - \beta). \quad (46.5)$$

Proof (manuscript 05). For indices below q , the functions $(1 - \ell\alpha)_+$ are affine in a neighborhood of $1/q$. The new density a_q has derivative $-q$ on the left and 0 on the right, and is continuous at the boundary. Locality excludes it from all earlier coefficients. Apply [theorem 44.1](#) to the first coefficient where it can appear. The exponential factor is smooth in α , and the coefficient itself is continuous, so its derivative jump is obtained by multiplying by $e^{-\lambda}$. $\square$

These boundaries are not singularities of the limiting Poisson law. They are kinks in successively smaller finite-size correction terms. That distinction prevents an artificial claim of a macroscopic phase transition where the limiting distribution remains smooth.

46.1 Sharpness at every order

Proposition 46.3 (A nonvanishing family; manuscript 05). *In the directed model, for $q \geq 2$ and $0 < \beta \leq 1/q$,*

$$H_q^{(1)}(\mathbf{b}(\beta)) = -\beta^2(1 + \beta)^{q-2} \neq 0. \quad (46.6)$$

For each $\theta \in \{1, 2\}$, the locality cutoff $M + 1$ in [theorem 38.1](#) cannot in general be decreased to M .

Proof (manuscript 05). Through degree q , $b_\ell = 1 - \ell\beta$. Thus for the coefficient in question we may replace $\mathcal{B}(t)$ by

$$\frac{t}{1 - t} - \frac{\beta t}{(1 - t)^2}.$$

A direct simplification gives

$$\frac{\mathcal{B}(t)}{1 + \mathcal{B}(t)} = \frac{t(1 - \beta - t)}{1 - (1 + \beta)t} \quad \text{through degree } q. \quad (46.7)$$

For $q \geq 2$ its coefficient is $(1 - \beta)(1 + \beta)^{q-1} - (1 + \beta)^{q-2}$, equal to (46.6). For a stronger, information-theoretic form of sharpness valid in both models, first note that a matching target with $b_1 = 1/2$ and $b_j = 0$ for $j \geq 2$ gives $H_q^{(\theta)} = (-1)^{q-1} \theta^q / 2^q \neq 0$. Compare forests made from repeated two-path blocks of vertex lengths $(q - 1, q + 1)$ and (q, q) . Each block has $2q$ vertices, and both have $S_j = 2q - 2j$ for $1 \leq j \leq q - 1$, but their S_q values are 1 and 0. Thus the normalized lower densities coincide exactly, while their a_q values differ by $1/(2q)$. Use this common matching target, and let n run through multiples of $2q$. The sensitivity identity gives a nonzero difference $u^q H_q(\mathbf{b})/(2q)$ between their order- $q - 1$ correction polynomials. No rule using only the lower densities can distinguish these two pairs. In particular the cutoff is sharp even among actual path forests. $\square$

46.2 An exact cancellation at an irrational ratio

Nonvanishing is generic, not universal. In the directed model, take $q = 3$ and $1/3 < \beta < 1/2$. Then $b_3 = 0$ and

$$H_3^{(1)} = b_3 - 2b_1b_2 + b_1^3 = (1 - \beta)(\beta^2 + 2\beta - 1). \quad (46.8)$$

Hence

$$\boxed{\beta = \sqrt{2} - 1} \quad (46.9)$$

annuls the $\alpha = 1/3$ boundary in the *entire* second correction polynomial $B_2(u)$, not only in the avoidance specialization $u = -1$. For this fixed macroscopic target ratio, B_2 is independent of a_3 and is smooth across $\alpha = 1/3$.

This does not imply cancellation at every later order. Nor can an integer offset ratio equal this irrational value exactly: with $s = \lfloor (\sqrt{2} - 1)n \rfloor$, the discrepancy in the ratio is $O(1/n)$ and can reintroduce an effect one order later. Determining the next surviving boundary is an explicit further question. [Note of the batch-97 write: “can reintroduce an effect one order later” is a conjectural remark, not a proved statement; it is recorded with [question 50.2](#) in [section 50](#).]

47 Floor effects and critical windows

A uniform formula can be weakened accidentally by replacing a moving parameter with its limit too early. Let $r_n = \lfloor \alpha n \rfloor$ and $s_n = \lfloor \beta n \rfloor$. Exactly,

$$a_q(n) = \left(1 - q\alpha + \frac{q\{\alpha n\}}{n}\right)_+, \quad b_q(n) = \left(1 - q\beta + \frac{q\{\beta n\}}{n}\right)_+. \quad (47.1)$$

The fractional parts already influence ordinary inverse powers of n . Thus a constant-coefficient expansion based only on α, β is not generally correct beyond leading order.

For rational ratios, restricting n to one residue class modulo a common denominator makes these phases constant. Restricting to multiples of that denominator makes them zero. One then obtains ordinary constant-coefficient expansions, to which [section 49](#) applies. For irrational ratios the bounded phases need not converge; the exact uniform expression (38.4) remains valid with the densities in (47.1). No assertion that the resulting coefficients converge is needed.

At a reciprocal boundary $q\alpha = 1$, the density becomes

$$a_q(n) = \frac{q\{\alpha n\}}{n}. \quad (47.2)$$

Since it first enters B_{q-1} , this boundary-floor effect first contributes at order n^{-q} , one order after the macroscopic kink. More generally, in a critical window $r_n/n = 1/q + \zeta/n$,

$$a_q(n) = \frac{(-q\zeta)_+}{n}. \quad (47.3)$$

The same sensitivity polynomial H_q gives its leading contribution. There is no omitted smoothing layer in this statement: the positive-part formula is exact for the discrete path count. Any additional dependence of ζ on the integer rounding should be retained explicitly.

Remark 47.1 (Why fixed-gap substitution can look plausible and still fail; manuscript 05). For small fixed q , the identity $S_q = n - qr$ holds only while $qr \leq n$. The first order at which replacing it by an unrestricted linear polynomial can fail is exactly the order predicted by [theorem 44.1](#). A formal substitution of $r \asymp n$ into a long-path expression may therefore reproduce a few early terms and then become wrong at a specific, geometrically determined order.

48 Exact coefficient generation and certified finite checks

48.1 A path-polynomial implementation

The cluster proof supplies a finite algorithm, but direct enumeration of all labeled connected graphs is unnecessary. Introduce variables z_ℓ marking rods of edge length ℓ , and let $T_N(\mathbf{z})$ be the rod-tiling polynomial for one path on N vertices. Then

$$T_0 = T_1 = 1, \quad T_N = T_{N-1} + \sum_{\ell=1}^{N-1} z_\ell T_{N-\ell-1}. \quad (48.1)$$

The alternatives are an unused first vertex or a rod beginning there. For a forest, $T_F = \prod_{L \in F} T_L$. Define formal local logarithms

$$L_q(\mathbf{z}) = \log T_{q+1}(\mathbf{z}) - 2 \log T_q(\mathbf{z}) + \log T_{q-1}(\mathbf{z}). \quad (48.2)$$

The logarithm here is its formal expansion at constant term 1.

Proposition 48.1 (Exact logarithmic local decomposition; manuscript 05). *One has the formal identity*

$$\log T_F(\mathbf{z}) = \sum_{q \geq 1} S_q(F) L_q(\mathbf{z}). \quad (48.3)$$

If z_ℓ is given weight ℓ , L_q contains no term of weighted degree below q .

Proof (manuscript 05). For a single path the identity is discrete twice-summation of (48.2): $\log T_N = \sum_{q=1}^{N-1} (N-q) L_q$, because $\log T_0 = \log T_1 = 0$. Sum over components to obtain (48.3).

For the weight assertion, the connected-graph expansion of the logarithm is the connected part of the graph expansion in [proposition 40.2](#). A connected rod configuration of total edge weight k spans an interval of at most k edges, by [lemma 40.1](#). For a fixed profile, its coefficient in $\log T_N$ is therefore a linear combination of $(N-p)_+$ with $p \leq k$. The second difference in (48.2) vanishes if $q > k$. All formal operations at fixed weight are finite. $\square$

Writing $x = 1/n$, define

$$\bar{C}_a(\mathbf{m}, x) = [\mathbf{z}^{\mathbf{m}}] \exp \left(x^{-1} \sum_{q \geq 1} a_q L_q(xz_1, xz_2, \dots) \right). \quad (48.4)$$

For actual forest densities this equals $n^{-c}C_F(\mathbf{m})$ at $x = 1/n$. A logarithmic monomial containing c rods acquires a factor x^{c-1} , which is the merger cost. In terms of the preceding construction, $F_\ell(\mathbf{a}, x) = \mathbf{m}! \overline{C}_\mathbf{a}(\mathbf{m}, x)$. The contribution to a factorial moment is therefore

$$\theta^c \mathbf{m}! x^{k-c} \frac{\overline{C}_\mathbf{a}(\mathbf{m}, x) \overline{C}_\mathbf{b}(\mathbf{m}, x)}{D_{k+c}(x)}. \quad (48.5)$$

To compute $B_0, \dots, B_M$, retain rod weight through $2M$ and powers of x through M , then apply (41.7).

The standard-library engine `code/local_expansions.py` implements these identities with sparse polynomial dictionaries and `fractions.Fraction`; no floating point enters this calculation. It also has an independent product-of-path-polynomials enumerator for finite forest counts. [The engine is shipped as `code/04-moving-gaps-local_expansions.py`.] For illustration, the initial local logarithmic coefficients are

$$\begin{aligned} [z_1] \sum_q a_q L_q &= a_1, & [z_1^2] \sum_q a_q L_q &= -a_1/2 - a_2, \\ [z_1^3] \sum_q a_q L_q &= a_1/3 + 2a_2 + a_3, & [z_2] \sum_q a_q L_q &= a_2, \\ [z_1 z_2] \sum_q a_q L_q &= -2a_2 - 2a_3. \end{aligned} \quad (48.6)$$

These reproduce the small-profile formulas in section 43.

48.2 Coefficients along rational rays

Let $r = \alpha n$ and $s = \beta n$ be integers, and write $c_j = B_j(-1; \mathbf{a}(\alpha), \mathbf{b}(\beta))$. Then

$$\frac{\#\{\pi : X(\pi) = 0\}}{n!} = e^{-\lambda} \left(\sum_{j=0}^M \frac{c_j}{n^j} + O_M(n^{-M-1}) \right), \quad c_0 = 1. \quad (48.7)$$

Tables 16 and 17 give exact rational coefficients, not fitted numerical estimates. The supplementary JSON (`data/04-moving-gaps-coefficients.json`) also records all coefficients of the full polynomials $B_j(u)$ through order six for the rays $(1/2, 1/2)$, $(1/3, 1/2)$, and $(2/5, 1/2)$, in both models.

j	c_j , directed ($\lambda = 1/4$)	c_j , absolute ($\lambda = 1/2$)
0	1	1
1	$-1/4$	$-1/2$
2	$-5/32$	$-1/8$
3	$11/128$	$11/16$
4	$1299/2048$	$235/128$
5	$12289/8192$	$129/256$
6	$60271/65536$	$-18117/1024$

Table 16: The ray $r = s = n/2$, for even n .

j	c_j , directed ($\lambda = 1/3$)	c_j , absolute ($\lambda = 2/3$)
0	1	1
1	$-7/18$	$-8/9$
2	$-77/216$	$-50/81$
3	$-7099/34992$	$1280/2187$
4	$1276321/2519424$	$109052/19683$
5	$30642947/8398080$	$20579656/885735$
6	$405923643097/24488801280$	$1566255886/23914845$

Table 17: The ray $(r, s) = (n/3, n/2)$, for $6 \mid n$.

48.3 Rational Bonferroni enclosures

For any nonnegative integer-valued X the alternating factorial-moment sums obey

$$\sum_{k=0}^{2K+1} (-1)^k \mu_k \leq \mathbb{P}(X = 0) \leq \sum_{k=0}^{2K} (-1)^k \mu_k. \quad (48.8)$$

For a direct proof, when $X = j \geq 1$ the truncated binomial sum is $(-1)^m \binom{j-1}{m}$ at cutoff m ; when $j = 0$ it is 1. Taking expectations gives the inequalities.

When the target forest is a matching, only source matchings contribute. If $M_k(\mathcal{P})$ counts k disjoint source edges, then

$$\mu_k = \frac{\theta^k M_k(\mathcal{P}) \binom{n-s}{k} k!}{n^{2k}}. \quad (48.9)$$

For a source path of L vertices its matching polynomial has coefficient $\binom{L-k}{k}$; multiplying these polynomials computes M_k exactly. All three numerical rays above have a matching target, so the program obtains exact rational lower and upper avoidance probabilities using cutoffs 31 and 30 in (48.8).

The normalization e^λ is also enclosed rationally. For $0 \leq z < 102$, put

$$E_{100}(z) = \sum_{j=0}^{100} \frac{z^j}{j!}. \quad \text{Then} \quad E_{100}(z) \leq e^z \leq E_{100}(z) + \frac{z^{101}/101!}{1 - z/102}. \quad (48.10)$$

The upper bound is the geometric bound on the positive Taylor tail. The printed decimal endpoints below are rounded outward from the resulting rational endpoints.

n	certified lower endpoint	certified upper endpoint
60	64.934296625701	64.934296625702
120	63.631305378671	63.631305378672
240	62.959011351875	62.959011351876
480	62.619451789834	62.619451789835

Table 18: Enclosures for $n^7(e^{1/3}\mathbb{P}(X=0) - \sum_{j=0}^6 c_j n^{-j})$ on the directed ray $(\alpha, \beta) = (1/3, 1/2)$. The rational endpoints, not just these decimals, are stored in `data/certificates.json` (shipped as `data/04-moving-gaps-certificates.json`).

The bounded scaled residual in [table 18](#) is consistent with the order-six theorem. These four intervals are not themselves a proof of an asymptotic remainder for all n ; that proof is [section 42](#).

For the short-path theorem, compare source lengths $(L, n - L)$ with $(n/2, n/2)$, and take a single target path. Use (45.7) to compute moments exactly and Bonferroni cutoffs 41 and 40. Subtract the two rational intervals and multiply by a rational enclosure of e^2 , reversing the appropriate endpoints for a negative interval. The result is table 19.

L	n	certified lower endpoint	certified upper endpoint
1	60	-2.179917857432	-2.179917857431
1	480	-2.021026163752	-2.021026163751
2	60	-2.338617980137	-2.338617980136
2	480	-2.038045558227	-2.038045558226
3	60	-2.556277235149	-2.556277235148
3	480	-2.059553402809	-2.059553402808

Table 19: Absolute-model enclosures for $n^{L+1}e^2(\mathbb{P}(X_{\text{short}} = 0) - \mathbb{P}(X_{\text{long}} = 0))$. The proved limit is -2 . The CSV (`data/04-moving-gaps-short_path_certificates.csv`) records all twelve tested pairs, also including $n = 120, 240$.

48.4 What was actually checked

The delivered run of `code/verify.py` (shipped as `code/04-moving-gaps-verify.py`) completed 3,209 exact assertions, recorded in `data/04-moving-gaps-verification.txt`. The breakdown is printed in table 20. Its permutation enumeration covers every $n = 2, \dots, 7$, every $1 \leq r, s < n$, and both orientations. The tiling checks range over every integer partition of $n \leq 9$, at rod weight at most five. The symbolic identities in section 43 are independently checked by `code/symbolic_checks.py` using SymPy (Remark 43.2). The standard-library verification script does not require SymPy.

Check category	Assertions
Local-log tilings versus independent finite path products	1618
Factorial moments versus exhaustive permutation histograms	1064
Avoidance versus the same exhaustive histograms	182
OEIS initial values, A189281 and A110128	12
First correction and second-order quartic identity	32
Forest-exchange symmetry and order-two locality	32
Highest-path sensitivity in exact rational specializations	108
Nonzero boundary family, $q = 2, \dots, 12$	11
Bonferroni and scaled-residual endpoint ordering	48
Strict sign of short-path intervals	12
Exact directed shape blindness with one target path	90
Total	3209

Table 20: Finite assertions recorded in `data/verification.txt`. They support, but do not replace, the infinite proofs.

The OEIS checks use only twelve initial values at $n = 2, \dots, 7$ from [1, 2]; no downloaded b-file is needed. The coefficient engine’s tests at abstract rational density vectors check polynomial identities even when those particular vectors do not come from a forest. The probabilistic estimates themselves are asserted only for actual forests.

49 Index inversion along rational moving-gap rays

Fix rational $0 < \alpha, \beta < 1$, and let D_0 be a common denominator. For positive multiples n of D_0 set

$$A(n) = \#\{\pi \in S_n : X_{\alpha n, \beta n, n}^{(\theta)}(\pi) = 0\}. \quad (49.1)$$

All densities and $\lambda = \theta(1 - \alpha)(1 - \beta)$ are now constant along the ray. We invert along the actual values of this discrete sequence. No unspecified interpolation of $A(n)$ is assumed.

49.1 Centered logarithmic coordinates

Write formally

$$\log \left(1 + \sum_{j \geq 1} c_j n^{-j} \right) = \sum_{j \geq 1} \chi_j n^{-j}. \quad (49.2)$$

Then

$$\chi_1 = c_1, \quad \chi_2 = c_2 - \frac{c_1^2}{2}, \quad \chi_3 = c_3 - c_1 c_2 + \frac{c_1^3}{3}. \quad (49.3)$$

Set $\hat{n} = n + 1/2$. Centering Stirling's expansion [7] gives

$$\log \Gamma(\hat{n} + 1/2) = \hat{n} \log \hat{n} - \hat{n} + \frac{1}{2} \log(2\pi) - \frac{1}{24\hat{n}} + \frac{7}{2880\hat{n}^3} + O(\hat{n}^{-5}). \quad (49.4)$$

The all-orders version and [theorem 38.1](#) imply

$$\log A(n) = \hat{n} \log \hat{n} - \hat{n} + \frac{1}{2} \log(2\pi) - \lambda + \sum_{j=1}^M h_j \hat{n}^{-j} + O_M(\hat{n}^{-M-1}), \quad (49.5)$$

where

$$h_1 = \chi_1 - \frac{1}{24}, \quad h_2 = \chi_2 + \frac{\chi_1}{2}, \quad h_3 = \chi_3 + \chi_2 + \frac{\chi_1}{4} + \frac{7}{2880}. \quad (49.6)$$

Higher h_j follow by the finite substitution $n = \hat{n} - 1/2$.

For a large input $y > 0$ define

$$\Lambda = \log y + \lambda - \frac{1}{2} \log(2\pi), \quad N = \frac{\Lambda}{W_0(\Lambda/e)}, \quad \Delta = \log N. \quad (49.7)$$

The principal Lambert function W_0 satisfies $W_0(z)e^{W_0(z)} = z$ [8], so $N \log N - N = \Lambda$ exactly. Absorbing the constant in this coordinate keeps the subsequent expressions short.

Theorem 49.1 (All-orders index reconstruction; manuscript 05). *There are rational functions $d_j(\Delta)$, computable recursively from $h_1, \dots, h_j$, such that for $y = A(n)$ on the chosen ray,*

$$n = N - \frac{1}{2} + \sum_{j=1}^J d_j(\Delta) N^{-j} + O_J \left(\frac{N^{-J-1}}{\Delta} \right). \quad (49.8)$$

The first three terms are

$$\begin{aligned} n = N - \frac{1}{2} - \frac{h_1}{N\Delta} - \frac{h_2}{N^2\Delta} \\ - \frac{1}{N^3} \left(\frac{h_3}{\Delta} + \frac{h_1^2}{\Delta^2} + \frac{h_1^3}{2\Delta^3} \right) + O \left(\frac{N^{-4}}{\Delta} \right). \end{aligned} \quad (49.9)$$

For sufficiently large sequence values, rounding a sufficiently accurate truncation to the nearest multiple of D_0 recovers the exact index. No numerical value for that eventual threshold is asserted.

Proof (manuscript 05). Substitute $\hat{n} = N + \delta$ into (49.5) and subtract $N \log N - N = \Lambda$. Taylor expansion gives

$$0 = \Delta\delta + \frac{\delta^2}{2N} - \frac{\delta^3}{6N^2} + \cdots + \sum_{j \geq 1} h_j(N + \delta)^{-j}. \quad (49.10)$$

Set $\delta = \sum_{j \geq 1} d_j(\Delta)N^{-j}$. The first three equations are

$$\begin{aligned} \Delta d_1 + h_1 &= 0, \\ \Delta d_2 + h_2 &= 0, \\ \Delta d_3 + \tfrac{1}{2}d_1^2 - h_1d_1 + h_3 &= 0. \end{aligned}$$

They yield (49.9). At every later order the new coefficient appears only as Δd_j , and all remaining terms have already been determined. This proves the formal recursive assertion.

For the remainder, use a smooth truncated carrier from (49.5). Substitution of the computed inverse truncation leaves a residual $O(N^{-J-1})$. Its derivative is $\log \hat{n} + O(1/\hat{n}^2)$ near N , hence is comparable to Δ . The mean value theorem divides the residual by this derivative. The actual sequence value differs from the same smooth carrier by the remainder in (49.5), so the same estimate applies along $y = A(n)$. The error tends to zero, and distinct admissible indices are separated by D_0 , proving eventual rounding recovery. $\square$

Remark 49.2 (The remainder estimate of [theorem 49.1](#), completed). [Added 4 October 2026, batch 97; the argument is the write's.] The last paragraph of the proof above is a sketch: it does not show that the coefficients $d_j(\Delta)$ stay bounded as $\Delta \rightarrow \infty$, on which the uniformity of the residual depends. The following completes it. Put $\varepsilon = 1/N$ and $\phi_M(x) = x \log x - x + \sum_{j \leq M} h_j x^{-j}$, so that (49.5) reads $\Lambda = \phi_M(\hat{n}) + O(\hat{n}^{-M-1})$ for $y = A(n)$, while $N \log N - N = \Lambda$.

(i) *Each d_j is a polynomial in $1/\Delta$ with zero constant term.* For any δ with $|\varepsilon\delta| < 1$,

$$\phi_M(N + \delta) - \Lambda = \Delta\delta + g(\delta, \varepsilon) + \sum_{j \leq M} h_j \varepsilon^j (1 + \varepsilon\delta)^{-j}, \quad g(\delta, \varepsilon) = \sum_{m \geq 2} \frac{(-1)^m \delta^m \varepsilon^{m-1}}{m(m-1)},$$

which is (49.10). Insert $\delta = \sum_j d_j \varepsilon^j$. In the coefficient of ε^j every term other than Δd_j contains only $d_1, \dots, d_{j-1}$ (each term of g carries the extra factor ε^{m-1} , $m \geq 2$, and each h_i -term the factor ε^i , $i \geq 1$), and none contains Δ . Hence $d_j = -P_j(d_1, \dots, d_{j-1})/\Delta$ with a polynomial P_j free of Δ , and induction proves the claim. In particular $|d_j| \leq C_j$ and $|\Delta d_j| \leq C_j$ for $\Delta \geq 1$.

(ii) *The residual is uniform.* Let $\delta_J = \sum_{j \leq J} d_j \varepsilon^j$ and take $M \geq J$. By (i), for $\Delta \geq 1$ the function $\varepsilon \mapsto \phi_M(N + \delta_J) - \Lambda$, given by the display above with ε as a complex variable, is holomorphic and bounded by a constant B on $|\varepsilon| \leq \varepsilon_0$, where $\varepsilon_0 > 0$ and B depend only on J , M and $h_1, \dots, h_M$ (choose ε_0 with $|\varepsilon\delta_J| \leq \frac{1}{2}$). Its Taylor coefficients through ε^J vanish by the choice of $d_1, \dots, d_J$. Cauchy's estimate gives $|\phi_M(N + \delta_J) - \Lambda| \leq 2B(\varepsilon/\varepsilon_0)^{J+1}$ for $0 < \varepsilon \leq \varepsilon_0/2$, that is, $O(N^{-J-1})$ uniformly in $\Delta \geq 1$.

(iii) *Inversion.* ϕ_M is eventually increasing, with $\phi'_M(x) = \log x + O(x^{-2})$. Since $\phi_M(\hat{n}) = \Lambda + o(1)$ and $\phi_M(N) - \Lambda = O(1/N)$, monotonicity and the mean value theorem give $\hat{n} - N = O(1/(N\Delta))$; also $\delta_J = O(1/N)$. Hence $\phi_M(\hat{n}) - \phi_M(N + \delta_J) = O(\hat{n}^{-M-1}) + O(N^{-J-1})$, and $\phi'_M = \Delta + O(1/N)$ between the two points, so $|\hat{n} - N - \delta_J| \leq C N^{-J-1}/\Delta$. As $n = \hat{n} - \frac{1}{2}$, this is (49.8), and eventual rounding recovery follows as stated. The constants come from (49.5), hence from the non-explicit constants of [theorem 38.1](#); no effective threshold follows ([question 50.7](#)).

This is a specialization of the familiar Lambert- W inversion strategy used in the existing report [20]; it is not a new definition or new property of W . The new input is the moving-gap coefficient sequence. Expanding W_0 itself in nested logarithms converts (49.8) into a logarithmic asymptotic transseries in y . Neither that formal expansion nor the direct W_0 carrier asserts Borel summability or controls exponentially small sectors. [Note of the batch-97 write: the strategy is Part I's [theorem 9.3](#) (Remark 9.1), in the coordinates of Part II's (20.1) and Part III's

(32.1) with λ in place of θ ; the coordinates Λ , N , Δ and the input y are manuscript 05's $\mathcal{L}$, Y , ℓ and T (table 14). No novelty is claimed for the method.]

Along the ray, $A(n + D_0)/A(n) \sim n^{D_0}$ by (48.7). Thus the counts are eventually strictly increasing. Inverting arbitrary thresholds between consecutive values would additionally require a stated staircase convention; the theorem above deliberately concerns index recovery from actual sequence values.

49.2 Numerical reconstruction from exact integer counts

For $r = s = n/2$ with even n , the exact integer count used in the numerical check is

$$A(n) = \sum_{k=0}^{n/2} (-\theta)^k \binom{n/2}{k}^2 k!(n-2k)!. \quad (49.11)$$

The input is generated by integer arithmetic, not supplied by an asymptotic approximation. The script then evaluates logarithms and W_0 with 100 decimal digits of working precision.

n	directed error after three terms	absolute error after three terms
20	$1.47805 \cdot 10^{-6}$	$6.24774 \cdot 10^{-6}$
40	$6.85472 \cdot 10^{-8}$	$3.15874 \cdot 10^{-7}$
80	$3.41178 \cdot 10^{-9}$	$1.64420 \cdot 10^{-8}$
160	$1.77936 \cdot 10^{-10}$	$8.78773 \cdot 10^{-10}$
320	$9.57420 \cdot 10^{-12}$	$4.79548 \cdot 10^{-11}$

Table 21: Computed inverse truncation minus the true index, using (49.9). These are *uncertified* high-precision diagnostics, unlike the rational probability intervals in tables 18 and 19.

50 Further questions and research

[Added 4 October 2026, batch 97.] This section collects every unproved claim of manuscript 05: its eight questions, printed as proposals, and the conjectural remark at the end of section 46. Manuscript 05 states no claim that this write found wrong. After each question a note of the write says which earlier questions of this report it continues and records the evidence the intake collected; evidence is not proof. The precise form of Remark 45.3, the completed remainder of Remark 49.2 and the computer verification of Remark 43.2 are not open.

The following questions are proposed directions, not statements proved by the preceding theorems. They deliberately separate algebraic-order extensions from beyond-all-orders problems.

Question 50.1 (The next directed short-path term; manuscript 05). For fixed $r, s \geq 2$ and one source path of fixed vertex length L , what is the first nonzero term of the directed shape difference? The leading absolute-model order cancels by theorem 45.1, while $s = 1$ gives exact equality. Determine the precise exponent and the dependence on r, s, L , including any further exceptional cancellations.

The finite coefficient engine provides a direct experimental route, but a useful theorem should isolate a connected-pattern mechanism rather than merely tabulate a fixed number of small L .

[Note of the batch-97 write.] This continues the fixed-length half of Part II's question 2 (section 22) in the directed model. *Evidence, not proof:* the batch-97 intake computed the directed and absolute differences exactly, with its own implementation of the profile identity

(39.2) (validated against brute force on eight forest pairs at $n = 7$; not shipped), for a two-path target $(n/2, n/2)$ and sources $(L, n - L)$ against $(n/2, n/2)$, so $r = s = 2$. The scaled differences $n^{L+1}e^2(\mathbb{P}(X_{\text{short}} = 0) - \mathbb{P}(X_{\text{long}} = 0))$ at $n = 20, 30, 40$ were

	$n = 20$	$n = 30$	$n = 40$
absolute, $L = 1$	-2.645	-2.391	-2.281
absolute, $L = 2$	-3.155	-2.672	-2.473
directed, $L = 1$	-0.0196	-0.0077	-0.0041
directed, $L = 2$	0.0295	0.0100	0.0049

The absolute rows approach the proved -2 like $-2 + O(1/n)$. The directed rows decay like n^{-2} , which suggests a directed difference of order n^{-L-3} at $(r, s) = (2, 2)$, one order below the proved $O(n^{-L-2})$; the sign appears to depend on L . A partial mechanism: for $b_q = 1 - sq/n$, (46.6) with $\beta = s/n$ gives $H_q^{(1)}(\mathbf{b}) = -(s/n)^2(1 + s/n)^{q-2}$, so the order- $(q-1)$ term is $O(n^{-q-2})$; the contributions of B_q must cancel at order n^{-q-1} as well, which is not proved. A larger run ($n = 60$) did not finish.

Question 50.2 (Cancellation loci and their next visible layer; manuscript 05). Classify the zeros of $H_q^{(\theta)}(\mathbf{b}(\beta))$ on each interval between reciprocal integers. At $q = 3$, $\theta = 1$, and $\beta = \sqrt{2} - 1$, determine the first later coefficient whose dependence on the $\alpha = 1/3$ boundary survives. How do irrational rounding phases interact with that delayed boundary?

Each fixed q reduces the first part to finitely many polynomial root problems. An all- q description would organize cancellations across the whole hierarchy.

[Note of the batch-97 write.] The last sentence of section 46, that the rounding discrepancy of $s = \lfloor (\sqrt{2} - 1)n \rfloor$ “can reintroduce an effect one order later”, belongs here: it is offered without proof. Its sketch is the floor formula (47.1), which moves b_q by $O(1/n)$; what is missing is the computation of the resulting order- n^{-3} term and a proof that it does not vanish.

Question 50.3 (A growing locality scale; manuscript 05). Can the constants in theorems 38.1 and 45.1 be controlled jointly in the order and in the short-path length? In particular, identify the first shape-dependent sector when $L \asymp c \log n / \log \log n$.

The current proof fixes M before letting n grow. Its moment cutoff and tree bound establish uniformity over geometry, but do not justify letting M or L grow without additional quantitative work.

[Note of the batch-97 write.] This is what remains of Part I’s question 5 (the transition when path lengths are comparable to $\log n / \log \log n$), of the growing half of Part II’s question 2, and of the beyond-all-orders questions of Part I (question 3) and Part III (question 2) for long paths. At each fixed order Remark 45.4 settles the threshold for fixed lengths.

Question 50.4 (Large-order structure; manuscript 05). What is the growth of $B_M(-1; \mathbf{a}, \mathbf{b})$ on compact subsets of a fixed proportional-gap cell? Is there a useful Gevrey bound, and what is the optimal truncation error? Are the dominant large-order mechanisms local cluster proliferation, finite-permutation constraints, or both?

An answer would be the necessary bridge from the present algebraic sector to any justified exponentially improved transseries.

[Note of the batch-97 write.] This is Part I’s question 4 for moving gaps. At the fixed directed offsets $(2, 2)$ Part V determines the growth of the coefficients c_J (theorem 60.2) and proves a truncation rule for a Borel completion (theorem 61.1); nothing of this kind is known on a proportional-gap cell.

Question 50.5 (Several gap pairs and cycles; manuscript 05). Extend the construction to two or more simultaneous gap constraints, including a joint law with separate marking variables. Which finite colored subgraph counts replace S_q ? At what order do cycles and orientation inconsistencies first affect the expansion?

The exact rod description relies on a single path structure in each forest. Merely adding the single-gap answers would miss shared vertices, coincident events, and new connected components. [Note of the batch-97 write: this continues Part I's question 6, Part II's question 4 and Part III's question 10.] [Added 5 October 2026, batch 101: answered for reciprocal pairs by Part VII. The joint law with separate marks is [theorem 78.1](#). The finite coloured subgraph counts that replace S_q are the products $i_S(H; n) i_T(H; n)$ of injective embedding counts of selected coloured graphs into the source and target graphs, eventual integer polynomials in n ([propositions 79.1](#) and [79.2](#)). Cycles first affect the expansion at order n^{-2} , through the alternating square, and incompatible cycles, the orientation inconsistencies, contribute exactly zero ([remark 80.2](#)). Part VII has fixed gaps and no local-density form; more colours, gap pairs sharing a gap, the directed model and moving gaps remain open ([section 87](#)).]

Question 50.6 (Growing observed counts; manuscript 05). For what ranges $j = j(n) \rightarrow \infty$ does the local probability expansion remain relatively accurate, rather than only absolutely accurate in ℓ^1 ? Can one derive uniform moderate-deviation estimates through $G(u)$ with a growing complex disk?

The present theorem fixes the disk radius. A growing saddle point would require new control of both the cluster sum and the factorial tail. [Note of the batch-97 write: this continues Part II's question 5.]

Question 50.7 (Explicit constants and certified inversion thresholds; manuscript 05). Turn the remainder proof into usable finite constants, then combine those constants with the inverse residual argument to certify a specific index-rounding threshold on a rational ray. Can structural cancellations reduce the conservative k^{3h} cluster majorant sufficiently to make such bounds practical?

The delivered Bonferroni certificates validate selected finite inputs; they do not yet constitute a uniform threshold theorem with a numerical cutoff. [Note of the batch-97 write: this continues Part I's question 8 and Part II's question 6.]

Question 50.8 (A small formal verification target; manuscript 05). Formalize the exact profile identity, the local-log recurrence, and the finite coefficient extraction in Lean. A first deliverable could be a verified checker for rational coefficient tables, followed by the cluster locality and factorial-moment inequalities.

This would separate finite algebraic certification from the analytic uniform-remainder layer. No formalization is claimed in this package. [Note of the batch-97 write: this continues Part I's question 7 and Part III's question 6; nothing in Part IV is formalized.]

51 Conclusion of manuscript 05

Growing gaps do not destroy the all-orders Poisson description; they change the correct geometric coordinates. Replacing fixed path counts by normalized short-subpath counts yields a theorem uniform over every path-forest pair, including the regime where most paths stay bounded. Connected-overlap cost explains why order M sees only $M + 1$ edge lengths, and the highest-path sensitivity formula identifies exactly how the new length enters.

The same mechanism resolves a concrete fixed-short-path question, gives sharp finite-size boundaries for proportional offsets, and exhibits an explicit cancellation that a leading Poisson

limit cannot detect. Exact coefficient generation and rational finite-input checks make the results reproducible, while the separation of algebraic asymptotics, rounding phases, and exponentially small effects keeps the remaining questions precise.

52 Manuscript 05’s reproducibility notes and shipped files

The main document is self-contained LaTeX with an internal bibliography. The PDF was compiled from the delivered source. The package contains the following files and directories:

File	Purpose
<code>article.tex</code> , <code>article.pdf</code>	Complete manuscript and compiled PDF
<code>README.md</code> , <code>SOURCES.md</code>	Run instructions, claims, and source audit
<code>Makefile</code> , <code>requirements-optional.txt</code>	PDF and verification commands; optional packages
<code>code/local_expansions.py</code>	Exact local-log coefficient engine and finite counting routines
<code>code/verify.py</code>	Standard-library exact checks and rational certificates
<code>code/symbolic_checks.py</code>	Symbolic verification of the first two corrections
<code>code/inverse_demo.py</code>	Uncertified high-precision inverse diagnostics
<code>data/</code>	Exact coefficients, interval endpoints, and recorded outputs

Run from the package root:

```
python code/verify.py
python code/symbolic_checks.py
python code/inverse_demo.py
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The first command requires only Python’s standard library. The second requires SymPy, and the third requires mpmath. Generated output is written under `data/` using paths relative to the scripts, not the caller’s working directory. The pseudo-random rational identity checks use a fixed seed; the main mathematical and numerical outputs are exact and deterministic. The recorded execution duration is machine-dependent and is not a mathematical datum.

The shipped files of manuscript 05. [Added 4 October 2026, batch 97.] Manuscript 05’s article, PDF, README and checksum ledger are not shipped (the article is this Part; the ledger was verified at placement, 20/20), and neither is its `requirements-optional.txt`, a byte copy of `data/02-fixed-gap-requirements.txt` (SymPy 1.14.0, mpmath 1.3.0). Its other sixteen files are shipped byte-identical with the prefix `04-moving-gaps-`: the four scripts and the `Makefile` in `code/`, the ten recorded outputs in `data/`, and `SOURCES.md` at the report root; four of the CSV files are CRLF as delivered. The scripts import their sibling by the delivered name (`from local_expansions import ...`) and write unprefix files into the `data/` directory beside their own, and the `Makefile` calls them by their delivered names; its `pdf` and `clean` targets would act on this report’s `article.tex`. The commands above therefore work only in a copy that restores the delivered layout; the report README gives that recipe, which the batch-97 write ran. Its source ledger cites the first author of [3] as “R. Spahn”; the arXiv record names George Spahn.

53 Proof dependencies and non-claims of manuscript 05

The logical dependency chain is as follows. The exact profile identity and the elementary moment bound are proved first. Connected graph expansion gives exact local cluster polynomials. Their

merger-cost filtration gives a finite coefficient formula and finite locality. The tree bound and the factorial tail then promote formal extraction to an all-orders theorem. Highest-path sensitivity is a separate refinement of the same filtration; it implies the fixed-short-path and correction-boundary results. The inverse theorem additionally uses the classical Stirling expansion and the defining identity for W_0 .

No theorem depends on a guessed recurrence, a high-precision fit, or the success of a finite test. The fixed-gap report is the source of the research questions and a prior-work comparison, not an unverified black box needed for the new proofs. Its mathematical infrastructure that is used here is reproved in the manuscript. The primary external analytic input in the inversion section is the standard all-orders Stirling expansion, with its fixed-order remainder.

The citations below support definitions, provenance, and established context. They do not establish publication priority for this manuscript. The source audit was restricted to the listed repository report, the two OEIS entries, the cited permutation papers, and the stated special-function references. In particular, failure to find a prior result in this audit is not evidence that none exists.

Part V

Beyond all algebraic orders in A189281: a Borel completion

54 Provenance, scope and notation of Part V

[Added 4 October 2026, batch 97.] Part V was added in the write of batch 97. It is built from manuscript 06 of that batch, which continues Part III’s [sections 29](#) and [30](#) for A189281, the directed offsets (2,2). It gives the entire Borel transform $\widehat{C}(t) = \mathcal{E}'(e^t - 1)$ of [theorem 30.1](#) an explicit representation through Dawson’s function, sums the algebraic sector along the positive ray into an analytic completion $C_*(n)$, shows that no angular sector admits an exponential bound, determines the sharp large-order growth of the coefficients c_J , proves a growing-order truncation rule for C_* , shows that the OEIS order-13 operator annihilates the completion $A_*(n) = \Gamma(n+1)C_*(n)/e$, and proves that the counts are *not* recovered: the flat discrepancy $\delta_n = e a_n/n! - C_*(n)$ is nonzero at least once in every four consecutive indices. Its inputs from Parts I and III are cited, not re-proved; its re-derivations are second routes ([table 23](#)). Manuscript 06’s symbols that clash with the rest of the report are renamed or declared in [table 24](#).

54.1 Provenance

Manuscript 06 of batch 97 is the archive `ProveIt_A189281_Borel_Completion.zip` (16 files, one of them the checksum ledger `SHA256SUMS.txt`, which lists the other 15), which arrived in commit `e88ed8bf6` and was placed in this report by commit `5a69916e8`. Its article is titled *Beyond all algebraic orders in OEIS A189281: An explicit Dawson kernel, sharp Borel growth, non-holonomic correction coefficients, and an unavoidable flat remainder* (21 pages, dated 4 October 2026); its author line reads “Prepared for Vladimir Reshetnikov, AI-assisted mathematical research report”, and its PDF metadata names “AI-assisted research report prepared for Vladimir Reshetnikov” as author. Its pinned repository commit is `79e7aab60ee856862b36c38cf32cfdb1be95313f` (2026-10-04, 18:07; [\[23\]](#)). It read this report’s `article.tex` as the Git blob `73a53d9784260c2f0d083f2baa62a5535f6d6105` and its README as the blob `59ad8b15b9522102e1f6d48f68c1db358a3ac044`; both are the blobs of

these files immediately before the batch-97 write, so manuscript 06 saw Parts I–III exactly as they stood. It was written independently of manuscript 05 (Part IV); the two share no theorem.

Manuscript 06 states correctly what Parts I–III contain: the all-orders expansion ([theorem 1.1](#)), the series Ψ and its Stirling transform (([29.4](#)), ([29.5](#))), the differential equation and recurrence of Ψ (([29.7](#)), ([29.8](#))), the entire Borel transform ([theorem 30.1](#)), the Lambert- W inversion, and the caveat that the Borel transform does not by itself give summability, growth or truncation.

Scope and non-claims of manuscript 06, kept. It is an AI-assisted, unrefereed research manuscript, not a Lean-checked formalization; its numerical quadrature is diagnostic and not interval-certified (its `data/05-borel-completion-numerical_status.json` records `"interval_certified": false`). “New” means an extension of the inspected report; historical priority beyond the sources reviewed is not asserted, and the absence of a proof on an OEIS page is not evidence that no published proof exists. It does *not* prove the conjectured recurrence of A189281 of order eight or the order-13 presentation: the operator identities and the fourth-order equation concern the analytic completion A_* , not the counts, and no minimal-order claim is made. Its non-P-recursiveness theorem concerns the correction coefficients c_J , not a_n and not $A_*(n)$. Its truncation theorem is for C_* , not for $e a_n/n!$, with no optimality, lower bound or explicit onset; the M column of its table only illustrates the rule. “Canonical” names the positive-ray prescription, not a uniqueness claim; C_* is not asserted to be a resurgent transseries, and the Gaussian sector changes of the kernel do not determine the Stokes data of the counting problem. The amplitude, scale and sign of δ_n are not determined; the flat-discrepancy theorem does not give $\delta_n \neq 0$ for every n , and the observed signs are numerical only. The Lambert- W inversion and the tiling formula are not new. The limsup law is an envelope law, not a termwise equivalent. No repository file or OEIS entry was modified.

54.2 What Part V adds

Source and novelty ledger of manuscript 06, restated. The OEIS entry, its values, its conjectured recurrences and Mark van Hoeij’s T – U – V factorization are external [[22](#)]; the matching-of-tilings method is Spahn and Zeilberger’s [[3](#)]; recurrence guessing is discussed by Kauers and Koutschan [[5](#)]; the special-function facts are from the DLMF [[24](#), [25](#), [26](#), [27](#), [28](#), [7](#)]. Manuscript 06 credits to this report the all-orders expansion, the rational collapse and integrality, the series Ψ with its equation and recurrence, the Stirling transform, the entire Borel transform and the Lambert- W inversion. It claims as its own: the Dawson–Volterra representation, the real asymptotics with both integration constants, positive-ray summation and meromorphic continuation of the completion, the strip threshold and the failure of sectorial bounds, the double-exponential growth and the root-growth law, the non-P-recursiveness of (c_J) , the truncation rule for C_* , the shift and fourth-order equations and the analytic realization of the OEIS factorization, the arithmetic proof of a nonzero flat discrepancy, and its propagation to the smooth inverse.

54.3 Notation

Throughout Part V, $\theta = 1$ and $(r, s) = (2, 2)$, so $c_J = c_J^{(1)}(2, 2)$, ψ_i , Ψ and $\mathcal{E}$ are Part III’s at $(2, 2)$, and $a_n = A189281(n)$, which Part III writes $A(n)$. Manuscript 06’s symbols that collide with the rest of the report are renamed as in [table 24](#); no normalization is changed. Real and imaginary parts are written $\Re$ and $\Im$ (manuscript 06: Re , Im), and Euler’s number in *italic*.

The following symbols keep manuscript 06’s meaning inside Part V and mean something else elsewhere in the report: a_n , the count (Part I’s a_j are logarithmic coefficients, Part IV’s a_q densities); $H = \mathcal{E}'$ (Part I’s Charlier $H_h(q; \theta)$, Part IV’s H_q); Dawson’s function D , the kernel K , the Green function g_v and K_0 (Part I’s D , $\mathcal{D}$ and K); E_1 , the exponential integral; δ_n , the flat discrepancy (in Part III’s proof of [proposition 32.1](#), δ is a shift); the operators T , U , V and

Topic	Parts I–III	Part V (manuscript 06 of batch 97), (r, s) = (2, 2), $\theta = 1$
Borel transform	$\widehat{C}(t) = \mathcal{E}'(e^t - 1)$ entire (theorem 30.1)	explicit Dawson–Volterra form (theorem 56.1); real asymptotics (theorem 57.2); order two and type 1/4 of $\mathcal{E}'$ (theorem 59.1)
Summation	not established (section 30; Part III, question 4)	positive-ray sum $C_*(n)$ exactly for $\Re n > 1$, meromorphic on $\mathbb{C}$ (theorem 58.2 and corollary 58.3); no sectorial 1-summability; maximal strip half-width $\pi/4$ (theorem 59.2 and corollary 59.4)
Coefficient growth	$ c_J \leq \gamma_A A^{J-1}(J-1)!$ for every $A > 0$ (30.3)	$\limsup(\log m) c_{m+1}/m! ^{1/m} = 2$; zero radius; not Gevrey $s < 1$ (theorem 60.2); (c_J) not P-recursive (theorem 60.4)
Truncation	open (Part I, question 4)	error $\exp(-\frac{1}{2}n \log n + O(n))$ at $M = \lfloor n(\frac{1}{2} \log n - 1) \rfloor$, for C_* only (theorem 61.1)
Recurrences	the OEIS recurrences unproved (Part I, question 2)	a shift equation for C_* (theorem 62.2); van Hoeij’s factorization VUT annihilates A_* , not proved for a_n (theorem 63.1)
Beyond all orders	no beyond-all-orders term identified (section 30)	δ_n flat and nonzero in every block of four indices (theorem 64.1); its effect on the inverse (proposition 65.1)

Table 22: Part V against Parts I–III.

Parts I–III	Manuscript 06’s use or route
theorem 1.1 at (2, 2), $\theta = 1$	cited as an input (55.2); used only where the completion is compared with the counts
(29.4), (29.5), (30.1), theorem 30.1	cited as inputs (55.3)–(55.5); the explicit kernel gives a second proof that $\mathcal{E}'$ is entire (theorem 56.1)
(29.7), (29.8)	the same identities (62.1), (62.2)
(30.3)	the type-zero Gevrey-one bound, re-derived inside theorem 60.2
proposition 32.1 at (2, 2); theorem 9.2, proposition 20.1	the same two inverse coefficients $-71/24$ and 1 (65.4)
Proposition 2.2 at $u = -1$ (Spahn–Zeilberger)	the matching-of-tilings formula (66.1), proved again (section 69)

Table 23: Results of Parts I–III that manuscript 06 uses or proves again, and where Part V records each.

Notion	Part V	Manuscript 06 (and the clash)
exponential generating function of ψ_i	$\mathcal{E}(x)$	$E(x)$ (Part III's $\mathcal{E}(\xi)$; Part I's $\mathbb{E}$ is an expectation)
Borel transform of $C(x) - 1$	$\widehat{C}(t)$	$\mathcal{B}(t)$ (Part III's $\widehat{C}$; Part IV's $\mathcal{B}, B_j$)
its Taylor coefficients $c_{m+1}/m!$	$\hat{c}_m$	b_m (Part I's profile b_ℓ ; Part IV's densities b_q)
completion, its Gamma form, inverse	$C_*, A_*, \mathcal{N}_*$	$C_*, A_*, \mathcal{N}_*$
Volterra primitive	$\mathcal{I}$	$\mathcal{I}$
Laplace transform	$\mathcal{L}$	$\mathcal{L}$ (Part I's $\mathcal{L}$ is a law)
quadrature function	$\mathcal{J}(a)$	$J(a)$ (the index J of c_J)
series in the proof of theorem 56.1	$\mathcal{Q}(z)$	$Q(z)$ (also its polynomial $Q(n)$)
truncation and split point	t_0	T (also its operator T)
integration constants in theorem 57.2	ν_1, ν_2	α, β (Part IV's ratios)
index in the proof of theorem 60.2	N_R	N (the Lambert coordinate)
logarithm of the Lambert coordinate	$\Delta = \log N$	L (Part II and III's Δ)
tile-size partition; profile count	$\varrho; \Theta_n(\varrho)$	$\lambda; T_n(\lambda)$
flat-term constants in question 1	A_1, A_2	α, β

Table 24: Notation of manuscript 06 against Part V. The coefficients $c_J, \psi_i, \Psi, \Lambda$ and N agree with Part III; the Stirling numbers and the factorial powers $x^{\underline{k}}, x^{\overline{k}}$ are written as in the rest of the report.

the polynomials $P, Q, \mathcal{R}$ of [section 63](#) (Part I's R_h , Part IV's $R_{J,k}$); the radius R , the order M and the maximum modulus $M_f(R)$; β_j , the coefficients of e^{-jt} in the proof of [corollary 58.3](#); and $L(p) = -\gamma - \log p$ in the proof of [theorem 57.2](#).

Two renamings carry a tempting false reading. $\widehat{C}$ is the Borel transform of the asymptotic coefficients, an entire function of t ; $C_*(n)$ is its Laplace integral, a function of n that has the same asymptotic expansion as $e a_n/n!$ but is not equal to it ([theorem 64.1](#)). And $\hat{c}_m = c_{m+1}/m!$ is not a coefficient of the inverse-power series.

55 Manuscript 06: the question selected from ProveIt

Let a_n be the number of permutations π of $\{1, \dots, n\}$ for which

$$\pi(i+2) - \pi(i) \neq 2 \quad (1 \leq i \leq n-2). \quad (55.1)$$

The convention at $n=0$ gives $a_0 = 1$. This is OEIS A189281 [22]. The initial values are

$$1, 1, 2, 5, 18, 75, 410, 2729, 20906, 181499, \dots$$

The OEIS entry displays a conjectured recurrence of order eight and degree eleven, an equivalent order-thirteen presentation, and a factorization of the latter. Its higher asymptotic coefficients were originally attached to those conjectural recurrences. The underlying counting problem and methods of recurrence guessing are discussed in [3, 5].

The ProveIt report *Unconditional all-orders asymptotics for two OEIS permutation sequences* develops a different route: exact path-forest moments, stabilization, and controlled inclusion–exclusion. Its Part III proves a rational-collapse identity and universal integrality for directed offsets [23]. We continue its Sections 29–30 rather than repeat its finite coefficient computations. [Note of the batch-97 write: these are [sections 29](#) and [30](#) of this report.]

55.1 The exact prior input

The following facts are inputs from [23]. They are not claimed as new. For every fixed integer $M \geq 0$,

$$\frac{ea_n}{n!} = \sum_{J=0}^M \frac{c_J}{n^J} + O_M(n^{-M-1}), \quad n \rightarrow \infty, \quad c_0 = 1. \quad (55.2)$$

Define the *formal* power series

$$\Psi(z) = \sum_{i \geq 0} \psi_i z^i = (1+z)^3 \sum_{j \geq 0} j! \left(-\frac{z^2}{1+z} \right)^j. \quad (55.3)$$

Then

$$c_J = \sum_{i=1}^J \psi_i \left\{ \begin{matrix} J-1 \\ i-1 \end{matrix} \right\} \quad (J \geq 1), \quad \mathcal{E}(x) = \sum_{i \geq 0} \frac{\psi_i}{i!} x^i \quad (55.4)$$

is entire, and the order-one Borel transform of the nonconstant asymptotic sector is

$$\widehat{C}(t) = \sum_{J \geq 1} \frac{c_J}{(J-1)!} t^{J-1} = \mathcal{E}'(e^t - 1). \quad (55.5)$$

We put $H = \mathcal{E}'$. The first auxiliary coefficients and correction coefficients are respectively

$$(\psi_0, \dots, \psi_{11}) = (1, 3, 2, -1, 1, 2, -6, 0, 24, -24, -96, 216), \quad (55.6)$$

$$(c_0, \dots, c_{10}) = (1, 3, 2, 1, 0, 3, 26, 101, 124, -1409, -13266). \quad (55.7)$$

No guessed recurrence for a_n is used in any proof below. [Note of the batch-97 write: (55.2) is [theorem 1.1](#) at $\theta = 1$, $(r, s) = (2, 2)$, $u = -1$, with $ea_n/n! = e\mathbb{P}(X = 0)$; (55.3) is (29.4) at $(2, 2)$, where $w_j = j!$; (55.4) is (29.5) and (30.1), with the variable x for Part III's ξ ; (55.5) is [theorem 30.1](#). The values (55.7) are those of [section 29](#) and (8.7).]

The distinction between the roles of these inputs is important. All analytic statements about H , $\widehat{C}$ and the completion constructed below follow directly from the explicit formal series (55.3). Only the statements comparing that completion with the original integer counts use (55.2). The full combinatorial stabilization proof is in [23]; this manuscript does not present that earlier proof as its own.

55.2 What was open, and what is resolved here

The earlier report explicitly warns that an entire Borel transform can grow too rapidly for a positive-axis Laplace integral. It also leaves large-order growth and a justified growing-order truncation rule open. The conclusions here are more differentiated than a single yes-or-no answer to summability.

Question	Conclusion in this manuscript
Does the positive-ray Borel integral exist?	Yes, exactly for $\Re n > 1$.
Is there an exponentially bounded open angular Borel sector at direction zero?	No; horizontal strips, not angular sectors, describe the admissible geometry.
Does the ordinary inverse-power series converge?	No; a sharp root-growth law proves zero radius.
Are the correction coefficients P-recursive?	No. This is not a claim about P-recursiveness of a_n .
Can the canonical sum be truncated at a growing order?	Yes, with error $\exp[-\frac{1}{2}n \log n + O(n)]$.
Does that sum recover all integer counts?	No; every block of four indices starting at $n \geq 2$ contains a discrepancy.
Is the full counting recurrence proved?	No. Its order-13 operator is proved to annihilate the analytic completion only.

55.3 Repository pin and audit boundary

The source inspected is [23]. The repository revision and main LaTeX blob are respectively

79e7aab60ee856862b36c38cf32cfdb1be95313f
73a53d9784260c2f0d083f2baa62a5535f6d6105.

The full path and source comparison are recorded in `SOURCES.md` (shipped as `05-borel-completion-SOURCES.md`). Parts I and II of that report are dated 1 October 2026; Part III was added on 3 October 2026. The specific summability caveat was verified at the pinned revision. No repository file was edited, no OEIS contribution was submitted, and no claim of independent peer review is implied by repository inclusion.

56 An explicit Dawson–Volterra representation

Dawson’s entire function is

$$D(z) = e^{-z^2} \int_0^z e^{u^2} du = \frac{\sqrt{\pi}}{2} e^{-z^2} \operatorname{erfi}(z). \quad (56.1)$$

The integral is path independent. Definitions and its relation with error functions are given in [24]. For an entire function f , write

$$\mathcal{I}f(x) = \int_0^x f(u) du, \quad \mathcal{I}^2 f(x) = \int_0^x (x-u)f(u) du. \quad (56.2)$$

Theorem 56.1 (Explicit kernel; manuscript 06). *Define*

$$K(x) = D\left(\frac{x+1}{2}\right) + e^{-x} D\left(\frac{x-1}{2}\right). \quad (56.3)$$

Then the entire function $H = \mathcal{E}'$ from (55.4) is exactly

$$H(x) = K''(x) + 4K'(x) + 6K(x) + 4\mathcal{I}K(x) + \mathcal{I}^2 K(x). \quad (56.4)$$

In particular, (56.4) and $\widehat{C}(t) = H(e^t - 1)$ give an explicit entire representation of the Borel transform, without summing a divergent hypergeometric series as an analytic power series.

56.1 A coefficientwise Gamma integral

Expanding first in the formal variable z gives

$$\Psi(z) = \int_0^\infty e^{-v} \frac{(1+z)^4}{1+z+ vz^2} dv. \quad (56.5)$$

Here the integral is *coefficientwise*: each coefficient of the rational integrand is a polynomial in v , integrated using $\int_0^\infty e^{-v} v^j dv = j!$. Equation (56.5) does not assert that Ψ has a convergent Taylor series near zero.

For $v \geq 0$, let g_v solve

$$g_v'' + g_v' + v g_v = 0, \quad g_v(0) = 0, \quad g_v'(0) = 1. \quad (56.6)$$

Its explicit entire solution is

$$g_v(x) = e^{-x/2} \frac{\sin(x\sqrt{v-1/4})}{\sqrt{v-1/4}}, \quad (56.7)$$

where the quotient is continued at $v = 1/4$ and is understood via its power series when $v < 1/4$. On every compact x -set, it and all its x -derivatives are bounded by a polynomial in v times $\exp(C\sqrt{v})$. Consequently

$$K_0(x) = \int_0^\infty e^{-v} g_v(x) dv \quad (56.8)$$

is entire, and differentiation under the integral is justified locally uniformly.

56.2 Evaluating the Green integral

For real x , splitting (56.8) at $v = 1/4$ and substituting $v = 1/4 \pm u^2$ gives

$$K_0(x) = 2e^{-1/4-x/2} \left\{ D(x/2) + \int_0^{1/2} e^{u^2} \sinh(xu) du \right\}. \quad (56.9)$$

In the first integral one uses $\int_0^\infty e^{-u^2} \sin(xu) du = D(x/2)$; this follows by differentiating the integral in x and solving the same first-order equation as Dawson's function. Completing squares in the second integral yields

$$K_0(x) = \frac{\sqrt{\pi}}{2} e^{-(x+1)^2/4} \left\{ \operatorname{erfi}\left(\frac{x+1}{2}\right) + \operatorname{erfi}\left(\frac{x-1}{2}\right) \right\}. \quad (56.10)$$

This is (56.3); equality for complex x follows by the identity theorem. It also follows from (56.9) that

$$K(x) > 0 \quad (x > 0), \quad K(0) = 0, \quad K'(0) = 1. \quad (56.11)$$

56.3 Recovering the exponential generating function

Proof of theorem 56.1 (manuscript 06). Put $d_m(v) = g_v^{(m)}(0)$. Equation (56.6) implies $d_{m+2} = -d_{m+1} - v d_m$, and hence, formally,

$$\sum_{m \geq 0} d_{m+1}(v) z^m = \frac{1}{1+z+ vz^2}.$$

Let $k_m = K^{(m)}(0)$. Integrating this identity coefficientwise gives

$$\mathcal{Q}(z) := \sum_{m \geq 0} k_{m+1} z^m = \int_0^\infty \frac{e^{-v}}{1+z+ vz^2} dv.$$

For the right side $\tilde{H}$ of (56.4), using $k_0 = 0$ and $k_1 = 1$ yields

$$\begin{aligned} \sum_{m \geq 0} \tilde{H}^{(m)}(0)z^m &= \frac{\mathcal{Q}(z) - 1}{z} + 4\mathcal{Q}(z) + 6z\mathcal{Q}(z) + 4z^2\mathcal{Q}(z) + z^3\mathcal{Q}(z) \\ &= \frac{(1+z)^4\mathcal{Q}(z) - 1}{z} = \frac{\Psi(z) - 1}{z}. \end{aligned}$$

Thus $\tilde{H}^{(m)}(0) = \psi_{m+1}$ for every m . Both sides of (56.4) are entire, so equality of all Taylor coefficients proves the claim. This also gives a direct proof of the entireness needed here. $\square$

57 Positive-axis asymptotics, including the integration constants

The two primitives in (56.4) make integration constants important: merely integrating a formal asymptotic series would not determine them. A Laplace-transform identity fixes both constants.

Lemma 57.1 (Laplace transform of the kernel; manuscript 06). *For real $p > 0$,*

$$\mathcal{L}K(p) := \int_0^\infty e^{-px} K(x) dx = \int_0^\infty \frac{e^{-v}}{p^2 + p + v} dv = e^{p^2+p} E_1(p^2 + p), \quad (57.1)$$

where $E_1(q) = \int_q^\infty e^{-u} du/u$ for $q > 0$.

Proof (manuscript 06). Taking the Laplace transform of (56.6) gives $\mathcal{L}g_v(p) = 1/(p^2 + p + v)$. Fubini is legitimate for $p > 0$: for $v \geq 1/4$, use (56.7) and a polynomial bound near $v = 1/4$; for $0 \leq v \leq 1/4$, the hyperbolic-sine expression is bounded by a constant times $1 + x$. Both bounds are integrable after multiplication by e^{-px-v} . The last equality is the substitution $u = p^2 + p + v$. $\square$

Theorem 57.2 (Exact leading real asymptotics; manuscript 06). *As $x \rightarrow +\infty$,*

$$K(x) = \frac{1}{x} - \frac{1}{x^2} + \frac{3}{x^3} - \frac{7}{x^4} + O(x^{-5}), \quad (57.2)$$

$$\mathcal{I}K(x) = \log x + \frac{1}{x} - \frac{3}{2x^2} + \frac{7}{3x^3} + O(x^{-4}), \quad (57.3)$$

$$\mathcal{I}^2 K(x) = x \log x - x + \log x + \frac{3}{2x} - \frac{7}{6x^2} + O(x^{-3}), \quad (57.4)$$

$$H(x) = (x + 5) \log x - x + \frac{23}{2x} - \frac{103}{6x^2} + O(x^{-3}). \quad (57.5)$$

Each has a full expansion in descending integer powers, with the logarithms displayed above. Consequently, as real $t \rightarrow +\infty$,

$$\widehat{C}(t) = e^t(t-1) + 4t + 7e^{-t} - 8e^{-2t} + O(e^{-3t}). \quad (57.6)$$

Proof (manuscript 06). The standard Dawson expansion, obtained from the error-function expansions in [25], is

$$D(z) \sim \sum_{j \geq 0} \frac{(2j-1)!!}{2^{j+1} z^{2j+1}}$$

on the positive real axis and uniformly on closed subsectors of $|\arg z| < \pi/4$. Its coefficients also follow from $D' + 2zD = 1$. Substitution in (56.3) proves (57.2) to every order; the second

term of (56.3) is exponentially small there. Integration first gives, with undetermined constants ν_1, ν_2 ,

$$\begin{aligned}\mathcal{I}K(x) &= \log x + \nu_1 + x^{-1} - \frac{3}{2}x^{-2} + \dots, \\ \mathcal{I}^2K(x) &= x \log x + (\nu_1 - 1)x + \log x + \nu_2 + \frac{3}{2}x^{-1} + \dots.\end{aligned}$$

These expansions with constants are rigorous: after subtracting sufficiently many terms, the remaining derivative is integrable at infinity.

Let $L(p) = -\gamma - \log p$. The convergent small-argument expansion of E_1 [26] and (57.1) give

$$\mathcal{L}K(p) = L(p) + pL(p) + p^2 \left(\frac{3}{2}L(p) + \frac{5}{4} \right) + O(p^3 |\log p|), \quad p \downarrow 0. \quad (57.7)$$

Since $\mathcal{L}(\mathcal{I}K) = \mathcal{L}K/p$, comparison with $\mathcal{L}(\log x) = L(p)/p$ shows $\nu_1 = 0$: a nonzero ν_1 would produce a term ν_1/p , whereas the remaining difference is only $O(|\log p|)$. Likewise

$$\mathcal{L}(x \log x - x) = L(p)/p^2, \quad \mathcal{L}(\mathcal{I}^2K) = \mathcal{L}K/p^2.$$

With $\nu_1 = 0$, subtracting the transforms of $x \log x - x + \log x$ leaves only $O(|\log p|)$ in (57.7), so $\nu_2 = 0$. The Abelian comparisons used here are justified directly by splitting each integral at $x = 1$; a remainder $O(1/x)$ contributes $O(|\log p|)$. No Tauberian inference is being made.

Substitution into (56.4), and differentiation of the Dawson expansions, now gives (57.5). Finally set $x = e^t - 1$, expand $\log x = t + \log(1 - e^{-t})$, and collect powers of e^{-t} . This gives (57.6) and its all-orders continuation. $\square$

58 The canonical positive-ray sum

Definition 58.1 (Canonical completion; manuscript 06). For $\Re n > 1$, define

$$C_*(n) = 1 + \int_0^\infty e^{-nt} \widehat{C}(t) dt = 1 + \int_0^\infty \frac{H(x)}{(1+x)^{n+1}} dx. \quad (58.1)$$

The real logarithm is used in $(1+x)^{n+1}$ for $x \geq 0$. Define also

$$A_*(n) = \frac{\Gamma(n+1)}{e} C_*(n). \quad (58.2)$$

The word canonical refers to this specified positive-ray prescription. It is not an assertion that the algebraic sector uniquely determines a counting function among all functions sharing its asymptotic expansion.

Theorem 58.2 (Existence and Poincaré expansion; manuscript 06). *The integral in (58.1) converges absolutely exactly in $\Re n > 1$ and is holomorphic there. It fails to converge as an ordinary improper integral at every n with $\Re n \leq 1$. For each fixed $M \geq 0$,*

$$C_*(n) = \sum_{J=0}^M \frac{c_J}{n^J} + O_M(n^{-M-1}) \quad (58.3)$$

as positive real $n \rightarrow \infty$. The corresponding expansion is uniform in closed cones $|\arg n| \leq \pi/2 - \varepsilon$ as $|n| \rightarrow \infty$.

Proof (manuscript 06). Equation (57.6) proves absolute convergence and locally uniform domination in the stated half-plane. At $\Re n = 1$ the leading integrand is $(t-1)e^{-i(\Im n)t}$, which has no convergent improper integral; for $\Re n < 1$ its antiderivative grows exponentially. The remaining terms cannot cancel this leading obstruction.

For the asymptotic expansion, take a fixed small $t_0 > 0$, Taylor-expand the entire function $\widehat{C}$ through degree $M-1$ on $[0, t_0]$, and use $\int_0^\infty e^{-nt} t^{J-1} dt = (J-1)!n^{-J}$. The local Taylor remainder contributes $O(n^{-M-1})$, and (57.6) bounds the tail by a polynomial times $e^{-(n-1)t_0}$. The same proof with $\Re n \geq |n| \sin \varepsilon$ gives the cone-uniform version. This is the usual Watson-lemma argument [27], written here to specify the actual integral and tail bound. $\square$

58.1 A single positive quadrature

Put $\mathcal{J}(a) = \int_0^\infty K(x)(1+x)^{-a} dx$, initially for $\Re a > 0$. Integration by parts in (58.1) gives

$$\begin{aligned} C_*(n) &= (n+1)(n+2)\mathcal{J}(n+3) + 4(n+1)\mathcal{J}(n+2) + 6\mathcal{J}(n+1) \\ &\quad + \frac{4}{n}\mathcal{J}(n) + \frac{1}{n(n-1)}\mathcal{J}(n-1). \end{aligned} \quad (58.4)$$

Indeed the K'' term contributes $-1 + (n+1)(n+2)\mathcal{J}(n+3)$, cancelling the initial 1; all other boundary terms vanish. The primitives are removed by Fubini, for example

$$\int_0^\infty \frac{\mathcal{I}^2 K(x)}{(1+x)^{n+1}} dx = \frac{\mathcal{J}(n-1)}{n(n-1)}.$$

Thus $C_*(n) > 0$ for real $n > 1$, by (56.9). The substitution $y = (1+x)^{-1}$ gives the numerically convenient formula

$$\begin{aligned} C_*(n) &= \int_0^1 K(y^{-1} - 1)y^{n-3} \left[(n+1)(n+2)y^4 + 4(n+1)y^3 + 6y^2 \right. \\ &\quad \left. + \frac{4y}{n} + \frac{1}{n(n-1)} \right] dy. \end{aligned} \quad (58.5)$$

No nested primitive or numerical differentiation is needed in this quadrature.

58.2 Meromorphic continuation

Corollary 58.3 (Pole structure of the completion; manuscript 06). *The function C_* continues meromorphically to the whole n -plane. Its only possible poles are $1, 0, -1, -2, \dots$. At the first four locations its principal parts are*

$$\begin{array}{l|l} n=1 & (n-1)^{-2} - (n-1)^{-1} \\ n=0 & 4n^{-2} \\ n=-1 & 7(n+1)^{-1} \\ n=-2 & -8(n+2)^{-1}. \end{array} \quad (58.6)$$

All possible poles at negative integers are simple. A coefficient may vanish, so no assertion that every possible pole occurs is needed.

Proof (manuscript 06). For any q , theorem 57.2 gives

$$\widehat{C}(t) = e^t(t-1) + 4t + \sum_{j=1}^q \beta_j e^{-jt} + O_q(e^{-(q+1)t}), \quad \beta_1 = 7, \quad \beta_2 = -8.$$

Split the Laplace integral at a fixed positive t_0 and subtract the displayed terms on its tail. The subtracted integral is holomorphic in $\Re n > -(q+1)$; integrating the finite explicit sum gives the claimed principal parts. These continuations agree on overlaps, and q is arbitrary. $\square$

59 Complex growth and the sharp strip geometry

The real asymptotics alone do not settle sectorial summability. The decisive feature is the Gaussian term in (56.10), which is small in one cone and very large in another.

Theorem 59.1 (Growth of the entire kernel; manuscript 06). *The entire function H has order two and type $1/4$. More explicitly,*

$$\log \max_{|x| \leq R} |H(x)| = \frac{1}{4}R^2 + O(R + \log R). \quad (59.1)$$

Along the positive imaginary axis,

$$H(iy) = -\frac{i\sqrt{\pi}}{4}e^{-1/4}y^2 \exp\left(\frac{y^2}{4} - \frac{iy}{2}\right) \left(1 + O(y^{-1})\right). \quad (59.2)$$

Uniformly on every closed cone in $\pi/4 < \arg x < 3\pi/4$,

$$H(x) = \frac{i\sqrt{\pi}}{4}x^2 e^{-(x+1)^2/4} \left(1 + O(|x|^{-1})\right). \quad (59.3)$$

Proof (manuscript 06). The integral form $D(z) = z \int_0^1 \exp[-z^2(1-u^2)] du$ gives the global bound $|D(z)| \leq |z| \exp(|z|^2)$. Applied to (56.3), it gives

$$|K(x)| \leq C(1 + |x|) \exp\left(\frac{1}{4}|x|^2 + C|x| + C\right).$$

Cauchy's derivative estimates on circles of radius one, and direct estimates on the primitives along $[0, x]$, give the same quadratic exponent for H . This proves the upper bound in (59.1).

In a closed cone inside $\pi/4 < \arg x < 3\pi/4$, the error-function asymptotics [25] give

$$\begin{aligned} \operatorname{erfi}((x+1)/2) &= i + \text{exponentially small terms}, \\ \operatorname{erfi}((x-1)/2) &= i + \text{exponentially small terms}. \end{aligned}$$

Consequently $K(x) \sim i\sqrt{\pi}e^{-(x+1)^2/4}$. Differentiating its explicit expression twice gives the leading factor $x^2/4$. The terms K' and K in (56.4) are lower by powers of x . Endpoint integration, or one integration by parts in the quadratic exponential, shows $\mathcal{I}K = O(K/x)$ and $\mathcal{I}^2K = O(K/x^2)$ in these cones; contributions from a fixed bounded initial segment are negligible. These bounds are uniform on each closed cone, since $-\Re x^2 \geq c|x|^2$ there. This proves (59.3). Its specialization to $x = iy$ is (59.2), which proves the lower bound in (59.1) and fixes the type. $\square$

Theorem 59.2 (Sharp horizontal-strip threshold; manuscript 06). *For every $0 < \delta < \pi/4$, there is a constant C_δ such that*

$$|\widehat{C}(s + i\theta)| \leq C_\delta(1 + s)e^s \quad (s \geq 0, |\theta| \leq \delta). \quad (59.4)$$

For each fixed $\theta \in (\pi/4, \pi/2)$,

$$\widehat{C}(s + i\theta) = \frac{i\sqrt{\pi}}{4}(e^{s+i\theta} - 1)^2 \exp\left(-\frac{e^{2s+2i\theta}}{4}\right) (1 + O(e^{-s})), \quad (59.5)$$

$$\log |\widehat{C}(s + i\theta)| = -\frac{\cos(2\theta)}{4}e^{2s} + O(s). \quad (59.6)$$

Thus no wider open horizontal strip centered on the positive real axis has an exponential bound of the form $C \exp(A\Re t)$. The threshold statement concerns open strips; no boundary claim at $|\Im t| = \pi/4$ is required.

Proof (manuscript 06). For $x = e^{s+i\theta} - 1$ and $|\theta| \leq \delta < \pi/4$, the algebraic Dawson expansion gives $K(x) = O(1/|x|)$. Its primitives are respectively $O(\log|x|)$ and $O(|x| \log|x|)$, uniformly in a slightly larger closed cone. For the second Dawson term in (56.3), the factor e^{-x} supplies exponential decay. Equation (56.4) then proves (59.4), with bounded s absorbed into the constant. For fixed $\theta > \pi/4$, apply (59.3) and use $x+1 = e^{s+i\theta}$ exactly. Any strip of half-width greater than $\pi/4$ contains such a horizontal line with $\theta < \pi/2$. $\square$

59.1 Why a positive-ray sum is not sectorial 1-summability

Definition 59.3 (The sectorial convention used here; manuscript 06). A formal series in $1/n$ is *sectorially 1-summable in direction zero* if its order-one Borel germ continues to an open angular sector $|\arg t| < \varepsilon$ about that direction and satisfies an exponential bound $C \exp(A|t|)$ on some smaller such sector for large $|t|$. Its positive-ray Borel–Laplace sum, by contrast, requires only an appropriate continuation and growth bound along the integration ray. These are distinct requirements in this manuscript.

Corollary 59.4 (Ray summation without an angular sector; manuscript 06). *The formal expansion (55.2) has the positive-ray sum (58.1), but is not sectorially 1-summable in direction zero under the stated convention.*

Proof (manuscript 06). Take $t_y = \log(1 + iy)$, with the principal logarithm. Then $\arg t_y \rightarrow 0$ and $\widehat{C}(t_y) = H(iy)$. By (59.2),

$$\log |\widehat{C}(t_y)| = \frac{1}{4}y^2 + O(\log y), \quad |t_y| \sim \log y.$$

Every open angular sector about the positive t -axis eventually contains these points, so no exponential bound in $|t|$ is possible there. The positive-ray assertion is [theorem 58.2](#). $\square$

Remark 59.5 (No finite Borel singularity is involved; manuscript 06). The function $\widehat{C}$ is entire and $2\pi i$ -periodic. The obstruction above is at infinity, not a finite singularity crossed by a contour. Gaussian sector changes in the explicit Dawson kernel explain the strip boundary, but do not by themselves determine the Stokes data or the nonperturbative terms of the original counting problem. General cautions about exponentially improved asymptotics and Stokes phenomena are discussed in [28].

60 Sharp large-order growth and non-holonomicity

For an entire function f , put $M_f(R) = \max_{|z| \leq R} |f(z)|$.

Theorem 60.1 (Double-exponential maximum growth; manuscript 06). *The Borel transform satisfies the sharp limit*

$$\boxed{\lim_{R \rightarrow \infty} e^{-2R} \log M_{\widehat{C}}(R) = \frac{1}{4}.} \quad (60.1)$$

In particular it is an entire function of infinite order.

Proof (manuscript 06). For $|t| \leq R$, one has $|e^t - 1| \leq e^R + 1$. The global estimate in the proof of [theorem 59.1](#) gives

$$\log M_{\widehat{C}}(R) \leq \frac{1}{4}(e^R + 1)^2 + O(e^R + R),$$

which proves the upper bound in (60.1). For the lower bound choose $y = y(R) > 0$ so that $|\log(1 + iy)| = R$. Such a y exists uniquely for large R , and

$$\log y = R - \frac{\pi^2}{8R} + o(R^{-1}), \quad y/e^R \rightarrow 1.$$

Applying (59.2) at $t = \log(1 + iy)$ gives $\log M_{\widehat{C}}(R) \geq y^2/4 + O(\log y)$, proving the lower bound. Thus $\log \log M_{\widehat{C}}(R) = 2R - \log 4 + o(1)$, so $\log \log M_{\widehat{C}}(R)/\log R \rightarrow \infty$. $\square$

Theorem 60.2 (Exact root-growth scale of the coefficients; manuscript 06). *With $\hat{c}_m = c_{m+1}/m!$, one has*

$$\boxed{\limsup_{m \rightarrow \infty} (\log m) |\hat{c}_m|^{1/m} = 2.} \quad (60.2)$$

The ordinary formal power series $\sum_{J \geq 0} c_J z^J$ has radius of convergence zero. Its coefficients are Gevrey one of type zero, meaning that for every $A > 0$ there is C_A with $|c_J| \leq C_A A^J J!$. They are not Gevrey of any order $s < 1$.

Proof (manuscript 06). Cauchy's inequality and [theorem 60.1](#) give $|\hat{c}_m| \leq M_{\hat{C}}(R)R^{-m}$. Choose $R = \frac{1}{2} \log m - \log \log m$ for sufficiently large m . Since $\log M_{\hat{C}}(R) = O(m/(\log m)^2)$, we obtain

$$|\hat{c}_m|^{1/m} \leq \frac{\exp(O((\log m)^{-2}))}{\frac{1}{2} \log m - \log \log m},$$

proving that the limsup in (60.2) is at most two.

For the reverse bound, suppose there were numbers $a < a' < 2$ such that $|\hat{c}_m| \leq (a/\log m)^m$ for all sufficiently large m . Put $N_R = \lceil e^{a'R} \rceil$. For $m > N_R$, one has $|\hat{c}_m|R^m \leq (a/a')^m$, so that part of the absolute Taylor sum is bounded by a geometric tail. For $m \leq N_R$, Cauchy's bound at radius one gives the uniform estimate $|\hat{c}_m| \leq M_{\hat{C}}(1)$; for $R > 1$ the sum of these terms is at most $M_{\hat{C}}(1)(N_R + 1)R^{N_R}$. Thus

$$\log \log M_{\hat{C}}(R) \leq a'R + o(R),$$

contradicting (60.1). This proves (60.2).

Along a subsequence, $|\hat{c}_m|^{1/m}$ is bounded below by a positive constant times $1/\log m$. Stirling's formula then gives coefficient roots for c_{m+1} of order at least $m/\log m$, which tend to infinity. This proves zero radius. The type-zero Gevrey-one bound follows from entireness of $\hat{C}$ by Cauchy's inequality at an arbitrary fixed radius. If a Gevrey- s bound held with $s < 1$, the same subsequence, divided by $((m+1)!)^s$, would have roots at least a constant times $m^{1-s}/\log m$, a contradiction. $\square$

[Note of the batch-97 write.] The type-zero Gevrey-one bound is Part III's (30.3), proved there by the same Cauchy estimate on the entire function $\hat{C}$; what is new here is the exact limsup and the failure of every Gevrey bound of order $s < 1$, which answer the growth half of Part III's question 3 and of Part I's question 4 for (2, 2). Signs, zeros and a termwise equivalent remain open ([section 67](#), question 3).

Remark 60.3 (What the limsup does not say; manuscript 06). Equation (60.2) is an exact envelope-scale result, not a termwise asymptotic equivalent for c_J . It does not describe the signs, zeros, or oscillatory cancellation of individual coefficients. In particular it is not legitimate to replace every c_J by $J!(2/\log J)^J$ in a numerical approximation.

Theorem 60.4 (The correction coefficients are not P-recursive; manuscript 06). *There is no nonzero finite-order linear recurrence with polynomial coefficients in J satisfied eventually by the sequence (c_J) . Equivalently, its formal ordinary generating series is not D-finite.*

Proof (manuscript 06). Recall that a sequence is P-recursive if it satisfies such a recurrence; its ordinary generating series is then D-finite, meaning that it satisfies a nonzero linear differential equation with polynomial coefficients. This equivalence follows by multiplying the recurrence by powers of the series variable and translating J into the Euler differential operator. The converse follows by comparing Taylor coefficients.

If c_J were P-recursive, then so would be $\hat{c}_m = c_{m+1}/m!$: substitute $c_{m+1+k} = (m+k)!\hat{c}_{m+k}$ in its recurrence and divide by $m!$; factorial ratios for fixed k are polynomials, after an index shift if necessary. Therefore the entire function $\hat{C}(t) = \sum \hat{c}_m t^m$ would be D-finite.

Every entire D-finite function has finite order. For completeness, take its polynomial differential equation, choose a circle outside every zero of the leading polynomial, and write the equation as a first-order system $Y' = A(z)Y$ for the function and its derivatives. Outside that circle, the rational matrix A satisfies $\|A(z)\| \leq C|z|^d$ for some integer $d \geq 0$. The initial vector

is uniformly bounded on the circle. Integration along radial segments and Gronwall's inequality give $\|Y(z)\| \leq C' \exp(C'|z|^{d+1})$ in every direction, uniformly. Thus its first component has finite order. This contradicts [theorem 60.1](#). $\square$

This theorem concerns the asymptotic correction array, not the original integer sequence a_n , and not the sampled analytic function $A_*(n)$. Indeed, [section 63](#) proves a polynomial recurrence for the latter. There is no rule asserting that asymptotic coefficient extraction preserves P-recursiveness.

61 A justified growing-order truncation

The fixed-order theorem ([58.3](#)) does not permit its truncation order to grow with n without further estimates. The explicit complex growth bound supplies them.

Theorem 61.1 (Superexponential truncation for the canonical completion; manuscript 06). *For positive real n sufficiently large, put*

$$R = \frac{1}{2} \log n, \quad t_0 = R - \frac{1}{2}, \quad M = \lfloor n(R - 1) \rfloor. \quad (61.1)$$

There are constants A, n_0 such that, for $n \geq n_0$,

$$\left| C_*(n) - \sum_{J=0}^M \frac{c_J}{n^J} \right| \leq \exp\left(-\frac{1}{2}n \log n + An\right). \quad (61.2)$$

The rule uses $M \sim \frac{1}{2}n \log n$ coefficients. The theorem does not claim an optimal lower bound or an explicit numerical onset n_0 .

Proof (manuscript 06). Write $P_{M-1}(t) = \sum_{m=0}^{M-1} \hat{c}_m t^m$. By the global estimate underlying [theorem 60.1](#), $M_{\hat{C}}(R) \leq \exp(Cn)$ for the value of R in (61.1). On $0 \leq t \leq t_0$, the geometric Taylor-tail bound is

$$|\hat{C}(t) - P_{M-1}(t)| \leq \frac{M_{\hat{C}}(R)}{1 - t_0/R} \left(\frac{t}{R}\right)^M.$$

Thus its Laplace integral over $[0, t_0]$ is at most

$$\frac{M_{\hat{C}}(R)}{1 - t_0/R} \frac{M!}{n^{M+1} R^M}. \quad (61.3)$$

Since $M = n(R - 1) + O(1)$, Stirling's formula gives

$$\log \left(\frac{M!}{n^{M+1} R^M} \right) = M \log \frac{M}{nR} - M + O(\log(nM)) = -nR + O(n).$$

The prefactor in (61.3) changes the exponent by only $O(n)$.

The actual Borel tail on $[t_0, \infty)$ is controlled by $|\hat{C}(t)| \leq C(1+t)e^t$, so its integral is at most

$$C \left(\frac{t_0 + 1}{n - 1} + \frac{1}{(n - 1)^2} \right) e^{-(n-1)t_0} = \exp(-nR + O(n)).$$

It remains to bound the tail of the Taylor polynomial, because the complete monomial Laplace integrals, not only their parts up to t_0 , produce the coefficients in (61.2). For $0 \leq m < M$, the inequality $\log(t/t_0) \leq (t - t_0)/t_0$ gives

$$\int_T^\infty e^{-nt} t^m dt \leq \frac{e^{-nt_0} t_0^m}{n - m/t_0}.$$

The denominator is at least $n/(2t_0)$, since $M \leq n(R-1)$ and $t_0 = R-1/2$. Summing Cauchy's bounds $|\hat{c}_m| \leq M_{\hat{C}}(R)R^{-m}$ yields

$$\int_T^\infty e^{-nt} |P_{M-1}(t)| dt \leq M_{\hat{C}}(R) e^{-nt_0} \frac{2t_0}{n} \frac{1}{1-t_0/R} = \exp(-nR + O(n)).$$

Adding these three bounds and using $nR = \frac{1}{2}n \log n$ proves the theorem. $\square$

The essential limitation. Equation (61.2) is proved for $C_*(n)$, not for $ea_n/n!$. The earlier combinatorial theorem supplies fixed- M bounds; it does not, merely by being true for every fixed M , supply the $M \asymp n \log n$ uniformity needed here. Transferring the exponential remainder estimate to the actual counts requires an independent estimate of the flat discrepancy studied below.

62 A differential equation and an exact shift equation

The prior formal series obeys

$$z^3(1+z)\Psi'(z) + (2+3z+3z^2-2z^3)\Psi(z) = (z+2)(z+1)^4. \quad (62.1)$$

This identity follows directly by substituting $F(w) = \sum_{j \geq 0} j! w^j$ and its formal equation $w^2 F'(w) + (w-1)F(w) = -1$ into (55.3); it is also recorded in [23]. [Note of the batch-97 write: it is (29.7), and F is Part III's $\mathcal{F}$ of (29.6); the next display is (29.8).] Coefficient comparison gives

$$2\psi_i + 3\psi_{i-1} + (i+1)\psi_{i-2} + (i-5)\psi_{i-3} = 0 \quad (i \geq 6). \quad (62.2)$$

Proposition 62.1 (Entire differential equations; manuscript 06). *The functions H and $\hat{C}$ satisfy*

$$2H''' + (x+3)H'' + (x+5)H' - H = 6 + x, \quad (62.3)$$

and

$$2\hat{C}''' + (-6 + 2e^t + e^{2t})\hat{C}'' + (4 - 2e^t + 3e^{2t} + e^{3t})\hat{C}' - e^{3t}\hat{C} = e^{4t} + 5e^{3t}. \quad (62.4)$$

The initial data are $\hat{C}(0) = 3$, $\hat{C}'(0) = 2$, $\hat{C}''(0) = 1$.

Proof (manuscript 06). The coefficient of $x^m/m!$ on the left of (62.3) is

$$2\psi_{m+4} + 3\psi_{m+3} + (m+5)\psi_{m+2} + (m-1)\psi_{m+1}.$$

It vanishes for $m \geq 2$ by (62.2); its values for $m = 0, 1$ are 6, 1, respectively. Since H is entire, this coefficient proof is an identity of entire functions. Apply $x = e^t - 1$ and the chain rule, then multiply by e^{3t} , to get (62.4). The initial data follow from (55.7). $\square$

Theorem 62.2 (Inhomogeneous shift equation; manuscript 06). *For $\Re n > 4$, and hence meromorphically wherever both sides are defined,*

$$\begin{aligned} & 2n(n-1)(n-2)C_*(n) + 2(n-1)(n-2)C_*(n-1) \\ & + (n-2)(n+1)C_*(n-2) + (n-4)C_*(n-3) = \mathcal{R}(n), \end{aligned} \quad (62.5)$$

$$\mathcal{R}(n) = 2n^3 + 3n^2 - 7n + \frac{1}{n-4} + \frac{5}{n-3}. \quad (62.6)$$

Proof (manuscript 06). Set $S(n) = C_*(n) - 1 = \mathcal{L}\hat{C}(n)$. The real asymptotics of $\hat{C}$ and its first three derivatives justify integration by parts for $\Re n > 4$. For example,

$$\mathcal{L}\hat{C}'(n) = nS(n) - 3, \quad \mathcal{L}\hat{C}''(n) = n^2S(n) - 3n - 2, \quad \mathcal{L}\hat{C}'''(n) = n^3S(n) - 3n^2 - 2n - 1.$$

Multiplication by e^{kt} shifts n to $n - k$. Applying these rules to (62.4) gives the four coefficients in (62.5), acting on S , together with the boundary polynomial $-6n^2 + 5n - 2$. The transform of the right side is $1/(n - 4) + 5/(n - 3)$. Replacing S by $C_* - 1$ and using

$$2n(n - 1)(n - 2) + 2(n - 1)(n - 2) + (n - 2)(n + 1) + (n - 4) = 2n^3 - 3n^2 - 2n - 2$$

proves (62.6). Meromorphic continuation follows from corollary 58.3. $\square$

63 The OEIS operator annihilates the analytic completion

The OEIS entry records the following factorization of its conjectured order-thirteen recurrence, attributed there to Mark van Hoeij [22]. For functions of n , define

$$(Tf)(n) = f(n + 3) + f(n + 2) + \left(\frac{n}{2} + 2\right)f(n + 1) + \frac{n - 1}{2}f(n), \quad (63.1)$$

$$(Ug)(n) = g(n + 4) + \left(\frac{n}{2} + 2\right)g(n + 2) - \frac{1}{2}g(n + 1) + \frac{1 - n}{2}g(n), \quad (63.2)$$

$$\begin{aligned} (Vh)(n) = & h(n + 6) - (n + 11)h(n + 5) - \left(2n + \frac{75}{4}\right)h(n + 4) \\ & + \frac{3 - n}{4}h(n + 3) - \frac{1}{2}h(n + 2) - \frac{7n + 22}{4}h(n + 1) - nh(n). \end{aligned} \quad (63.3)$$

The displayed conjecture is $(VUTA)(n) = 0$. We prove the corresponding identity for A_* , not for a .

Theorem 63.1 (Analytic realization of the factorization; manuscript 06). *Define*

$$Q(n) = 2n^5 + 19n^4 + 44n^3 - 5n^2 - 54n - 5, \quad (63.4)$$

$$\begin{aligned} P(n) = & 4n^8 + 132n^7 + 1745n^6 + 11926n^5 + 45249n^4 \\ & + 94552n^3 + 99549n^2 + 39956n + 4. \end{aligned} \quad (63.5)$$

For $\Re n > 1$,

$$(TA_*)(n) = \frac{\Gamma(n - 1)}{2e}Q(n), \quad (UTA_*)(n) = \frac{\Gamma(n)}{4e}P(n), \quad (VUTA_*)(n) = 0. \quad (63.6)$$

Moreover A_* satisfies the fourth-order polynomial-coefficient equation

$$Q(n)(TA_*)(n + 1) - (n - 1)Q(n + 1)(TA_*)(n) = 0. \quad (63.7)$$

No minimal-order assertion is made.

Proof (manuscript 06). Replace n by $n + 3$ in (62.5) and multiply by $\Gamma(n + 1)/(2e)$. Using (58.2) gives

$$(TA_*)(n) = \frac{\Gamma(n + 1)}{2e}\mathcal{R}(n + 3). \quad (63.8)$$

A direct expansion shows

$$\mathcal{R}(n + 3) = 2n^3 + 21n^2 + 65n + 60 + \frac{1}{n - 1} + \frac{5}{n} = \frac{Q(n)}{n(n - 1)}.$$

This proves the first formula in (63.6); initially $\Re n > 1$ follows from the half-plane used in theorem 62.2. Substituting that formula in (63.2), using the Gamma shift identity, and collecting powers of n gives the second formula with the polynomial P in (63.5).

For a transparent certificate of the last identity, write $\ell_k(n)$ for the coefficients of $h(n+k)$ in (63.3), $1 \leq k \leq 6$. After dividing by $\Gamma(n+1)/(4e)$ and clearing the remaining denominators, the desired identity is the polynomial identity

$$-4P(n) + \sum_{k=1}^6 4\ell_k(n)(n+1)^{\overline{k-1}}P(n+k) = 0. \quad (63.9)$$

The six polynomials $4\ell_k$ are, in order,

$$-(7n+22), \quad -2, \quad 3-n, \quad -(8n+75), \quad -4(n+11), \quad 4.$$

Expansion with the displayed P gives the zero polynomial. The accompanying standard-library program `code/factor_certificate.py` (shipped as `code/05-borel-completion-factor_certificate.py`) performs precisely this integer-polynomial identity, returning the single coefficient list `[0]`; no sampling or recurrence guessing is involved. Finally, applying $\Gamma(n) = (n-1)\Gamma(n-1)$ to the first formula proves (63.7). $\square$

Why this is not the counting recurrence. The same differential and shift equations that determine an analytic completion do not select the integer counts from all functions with the same algebraic asymptotics. The fourth-order equation above is an especially clear warning: it describes the analytic core, not a newly proved shorter recurrence for A189281. In fact the next section proves that the two objects cannot agree everywhere.

64 An unavoidable beyond-all-orders correction

For integers $n \geq 2$, define the normalized discrepancy

$$\delta_n = \frac{ea_n}{n!} - C_*(n). \quad (64.1)$$

Combining the prior counting theorem (55.2) with theorem 58.2 immediately gives

$$\delta_n = O_M(n^{-M}) \quad \text{for every fixed } M. \quad (64.2)$$

We call this *flat at inverse-power scale*. This does not mean that it is zero or that its precise exponential scale is known.

Theorem 64.1 (Arithmetic obstruction to exact recovery; manuscript 06). *For every integer $m \geq 2$, at least one of*

$$\delta_m, \quad \delta_{m+1}, \quad \delta_{m+2}, \quad \delta_{m+3} \quad (64.3)$$

is nonzero. In particular, the positive-ray Borel completion does not recover the original count at every integer, even after discarding any finite set of initial indices. The nonzero flat discrepancy has no gap of four consecutive indices.

Proof (manuscript 06). Suppose all four quantities in (64.3) vanished. Then $A_*(j) = a_j \in \mathbb{Z}$ for $j = m, m+1, m+2, m+3$, and (63.1) would make $(TA_*)(m)$ rational (indeed a half-integer). But theorem 63.1 gives

$$(TA_*)(m) = \frac{(m-2)!Q(m)}{2e}. \quad (64.4)$$

The numerator is a nonzero integer: $Q(m) = m(m-1)\mathcal{R}(m+3) > 0$ for $m \geq 2$, as seen from (62.6). Since e is irrational, the right side of (64.4) is irrational, a contradiction. $\square$

Remark 64.2 (The arithmetic input is elementary; manuscript 06). Only the irrationality of e , not its transcendence, is used. One standard proof is to assume $e = p/q$ with $q \geq 2$, multiply its factorial series by $q!$, and observe that the remaining tail is a strictly positive number less than one whereas the difference of the two other terms is an integer. Thus the obstruction does not rely on a numerical test of equality or on an unproved transcendence assertion about special-function values.

64.1 What information an algebraic sector loses

There are now three rigorously distinguished levels. The formal coefficients c_J determine the entire Borel transform uniquely. The chosen positive-ray Laplace prescription determines C_* uniquely. But neither of those facts forces $A_*(n)$ to be the original integer count. The difference is not merely a hypothetical freedom to add an arbitrary flat function: [theorem 64.1](#) proves that this specific counting problem actually needs such a correction.

The arithmetic theorem is stronger than proving one accidental discrepancy at a small index. It forbids eventual exact agreement and locates a discrepancy in every block of four. It is weaker than asserting $\delta_n \neq 0$ for every n , and it gives neither a universal sign nor a leading asymptotic equivalent. The numerical diagnostics in [section 66](#) even suggest a sign change within the tested range. No sign theorem is inferred from them. [Note of the batch-97 write: the table there samples only $n \equiv 0 \pmod{4}$ and $n = 30$; the full range $2 \leq n \leq 30$ shows fourteen sign changes, not one ([Remark 66.1](#)).]

The decomposition

$$a_n = \frac{n!}{e} (C_*(n) + \delta_n) \quad (64.5)$$

is therefore a genuine separation of a canonical analytic sector from unresolved beyond-all-orders information. It is not yet a full transseries for the exact sequence.

65 Index inversion and the propagation of flat information

The ProveIt transseries program and the earlier path-forest report already provide Lambert- W inversion of Gamma-scale counting functions [23]. We do not claim that inversion method as new. The completion allows a precise comparison of its smooth inverse with inversion along the exact counts.

Proposition 65.1 (Smooth inverse and flat displacement; manuscript 06). *The real function $A_*(x)$ is positive and strictly increasing for all sufficiently large x . Let $\mathcal{N}_*$ denote its inverse on that final interval. Then*

$$\mathcal{N}_*(a_n) - n = \frac{\delta_n}{C_*(n)(\log n + O(n^{-1}))} + O\left(\frac{\delta_n^2}{\log n}\right). \quad (65.1)$$

In particular $\mathcal{N}_(a_n) - n = O_M(n^{-M})$ for every fixed M , and eventual nearest-integer recovery of n holds. The discrepancy in the smooth inverse is nonzero at some index in every block of four sufficiently large indices.*

Proof (manuscript 06). Positivity follows from (58.4). Differentiating the Laplace integral, with the same local Taylor argument as in [theorem 58.2](#), gives $C'_*(x) = -3x^{-2} + O(x^{-3})$. The differentiated Stirling expansion [7] gives

$$\frac{A'_*(x)}{A_*(x)} = \frac{\Gamma'(x+1)}{\Gamma(x+1)} + \frac{C'_*(x)}{C_*(x)} = \log x + O(x^{-1}). \quad (65.2)$$

Thus an increasing inverse exists eventually. Since $a_n/A_*(n) = 1 + \delta_n/C_*(n)$, Taylor expansion of that inverse about $A_*(n)$ gives (65.1). The second derivative estimate needed for the error

follows by differentiating (65.2), or directly from the Gamma and Laplace representations: $A''_*(x) = A_*(x)((\log x)^2 + O((\log x)/x))$. The error is therefore $O(\delta_n^2/\log n)$. The flatness and rounding conclusions follow from (64.2); inverse injectivity and [theorem 64.1](#) give the last assertion. $\square$

65.1 The familiar explicit Lambert coordinate

For completeness, let

$$\Lambda = \log y + 1 - \frac{1}{2} \log(2\pi), \quad N = \frac{\Lambda}{W_0(\Lambda/e)}, \quad \Delta = \log N. \quad (65.3)$$

The branch W_0 is real and positive for the large arguments used here. Then $N(\log N - 1) = \Lambda$ and

$$\mathcal{N}_*(y) = N - \frac{1}{2} - \frac{71}{24N\Delta} + \frac{1}{N^2\Delta} + O\left(\frac{1}{N^3\Delta}\right). \quad (65.4)$$

This is the specialization already available in [23]. [Note of the batch-97 write: it is [proposition 32.1](#) at $(2, 2)$, where $1/24 - (\sigma - 1) = -71/24$ and $(r - 1)(s - 1) = 1$, and Part II's [proposition 20.1](#) and Part I's [theorem 9.2](#) in other coordinates; the coefficients below, $37/12 = c_1 + \frac{1}{12}$ and $-5/2 = c_2 - \frac{1}{2}c_1^2$, are Part III's a_1 and a_2 .] To verify the coefficients, combine (55.7) with Stirling's formula to obtain

$$\log A_*(x) + 1 - \frac{1}{2} \log(2\pi) = x(\log x - 1) + \frac{1}{2} \log x + \frac{37}{12x} - \frac{5}{2x^2} + O(x^{-3}).$$

Insert $x = N - 1/2 + u/N + v/N^2$. The coefficients at orders N^{-1} and N^{-2} vanish for $u = -71/(24\Delta)$ and $v = 1/\Delta$. Higher terms follow recursively. The new information in [proposition 65.1](#) is not these algebraic coefficients but the propagation law for the unresolved flat correction.

66 Exact checks and numerical diagnostics

66.1 Three independent exact constructions

The supplied code keeps exact arithmetic separate from quadrature. `code/core.py` (shipped as `code/05-borel-completion-core.py`) generates ψ_i from (62.2) and applies the shifted Stirling transform. A separate construction expands (55.3) directly by generalized binomial coefficients. A third construction produces the derivative polynomials of g_v from (56.6), integrates them using $v^j \mapsto j!$, and checks (56.4). These independent routes catch coefficient, sign, and normalization errors that a single recurrence implementation would not.

The integer counts are computed independently by matching tilings of two paths, with no use of the asymptotic coefficients or any guessed recurrence. For a nondecreasing tuple of tile sizes ϱ , let $m_j(\varrho)$ be the multiplicity of size j , and let $\Theta_n(\varrho)$ count tilings with that combined profile on paths of lengths $\lfloor n/2 \rfloor$ and $\lceil n/2 \rceil$. Then the exact formula is

$$a_n = \sum_{\varrho \vdash n} (-1)^{n-\ell(\varrho)} \Theta_n(\varrho)^2 \prod_{j \geq 1} m_j(\varrho)!. \quad (66.1)$$

For clarity, a proof is included in [section 69](#). This is the known matching-of-tilings mechanism of [3, 23], not a new enumeration formula. [Note of the batch-97 write: it is Part I's Proposition 2.2 at $u = -1$ for the two paths of each forest, with tiles counted by vertices.] Our partition-based implementation is intended for independent finite validation, not as an asymptotically fast counting algorithm.

Exact check	Number
Auxiliary coefficients from the recurrence and direct formal expansion	401
Dawson–Green–Volterra Taylor derivative identities	399
Coefficient identities in the third-order equation for H	396
Displayed OEIS asymptotic coefficients reproduced	11
OEIS integer values checked against independent tilings, $n = 0, \dots, 21$	22
Permutations exhaustively tested, sizes $0, \dots, 8$	46,234
Full cleared polynomial certificate (63.9)	1

The program additionally generates exact a_n through $n = 30$ and both coefficient arrays through order 400. Finite checks do not prove theorems for all orders; those proofs are in the preceding sections. The exact count checks and the analytic statements are not certified by a proof assistant.

66.2 Canonical sum versus the count

The following values use the single quadrature (58.5), at 65-decimal working precision. They are high-precision diagnostics, not outward-rounded interval enclosures. The sign and displayed digits in a row are numerical observations, not a substitute for [theorem 64.1](#).

n	$C_*(n)$	$ea_n/n! - C_*(n)$	M from (61.1)
8	1.40842753186424087	$1.00698960810 \cdot 10^{-3}$	1
12	1.26449094066174202	$2.88304949032 \cdot 10^{-6}$	2
16	1.19556142153092809	$4.16716348566 \cdot 10^{-9}$	6
20	1.15512642321212123	$3.58868930538 \cdot 10^{-12}$	9
24	1.12854509536822108	$2.04151577943 \cdot 10^{-15}$	14
30	1.10225942309135723	$-1.47403467151 \cdot 10^{-20}$	21

The M column illustrates the explicit rule, not a claim that these small indices have reached the unknown onset n_0 in [theorem 61.1](#). The numerical error of the corresponding truncated series is separately recorded in `data/numerical_diagnostics.csv` (shipped as `data/05-borel-completion-numerical_diagnostics.csv`); it is not the same quantity as the counting discrepancy.

The shift equation was also evaluated at $n = 7.5$ and $n = 10$. At the recorded precision its relative residual was about $1.3 \cdot 10^{-66}$ in the first case and rounded to zero in the second. This is a numerical consistency check of the integral and shift formula, not an exact numerical certificate. The imaginary-axis ratios to the leading term of (59.2) at $y = 8, 12, 20$ are recorded as well; their visible $O(1/y)$ complex correction warns against expecting an early termwise limiting pattern.

Remark 66.1 (Two corrections to the table above). [Added 4 October 2026, batch 97; the computations are the write's.] Two statements about the table are wrong as printed; the table and the delivered files are kept as delivered.

(a) *The M column at $n = 8$.* The rule (61.1) gives $M = \lfloor 8(\frac{1}{2} \log 8 - 1) \rfloor = \lfloor 0.3178\dots \rfloor = 0$, not 1. The program that produced the column, shipped as `code/05-borel-completion-verify.py`, computes `max(1, int(n*(log(n)/2-1)))`: it clamps the order at 1 and truncates towards zero, so it also records $M = 1$ at $n = 4$, where the rule gives $\lfloor -1.227\dots \rfloor = -2$, in `data/05-borel-completion-verification.txt` and `data/05-borel-completion-numerical_diagnostics.csv`, whose column `C_star_minus_partial` at $n = 4$ and $n = 8$ uses that clamped order. The other rows agree with the rule

($12 \cdot 0.2425 = 2.91$, $16 \cdot 0.3863 = 6.18$, $20 \cdot 0.4979 = 9.96$, $24 \cdot 0.5890 = 14.14$, $30 \cdot 0.7006 = 21.02$). No theorem is affected: [theorem 61.1](#) concerns large n only.

(b) *The signs of δ_n .* [Section 64](#) says that these diagnostics “suggest a sign change within the tested range”. The rows printed above show one change, between $n = 24$ and $n = 30$, but they sample only $n \equiv 0 \pmod{4}$. Over the whole range the signs follow the period-four pattern $(-, -, +, +)$, beginning at $n = 2$: $\delta_n < 0$ for $n \equiv 2, 3$ and $\delta_n > 0$ for $n \equiv 0, 1 \pmod{4}$, for every $2 \leq n \leq 30$. In particular $\delta_6, \delta_7 < 0 < \delta_8$ and $\delta_{30} < 0$, and there are fourteen sign changes, not one:

n	δ_n	n	δ_n	n	δ_n
2	$-7.9586 \cdot 10^{-1}$	12	$2.8830 \cdot 10^{-6}$	22	$-8.9704 \cdot 10^{-14}$
3	$-3.8025 \cdot 10^{-2}$	13	$1.9996 \cdot 10^{-7}$	23	$-5.8423 \cdot 10^{-15}$
4	$1.3831 \cdot 10^{-1}$	14	$-1.1792 \cdot 10^{-7}$	24	$2.0415 \cdot 10^{-15}$
5	$7.9473 \cdot 10^{-3}$	15	$-8.0879 \cdot 10^{-9}$	25	$1.3131 \cdot 10^{-16}$
6	$-1.3380 \cdot 10^{-2}$	16	$4.1672 \cdot 10^{-9}$	26	$-4.2634 \cdot 10^{-17}$
7	$-9.1181 \cdot 10^{-4}$	17	$2.8222 \cdot 10^{-10}$	27	$-2.7091 \cdot 10^{-18}$
8	$1.0070 \cdot 10^{-3}$	18	$-1.2950 \cdot 10^{-10}$	28	$8.2246 \cdot 10^{-19}$
9	$7.0236 \cdot 10^{-5}$	19	$-8.6557 \cdot 10^{-12}$	29	$5.1654 \cdot 10^{-20}$
10	$-5.9519 \cdot 10^{-5}$	20	$3.5887 \cdot 10^{-12}$	30	$-1.4740 \cdot 10^{-20}$
11	$-4.1595 \cdot 10^{-6}$	21	$2.3675 \cdot 10^{-13}$		

The write computed these values from the quadrature identity [\(58.4\)](#) with its own implementation (mpmath 1.3.0, 50 digits, Dawson’s function as $z {}_1F_1(1; \frac{3}{2}; -z^2)$), with the counts of `data/05-borel-completion-exact_values.json`, which equal the b-file excerpt `data/03-rational-collapse-b189281_0_30.txt`. Two independent checks support them: the computed $C_*(n)$ satisfy the shift equation [\(62.5\)](#) at every $5 \leq n \leq 30$ with relative residual below $1.3 \cdot 10^{-48}$, and at $n = 6, 7, 8$ a direct Laplace integral of the Taylor series of H (exact ψ_i) with an asymptotic tail agrees with them to $4 \cdot 10^{-11}$ or better, far below $|\delta_n|$. The values at $n = 8, 12, \dots, 30$ agree with the table above to all printed digits. They are floating-point diagnostics, not interval enclosures, and the sign pattern is an observation, not a theorem ([section 67](#), questions 1 and 6). The program is not shipped.

66.3 Reproduction

From the extracted archive root, run [note of the batch-97 write: in the delivered layout only; see the end of this paragraph]

```
python code/verify.py --order 400
python code/factor_certificate.py
python -m pip install -r requirements-optional.txt
python code/verify.py --order 400 --numerical --dps 65
./build.sh
```

The first two commands use only the Python standard library. Numerical functions require `mpmath`; they are not used by the exact checks. All paths are anchored at the scripts, and every output is written into `data/`. Running the verification intentionally regenerates the corresponding output files. The build script is anchored at the archive root and does not depend on the calling working directory.

The shipped files of manuscript 06. [Added 4 October 2026, batch 97.] Manuscript 06’s article, PDF, README and checksum ledger are not shipped (the article is this Part; the ledger was verified at placement, 15/15). Its other twelve files are shipped byte-identical with the prefix `05-borel-completion-:` `build.sh` and the three scripts in `code/`, the six recorded outputs

and `requirements-optional.txt` (the line `mpmath==1.3.0`) in `data/`, and `SOURCES.md` at the report root; `numerical_diagnostics.csv` is CRLF as delivered. `verify.py` imports its sibling by the delivered name (`from core import ...`), both scripts write files under their delivered names into the `data/` directory beside their own, and `build.sh` builds an `article.tex` in its own directory, which in `code/` does not exist. The commands above therefore work only in a copy that restores the delivered layout; the report README gives that recipe, which the batch-97 write ran (all checks passed, outputs equal to the shipped ones up to line endings).

67 Further questions and research

[Added 4 October 2026, batch 97.] This section keeps manuscript 06’s ten questions, under its title “Further research questions and concrete next targets”. Together with the two corrections of Remark 66.1 they are every unproved or refuted claim of manuscript 06 that this write found; its heuristic remark on complex saddle pairs (question 3) is one of them. Notes of the write after the list say which earlier questions of this report each continues, and record evidence; evidence is not proof.

The results narrow the missing information rather than end the problem. The following questions are ordered roughly from the most direct continuation to wider applications. None is represented as solved in this manuscript.

- Q1. The leading flat term of the exact count.** Find an asymptotic equivalent, or a finite oscillatory sum of equivalents, for δ_n . Determine whether its scale is $\exp[-A_1 n \log n + A_2 n + o(n)]$, whether parity matters, and whether the observed sign change reflects a systematic oscillation. A starting point is the boundary correction between stable moments and the finite path tilings.
- Q2. A proof of the counting recurrence.** Prove or disprove $(VUTa)(n) = 0$ by an exact finite-tiling or dynamic-programming identity. Since $VUTA_* = 0$ is now proved, this is equivalent, after the Gamma normalization, to an exact recurrence problem for the flat sector. The equivalence does not reduce to matching finitely many asymptotic terms.
- Q3. A sharper large-order formula.** Strengthen (60.2) to a saddle-point description of c_J , including phases, cancellations, zeros, and sign blocks. The explicit kernel suggests complex saddle pairs of the exponential composition rather than a single positive real saddle. [Note of the batch-97 write: this suggestion is heuristic; no saddle-point analysis is given.]
- Q4. Optimal truncation and its prefactor.** Determine the true least-term index and the leading remainder for C_* . The proven rule $M \sim \frac{1}{2}n \log n$ has the right natural scale suggested by the coefficient envelope, but the theorem does not prove optimality, a limiting prefactor, or a sharp lower bound.
- Q5. An exponential error bound for the actual sequence.** Obtain effective uniform bounds on the path-forest stabilization error strong enough to compare δ_n with the canonical truncation error. This would turn a theorem about an analytic completion into a controlled exponentially improved approximation of the integer count.
- Q6. Exact recovery at individual indices.** Can δ_n vanish at even one integer $n \geq 2$? The arithmetic argument only excludes four consecutive zeros. Investigate shorter zero blocks, quantitative separation from zero, and whether irrationality estimates for e can give useful lower bounds for combinations of discrepancies.
- Q7. Other directed offsets.** For fixed $r, s \geq 2$, the earlier kernel is $(1+z)^{r+s-1} {}_2F_0(r-1, s-1; -; -z^2/(1+z))$. Use Gamma-integral weights or differential operators in the kernel

parameter to seek analogues of (56.4). Determine whether the order-two inner kernel, strip threshold $\pi/4$, and logarithmic root constant two are universal. The degenerate one-path cases must be separated, since their earlier kernels terminate.

- Q8. The absolute-difference model A110128.** Construct an equally explicit Borel representation when $|\pi(i+2) - \pi(i)| = 2$ is forbidden. The directed rational collapse does not apply automatically: each nontrivial selected component has two possible orientations, altering the moment weights.
- Q9. Summation geometry at infinity.** Give a precise acceleration or strip-based summation theory adapted to $H(e^t - 1)$. Distinguish growth boundaries from genuine singularity-driven Stokes jumps, and specify exactly which analytic continuations and contour changes remain admissible. The fact that $\hat{C}$ is entire does not remove all summation geometry.
- Q10. Formal verification.** A manageable first target is a Lean proof of the finite coefficient identities, the shift-operator polynomial certificate, and the arithmetic four-index obstruction. A later analytic development would need entire function growth, Dawson asymptotics, and uniform Laplace estimates. The present Python assertions are not a substitute for that formalization.

Notes of the batch-97 write on these questions. Question 1 continues Part I's question 3 and Part III's question 2 for A189281: [theorem 64.1](#) is the first proof in this report that the count has a nonzero beyond-all-orders correction, relative to C_* ; its amplitude, scale and sign are open. The values of Remark 66.1 bear on the parity question: for $2 \leq n \leq 30$ the sign has period four, $(-, -, +, +)$ from $n = 2$, and $|\delta_n|$ decreases faster than every power of n , consistent with a scale $\exp(-A_1 n \log n + A_2 n)$ but not a proof of it. Question 2 continues Part I's question 2 and Part III's question 1: [theorem 63.1](#) reduces the recurrence of the counts to a recurrence for δ_n . Questions 3 and 4 continue Part III's questions 3 and 4 and Part I's question 4, which Part V answers in part (the dated notes there say how). Question 5 is what Part I's question 4 still asks for the counts themselves. For question 6, no δ_n with $2 \leq n \leq 30$ vanishes numerically; the smallest is $|\delta_{30}| \approx 1.5 \cdot 10^{-20}$, at 50-digit precision. Question 7 extends Part III's questions 3 and 4 to every directed pair of offsets, and question 8 continues Part III's question 7 (absolute differences); question 10 continues Part I's question 7 and Part III's question 6. Nothing in Part V is formalized.

68 Conclusion of manuscript 06

The explicit kernel resolves the summation question for the inspected A189281 algebraic sector, but with an important qualification: positive-ray summation works whereas angular-sector 1-summability fails. It also supplies a sharp large-order growth scale and proves that the correction coefficients are not P-recursive. Those conclusions require the global analytic structure of the Borel transform; they could not be inferred safely from its entireness alone.

The same representation defines a positive analytic completion, yields a meromorphic continuation and exact shift equations, and gives a justified superexponential truncation rule. Its compatibility with the conjectured OEIS operator is exact, yet does not identify it with the integer count. The irrational forcing in the rightmost factor proves that beyond-all-orders information is genuinely present in the original combinatorial problem. A full transseries for the count must recover that information rather than silently equate a preferred sum with the sequence.

69 Independent exact enumeration by path tilings (manuscript 06)

The directed position graph has edges $i \rightarrow i + 2$ and is the disjoint union of its odd and even vertex paths. The directed value graph has the same form. Inclusion–exclusion over selected forbidden events chooses a subset of the position edges. Its connected components, including isolated vertices, form a tiling of the two position paths by directed intervals.

If a selected component has j vertices, its image under a permissible bijection must be a directed interval of j vertices in the value graph. Because the map is injective, the images form a tiling of the value paths with the same multiset of interval sizes. Conversely, any two such tilings, with a matching of equal-sized intervals, define exactly one permutation satisfying all the selected events: orientation is forced by the positive signed difference. For profile ϱ , the matching count is $\prod_j m_j(\varrho)!$. The number of selected edges is $n - \ell(\varrho)$, hence the sign in (66.1).

For a single path of length L , a profile with k tiles and multiplicities m_j has $k! / \prod_j m_j!$ ordered tilings. Convolving these profile counts for the two paths gives $\Theta_n(\varrho)$. This proves (66.1), including empty paths and $n = 0$. The implementation enumerates integer partitions, not all permutations, while the separate brute-force check for $n \leq 8$ tests the original forbidden-difference condition directly. [Note of the batch-97 write: this is a second proof of the case $u = -1$, $(r, s) = (2, 2)$, $\theta = 1$ of Part I’s Proposition 2.2, the matching-of-tilings formula of Spahn and Zeilberger.]

70 Coefficient and certificate conventions of manuscript 06

All coefficient arrays in `data/coefficients.json` (shipped as `data/05-borel-completion-coefficients.json`) start at index zero. The ordinary series $\Psi(z)$ is formal; the exponential generating function $\mathcal{E}(x)$ and Borel series $\widehat{C}(t)$ are entire. The transform convention is

$$\frac{c_J}{n^J} \longleftrightarrow \frac{c_J}{(J-1)!} t^{J-1} \quad (J \geq 1),$$

with the constant coefficient kept outside the integral. Losing the shift by one in this convention changes both the kernel and the Laplace boundary terms.

Polynomial arrays in `code/factor_certificate.py` are stored in ascending degree order. That program multiplies, shifts, and adds integer polynomials exactly. Its certificate concerns the known analytic forcing $\Gamma(n)P(n)/(4e)$; the integer counts never enter it. Conversely, the exact enumeration in `code/core.py` never uses the analytic forcing. This separation is deliberate.

The numerical status file explicitly records `"interval_certified": false`. No theorem in this manuscript depends on rounding a floating-point residual to zero. The numerical table is useful for finding transcription errors and for suggesting further questions, not for certifying a Stokes multiplier, a zero of δ_n , or an onset for nearest-integer inversion.

Part VI

Fixed-displacement permutations: a third derivation and its additions

71 Provenance, scope and notation of Part VI

[Added 5 October 2026, batch 101.] Part VI was added in the write of batch 101. It is built from Report 83 of the session bundle and, on the precedent of Part II, prints only what that

report adds to Parts I–V. Report 83 proves, for the oriented model $\theta = 1$ at every fixed pair of offsets, Part I’s all-orders expansion of the avoidance probability, the stabilization of long path forests, a path-length comparison bound and a threshold inverse. Every one of these theorems is already in Parts I and II, in most cases in a stronger form; they are recorded as second (in places third or fourth) routes in [table 25](#) and not reprinted. What is new is printed: three exact Bonferroni enclosures of the A189281 probability at $n = 100, 200$ and 1000 ([section 72](#)); the two-sided ceiling bracket for the threshold, stated explicitly, and the smooth-model inverse in the exact digamma coordinate ([section 73](#)); and four literature items that the rest of the report does not cite ([section 74](#)).

71.1 Provenance

Report 83 is the archive `Fixed_Displacement_Permutation_Asymptotics_Source.zip` of the session bundle of Reports 1–243, which arrived in commit `60f54ea06`; batch 101 placed it in this report in commit `f7c612c72`. It holds 22 files in one directory with a subdirectory `independent_review/`, one of them the checksum manifest `manifest.json`, which lists the other 21 (all matched at the intake). Its article is titled *Complete Asymptotic Expansions for Fixed-Displacement Permutations* (7 pages, dated 1 October 2026); its author line and its PDF metadata both read “Research note prepared for Vladimir Reshetnikov with OpenAI”. Its pinned repository commit is `1512ef8356d10fef99e7caaa2753620f0c900c83` (2026-10-01, 08:38 –07:00; [29]), at which it inspected the transseries companion `Combinatorial_Transseries_Inverses` for its gamma-core and reversion tools.

An independent same-day derivation. The pin predates every source of this report. Manuscript 60 (Part I) arrived in commit `aa43cc555` at 13:54 on 1 October 2026, manuscript 03 of batch 74 (Part II) in `c664fc2f0` at 19:03, and Part I was written in `2f13bec21` at 19:56; none of these commits, nor the placement `6e193dd4f`, is an ancestor of the pin. The archive of Report 83 was created at 16:33 UTC, before manuscript 60 even arrived. Report 83 is therefore a third derivation of Part I’s main theorem, written on the same day as the first two and independently of both; its text shares 76 of 3,425 word 8-grams (2.2%) with this article, all of them bibliography strings, coefficient digits and the macro text of falling and rising factorials.

Scope and non-claims of Report 83, kept. It does not prove the guessed order-eight, degree-eleven recurrence of A189281 and uses no guessed recurrence: the work “neither settles nor relies on” it. The expansion is asymptotic, with “no assertion of convergence”. It does not identify every flat or exponentially small sector, nor Stokes data, and its comparison theorem “does not select a preferred analytic interpolation”. Its exact Bonferroni intervals are stored first; their decimal summaries are “regression evidence, not the remainder proof”. It does not need or audit the global closure argument of Padhi’s holonomicity note, which by its own README is not submitted or externally reviewed and contains no asymptotic expansion. Tauraso’s and Mendelsohn’s work are “established antecedent methods, not new principles here”; Mendelsohn “assumes the complete expansion exists”; and the bounded literature search “is not a claim that no such source exists”. The proofs “have been checked by direct derivation and exact computations” and “have not been externally refereed or formally verified”; its plain-text `06-fixed-disp-proof.md` is marked “awaiting independent review”. The gamma-core and reversion tools are the repository companion’s, “reused with credit”.

Stale statements of Report 83. [Added 5 October 2026, batch 101.] Report 83 states that a bounded search “located no source containing the complete directed-family expansion, quantitative path-length comparison, or threshold inverse stated here”, and its `06-fixed-disp-source-status.md` that the ProveIt companion “does not name A189281 or the fixed-displacement

permutation family”. Both were true at its pin. They are not true of the repository today: all three results are in Part I of this report ([theorem 1.1](#), [theorem 4.2](#) and the last subsection of [section 9](#)), placed in the evening of the same day. The delivered texts are kept; this note corrects them.

71.2 What Part VI adds

[Table 25](#) lists every result of Report 83 against the place where Parts I–V already prove it. Report 83’s coefficients of A189281 through n^{-10} , 1, 3, 2, 1, 0, 3, 26, 101, 124, -1409 , -13266 , are Part I’s ([8.7](#)) and are not new; Part I’s [table 1](#) goes to order 16 and Part III to order 100. Report 83’s closed forms of the first three coefficients for general (r, s) ,

$$c_1^{(1)} = r + s - 1, \quad c_2^{(1)} = \frac{r^2 - r + s^2 - s}{2}, \quad c_3^{(1)} = \frac{(r + s - 1)(r^2 - 4rs + 4r + s^2 + 4s - 6)}{6},$$

are Part II’s ([proposition 17.2](#)) and Part III’s ([section 28](#)). The new material is:

- three exact rational Bonferroni enclosures of the A189281 avoidance probability, at $n = 100$, 200 and 1000, the largest index certified anywhere in this report ([table 27](#));
- the two-sided ceiling bracket for the threshold $N(y) = \min\{n : A_{r,s}^{(1)}(n) \geq y\}$ stated as a theorem ([theorem 73.1](#)); Part I describes the mechanism in words and names `p0:thm:staircase`, of which the bracket is an instance;
- the smooth-model inverse of A189281 in the exact digamma coordinate, through relative order X^{-3} ([proposition 73.2](#));
- Mendelsohn’s 1956 paper, Tegua Tabuguia’s 2025 guessing paper, Padhi’s holonomicity note in its repository form, and the August 2026 update of the Spahn–Zeilberger author page ([section 74](#));
- a third coefficient checker, by Newton forward differences, written independently of both Report 83’s own generator and Part I’s ([section 75](#)).

The comparison theorem is weaker, not wrong. Report 83’s Theorem 5.1 takes $J = \lfloor \gamma n \rfloor$ Bonferroni terms, which agree for the two forest pairs by stabilization when every path has at least $2J \leq \delta n$ vertices, and bounds each truncation error by its Lemma 2.2 at $J + 1 \leq n/4$. The ranges are consistent and the proof is complete as far as the intake checked it. Part I’s [theorem 4.2](#) proves more: a total-variation bound for the whole law, with a larger range of exponents.

71.3 Notation

Part VI writes Report 83’s objects in Part I’s notation, as listed in [table 26](#). No normalization is changed: Report 83’s $d_j(r, s)$ is exactly $c_j^{(1)}(r, s)$, and its factorial macros mean what this report’s $x^{\underline{k}}$ and $x^{\overline{k}}$ mean.

72 Exact Bonferroni enclosures at $n = 100, 200$ and 1000

Report 83’s `code/06-fixed-disp-check_bonferroni.py` evaluates the alternating partial sums S_{J_B} and S_{J_B+1} of [\(10.1\)](#) for A189281 exactly, from the stable profile counts [\(3.2\)](#). It refuses to run outside the stable range: it requires $\lfloor n/2 \rfloor \geq 2(J_B + 1)$, which is more than [theorem 3.1](#) needs. Its output `data/06-fixed-disp-check_bonferroni.json` stores both

Report 83	Where Parts I–V already have it, and how the routes compare
Theorem 1.1 (complete expansion, $\theta = 1$, $u = -1$)	theorem 1.1 at $\theta = 1$, $u = -1$, which also gives the whole law on every disk $ u \leq U$ and the absolute case; polynomiality of c_J in (r, s) is proposition 17.1 (total degree at most J , a stronger statement than Report 83’s “polynomial”); symmetry is Part I’s (1.9); independence of the residue class of n is explained in section 1 and by theorem 3.1
c_1, c_2, c_3 in closed form; coefficients of A189281 through n^{-10}	proposition 17.2 , (7.3); (8.7) and table 1 (identical values)
Lemma 2.1 (paired tilings)	Proposition 2.2, the Spahn–Zeilberger formula [3] at $u = -1$; also Part V’s (66.1)
Lemma 2.2: $\mathbb{E}\binom{X}{j} \leq \exp(Cj^2/n)/j!$ for $2j \leq n/2$ cutoff $J \sim 2(K+3)\log n/\log \log n$	Lemma 4.1 gives $(\theta e^2)^k/k!$ for every k ; lemma 16.1 gives $\theta^k(1-2k/n)^{-2k}/k!$, the same device as Report 83’s with an explicit constant Part II’s $K_n = \lfloor (M+4)\log n/\log \log n \rfloor$ (16.7)
Lemma 3.1 and (3.3)–(3.6): stable forest coefficient, finite rising/falling-factorial formula, paths of at least $2J$ vertices	theorem 3.1 with the same formula (3.2) and the same Lagrange identity (3.6); Part I needs only paths of at least k vertices
Section 4: defect filtration, $\exp(u/(1-u))$, Touchard sums	Part I’s section 5 (defect filtration, finite differences) and lemma 16.5 ; Part II also evaluates the dimer sums by Touchard polynomials (table 5)
Theorem 5.1: lengths at least δn ; difference at most $\exp(-\gamma n \log n + O_\gamma(n))$ for $\gamma < \min(\delta/2, 1/4)$	<i>weaker</i> than theorem 4.2 , which bounds the total variation of the whole law with every exponent $\eta' < \min(\eta, 1/2)$; uniformity in the shapes is theorem 16.6 and corollary 16.7 (shortest path $\gg \log n/\log \log n$) and theorem 38.1 (no length hypothesis)
Theorem 6.1 (threshold inversion)	the smooth model $\mathcal{A}_M$ and its inverse, last subsection of section 9 ; stated explicitly here as theorem 73.1 , an instance of <code>p0:thm:staircase</code> [12]
(6.3)–(6.4): all-orders formal Lagrange reversion around the gamma core	the transseries volume’s gamma-carrier inversion (<code>p6:thm:gamma</code> , <code>p0:thm:perturbed-inversion</code> [12]), as Remark 9.1 records for every Part; Report 83 credits the companion
(6.6): A189281 inverse in the exact digamma coordinate	new as a display (proposition 73.2); Part I uses the Lambert coordinates z, w (theorems 9.2 and 9.3)

Table 25: Report 83’s results against Parts I–V. Only the last row and the items listed in [section 71.2](#) are new.

Notion	Part VI (Parts I–II)	Report 83 (and the clash)
oriented counts	$A_{r,s}^{(1)}(n)$	$a_{r,s}(n)$
avoidance coefficients	$c_j^{(1)}(r, s)$	$d_j(r, s)$ (Part I's d is the defect, its $d_j(w)$ inverse coefficients)
expansion order	K (Part I: M)	K (Part I's K is an edge degree)
Bonferroni order	J_B	J (Part I's J indexes B_J, c_J)
selected edges; components; defect	$k; c; d = k - c$	$j; c; D$
profile (paths with ℓ vertices)	$\mathbf{b} = (b_\ell)$	$\alpha = (\alpha_i)$
tiling count	$C_{\mathcal{P}}(\mathbf{b})$	$C_r(n; \alpha)$
marks; formal root; amplitude	$z_\ell; 1 + W; \mathcal{D}^{-1}$	$x_i; \lambda = 1 + w; C_0$
smooth model; its inverse	$\mathcal{A}_K$ at $\theta = 1; x_K = \mathcal{A}_K^{-1}$	$f_K; x_K$
root of the gamma core	$\mathfrak{X}$	X (also its violation count, Part I's X)
digamma, trigamma at $\mathfrak{X} + 1$	$\psi_{\mathfrak{X}}, \psi'_{\mathfrak{X}}$	ψ, ψ' (Part III's ψ_i)
logarithm of the correction series	$\ell_K(x)$, coefficients ℓ_j (9.2)	$q(X)$ (Part V's q)
length fraction; flat exponent	$\delta; \gamma$	$\delta; \gamma$ (Part V's δ_n , Part III's γ_A)
Touchard polynomials	T_v (this table only)	T_v (Part V's $T_n(\lambda)$)

Table 26: Notation of Report 83 against the report. The threshold $N(y)$ and the factorials agree.

endpoints as exact rationals. By the pointwise Bonferroni inequalities recalled after (10.1), the smaller and the larger of the two sums enclose $\mathbb{P}(X = 0) = \text{A189281}(n)/n!$.

Table 27 prints these enclosures in Part I's normalization (10.2), at order $M = 10$. The write recomputed the table from the shipped rationals with e bracketed by rational Taylor sums, so every endpoint is rounded outward; the decimals stored in the JSON file, computed with 70-digit decimal arithmetic, lie inside these intervals.

n	J_B	Certified interval for $e \text{A189281}(n)/n!$	Certified interval for $E_{10}(n)$
100	23	[1.030201000327020, 1.030201000327021]	[−58916.80, −58916.77]
200	27	[1.015050125009789, 1.015050125009790]	[−59083.680, −59083.679]
1000	29	[1.003002001000003, 1.003002001000004]	[−59118.96, −59109.60]

Table 27: Report 83's exact Bonferroni enclosures of the A189281 avoidance probability, rounded outward by the batch-101 write. $E_{10}(n)$ is Part I's scaled residual (10.2) at $M = 10$, whose limit is $c_{11} = -59103$. The intervals certify these finite residuals only.

Checks against Parts I and III. [Note of the batch-101 write.] The intervals are consistent with the coefficients printed elsewhere in the report, which they did not use. The truncation $\sum_{j \leq 40} c_j n^{-j}$ of Part III's coefficients (`data/03-rational-collapse-coefficients.json`) lies inside all three enclosures. The truncation at order 16 of Part I's table 1 lies inside at $n = 1000$ and just outside at $n = 100$ and $n = 200$, as the omitted term of order n^{-17} allows. At $n = 1000$ the E_{10} interval, of width 9.35, contains $c_{11} + c_{12}/n = -59103 - 9.448$ but not c_{11} alone: it is a check of Part I's c_{11} that also requires a correction c_{12}/n of the printed sign and rough size, beyond the order through which Report 83 itself computed coefficients. (Dated note, 5 October 2026: this sentence first called the containment “an independent check of Part I's c_{11} and c_{12} ”. With width 9.35 the interval fixes $c_{11} + c_{12}/n$ only to within about ± 4.7 , so it checks c_{11} to

about 0.01 % and excludes $c_{12} = 0$, but it does not check the value of c_{12} ; see the independent check recorded at the end of [section 87](#).) As in Part I, the widening of the E_{10} interval at $n = 1000$ is the factor n^{11} magnifying the fixed Bonferroni width; it is a precision limit of this finite computation, and the convergence of $E_{10}(n)$ to c_{11} is a consequence of [theorem 1.1](#), not of three samples.

73 The threshold inverse: a ceiling bracket and the exact-digamma display

Fix $r, s \geq 1$ and $K \geq 0$, put $\theta = 1$, and write $A(n) = A_{r,s}^{(1)}(n)$, $c_j = c_j^{(1)}(r, s)$ and

$$\mathcal{A}_K(x) = e^{-1}\Gamma(x+1) \sum_{j=0}^K c_j x^{-j},$$

which is Part I's smooth model of the last subsection of [section 9](#) with $M = K$. It is positive and strictly increasing on a half-line $[x_*, \infty)$; let $x_K = \mathcal{A}_K^{-1}$ there, and let $N(y) = \min\{n \geq 0 : A(n) \geq y\}$.

Theorem 73.1 (Threshold bracket; Report 83, Theorem 6.1). *There are constants $C = C_{r,s,K}$ and y_0 such that for every $y \geq y_0$*

$$\lceil x_K(y) - \varepsilon_K(y) \rceil \leq N(y) \leq \lceil x_K(y) + \varepsilon_K(y) \rceil, \quad \varepsilon_K(y) = \frac{C x_K(y)^{-K-1}}{\log x_K(y)}. \quad (73.1)$$

In particular $N(y) = \lceil x_K(y) \rceil$ whenever the distance of $x_K(y)$ from the integers exceeds $\varepsilon_K(y)$.

This is the staircase theorem `p0:thm:staircase` of the transseries volume [12] applied to $\mathcal{A}_K$, as Part I's remark at the end of [section 9](#) already says in words; neither the constant C nor the onset y_0 is explicit. Report 83 proves it in four sentences. The following proof spells out the two endpoint comparisons.

Proof (completed by the batch-101 write). Put $g = \log \mathcal{A}_K$, so $g(x) = \log \Gamma(x+1) - 1 + \ell_K(x)$ with $\ell_K(x) = \log \sum_{j \leq K} c_j x^{-j}$, a convergent power series in $1/x$ for large x with $\ell_K(x) = O(1/x)$ and $\ell'_K(x) = O(x^{-2})$. Since $\psi(x+1) = \log x + O(1/x)$, there is $x_1 \geq \max(x_*, 4)$ with $g'(x) \geq \frac{1}{2} \log x$ for $x \geq x_1$. By [theorem 1.1](#) at $u = -1$ and $\theta = 1$ there are C_1 and $n_1 \geq x_1$ such that $\eta_n = \log A(n) - g(n)$ satisfies $|\eta_n| \leq C_1 n^{-K-1}$ for $n \geq n_1$. Moreover $A(n+1)/A(n) = (n+1) \mathbb{P}_{n+1}(X=0)/\mathbb{P}_n(X=0) \rightarrow \infty$, so there is $n_2 \geq n_1$ with A strictly increasing on $[n_2, \infty)$. We may take $C_1 > 0$. Choose $C = 2^{K+4}C_1$, and choose y_0 so large that for $y \geq y_0$ the point $x = x_K(y)$ satisfies $x \geq 2n_2 + 2$ and $\varepsilon_K(y) \leq 1$, and $y > \max_{n < n_2} A(n)$.

Upper bound. Let $m = \lceil x + \varepsilon_K \rceil$, so $m \geq x + \varepsilon_K$ and $x \leq m \leq x + 2 \leq 2x$. By the mean value theorem and monotonicity of $\log$, $g(m) - g(x) \geq \frac{1}{2} \log x \cdot (m - x) \geq \frac{1}{2} C x^{-K-1}$. Hence, since $m \geq x$ gives $m^{-K-1} \leq x^{-K-1}$,

$$\log A(m) - \log y = g(m) - g(x) + \eta_m \geq (\tfrac{1}{2}C - C_1)x^{-K-1} > 0.$$

So $A(m) \geq y$ and $N(y) \leq m$.

Lower bound. Let n be an integer with $n < \lceil x - \varepsilon_K \rceil$, that is, $n < x - \varepsilon_K$. If $n < n_2$ then $A(n) < y$ by the choice of y_0 . Otherwise let n' be the largest integer with $n' < x - \varepsilon_K$; then $n_2 \leq n \leq n'$, $x - n' > \varepsilon_K$ and $n' \geq x - 2 \geq x/2$. For $\xi \in (n', x)$ we have $g'(\xi) \geq \frac{1}{2} \log n' \geq \frac{1}{2} \log(x/2) \geq \frac{1}{4} \log x$, because $x \geq 4$. Hence $g(x) - g(n') > \frac{1}{4} \log x \cdot \varepsilon_K = \frac{1}{4} C x^{-K-1} = 2^{K+2}C_1 x^{-K-1}$, while $\eta_{n'} \leq C_1(x/2)^{-K-1} = 2^{K+1}C_1 x^{-K-1}$. Therefore

$$\log A(n') - \log y = \eta_{n'} - (g(x) - g(n')) < (2^{K+1} - 2^{K+2})C_1 x^{-K-1} < 0,$$

and by monotonicity on $[n_2, \infty)$, $A(n) \leq A(n') < y$. Thus every $n < \lceil x - \varepsilon_K \rceil$ has $A(n) < y$, and $N(y) \geq \lceil x - \varepsilon_K \rceil$.

The last assertion holds because, when x is farther than ε_K from the integers, both ceilings in (73.1) equal $\lceil x \rceil$. $\square$

Report 83 also writes the model inverse as an all-orders formal Lagrange reversion around the exact gamma core. Let $\mathfrak{X} = \mathfrak{X}(y)$ be the root of

$$\log \Gamma(\mathfrak{X} + 1) - 1 = \log y, \quad (73.2)$$

and put $\psi_{\mathfrak{X}} = \psi(\mathfrak{X} + 1)$ and $\psi'_{\mathfrak{X}} = \psi'(\mathfrak{X} + 1)$. With $\ell(x) = \sum_{j \geq 1} \ell_j x^{-j}$ the formal logarithm of the correction series (Part I's (9.2)), Report 83's (6.4) reads

$$x(y) \sim \mathfrak{X} + \sum_{k \geq 1} \frac{(-1)^k}{k!} \left(\frac{1}{\psi_{\mathfrak{X}}} \frac{d}{d\mathfrak{X}} \right)^{k-1} \left(\frac{\ell(\mathfrak{X})^k}{\psi_{\mathfrak{X}}} \right). \quad (73.3)$$

Report 83 justifies only finite truncations of this series, by Taylor's theorem for $\mathcal{A}_K$; the series itself is formal (section 76, question 2). Its first terms give a display that Parts I–V do not contain.

Proposition 73.2 (A189281 inverse in the digamma coordinate; Report 83, (6.6)). *Let $(r, s) = (2, 2)$, $K \geq 3$, and $x = x_K(y)$. As $y \rightarrow \infty$,*

$$x = \mathfrak{X} - \frac{3}{\mathfrak{X}\psi_{\mathfrak{X}}} + \frac{5}{2\mathfrak{X}^2\psi_{\mathfrak{X}}} - \frac{4}{\mathfrak{X}^3\psi_{\mathfrak{X}}} - \frac{9}{\mathfrak{X}^3\psi_{\mathfrak{X}}^2} - \frac{9\psi'_{\mathfrak{X}}}{2\mathfrak{X}^2\psi_{\mathfrak{X}}^3} + O\left(\frac{\mathfrak{X}^{-4}}{\log \mathfrak{X}}\right). \quad (73.4)$$

Proof (batch-101 write). Report 83 states the display with its derivation from (73.3); here is the error bound. Write $G(x) = \log \Gamma(x + 1)$ and ℓ_K as in the previous proof. For A189281, $\ell_K(x) = 3x^{-1} - \frac{5}{2}x^{-2} + 4x^{-3} + O(x^{-4})$ when $K \geq 3$ (Part I's ℓ_1, ℓ_2, ℓ_3 ; they depend only on c_1, c_2, c_3), and since $\ell_K(x) = L(1/x)$ with L analytic at 0, also $\ell'_K(x) = -3x^{-2} + O(x^{-3})$ and $\ell''_K(x) = O(x^{-3})$. On the other side $G'(x) = \psi(x + 1) = \log x + O(1/x)$, $0 < G''(x) = \psi'(x + 1) \leq 1/x$ and $G'''(x) = O(x^{-2})$. By (73.2) the model equation $\log \mathcal{A}_K(x) = \log y$ is $G(x) + \ell_K(x) = G(\mathfrak{X})$. Put $\delta = x - \mathfrak{X}$ and

$$\Phi(t) = G(\mathfrak{X} + t) - G(\mathfrak{X}) + \ell_K(\mathfrak{X} + t).$$

For large $\mathfrak{X}$, Φ is increasing on $[-1, 1]$ with $\Phi' \geq \frac{1}{2} \log \mathfrak{X}$ there, $0 < \Phi(0) = \ell_K(\mathfrak{X}) \leq 4/\mathfrak{X}$ and $\Phi(-1) \leq -\frac{1}{2} \log \mathfrak{X} + \Phi(0) < 0$; hence its root δ lies in $(-1, 0)$ and, by the mean value theorem, $|\delta| \leq \Phi(0)/\inf \Phi' \leq 8/(\mathfrak{X} \log \mathfrak{X})$.

Taylor's theorem to second order in G and to first order in ℓ_K gives

$$0 = \psi_{\mathfrak{X}}\delta + \frac{1}{2}\psi'_{\mathfrak{X}}\delta^2 + \ell_K(\mathfrak{X}) + \ell'_K(\mathfrak{X})\delta + \rho, \quad |\rho| \leq \sup |G'''| \frac{|\delta|^3}{6} + \sup |\ell''_K| \frac{\delta^2}{2} = O(\mathfrak{X}^{-5}),$$

the suprema over $[\mathfrak{X} - 1, \mathfrak{X} + 1]$. Abbreviate $\ell = \ell_K(\mathfrak{X})$, $\ell' = \ell'_K(\mathfrak{X})$. Dividing by $\psi_{\mathfrak{X}} \asymp \log \mathfrak{X}$,

$$\delta = -\frac{\ell}{\psi_{\mathfrak{X}}} - \frac{\ell'\delta}{\psi_{\mathfrak{X}}} - \frac{\psi'_{\mathfrak{X}}\delta^2}{2\psi_{\mathfrak{X}}} + O\left(\frac{\mathfrak{X}^{-5}}{\log \mathfrak{X}}\right).$$

The last three terms are $O(\mathfrak{X}^{-3}/\log^2 \mathfrak{X})$, so $\delta = -\ell/\psi_{\mathfrak{X}} + \epsilon$ with $\epsilon = O(\mathfrak{X}^{-3}/\log^2 \mathfrak{X})$. Substituting this on the right,

$$-\frac{\ell'\delta}{\psi_{\mathfrak{X}}} = \frac{\ell\ell'}{\psi_{\mathfrak{X}}^2} + O\left(\frac{\mathfrak{X}^{-5}}{\log^3 \mathfrak{X}}\right), \quad -\frac{\psi'_{\mathfrak{X}}\delta^2}{2\psi_{\mathfrak{X}}} = -\frac{\psi'_{\mathfrak{X}}\ell^2}{2\psi_{\mathfrak{X}}^3} + O\left(\frac{\mathfrak{X}^{-5}}{\log^4 \mathfrak{X}}\right),$$

so that $\delta = -\ell/\psi_{\mathfrak{X}} + \ell\ell'/\psi_{\mathfrak{X}}^2 - \psi'_{\mathfrak{X}}\ell^2/(2\psi_{\mathfrak{X}}^3) + O(\mathfrak{X}^{-5}/\log \mathfrak{X})$; these are the terms $k = 1, 2$ of (73.3). Finally $-\ell/\psi_{\mathfrak{X}}$ gives the first three correction terms of (73.4) with error $O(\mathfrak{X}^{-4}/\log \mathfrak{X})$; $\ell\ell' = -9\mathfrak{X}^{-3} + O(\mathfrak{X}^{-4})$ gives $-9/(\mathfrak{X}^3\psi_{\mathfrak{X}}^2)$ with error $O(\mathfrak{X}^{-4}/\log^2 \mathfrak{X})$; and $\ell^2 = 9\mathfrak{X}^{-2} + O(\mathfrak{X}^{-3})$ with $\psi'_{\mathfrak{X}} \leq 1/\mathfrak{X}$ gives $-9\psi'_{\mathfrak{X}}/(2\mathfrak{X}^2\psi_{\mathfrak{X}}^3)$ with error $O(\mathfrak{X}^{-4}/\log^3 \mathfrak{X})$. $\square$

The display concerns the smooth model. For the counts it combines with [theorem 73.1](#): if $\hat{x}(y)$ denotes the right side of (73.4) without its remainder, then, since $x_K - \mathfrak{X} = O(1/(\mathfrak{X} \log \mathfrak{X}))$ and $K \geq 3$, there is a constant E with $\lceil \hat{x} - E\mathfrak{X}^{-4}/\log \mathfrak{X} \rceil \leq N(y) \leq \lceil \hat{x} + E\mathfrak{X}^{-4}/\log \mathfrak{X} \rceil$ for all large y ; and at the data $x_K(\text{A189281}(n)) = n + O(n^{-K-1}/\log n)$. Since $\psi'_{\mathfrak{X}} \sim 1/\mathfrak{X}$, the last two displayed terms are both of order $\mathfrak{X}^{-3}/\log^2 \mathfrak{X}$ or smaller; keeping the digamma function exact, rather than expanding it in $\log \mathfrak{X}$, is what makes the display this short. Report 83 adds that the gamma core starts at $\log(ey)/W(\log(ey)/e)$; that is its leading order only, the next Stirling term being of order one (Part I's z of (9.6) with $\log y$ shifted by one). [Note of the batch-101 write: the intake evaluated (73.4) against the exact model root at 60 digits with $K = 10$: the error times $\mathfrak{X}^4 \log \mathfrak{X}$ is 13.29, 12.72, 12.14, 11.60, 11.14, 10.75 at $\mathfrak{X} = 20, 50, 100, 200, 400, 800$, bounded as the proposition asserts. This is evidence, not part of the proof.]

74 Literature that the rest of the report does not cite

Report 83's `06-fixed-disp-source_status.md` records a primary-source check made on 1 October 2026. Four of its items are not cited elsewhere in this report; they are Report 83's readings, and the intake did not open the sources.

- *Mendelsohn (1956)* [30] develops a calculus of Poisson and falling-factorial corrections for permutation problems under a constant-coefficient operator recurrence. By Report 83's reading of its printed page 240, the coefficient calculation *assumes* that the complete asymptotic series exists; its examples are rencontres, ménages, rook-king successions and adjacent forbidden positions. It is an antecedent of the correction calculus, not a proof of the present remainder theorem. Part VII cites it for the same reason.
- *Tegui Tabuguia (2025)* [31], Section 5.2, re-obtains the A189281 recurrence by D-algebraic guessing and states it as Conjecture 5.1; it does not prove it.
- *Padhi's repository note* [32], *Holonomicity of the restricted permutation counts $a_{r,s}$ and $b_{r,s}$* , August 2026 (source blob `ec04817377b7531d19cbc0d9b0243242de682a95`), claims holonomicity for every fixed (r, s) through a bounded-seam construction. It contains no asymptotic expansion and does not derive the guessed operator; its README says it is not submitted or externally reviewed. Part I cites Padhi's arXiv preprint [6], which the intake had confirmed only as a record; the note is its repository form.
- *The Spahn–Zeilberger author page* [33] records, in an update of 14 August 2026, Padhi's proposed solution of the general-holonomicity challenge. The paper itself, cited by this report as [3], appeared as *Enumerative Combinatorics and Applications* 3:2 (2023), S2R10.

Report 83 also reads Kauers and Koutschan [5] as calling the order-eight recurrence conjectural (their Conjecture 10), and Tauraso as proving the leading and first corrections of the absolute family; both are cited in Part I.

75 Report 83's computations and shipped files

Report 83 ships five programs and seven recorded outputs, all prefixed `06-fixed-disp-`; below, the prefix is omitted.

- `code/compute_coefficients.py` (SymPy) implements only the finite formula (3.2), the expansion of logarithms of falling factorials and Touchard sums, and writes `data/coefficients_2_2_10.json` and `data/coefficients_2_2_5.json` (orders 10 and 5 at $(2, 2)$).

- `code/check_stable_tilings.py` (standard library) checks 180 stable-profile identities by a path-composition dynamic program, 48 brute-force permutation cases through $n = 8$ and the first 13 values of A189281 (OEIS data, CC BY-SA 4.0), and writes `data/check_stable_tilings.json`.
- `code/check_bonferroni.py` produces `data/check_bonferroni.json` and [table 27](#).

The two programs of its `independent_review/` directory are a third coefficient checker, independent of Report 83’s generator and of Part I’s `code/path_forests.py`: it expands normalized falling products as truncated integer polynomials, inverts the denominator by a complete-homogeneous recurrence, and sums the polynomial dependence on the dimer count by Newton forward differences,

$$e \sum_{k \geq 0} \frac{(-1)^k P(k)}{k!} = \sum_{j \geq 0} \frac{(-1)^j \Delta^j P(0)}{j!},$$

checking two interpolation points beyond the degree bound 2ℓ at retained order ℓ rather than fitting them. Its first program (`code/independent_review-check_coefficients.py`, prefix omitted as above) reconstructs the A189281 coefficients through order six, 30 higher-defect profiles, 299 finite-difference checks, all 16 pairs $1 \leq r, s \leq 4$ through order three and 25 profile counts by fresh edge-subset enumeration; the second (`..._check_order_ten.py`) extends the reconstruction to order ten (139 profiles, 423 checks). Their outputs are `data/independent_review-check_coefficients.json`, `data/independent_review-order_ten.json` and the log `..._order_ten_replay.log`, which holds both programs’ output because the second imports the first, whose module code reruns. The audit `independent_review-AUDIT.md` at the report root reviews `proof.md`, the plain-text version of the proofs.

All checks raise explicit exceptions, so Python’s `-O` flag does not disable them. Every program writes its output beside itself under its delivered name and imports its sibling by that name; the report README gives a recipe that restores the delivered layout in a scratch copy. The intake ran the whole suite on such a copy in under three minutes: all outputs equal the shipped ones after removing the carriage returns that Windows adds.

Remark 75.1 (Text drift in Report 83, recorded). [Batch-101 write.] Three sentences of the delivery are inaccurate, none of them mathematically. Its article says that the independent checker reaches order six; its audit says the reconstruction was “subsequently extended” to order ten, and the shipped `..._check_order_ten.py` does reach ten, as its delivered README says. Its delivered README says that “the delivered PDF is the same manuscript under a descriptive filename”, but the archive holds only `article.pdf`. And the gamma-core start $\log(ey)/W(\log(ey)/e)$ is a leading order only (see after [proposition 73.2](#)).

76 Further questions and research

[Added 5 October 2026, batch 101.] Report 83 poses no research questions. The following are its unproved or unchecked claims, moved here under the intake’s standing rule, with what is missing.

1. *Explicit constants in the threshold bracket.* [Theorem 73.1](#) asserts the existence of $C_{r,s,K}$ and of an onset y_0 ; neither is computed. Explicit values need an effective remainder in [theorem 1.1](#). This is Part III’s open onset question for nearest-integer rounding and Part I’s question 8; Part VII’s effective Newton certificate ([proposition 84.3](#)) covers its smooth model only.
2. *The all-orders reversion.* (73.3) is formal. Each finite truncation is justified, as in the proof of [proposition 73.2](#), but no statement uniform in the number of terms is proved, and the series may diverge. Compare `p0:thm:perturbed-inversion` of [12].

3. *Constants and sharpness of the flat comparison.* The constant C of Report 83’s Lemma 2.2 and the $O_\gamma(n)$ of its Theorem 5.1 are not explicit, and its range $\gamma < \min(\delta/2, 1/4)$ is narrower than Part I’s. Whether a bound $\exp(-\gamma n \log n)$ is sharp in γ for some pair of forests is open; this is part of Part I’s question 3.
4. *Reported, not verified.* Padhi’s proposed general holonomicity proof and the August 2026 update of the Spahn–Zeilberger page are reported by Report 83 and were not checked by the intake.

No claim of Report 83 was found to be wrong. Nothing in Part VI is formalized.

Part VII

Reciprocal coloured gaps: a joint expansion, the camel and knight counts, and threshold inverses

77 Provenance, scope and notation of Part VII

[Added 5 October 2026, batch 101.] Part VII was added in the write of batch 101. It is built from Report 232 of the session bundle, which treats two forbidden gaps at once: for fixed $1 \leq r < s$, the violations of the absolute gap (r, s) (positions r apart, values s apart) and of the reciprocal gap (s, r) , marked separately. It proves an all-orders expansion of their joint marked generating function, uniform on compact sets of $\mathbb{C}^2$, in which selected components may contain cycles; computes the second correction in full for the camel and knight sequences A189358 and A137774; and inverts both counts. This answers, for reciprocal pairs, the question of several forbidden gaps that the report asks four times: Part I’s question 6 ([section 12](#)), Part II’s question 4 ([section 22](#)), Part III’s question 10 ([section 34](#)) and Part IV’s [question 50.5](#); dated notes there say what remains open. Each single-gap marginal of the joint expansion is Part I’s absolute expansion ($\theta = 2$), and the write checks the printed coefficients against it ([remark 78.3](#)). Report 232 is new throughout apart from those marginals, so it is printed nearly in full, with its symbols renamed where they clash ([table 30](#)), its headings marked, and bracketed notes of the write.

77.1 Provenance

Report 232 is the archive `Report232.zip` of the session bundle of Reports 1–243, which arrived in commit `60f54ea06` (archive created 2026-10-05, 09:25 UTC); batch 101 placed it in this report in commit `f7c612c72`, reversing the triage’s proposal of a new report because this report names the problem four times. The archive holds ten files under `Report232/`, one of them the checksum manifest `SHA256SUMS`, which lists the other nine (all matched at the intake). Its article, `report.tex`, is titled *Reciprocal coloured gap avoidance: Uniform expansions and factorial growth inverses* (27 pages, dated 5 October 2026, headed “Report 232”); its author line reads “A self-contained mathematical and computational report” and its PDF metadata names “Research report” as author. It pins no repository commit and names no ProveIt path or file. One sentence of its literature section (“treat the present family separately from a one-source/one-target forest calculation”) echoes the non-claim of Part IV that its uniform theorem “concerns one source forest and one target forest, not unions of differently coloured gap constraints”; Part IV was written in commit `eea159818` about five hours before the archive was made, so Report 232 probably saw it, but nothing records this. It does not cite this report.

Scope and non-claims of Report 232, kept. The leading e^{-4} limits of the two chess-piece counts and the general methods (graph inclusion–exclusion, partition-lattice inversion, finite transfer matrices, analytic perturbation, Poisson correction polynomials) are classical and “not claimed as inventions”; there is no worldwide priority claim, and its source comparison is “bounded”, “not a proof that no equivalent theorem exists elsewhere”; renaming a model or changing one gap “would not establish a new theorem”. “Fixed order” asserts neither convergence of the $1/n$ series, nor uniformity for growing gaps, nor an effective error bound at a stated small n . The polynomial-identity thresholds ($n \geq 12$ camel, $n \geq 8$ knight) are not the onset of asymptotic accuracy, and the camel counterexample at $n = 9$ is kept on purpose. The inverse constants and onset are not evaluated; $N(y) = \xi_K(y) + o(1)$ is not uniform in y ; the effective Newton certificate for $\tilde{z} \geq 32$ concerns the smooth model only; no exact rounding of a centre is asserted. The core-edge lemma uses the reciprocal two-gap structure and “should not be silently extended” to general unions of gaps. Third and higher coefficients exist, but no value is claimed; the higher coefficients of its inverse fixture are a method fixture, not coefficients of the sequences. Its numerical tests are “consistency checks, not a substitute for the analytic remainder or an effective sequence onset”; it ships no third-party paper and no private review material, used no network access, and recommends external peer review for any publication claim.

77.2 What Part VII adds

Topic	Parts I–VI	Part VII (Report 232 of batch 101)
Model	one gap (r, s) , one source and one target forest (Parts I–IV)	two reciprocal gaps (r, s) and (s, r) , $r < s$, absolute differences, marked separately (section 78)
Selected graphs	unions of paths; profiles with a defect (section 2)	two-coloured graphs, cycles allowed; exact identity (proposition 79.1); compatible graphs triangle-free and 2-degenerate, with a <i>deficit</i> (lemma 80.1)
All-orders law	$e^{\theta u} \sum B_J(u) n^{-J}$ (theorem 1.1)	$e^{2u+2v} \sum \mathfrak{B}_j(u, v) n^{-j}$, uniform on compact sets of $\mathbb{C}^2$, $\deg \mathfrak{B}_j \leq 2j$ (theorem 78.1)
Cycles	asked for (Part I, question 6; question 50.5)	first at order n^{-2} , through the alternating square $4u^2v^2$; incompatible cycles vanish exactly (section 79.3, remark 80.2)
Explicit terms	B_1, B_2 for one gap (1.10), (7.5)	$\mathfrak{B}_1, \mathfrak{B}_2$ for camel and knight; $A189358 = e^{-4}n!(1 + 8/n + 48/n^2 + \dots)$, $A137774 = e^{-4}n!(1 + 4/n + 28/n^2 + \dots)$ (theorem 78.2); covariance and interaction polynomials (section 83)
Inverse	Lambert- W inverse; ceiling bracket (theorem 9.3, theorem 73.1)	the same apparatus for these counts, with an effective Newton certificate for the quadratic models and $N_{\text{camel}} \leq N_{\text{knight}} \leq N_{\text{camel}} + 1$ (section 84)

Table 28: Part VII against Parts I–VI.

Source and novelty ledger of Report 232, restated. Report 232 credits the leading limits to Kotěšovec’s compilation [34], higher-order Poisson corrections for adjacency problems to Aspvall and Liang [35], the existence-assuming correction calculus to Mendelsohn [30], the occupancy-mask transfer construction and the Poisson limit for fixed forbidden exceedances

Parts I–VI	Report 232’s use or route
theorem 1.1 at $\theta = 2$, offsets (r, s) and (s, r)	the marginals $v = 0$ and $u = 0$ of theorem 78.1 ; the printed $\mathfrak{B}_1$, $\mathfrak{B}_2$ restrict to (1.10) and (7.5), and the marginal avoidance coefficients 4, 10 (camel) and 2, 2 (knight) are Part I’s $c_1^{(2)}, c_2^{(2)}$ and Part II’s general C_1, C_2 (data/02-fixed-gap-general_pgf_first_three.txt); remark 78.3
Proposition 2.2	generalized from one pair of path forests to two-coloured graphs with cycles (proposition 79.1); the method is classical, as Report 232 says
Lemma 4.1	its role is played by lemma 80.3 , proved differently, by four matchings and Hölder’s inequality
the defect filtration (section 5)	the deficit filtration and the geometric majorant (lemmas 80.1 and 80.4); on cycles the two notions differ (section 77.3)
theorem 9.3 , theorem 49.1 ; theorem 73.1	the same Lambert- W apparatus and the same ceiling bracket, an instance of <code>p0:thm:staircase</code> [12], with new coefficient input (section 84); Report 232’s threshold envelope is more explicit than Part IV’s, which asserts no rounding threshold

Table 29: Results of Parts I–VI that Report 232 uses, generalizes or proves again.

to Flajolet and Sedgewick [36], small-subgraph asymptotics for static boards to Godsil and McKay [37], a one-direction generating function for the knight objects to Bagno, Eisenberg, Reches and Sigron [38], and mod-Poisson methodology to Méliot, Nikeghbali and Visentin [39]. It claims as its own the two-sided reciprocal coloured finite-core expansion, its uniform factorial-scale summation bound, and the explicit mixed and cycle corrections for the two named sequences. It does not know this report; its marginal statements are Part I’s.

77.3 Notation

Report 232 uses several letters for two or three objects, and several that mean something else in Parts I–VI. [Table 30](#) lists every renaming; no normalization is changed. The factorial power is written $n^{\underline{h}} = n(n-1) \cdots (n-h+1)$, as in the rest of the report (Report 232: $(n)_h$).

Defect and deficit count different things. Part I’s *defect* of a selected set is $d = k - c$, the number of selected edges minus the number of components; for a forest it is $v - 2c$, the number of vertices minus twice the number of components. Report 232’s *deficit* is $\delta(H) = v(H) - 2c(H)$ by definition. The two agree on forests and differ on graphs with cycles: the alternating square of [section 79.3](#) has deficit 2 but $e - c = 3$. It is the deficit that measures the order $n^{-\delta}$ at which a coloured graph enters, since each component has one translational freedom on each side while every vertex costs one power of n . The tempting false reading is that a selected graph with e edges and c components enters at order $n^{-(e-c)}$, as a path forest does; for the square that would be n^{-3} , and it enters at n^{-2} .

The following symbols keep Report 232’s meaning inside Part VII and mean something else elsewhere in the report: the marks u, v (Part I has one mark u); X_n, Y_n , the two violation counts (Part I’s $X_{r,s,n}^{(\theta)}$); the colours A and B ; the source and target graphs S_n, T_n (Part IV’s S_q); the deficit δ ; $h = |V(H)|$, h_0 and the expansion variable h of (81.6) (Part I’s lengths); $c(H)$, the number of components; L , the edge cutoff $C \log n$ (Part IV’s path length); D , the largest support diameter in [section 81](#); x, y, w_i and X, Y, W_i , activities in [section 81](#) (elsewhere y is a threshold level); Z_n , the packing polynomial, and Z , a matching count; $\lambda(\mathbf{z}), P(\mathbf{z}), Q(\mathbf{z}), F, F_1, \Phi, P_j$ of the spectral argument (Part IV’s λ , Part I’s F_r); $M_{k,l}$, mixed binomial moments; N_S ,

Notion	Part VII	Report 232 (and the clash)
correction polynomials	$\mathfrak{B}_j(u, v; r, s), \mathfrak{B}_{j,R}$	$B_j(u, v; r, s), B_{j,R}$ (Part I's $B_J^{(\theta)}(u; r, s)$, one mark)
counts	$A_{r,s}^{\text{cg}}(n); A(n)$ in section 84	$a_{r,s}(n); a_n$ (Part I's a_j , Part IV's a_q)
incident and disjoint mixed pairs	$\nu_{\text{inc}}, \nu_{\text{dis}}$	S, D (also its source graph S_n ; Part I's D)
transfer matrix	$\mathbf{T}(\mathbf{z})$	$T(\mathbf{z})$ (also its target graph T_n)
auxiliary radius in lemma 80.3	U'	T
connected homomorphism polynomial	$\alpha_Q n - \beta_Q$	$A_Q n - B_Q$ (also its colours A, B)
log-amplitude; spectral remainder	$\Omega(\mathbf{z}); \varepsilon_n(\mathbf{z})$	$A(\mathbf{z}); B_n(\mathbf{z})$
core automorphism factor	$\text{aut}_{\mathbf{t}}$	$A_{\mathbf{t}}$
species support	$\mathbf{P}$	P (also its spectral projection $P(\mathbf{z})$)
avoidance coefficients of A189358, A137774	c_j	c_j (not Part I's $c_J^{(\theta)}(r, s)$)
logarithmic coefficients	ℓ_j, a_j (Part I's (9.2), (9.4))	$d_j; b_j$ and β_j
model; its inverse; envelope roots	$\mathcal{F}_K; \xi_K(y); \xi_K^{\pm}(y)$	$F_K; h_K(y); r_K^{\pm}(y)$ (its $h = V(H) $, offset r)
correction series	$C_K(t) = 1 + \sum_{j \leq K} c_j t^j$	$Q_K(t)$ (its quotient Q , spectral part $Q(\mathbf{z})$)
Lambert seed	$\tilde{L} = \log y + 4, \tilde{z}, \tilde{w} = \log \tilde{z}$	S, X, L (Part I's $L = \log y, z, w$; its X_n , cutoff L)
inverse shift; its coefficients	$\tilde{t}; D(1/\tilde{z}) = \sum d_j \tilde{z}^{-j}$	$t; \delta(z) = \sum u_j z^j$ (its deficit δ , mark u)
Newton intermediate point	η_j	ξ_j

Table 30: Notation of Report 232 against Part VII. Part I's $d_j(w)$ and Report 232's u_j are the same kind of inverse coefficient, in different coordinates.

N_T , $N_{k,l,t}$, counts; the constants C , C_K , D_K , E_K , n_K ; and the digamma function ψ , always written with its argument (Part III's ψ_i).

78 The model and the conclusions (Report 232)

78.1 Two reciprocal constraints

Fix integers $1 \leq r < s$. Write $[n] = \{1, \dots, n\}$ and let π be uniform on the permutations of $[n]$. Define

$$X_n = \#\{i \leq n - r : |\pi(i + r) - \pi(i)| = s\}, \quad (78.1)$$

$$Y_n = \#\{i \leq n - s : |\pi(i + s) - \pi(i)| = r\}, \quad (78.2)$$

$$G_n(u, v; r, s) = \mathbb{E}[(1 + u)^{X_n} (1 + v)^{Y_n}]. \quad (78.3)$$

Empty index sets contribute zero. A mark is the increment from 1, so that $u = v = 0$ gives $G_n = 1$ and $u = v = -1$ gives the avoidance probability. In particular, the exponential factor below is e^{2u+2v} , not $e^{2(u-1)+2(v-1)}$. [Note of the batch-101 write: in Part I's notation $X_n = X_{r,s,n}^{(2)}$ and $Y_n = X_{s,r,n}^{(2)}$, so $G_n(u, 0; r, s)$ and $G_n(0, v; r, s)$ are Part I's $G_{r,s,n}^{(2)}(u)$ and $G_{s,r,n}^{(2)}(v)$.]

There are two coloured graphs on $[n]$. The source graph S_n has colour A for edges of length r and colour B for edges of length s . In the target graph T_n , those lengths are reversed. A source edge is a violation exactly when its image is an edge of the same colour in T_n . A selected subgraph means a selected *edge set*: other edges between its vertices are irrelevant. This distinction matters for component counting and for the packing construction.

Write $A_{r,s}^{\text{cg}}(n) = n! G_n(-1, -1; r, s)$. For $r = 1$, put $R = s$ and abbreviate $\mathfrak{B}_j(u, v; 1, R)$ by $\mathfrak{B}_{j,R}(u, v)$. A permutation matrix already has one point in each row and column. Its points additionally avoid camel attacks when $(r, s) = (1, 3)$, and knight attacks when $(r, s) = (1, 2)$. Thus

$$A_{1,3}^{\text{cg}}(n) = \text{A189358}(n), \quad A_{1,2}^{\text{cg}}(n) = \text{A137774}(n)$$

[40, 41]. No cyclic wraparound is imposed on positions or values.

Theorem 78.1 (Uniform joint expansion; Report 232). *For fixed positive distinct integers $r < s$ and each integer $K \geq 0$, there are rational polynomials $\mathfrak{B}_j(u, v; r, s)$, $0 \leq j \leq K$, such that $\mathfrak{B}_0 = 1$, $\deg \mathfrak{B}_j \leq 2j$, and for every fixed $U < \infty$,*

$$G_n(u, v; r, s) = e^{2u+2v} \sum_{j=0}^K \frac{\mathfrak{B}_j(u, v; r, s)}{n^j} + O_{r,s,K,U}(n^{-K-1}) \quad (78.4)$$

uniformly for complex $|u|, |v| \leq U$. The coefficient polynomials admit a finite exact-arithmetic construction. Equivalently, $\mathfrak{B}_j$ is determined by the mixed binomial moments through total marked degree $2j$, once the lower orders have been determined.

The quantifier “fixed order” permits every prescribed K , with constants and an eventual onset depending on K . It does not assert convergence of the infinite $1/n$ series, uniformity for growing gaps, or an effective numerical error bound at a stated small n .

Theorem 78.2 (The first two corrections; Report 232). *For $R \in \{2, 3\}$,*

$$\mathfrak{B}_{1,R}(u, v) = -2R(u + v) - 2(u^2 + v^2). \quad (78.5)$$

The second polynomials are

$$\begin{aligned} \mathfrak{B}_{2,3}(u, v) &= 2(u^4 + v^4) + 14(u^3 + v^3) + 22(u^2 + v^2) \\ &\quad + 8u^2v^2 + 12(u^2v + uv^2) + 44uv, \end{aligned} \quad (78.6)$$

$$\begin{aligned} \mathfrak{B}_{2,2}(u, v) &= 2(u^4 + v^4) + 10(u^3 + v^3) + 10(u^2 + v^2) \\ &\quad + 8u^2v^2 + 4(u^2v + uv^2) + 24uv. \end{aligned} \quad (78.7)$$

Consequently,

$$A_{1,3}^{\text{cg}}(n) = e^{-4}n! \left(1 + \frac{8}{n} + \frac{48}{n^2} + O(n^{-3}) \right), \quad (78.8)$$

$$A_{1,2}^{\text{cg}}(n) = e^{-4}n! \left(1 + \frac{4}{n} + \frac{28}{n^2} + O(n^{-3}) \right). \quad (78.9)$$

The coefficient theorem is not a conclusion drawn by fitting a few numerical moments. Its logical order is: prove the uniform expansion and degree bound; construct every necessary exact embedding polynomial; then extract the finite coefficient data. [Sections 79](#) to [82](#) give the proof and [section 83](#) gives the finite calculation.

Remark 78.3 (The marginals are Part I's absolute expansions). [Batch-101 write.] For every j and every $1 \leq r < s$,

$$\mathfrak{B}_j(u, 0; r, s) = B_j^{(2)}(u; r, s), \quad \mathfrak{B}_j(0, v; r, s) = B_j^{(2)}(v; s, r) = B_j^{(2)}(v; r, s).$$

Proof. By the note after (78.3), $G_n(u, 0; r, s) = G_{r,s,n}^{(2)}(u)$. [Theorem 78.1](#) at $v = 0$ and Part I's [theorem 1.1](#) both expand this function, for each fixed complex u , as $e^{2u} \sum_{j \leq K} P_j(u) n^{-j} + O(n^{-K-1})$. Subtracting, $e^{2u} \sum_{j \leq K} (\mathfrak{B}_j(u, 0) - B_j^{(2)}(u)) n^{-j} = O(n^{-K-1})$; multiplying by $n^0, n^1, \dots$ in turn and letting $n \rightarrow \infty$ shows that every difference vanishes at every u , so the polynomials agree. The same argument at $u = 0$ gives the second identity, and the last equality is the symmetry in (1.9). $\square$

At $(r, s) = (1, R)$, (78.5) restricts to $-2Ru - 2u^2 = B_1^{(2)}(u; 1, R)$ of (1.10), and (78.6), (78.7) restrict to $2u^4 + 14u^3 + 22u^2$ and $2u^4 + 10u^3 + 10u^2$, which are (7.5) at (1, 3) and (1, 2). The marginal avoidance coefficients are therefore $c_1^{(2)}, c_2^{(2)} = 4, 10$ (camel) and 2, 2 (knight), as Report 232 states in [section 83.4](#); they are Part I's and Part II's values, a second route, and the printed polynomials pass this check exactly.

78.2 What is proved and what is numerical

The proofs establish (78.4) and the inverse statements for all sufficiently large n or thresholds. The code supplies finite checks and exact certificates, rather than proving an asymptotic remainder from numerical agreement. It uses rational arithmetic throughout its certificate path.

The eventual embedding polynomials for graphs with at most four edges are safe for $n \geq 12$ in the camel case and $n \geq 8$ in the knight case. These are *polynomial-identity thresholds*, not the onset of asymptotic accuracy. In fact, the camel fourth factorial-moment polynomial fails at $n = 9$; the report preserves that counterexample explicitly (83.11). Small- n checks below the safe thresholds use actual edge-set counts instead of extrapolating the eventual polynomials.

79 Exact graph identities (Report 232)

79.1 Selected edges and injective embeddings

A coloured graph has no isolated vertices unless it is the empty graph. For such a graph H , let $h = |V(H)|$, $e_A(H), e_B(H)$ be the colour counts, and $\text{aut}(H) = |\text{Aut}(H)|$ be the colour-preserving automorphism count. Write $i_S(H; n)$ and $i_T(H; n)$ for the numbers of injective colour-preserving homomorphisms from H into S_n and T_n . These homomorphisms need not be induced.

Proposition 79.1 (Exact coloured graph identity; Report 232). *For every n ,*

$$G_n(u, v; r, s) = \sum_{[H]} \frac{i_S(H; n) i_T(H; n)}{\text{aut}(H) n^{|V(H)|}} u^{e_A(H)} v^{e_B(H)}. \quad (79.1)$$

The sum is over colour-preserving isomorphism types of selected source-edge sets, including the empty graph with contribution 1. In particular only $|V(H)| \leq n$ occurs.

Proof. For a source edge e let I_e indicate that its image is a target edge of the same colour. Expanding the finite product

$$\prod_{e \in E_A(S_n)} (1 + u I_e) \prod_{e \in E_B(S_n)} (1 + v I_e)$$

selects an arbitrary coloured source-edge subset F . If its isomorphism type is H , the restriction of π to its h vertices is uniform over the n^h injections into $[n]$. Exactly $i_T(H; n)$ restrictions satisfy every selected event. There are $i_S(H; n)/\text{aut}(H)$ source-edge sets of type H : every such edge set has precisely $\text{aut}(H)$ isomorphisms from a fixed representative. Grouping the expectation by type gives (79.1). The argument neither deletes cycles nor assumes the two colours independent. $\square$

[Note of the batch-101 write: for a single gap the selected graphs are path forests and (79.1) specializes to Part I's Proposition 2.2.]

Put $N_S(H; n) = i_S(H; n)/\text{aut}(H)$, and similarly for T . Then the contribution of H may also be written

$$\frac{\text{aut}(H) N_S(H; n) N_T(H; n)}{n^{|V(H)|}} u^{e_A(H)} v^{e_B(H)}. \quad (79.2)$$

Because T_n is obtained by swapping colours, $N_T(H; n) = N_S(H^*; n)$, where H^* is H with its colours exchanged. This will simplify the exact checks.

79.2 Partition-lattice certificates

Proposition 79.2 (Exact eventual embedding polynomials; Report 232). *For every finite coloured graph H , its injection counts into the two-gap source and target graphs are eventual polynomials in n with integer coefficients. The unlabelled counts N_S, N_T are eventual rational polynomials. An exact finite construction is*

$$i_S(H; n) = \sum_{\rho \in \Pi(V(H))} \mu(\rho) \text{hom}(H/\rho, S_n), \quad \mu(\rho) = \prod_{\mathbf{b} \in \rho} (-1)^{|\mathbf{b}|-1} (|\mathbf{b}|-1)!. \quad (79.3)$$

Here $\Pi(V(H))$ is the set of partitions of the vertex set, and $\mathbf{b}$ runs over the blocks of ρ . The polynomials are valid for $n \geq s e(H)$; a more precise safe bound is the sum of the specified edge lengths. The same assertion holds for T_n .

Proof. A map from $V(H)$ to $[n]$ has a kernel partition recording equal images. Möbius inversion on the partition lattice isolates the discrete kernel, giving (79.3). A quotient with a loop contributes zero, since all gaps are positive. If two constraints on the same quotient pair demand distinct lengths, that quotient also contributes zero. Repeated identical constraints may be reduced to one.

Consider a connected quotient Q . Fix a root and a spanning tree. Give each tree edge one of its two signs and propagate relative integer coordinates from root coordinate 0. There are finitely many sign assignments. Retain an assignment precisely when every non-tree edge has its required absolute length. At this homomorphism stage coincident coordinates at distinct quotient vertices are allowed. Each valid relative assignment has span

$$d = \max_{w \in V(Q)} x_w - \min_{w \in V(Q)} x_w$$

and exactly $\max(0, n - d)$ translations into $[n]$. Its span is at most the sum of the tree-edge lengths, hence at most the sum of the original edge lengths. For n at least that bound its

contribution is $n - d$. Thus the connected homomorphism polynomial is $\alpha_Q n - \beta_Q$, with α_Q, β_Q nonnegative integers. For a disconnected quotient its components map independently, so their homomorphism polynomials multiply. The finite integer linear combination (79.3) is the claimed polynomial. Division by $\text{aut}(H)$ gives the unlabelled count. All target lengths are also at most s , so the same safe threshold $se(H)$ applies to both sides. $\square$

The formula is an arithmetic certificate, not interpolation. Every sign assignment is tested against every constraint. In particular an incompatible cycle vanishes for a mathematical reason, not because it happens to be absent from a finite numerical sample.

79.3 The alternating square

For $(r, s) = (1, R)$, consider vertices at $0, 1, R, R + 1$, with A edges $0-1$ and $R-(R + 1)$ and B edges $0-R$ and $1-(R + 1)$. The permutation of these coordinates

$$(0, 1, R, R + 1) \mapsto (0, R, 1, R + 1)$$

interchanges the required lengths, so this coloured square embeds on both sides. Its colour-preserving automorphism group has order 4. For $R = 2, 3$ all injective realizations are the translated alternating rectangle and its four labellings; the exact counts are

$$i_S(H; n) = i_T(H; n) = 4 \max(0, n - R - 1). \quad (79.4)$$

To check exhaustiveness, the two equal-colour opposite sides have signed length 1 and the other two have signed length R . Closing the square forces the two 1 steps to cancel and the two R steps to cancel; otherwise one would have $2 = 2R$. Distinctness forces the four corners of the stated rectangle. Hence for $n \geq \max(4, R + 1)$ its exact graph contribution is

$$\frac{4(n - R - 1)^2}{n^4} u^2 v^2 = \frac{4u^2 v^2}{n^2} + O(n^{-3}). \quad (79.5)$$

For the camel there is also a source four-cycle using three length-1 edges and one length-3 edge. In the target, three signed 3 steps cannot sum to a signed 1 step. That cycle is incompatible and contributes zero. Both examples show why cycles must be tested rather than excluded wholesale.

80 Connected cores and positive tail bounds (Report 232)

80.1 Deficit measures the asymptotic cost

Let $c(H)$ be the number of connected components and $v(H) = |V(H)|$. Define the *deficit*

$$\delta(H) = v(H) - 2c(H). \quad (80.1)$$

[Note of the batch-101 write: not Part I's defect $k - c$; the two agree on forests only (section 77.3).] A component consisting of one edge has deficit zero. Every other connected component has at least three vertices and positive deficit. The *core* of H is the union of its nontrivial components; throughout, “nontrivial” means having at least three vertices. If the core has t components and h_0 vertices, then

$$h_0 = \delta + 2t \leq 3\delta, \quad (80.2)$$

since $t \leq \delta$. Both source and target placements of each component have one translational freedom, whereas division by $n^{\overline{v(H)}}$ costs one power of n per vertex. Thus its leading scale is $n^{-\delta}$. We next make the required finite-core and tail claims precise.

Lemma 80.1 (Compatible cores have few edges; Report 232). *Every graph admitting embeddings in both S_n and T_n is triangle-free and 2-degenerate. Each of its connected components with $v \geq 3$ vertices has at most $2v - 4$ edges. Consequently its nontrivial core satisfies*

$$e_{\text{core}} \leq 2\delta, \quad h_0 \leq 3\delta. \quad (80.3)$$

For fixed K , only finitely many compatible core isomorphism types and component multiplicity vectors have deficit at most K .

Proof. In any subgraph of S_n , the vertex with least coordinate has at most two neighbours to its right, at distances r and s . Remove it and repeat. This is a degeneracy ordering with at most two later neighbours per vertex.

Suppose a triangle is realizable in S_n . Order its three coordinates. The largest gap must equal the sum of the other two. With only $r < s$ available, this forces $s = 2r$ and length multiset (r, r, s) . After colour reversal, the target triangle would require length multiset (s, s, r) . The largest of these lengths cannot be the sum of the other two, so no three distinct real coordinates can realize it. A compatible triangle therefore cannot exist.

For a triangle-free 2-degenerate graph on $v \geq 3$ vertices, remove $v - 3$ vertices in a degeneracy ordering. At most $2(v - 3)$ edges are removed and at most two remain among the final three vertices. Hence $e \leq 2v - 4$. Summing this inequality over core components gives $e_{\text{core}} \leq 2\delta$. Equation (80.2) gives the vertex bound. There are only finitely many coloured finite graphs on at most $3K$ vertices, proving finiteness. $\square$

This lemma uses the reciprocal two-gap structure. A general union of many coloured gaps can admit triangles and has a different degree bound; the present theorem should not be silently extended to that setting.

Remark 80.2 (Cycles enter first at order n^{-2}). [Batch-101 write; this answers the cycle part of [question 50.5](#) for reciprocal pairs.] Let $1 \leq r < s$. A compatible graph containing a cycle has a component with a cycle; by [lemma 80.1](#) that cycle has at least four vertices, so the component has deficit $v - 2 \geq 2$. Graphs containing cycles therefore enter (79.1) only at order n^{-2} or later. At order n^{-2} exactly one cyclic core can occur, a single component with four vertices and a cycle, that is a coloured four-cycle (a fifth edge would create a triangle). Among the six colour patterns of a four-cycle only the alternating one, $ABAB$, embeds on both sides:

- $AAAA$ and $BBBB$: four signed steps of one length return to the start only by revisiting a vertex, so there is no source embedding;
- $AAAB$: three signed r steps must sum to $\mp s$, which needs $s \in \{r, 3r\}$; in the target three signed s steps must sum to $\mp r$, impossible since $r < s$; $ABBB$ is impossible in the source for the same reason;
- $AABB$ (adjacent equal colours): the two A steps must be equal (else a vertex repeats), so their sum is $\pm 2r$, which must equal minus the sum of the two B steps, in $\{0, \pm 2s\}$; this is impossible;
- $ABAB$: the steps close only as $a, b, -a, -b$, since $\pm 2r = \pm 2s$ is impossible, which is the rectangle $\{0, r, s, r + s\}$ of [section 79.3](#), with $i_S = i_T = 4 \max(0, n - r - s)$ and $\text{aut} = 4$, exactly as there with $R + 1$ replaced by $r + s$.

In the resummation of [section 82](#), the alternating-square core has one source and one target species, so $p_{0,t} = 1$ in (81.2), and (82.1), (82.2) and (82.6) give its sector as $4u^2v^2e^{2u+2v}n^{-2} + O(n^{-3})$. Hence the selected graphs with cycles contribute exactly $4u^2v^2$ to $\mathfrak{B}_2(u, v; r, s)$ for every $1 \leq r < s$, and nothing earlier; incompatible cycles, such as the camel's $AAAB$ square, contribute exactly zero.

80.2 A uniform exponential moment

Lemma 80.3 (Long selected-edge tail; Report 232). *For every real $s_0 \geq 0$ and every n ,*

$$\mathbb{E}e^{s_0(X_n+Y_n)} \leq \exp\{2(e^{4s_0} - 1)\}. \quad (80.4)$$

For any fixed $U > 0$, any $U' > U$, and any nonnegative integer L ,

$$\sum_{k+l>L} U^{k+l} \mathbb{E} \left[\binom{X_n}{k} \binom{Y_n}{l} \right] \leq (U/U')^L \exp\{2((1+U')^4 - 1)\}. \quad (80.5)$$

In particular, the joint marked expansion can be truncated at $L = C_K \log n$ with uniform error $O(n^{-K-1})$ on $|u|, |v| \leq U$, by choosing C_K sufficiently large.

Proof. A single-length source graph is a disjoint union of paths, one for each residue class modulo its length. Alternately colouring each path's edges splits it into two matchings. Both source lengths together therefore split into four matchings. Each matching has at most $M = \lfloor n/2 \rfloor$ edges. Let Z count its violations.

Select k specified matching edges. After assigning the image vertices of the preceding j edges, there are $n - 2j$ unused target vertices. Whatever the preceding assignment, there are at most $2(n - 2j)$ ordered pairs of unused vertices at a specified absolute difference: the graph of that length has maximum degree two. Dividing by $(n - 2j)(n - 2j - 1)$ bounds the conditional probability by $2/(n - 2j - 1)$. Thus the probability of all k events is at most

$$\frac{2^k}{\prod_{j=0}^{k-1} (n - 2j - 1)}.$$

The coefficient $\mathbb{E} \binom{Z}{k}$ sums over at most $\binom{M}{k}$ selections. For $0 \leq j < k \leq M$, $M - j \leq n - 2j - 1$; hence

$$\mathbb{E} \binom{Z}{k} \leq \frac{2^k}{k!}.$$

For $t \geq 0$, the binomial expansion gives $\mathbb{E}(1+t)^Z \leq e^{2t}$. Take $t = e^{4s_0} - 1$, so that $\mathbb{E}e^{4s_0 Z} \leq \exp\{2(e^{4s_0} - 1)\}$, and apply Hölder's inequality with four exponents equal to four to the four matching counts. This proves (80.4). No independence between the matchings was used.

All mixed binomial moments are nonnegative. In the terms $k + l > L$, replace U^{k+l} by at most $(U/U')^L U'^{k+l}$, sum the full binomial expansion, and apply (80.4) with $s_0 = \log(1 + U')$. This proves (80.5). Choosing C_K with $C_K \log(U'/U) > K + 1$ gives the asserted error. At $U = 0$ the statement is immediate, and a larger positive U can always bound a given compact set. $\square$

The bound controls the sum of absolute marked contributions, so it applies to complex marks and to $u = v = -1$ without invoking cancellation in inclusion–exclusion. [Note of the batch-101 write: Report 232 writes “Take $t = e^{s_0} - 1$ and apply Hölder”; the exponent $4s_0$ is what Hölder's inequality with four factors requires, and it is the exponent in (80.4). The write states the step that way. In the role it plays, this lemma replaces Part I's Lemma 4.1; the proof is different.]

80.3 A geometric majorant for large deficits

Lemma 80.4 (High-deficit core tail; Report 232). *Fix U, K , and a cutoff $L = C \log n$. Within the exact selected-edge sum restricted to at most L edges, the total absolute contribution of cores with deficit $\delta \geq K + 1$ is $O(n^{-K-1})$ uniformly on $|u|, |v| \leq U$.*

Proof. The number of vertices of any retained graph is at most $2L$. For all sufficiently large n ,

$$\frac{n^v}{n^{\underline{v}}} = \prod_{i=0}^{v-1} (1 - i/n)^{-1} \leq 2 \quad (0 \leq v \leq 2L), \quad (80.6)$$

since the logarithm is $O(L^2/n)$. This is a uniform positive bound.

A connected source component on v vertices, anchored at its smallest coordinate, has support in $[0, s(v-1)]$: follow paths from the anchor, whose edge lengths are at most s . The number of possible supports of size v in this interval is at most an exponential C_1^v depending only on s . There are at most $2v$ candidate source edges between the occupied vertices, so the possible selected edge sets cost another factor 2^{2v} . Thus there are at most C^v anchored connected source shapes (remark 80.5). Given one fixed source component, choose the image of a fixed root in at most n ways and the signs on a rooted spanning tree in at most 2^{v-1} ways. Ignoring target collisions and cycle consistency increases this count, so the component has at most $n2^{v-1}$ target maps.

Suppose the core has deficit δ , t components, and $h_0 = \delta + 2t \leq 3\delta$ vertices. Write their positive deficits as an ordered composition of δ into t parts. There are $\binom{\delta-1}{t-1}$ such compositions. For each one, ordered core source placements cost at most $n^t C^{h_0}$. Divide by $t!$ for unordered components. This division remains valid when abstract component types repeat: in an actual valid packing the placed nonempty vertex supports are disjoint and therefore the placed components are distinct; every unordered packing has exactly $t!$ orderings. Allowing repeated or intersecting placements in the ordered upper count only makes it larger.

Let j be the number of remaining isolated edges, with either colour allowed. Their source selections cost at most $(2n)^j/j!$, because the source has at most $2n$ edges. For each of them there are at most $2n$ target assignments, giving $(2n)^j$. Core target assignments cost at most $n^t C^{h_0}$ after enlarging C . Divide by the falling factorial for $v = 2j + h_0$ vertices and use (80.6). The absolute marking factor is at most $U^j \max(1, U)^{2h_0}$, a deliberately loose bound valid because the core has at most $2h_0$ edges. We obtain, with constants depending only on r, s, U ,

$$2 \frac{(4U)^j}{j!} C_U^{3\delta} n^{-\delta} \frac{\binom{\delta-1}{t-1}}{t!}. \quad (80.7)$$

This bounds the contribution directly in the marked graph sum; no momentwise cancellation or collision expansion is being assumed.

Sum first over $j \geq 0$, which contributes e^{4U} , and then over $1 \leq t \leq \delta$. The latter sum is at most $\sum_t \binom{\delta-1}{t-1} = 2^{\delta-1}$. Thus the full deficit- δ sector is bounded by $C'_U (C''_U/n)^\delta$. Once $n > 2C''_U$, summing over $\delta \geq K+1$ is a convergent geometric tail of order n^{-K-1} . The empty core is handled separately in the finite-core part of the proof. $\square$

The absence of a factor growing with L in the final error is important. Bounding each fixed-deficit moment crudely and then retaining a logarithmic loss would not by itself prove the remainder in theorem 78.1.

Remark 80.5 (The count of anchored shapes). [Batch-101 write: Report 232 asserts the bound C_1^v without computing it; here is a value.] A support of size v contains the anchor 0 and $v-1$ further points of $\{1, \dots, s(v-1)\}$, so there are at most $\binom{s(v-1)}{v-1} \leq (es(v-1)/(v-1))^{v-1} = (es)^{v-1}$ supports, by $\binom{m}{k} \leq (em/k)^k$. Each occupied vertex has at most two source neighbours to its right, so at most $2v$ candidate edges join occupied vertices and at most 4^v edge sets can be selected. Hence there are at most $(es)^{v-1} 4^v \leq (4es)^v$ anchored connected source shapes on v vertices, and the proof above holds with $C_1 = es$ and $C = 4es$ before the enlargement made for the target maps.

81 A uniform finite-range packing lemma (Report 232)

81.1 Species and the exact finite matrix

A polymer species consists of a finite nonempty support $P \subseteq \{0, \dots, D\}$ with $\min P = 0$, together with a species identity and a type label. A placement translates its occupied support into $[n]$. A packing is an unordered set of placements whose occupied supports are pairwise disjoint. Their containing intervals are allowed to overlap. Distinct species may have the same support, and distinct species may have the same type. The identity can encode a selected coloured edge pattern, not just the occupied vertices.

Assume exactly one species has type x , exactly one has type y , and a_i species have type w_i , for $1 \leq i \leq q$. Let $Z_n(x, y, \mathbf{w})$ be the packing polynomial, including the empty packing, and for a fixed multi-index $\mathbf{t} \in \mathbb{N}^q$ put

$$N_{k,l,\mathbf{t}}(n) = [x^k y^l \mathbf{w}^{\mathbf{t}}] Z_n(x, y, \mathbf{w}), \quad |\mathbf{t}| = t_1 + \dots + t_q.$$

An absent type with $a_i = 0$ and $t_i > 0$ makes this coefficient identically zero and may be discarded.

Theorem 81.1 (Factorial-scale packing expansion; Report 232). *For every fixed finite species dictionary, multi-index $\mathbf{t}$, order $J \geq 0$, and $C > 0$, uniformly in nonnegative k, l with $k + l \leq C \log n$,*

$$n^{-(k+l+|\mathbf{t}|)} N_{k,l,\mathbf{t}}(n) = \frac{1}{k! l!} \left\{ \sum_{j=0}^J \frac{p_{j,\mathbf{t}}(k, l)}{n^j} + O\left(\frac{(1+k+l)^{2J+3}}{n^{J+1}}\right) \right\}. \quad (81.1)$$

The $p_{j,\mathbf{t}}$ are rational polynomials of total degree at most $2j$, and

$$p_{0,\mathbf{t}} = \prod_{i=1}^q \frac{a_i^{t_i}}{t_i!}. \quad (81.2)$$

Constants may depend on the displayed fixed parameters, but not on n, k, l .

Proof. We give the finite matrix, the uniform analytic estimate, and the coefficient extraction separately. These are all needed: an expansion at fixed k, l alone would not justify the final resummation.

The occupancy matrix and both boundaries. Use a D -bit mask $b = (b_0, \dots, b_{D-1})$ immediately before scanning the current site. A set bit records a site already occupied by a polymer that began earlier, from the current site through $D - 1$ sites ahead. If $b_0 = 1$, no species can begin here; shift the mask one site left. If $b_0 = 0$, either leave this site empty and shift, or begin one species P whose occupied sites avoid the preexisting mask. Insert all of P , including its possible site D , then shift. This gives a D -bit next mask. Assign weight 1 for no start and the species' type variable for a start, adding weights when transitions coincide. All supports contain their anchor 0, so two species cannot begin at the same site in a valid packing.

These rules define a finite matrix $\mathbf{T}(\mathbf{z})$ polynomial in the activities $\mathbf{z} = (x, y, w_1, \dots, w_q)$. With e_0 the empty-mask vector,

$$Z_n(\mathbf{z}) = e_0^\top \mathbf{T}(\mathbf{z})^n e_0. \quad (81.3)$$

The initial empty mask enforces the left boundary. After n sites have been scanned, a polymer extending beyond $[n]$ leaves a nonempty mask. Requiring the final empty mask therefore enforces the right boundary exactly. Disjoint supports such as $\{0, 3\}$ and $\{1, 2\}$ are permitted: the mask tests occupied sites, not containing intervals.

At zero activity every state shifts to zero within D steps. Hence $\mathbf{T}(0)^D$ has all its rows concentrated in column zero, has rank one, and is idempotent. The eigenvalue 1 of $\mathbf{T}(0)$ is algebraically simple, and all other eigenvalues are zero. The case $D = 0$, if present, is a scalar matrix and follows by the same argument without the flushing step.

Analytic spectral decomposition. Finite-dimensional analytic perturbation at this isolated simple eigenvalue gives a complex polydisc about 0 on which an analytic eigenvalue $\lambda(\mathbf{z})$ and analytic spectral projection $P(\mathbf{z})$ exist, with $\lambda(0) = 1$. Put

$$Q(\mathbf{z}) = \mathbf{T}(\mathbf{z}) - \lambda(\mathbf{z})P(\mathbf{z}).$$

Then $PQ = QP = 0$. On a sufficiently small closed polydisc the spectrum of Q is inside a circle $|\zeta| = \rho < 1$. The contour formula for powers,

$$Q(\mathbf{z})^n = \frac{1}{2\pi i} \int_{|\zeta|=\rho} \zeta^n (\zeta I - Q(\mathbf{z}))^{-1} d\zeta,$$

with a fixed uniformly bounded resolvent, gives $\|Q(\mathbf{z})^n\| \leq C_0 \rho^n$. Let $\alpha(\mathbf{z}) = e_0^T P(\mathbf{z}) e_0$. It satisfies $\alpha(0) = 1$. Shrinking the polydisc again gives analytic logarithms $F = \log \lambda$ and $\Omega = \log \alpha$, and therefore

$$Z_n(\mathbf{z}) = \exp\{nF(\mathbf{z}) + \Omega(\mathbf{z})\} + \varepsilon_n(\mathbf{z}), \quad |\varepsilon_n(\mathbf{z})| \leq C_0 \rho^n. \quad (81.4)$$

Both F and Ω vanish at 0.

The linear part of F is

$$F_1(\mathbf{z}) = x + y + \sum_i a_i w_i. \quad (81.5)$$

Indeed a single fixed species of diameter d has $n - d$ placements for $n \geq D$. Compare its linear activity coefficient in (81.4), divide by n , and let $n \rightarrow \infty$; Cauchy's estimate makes the corresponding coefficient of ε_n exponentially small. Each species contributes 1 to F_1 , and grouping species by type gives a_i .

Scale every activity. Write now $\mathbf{z} = (X, Y, W_1, \dots, W_q)$ and scale *all* activities by $1/n$, including the core markers W_i . Define

$$\Phi(h, \mathbf{z}) = \frac{F(h\mathbf{z}) - hF_1(\mathbf{z})}{h} + \Omega(h\mathbf{z}). \quad (81.6)$$

The apparent singularity at $h = 0$ is removable. If F_d, Ω_d are homogeneous parts, then the coefficient of h^j in Φ is $F_{j+1}(\mathbf{z}) + \Omega_j(\mathbf{z})$, of degree at most $j + 1$. Consequently

$$e^{\Phi(h, \mathbf{z})} = \sum_{j=0}^J P_j(\mathbf{z}) h^j + \text{remainder}, \quad P_0 = 1, \quad \deg P_j \leq 2j. \quad (81.7)$$

For the degree bound, a product of exponent terms with positive orders $j_1, \dots, j_a$ summing to j has degree at most $\sum_i (j_i + 1) = j + a \leq 2j$.

Put $s_* = 1 + \sum_i |z_i|$. On the circle $|h| = c/s_*^2$, with one sufficiently small fixed $c > 0$, the arguments $h\mathbf{z}$ lie in the analytic polydisc. The quadratic-and-higher part of F contributes $O(|h|s_*^2)$ to (81.6), and $\Omega(h\mathbf{z})$ contributes $O(|h|s_*)$. Hence $|\Phi|$ is bounded on the whole circle independently of $\mathbf{z}$. Cauchy's Taylor estimate in h , evaluated at $h = 1/n$ when $n \geq 2s_*^2/c$, gives

$$\left| e^{\Phi(1/n, \mathbf{z})} - \sum_{j=0}^J P_j(\mathbf{z}) n^{-j} \right| \leq C_J n^{-J-1} s_*^{2J+2}. \quad (81.8)$$

Combining (81.4)–(81.6),

$$Z_n(\mathbf{z}/n) = e^{F_1(\mathbf{z})} e^{\Phi(1/n, \mathbf{z})} + \varepsilon_n(\mathbf{z}/n).$$

Thus the analytic remainder from the first term is bounded by the right side of (81.8) times $e^{\Re F_1(\mathbf{z})}$.

Extract coefficients at factorial-scale radii. Use the Cauchy circles

$$|X| = \max(1, k), \quad |Y| = \max(1, l), \quad |W_i| = 1.$$

If $m = k + l \leq C \log n$, then $s_* \leq m + q + 3$, so the previous estimate applies uniformly for all sufficiently large n . On the circles,

$$|e^{F_1(z)}| \leq C_1 e^{\max(1, k) + \max(1, l)}.$$

Stirling's upper bound for factorials gives, for every $k \geq 0$,

$$\frac{e^{\max(1, k)}}{\max(1, k)^k} \leq C_2 \frac{\sqrt{k+1}}{k!}. \quad (81.9)$$

Cauchy's coefficient bound applied to (81.8), followed by (81.9) for k and l , now yields

$$O\left(\frac{(1+m)^{2J+3}}{k! l! n^{J+1}}\right).$$

The two square roots cost at most one extra power of $1 + m$. The fixed W_i coefficient circles have radius one and add only a constant. The spectral remainder contributes at most $C_0 \rho^n / (\max(1, k)^k \max(1, l)^l)$, which is smaller than the same bound uniformly for $m \leq C \log n$.

Finally,

$$[X^k Y^l \mathbf{W}^t] Z_n(\mathbf{z}/n) = n^{-(k+l+|t|)} N_{k,l,t}(n).$$

Since

$$[X^k Y^l] e^{X+Y} X^a Y^b = \frac{k^a l^b}{k! l!}, \quad (81.10)$$

coefficient extraction from $e^{F_1} P_j$ yields a polynomial in k, l of degree at most $2j$. The W coefficients at $j = 0$ give (81.2). Rationality follows by expanding the finite matrix's characteristic equation at its simple eigenvalue 1: its coefficients are rational and the nonzero linear derivative permits a rational recursive solution. The spectral projection and its amplitude also have rational Taylor recursions, hence so do F, Ω, P_j and the extracted coefficients (remark 81.2). This completes the proof. $\square$

Remark 81.2 (Rationality of the packing coefficients). [Batch-101 write: the last step of the proof above is stated in two sentences; here it is in full.] Let $\chi(\zeta, \mathbf{z}) = \det(\zeta I - \mathbf{T}(\mathbf{z})) \in \mathbb{Z}[\zeta, \mathbf{z}]$. Since 1 is a simple root of $\chi(\cdot, 0)$, $\partial_\zeta \chi(1, 0) \neq 0$, and the formal implicit-function theorem gives a unique power series $\lambda(\mathbf{z})$ with $\lambda(0) = 1$ and $\chi(\lambda, \mathbf{z}) = 0$; comparing coefficients degree by degree determines each coefficient by rational operations from earlier ones and division by $\partial_\zeta \chi(1, 0) \in \mathbb{Z} \setminus \{0\}$, so $\lambda \in \mathbb{Q}[[\mathbf{z}]]$, and it is the Taylor series of the analytic eigenvalue. For a simple eigenvalue the spectral projection is $P = \text{adj}(\lambda I - \mathbf{T}) / \partial_\zeta \chi(\lambda, \mathbf{z})$ near 0. Indeed, at each $\mathbf{z}$ near 0 the matrix $\lambda I - \mathbf{T}$ has rank one less than its size, so its adjugate is a multiple of $xy^\top$, where x and y span its right and left kernels; so is $P = xy^\top / (y^\top x)$; and the traces are $\text{tr adj}(\lambda I - \mathbf{T}) = \partial_\zeta \chi(\lambda, \mathbf{z}) \neq 0$ and $\text{tr } P = 1$. Its entries are therefore in $\mathbb{Q}[[\mathbf{z}]]$, hence so is $\alpha = e_0^\top P e_0$ with $\alpha(0) = 1$, and $\log \lambda, \log \alpha, \Phi, e^\Phi$, the P_j and, by (81.10), the $p_{j,t}$ have rational coefficients. Independently, rationality of the $\mathfrak{B}_j$ themselves also follows from (82.7): each mixed binomial moment is, for $n \geq s(k+l)$, a finite sum of terms $i_S i_T / (\text{aut } n^u)$ with integer polynomials i_S, i_T (proposition 79.2), so its expansion in $1/n$ has rational coefficients.

82 Assembly of the uniform expansion (Report 232)

We now prove theorem 78.1, including the normalization and the infinite isolated-edge summation.

82.1 Two finite polymer dictionaries for each core

Fix a compatible multiset of nontrivial connected coloured graph types. Let t_Q be the multiplicity of type Q , let $\mathbf{t} = (t_Q)$, $t = \sum_Q t_Q$, and let $h_0 = \sum_Q t_Q v(Q)$. Its deficit is $\delta = h_0 - 2t$. At fixed $\delta \leq K$, there are finitely many such vectors by [lemma 80.1](#).

For each connected type Q , list its normalized source realizations: the selected coloured edge patterns with minimum occupied coordinate 0. Each has bounded diameter at most $s(v(Q) - 1)$ and there are finitely many. Treat every distinct selected edge pattern as a separate species, marked by a common variable w_Q . Include one isolated A -edge species with marker x and one isolated B -edge species with marker y . This is the source dictionary. Construct the target dictionary in the same fashion using the reversed lengths. The two dictionaries may have different numbers of species of a given type, but have the same finite type list.

Packings correspond bijectively to selected edge sets with the prescribed component multiset. A component's minimum vertex is its unique anchor. Separate component supports must be vertex-disjoint. Unselected ambient edges joining distinct supports do not join components of the selected graph, so no further restriction is imposed. Species identity retains the selected edges even when distinct species have the same occupied support. Thus the hypotheses of [theorem 81.1](#) apply on both sides.

82.2 Automorphisms and the denominator

Add k isolated A edges and l isolated B edges. Write $m = k + l$. The resulting abstract graph H has $2m + h_0$ vertices and

$$\text{aut}(H) = 2^m k! l! \text{aut}_{\mathbf{t}}, \quad \text{aut}_{\mathbf{t}} = \prod_Q \text{aut}(Q)^{t_Q} t_Q!. \quad (82.1)$$

The factors 2^m orient the isolated edges, $k! l!$ permute identical isolated components within each colour, and $\text{aut}_{\mathbf{t}}$ accounts for the core components. This is the automorphism factor in (79.2); using the reciprocal factor here would be an error.

Apply [theorem 81.1](#) separately to source and target with $J = K - \delta$. Each unlabelled count has normalization $n^{m+t}/(k! l!)$. After multiplying the two counts, multiplying by (82.1), and dividing by n^{2m+h_0} , the leading scale is

$$n^{2t-h_0} \frac{2^m}{k! l!} = n^{-\delta} \frac{2^m}{k! l!}. \quad (82.2)$$

All fixed core constants are absorbed in the coefficient polynomials.

For completeness, the remaining denominator factor has its own uniform expansion. Put $v = 2m + h_0$. Then

$$\log \frac{n^v}{n^v} = \sum_{a \geq 1} \frac{n^{-a}}{a} \sum_{i=0}^{v-1} i^a. \quad (82.3)$$

The inner power sum is a polynomial in v of degree $a + 1$. Its coefficient at order n^{-j} after exponentiation has degree at most $2j$ in m : a product with total order j has degree at most j plus its number of factors. For a uniform remainder, regard the product $\prod_{i=0}^{v-1} (1 - ih)^{-1}$ as analytic in h on $|h| \leq c/(1+m)^2$, with h_0 fixed and c sufficiently small. Its logarithm is bounded on that circle, since $\sum_i i|h| = O(1)$ and every factor stays away from zero. Cauchy's estimate gives a remainder

$$O\left((1+m)^{2J+2} n^{-J-1}\right) \quad (82.4)$$

uniformly for $m \leq C \log n$.

Multiplying the two packing expansions and this denominator expansion is legitimate on that range. Their fixed-order coefficients are polynomials and their errors have a uniform polynomial majorant. The core sector is therefore

$$u^{e_A(\text{core})} v^{e_B(\text{core})} n^{-\delta} \frac{2^{k+l} u^k v^l}{k! l!} \times \left\{ \sum_{j=0}^{K-\delta} \frac{q_{j,t}(k, l)}{n^j} + O\left(\frac{(1+k+l)^{d_K}}{n^{K-\delta+1}}\right) \right\}, \quad (82.5)$$

with $\deg q_{j,t} \leq 2j$ and some fixed exponent d_K . The precise value of d_K is immaterial because the factorially weighted polynomial series is summable.

82.3 Resummation and the degree bound

Choose the logarithmic edge cutoff from [lemma 80.3](#). The actual constraint in the fixed-core sector is

$$k + l + e_{\text{core}} \leq L,$$

not merely $k + l \leq L$. Since $e_{\text{core}} \leq 2K$ is fixed, this is only a bounded shift in the cutoff. For any fixed polynomial q , extending its factorially weighted sum from that range to all $k, l \geq 0$ makes a superpolynomially small error. One direct bound groups by $m = k + l$:

$$\sum_{k+l=m} \frac{(2U)^{k+l}}{k! l!} (1+k+l)^d = \frac{(4U)^m}{m!} (1+m)^d.$$

Its tail from $C \log n - O(1)$ has logarithm at most $-C \log n \log \log n + O(\log n)$, so is smaller than every prescribed inverse power of n eventually.

The remainders in (82.5) sum absolutely to $O(n^{-K-1})$ because

$$\sum_{k,l \geq 0} \frac{(2U)^{k+l}}{k! l!} (1+k+l)^{d_K} < \infty.$$

Expand $q_{j,t}$ in the falling-factorial basis $k^a l^b$. The exact Poisson summation identity is

$$\sum_{k,l \geq 0} \frac{2^{k+l} u^k v^l}{k! l!} k^a l^b = e^{2u+2v} (2u)^a (2v)^b. \quad (82.6)$$

Hence the fixed-core order- $(\delta + j)$ contribution is e^{2u+2v} times a rational polynomial of degree at most

$$e_{\text{core}} + 2j \leq 2\delta + 2j.$$

There are finitely many core vectors with $\delta \leq K$. [Lemma 80.4](#) bounds all omitted deficits, and [lemma 80.3](#) bounds all omitted selected-edge counts. Together these prove (78.4), uniformly for complex marks in the stated compact set. The empty core at leading order has one isolated-edge species of each colour on both sides. Its leading sum is exactly e^{2u+2v} , so $\mathfrak{B}_0 = 1$. This proves [theorem 78.1](#).

82.4 Why finitely many moments suffice

The theorem's degree bound gives a finite extraction rule, not an extra hypothesis. Let $[n^{-j}]$ denote the coefficient in the proved expansion, and truncate a bivariate formal series to total marked degree at most $2j$. Then

$$\mathfrak{B}_j(u, v) = \left[e^{-2u-2v} [n^{-j}] G_n(u, v) \right]_{\deg \leq 2j}. \quad (82.7)$$

The coefficients of $u^k v^l$ in G_n are the mixed binomial moments. Moments through total degree $2j$ therefore determine every coefficient of $\mathfrak{B}_j$. Each such moment is obtained by the exact graph identity from graphs with at most $2j$ edges, and [proposition 79.2](#) computes every embedding polynomial. This supplies a second explicit finite construction besides expanding the occupancy matrices. It is computationally convenient at order two and remains finite at every fixed order.

83 Explicit mixed corrections and exact checks (Report 232)

83.1 The first correction by component sectors

Let $(r, s) = (1, R)$ with $R = 2$ or 3 , and write $M_{k,l} = \mathbb{E}[(X_n^k)(Y_n^l)]$, $m = k + l$. The matching sector has source count

$$\frac{n^m}{k!l!} \left[1 - \frac{1}{n} \left\{ k + Rl + 3 \binom{k}{2} + 3 \binom{l}{2} + 4kl \right\} + O\left(\frac{(1+m)^d}{n^2}\right) \right]. \quad (83.1)$$

Here and below these expansions are uniform for $m \leq C \log n$, with a fixed polynomial remainder as guaranteed by the packing lemma. The terms can also be read directly: a length- d edge has $n-d$ placements; choosing distinct same-colour edges costs $\binom{k}{2}/n$; excluding incident same-colour pairs costs another $2\binom{k}{2}/n$; and cross-colour incidence costs $4kl/n$. Multiple exclusions affect only the next order. In the target count the linear boundary term is $Rk + l$.

Multiply the two matching counts, the automorphism factor $2^m k! l!$, and the denominator expansion $n^{2m}/n^{2m} = 1 + m(2m-1)/n + O((1+m)^4/n^2)$. The relative first coefficient in this sector simplifies to

$$-(R-1)m - k^2 - l^2 - 4kl. \quad (83.2)$$

The only other deficit-one graphs are one two-edge path plus isolated edges. An AA path contributes the relative coefficient $\binom{k}{2}$; a BB path contributes $\binom{l}{2}$. A mixed AB path has $4n + O(1)$ unlabelled source realizations and $4n + O(1)$ target realizations, and contributes $4kl$. Triangles are incompatible by [lemma 80.1](#), so no further deficit-one term is missing. Combining with (83.2) gives

$$M_{k,l} = \frac{2^m}{k!l!} \left[1 - \frac{Rm + \binom{k}{2} + \binom{l}{2}}{n} + O_{k,l}(n^{-2}) \right]. \quad (83.3)$$

This last display can be read for fixed k, l ; the uniform resummation is supplied by [section 82](#). Applying (82.6) gives (78.5).

83.2 An exact cross moment

For $R = 2, 3$ and $n \geq 4$, the number of incident source A, B edge pairs is

$$\nu_{\text{inc}} = 4n - 4R - 2. \quad (83.4)$$

This follows by summing the four possible placements of a short edge at the endpoints of a long edge, with the boundary losses, or by a direct intersection count. The number of disjoint pairs is $\nu_{\text{dis}} = (n-1)(n-R) - \nu_{\text{inc}}$. An incident mixed path has automorphism count one and ν_{inc} source and target realizations. A disjoint pair has automorphism count four and ν_{dis} realizations on each side. Therefore

$$\mathbb{E}[X_n Y_n] = \frac{\nu_{\text{inc}}^2}{n^3} + \frac{4\nu_{\text{dis}}^2}{n^4}. \quad (83.5)$$

For the camel, $\nu_{\text{inc}} = 4n - 14$, $\nu_{\text{dis}} = n^2 - 8n + 17$, and for $n \geq 4$ this becomes

$$\mathbb{E}[X_n Y_n] = \frac{4(n^4 - 12n^3 + 58n^2 - 139n + 142)}{n^4}. \quad (83.6)$$

Also $\mathbb{E}X_n = \mathbb{E}Y_n = 2 - 6/n$, so

$$\begin{aligned} \text{Cov}(X_n, Y_n) &= \frac{4(2n^3 - 13n^2 + 7n + 54)}{n^2(n-1)(n-2)(n-3)} \\ &= \frac{8}{n^2} - \frac{4}{n^3} + O(n^{-4}). \end{aligned} \quad (83.7)$$

The first-order expansion has no mixed term, yet the variables are not independent. Their covariance appears at second order.

83.3 The complete second polynomial

The degree bound $\deg \mathfrak{B}_2 \leq 4$ reduces the computation to selected graphs with one through four edges. The exact catalogue has 83 source-realizable coloured types for $R = 3$ and 87 for $R = 2$, including types whose target count is zero. To generate it without choosing a large arbitrary n , grow every connected selected edge set by adjoining an edge incident to its support, normalize its minimum coordinate to zero, and stop at four edges. Every connected graph has a connected edge ordering obtained from a spanning tree followed by its remaining edges, so this process is complete. Canonicalize the connected types and take all multisets with total edge count at most four. Disjoint translations realize every such multiset in a sufficiently large interval.

For each type compute N_S, N_T by (79.3). If $v = |V(H)|$, expand its denominator by

$$\frac{n^v}{n^{\underline{v}}} = 1 + \frac{\sigma_1(v)}{n} + \frac{\sigma_1(v)^2 + \sigma_2(v)}{2n^2} + O(n^{-3}), \quad (83.8)$$

where $\sigma_1(v) = v(v-1)/2$ and $\sigma_2(v) = v(v-1)(2v-1)/6$ (Report 232: S_1, S_2). Multiply $\text{aut}(H)N_S(H; n)N_T(H; n)n^{-v}$ by this expression, retain orders through two, attach the colour monomial, and sum. Multiplication by the marked Taylor polynomial of e^{-2u-2v} through degree four gives (78.6) and (78.7). This is a finite rational calculation whose full graph-by-graph data are shipped in `data/07-coloured-gaps-fixtures-exact.json`.

As compact checks on the coefficient extraction, table 31 lists every nonzero orbit of marked monomials under exchanging u, v . Its entries are coefficients of $\mathfrak{B}_2$. The constant and pure linear

Monomial or symmetric pair	Camel $R = 3$	Knight $R = 2$
u^4, v^4	2	2
u^3, v^3	14	10
u^2, v^2	22	10
u^2v^2	8	8
u^2v, uv^2	12	4
uv	44	24

Table 31: The nonzero coefficients of $\mathfrak{B}_{2,R}(u, v)$, one per orbit under $u \leftrightarrow v$ (Report 232's table).

coefficients vanish. Evaluating at $u = v = -1$ gives 48 and 28, respectively. The logarithms of the avoidance probabilities consequently satisfy

$$\begin{aligned} \log(A_{1,3}^{\text{cg}}(n)/n!) &= -4 + \frac{8}{n} + \frac{16}{n^2} + O(n^{-3}), \\ \log(A_{1,2}^{\text{cg}}(n)/n!) &= -4 + \frac{4}{n} + \frac{20}{n^2} + O(n^{-3}). \end{aligned} \quad (83.9)$$

83.4 The mixed interaction missed by marginal multiplication

Let $G_n(u, 0)G_n(0, v)$ be the product of the two marginal marked laws. Subtract its second correction polynomial from the actual $\mathfrak{B}_2$. The results are

$$\begin{aligned}\Delta_3(u, v) &= 4u^2v^2 + 8uv, \\ \Delta_2(u, v) &= 4u^2v^2 - 4u^2v - 4uv^2 + 8uv.\end{aligned}\tag{83.10}$$

The alternating square in (79.5) supplies the $4u^2v^2$ term. The remaining terms come from mixed component and exclusion interactions. It would therefore also be wrong to ascribe the whole second-order discrepancy solely to the square.

For camel marginal avoidance, the first two coefficients are 4 and 10. Their product predicts $10 + 10 + 4^2 = 36$, whereas the actual joint coefficient is 48, a difference of 12. For knight they are 2 and 2; their product predicts $2 + 2 + 2^2 = 8$, whereas the actual coefficient is 28, a difference of 20. These are two instances of one coloured-gap theorem, with genuinely different mixed corrections. [Note of the batch-101 write: the marginal pairs 4, 10 and 2, 2 are Part I's $c_1^{(2)}, c_2^{(2)}$ at offsets (1, 3) and (1, 2) (remark 78.3). The write recomputed Δ_3 and Δ_2 from (78.5), (78.6), (78.7) and (7.5), and the values 48, 28, 16, 20, by hand.]

83.5 Independent finite enumeration and the threshold warning

The checker separately enumerates all permutations for every $n = 4, \dots, 9$ and both $R = 2, 3$. It records the complete distribution of (X_n, Y_n) and computes every mixed binomial moment through degree four. A different computation enumerates selected source-edge subsets and uses (79.2) with actual finite- n subgraph counts. All these moments agree. This test uses no eventual polynomial below its threshold. At $n = 9$ the avoidance counts are 19480 for knight and 22854 for camel.

The camel fourth falling-factorial moment illustrates why this distinction is necessary. At $n = 9$ direct enumeration gives

$$9! \mathbb{E}(X_9 + Y_9)^{\underline{4}} = 12829728, \tag{83.11}$$

while evaluating the eventual graph-polynomial expression at 9 gives 13044000, a discrepancy of -214272 . Its safe validity threshold is $n \geq 12$, so this is neither a contradiction nor a negligible rounding error. It is a concrete counterexample to an all- n reading of that certificate. Knight's fourth-moment certificate is already valid at 9, which exceeds its safe threshold 8.

The source and target embedding polynomials are also checked by direct selected-edge enumeration at their safe threshold, the next integer, and $n = 16$. At the threshold some types can have zero realizations; at 16 every catalogue type occurs. The independent packing tests compare the occupancy scan with direct enumeration of disjoint translated placements for two dictionaries, $n = 0, \dots, 9$, and all terms through four polymers. The dictionaries include equal supports with distinct species, distinct species of a common type, interlacing supports, and right-boundary cases. These finite checks target the proof mechanisms but do not replace their general proofs.

84 Controlled finite-model inverses for the coloured-gap counts (Report 232)

This section is a consequence of the all-fixed-order coloured-gap expansion. It does not posit a differentiable interpolation of the counting sequence. All comparisons with that sequence are made at integer arguments. The Lambert W function and Stirling–Newton inversion are standard tools; no priority claim for those methods, or global historical priority for this

specialization, is intended. [Note of the batch-101 write: the apparatus is Part I's (section 9, Remark 9.1), an instance of the transseries volume's gamma-carrier inversion and staircase theorem [12]; the coordinates below are Part I's with $\log y$ shifted by 4 (table 30).]

84.1 Input, logarithmic coefficients, and eventual monotonicity

Let $A(n)$ denote either $A189358(n)$ (camel) or $A137774(n)$ (knight). The previously proved expansion says that for every fixed $K \geq 0$,

$$A(n) = e^{-4}\Gamma(n+1) \left(C_K(1/n) + O_K(n^{-K-1}) \right), \quad C_K(t) = 1 + \sum_{j=1}^K c_j t^j. \quad (84.1)$$

The coefficients are consistent as K varies and are computable by the finite coloured-core construction. In particular,

$$(c_1, c_2) = \begin{cases} (8, 48), & \text{camel,} \\ (4, 28), & \text{knight.} \end{cases}$$

The logarithmic coefficients ℓ_j are determined, as in Part I's (9.2), by

$$\log \left(1 + \sum_{j \geq 1} c_j t^j \right) = \sum_{j \geq 1} \ell_j t^j, \quad \ell_j = c_j - \frac{1}{j} \sum_{i=1}^{j-1} i \ell_i c_{j-i}.$$

Thus $(\ell_1, \ell_2) = (8, 16)$ for camel and $(4, 20)$ for knight. Since $A(n)/n! \rightarrow e^{-4} > 0$, the counts are positive eventually, tend to infinity, and

$$\frac{A(n+1)}{A(n)} = (n+1) \frac{A(n+1)/(n+1)!}{A(n)/n!} \sim n+1.$$

Consequently $A(n)$ is strictly increasing eventually. This proof needs no interpolation or differentiation. The second-order expansion also gives

$$\frac{A(n+1)}{A(n)} = n+1 - \frac{c_1}{n} + O(n^{-2}).$$

Indeed the difference of the two logarithmic expansions is $-c_1 n^{-2} + O(n^{-3})$; an uncontrolled derivative of a remainder has not been taken.

Define the upper threshold by

$$N(y) = \min\{n \geq 0 : A(n) \geq y\}.$$

For large y , this threshold lies in the strictly increasing tail. Any finitely many exceptional initial terms are excluded by taking y above their maximum. Equalities at $A(n) = y$ are included in this definition.

84.2 Finite smooth model and a threshold-envelope theorem

For a fixed K , set $p = K + 1$ and define, on a sufficiently large real half-line,

$$\mathcal{F}_K(x) = \log \Gamma(x+1) - 4 + \log C_K(1/x). \quad (84.2)$$

Here $C_K(1/x) > 0$ eventually because it tends to 1. The logarithm is the real logarithm. Direct differentiation of this explicitly given function and standard gamma-function estimates give

$$\begin{aligned} \mathcal{F}'_K(x) &= \psi(x+1) - \frac{C'_K(1/x)}{x^2 C_K(1/x)} = \log x + O(x^{-1}), \\ \mathcal{F}''_K(x) &= \psi'(x+1) + \frac{2C'_K(1/x)}{x^3 C_K(1/x)} + \frac{C''_K(1/x)}{x^4 C_K(1/x)} - \frac{C'_K(1/x)^2}{x^4 C_K(1/x)^2} = x^{-1} + O(x^{-2}). \end{aligned}$$

In particular, $\mathcal{F}_K$ is eventually increasing and convex, tends to infinity, and has a unique inverse $\xi_K(y)$ there, defined by $\mathcal{F}_K(\xi_K(y)) = \log y$.

Theorem 84.1 (Threshold envelope; Report 232). *There exist $C_K > 0$ and n_K such that*

$$|\log A(n) - \mathcal{F}_K(n)| \leq C_K n^{-p} \quad (n \geq n_K). \quad (84.3)$$

After enlarging the real lower endpoint if needed, the functions $\mathcal{F}_K(x) + C_K x^{-p}$ and $\mathcal{F}_K(x) - C_K x^{-p}$ are both strictly increasing. Let $\xi_K^-(y)$ and $\xi_K^+(y)$ be their respective inverse roots:

$$\mathcal{F}_K(\xi_K^-) + C_K(\xi_K^-)^{-p} = \log y, \quad \mathcal{F}_K(\xi_K^+) - C_K(\xi_K^+)^{-p} = \log y.$$

Then, for all sufficiently large y ,

$$\lceil \xi_K^-(y) \rceil \leq N(y) \leq \lceil \xi_K^+(y) \rceil. \quad (84.4)$$

Moreover,

$$\xi_K^\pm(y) = \xi_K(y) \pm \frac{C_K \xi_K(y)^{-p}}{\mathcal{F}'_K(\xi_K(y))} (1 + o(1)). \quad (84.5)$$

Consequently, for some $D_K > 0$ and every sufficiently large y ,

$$\left\lceil \xi_K(y) - D_K \frac{\xi_K(y)^{-p}}{\log \xi_K(y)} \right\rceil \leq N(y) \leq \left\lceil \xi_K(y) + D_K \frac{\xi_K(y)^{-p}}{\log \xi_K(y)} \right\rceil. \quad (84.6)$$

Proof. Since $C_K(1/n) \rightarrow 1$, taking the logarithm in (84.1) proves (84.3). The envelope derivatives differ from $\mathcal{F}'_K$ by only $O(x^{-p-1})$, so both envelopes eventually increase. If an integer $n < \lceil \xi_K^- \rceil$, then $n < \xi_K^-$ and $\log A(n) \leq \mathcal{F}_K(n) + C_K n^{-p} < \log y$, whenever n is in the tail. The finitely many preceding indices are excluded by the choice of y . At $m = \lceil \xi_K^+ \rceil$, instead, $\log A(m) \geq \mathcal{F}_K(m) - C_K m^{-p} \geq \log y$. These facts prove (84.4), including its endpoint convention. The mean-value theorem, $\mathcal{F}'_K \sim \log x$, and then local comparison with ξ_K give (84.5) and (84.6). For example, evaluating either envelope at $\xi_K \pm A' \xi_K^{-p} / \log \xi_K$, with $A' > C_K$ fixed, first traps its root in that interval and then permits the stated asymptotic evaluation. $\square$

[Note of the batch-101 write: (84.6) is the bracket of Part VI's [theorem 73.1](#) for these counts, and (84.4) its sharper form with explicit envelope roots; both are instances of `p0:thm:staircase` [12]. In the lower-bound step, “ n in the tail” must include that the envelope $\mathcal{F}_K + C_K x^{-p}$ is increasing on $[n, \xi_K^-]$, which the choice of the lower endpoint provides.]

The constants and sequence onset in this theorem are not numerically certified by the present argument. The all-order proof supplies existence of remainder bounds; this section has not evaluated those constants or the associated onset. Thus these are eventual rigorous envelopes, not a claim that a particular finite y has certified rounding merely because it is large in a numerical experiment. This is an absence of an explicit onset certificate, not a claim that effective constants cannot in principle be extracted from the finite proof.

84.3 Lambert–Newton inversion to any fixed order

Put

$$\tilde{L} = \log y + 4, \quad \tilde{z} = \frac{\tilde{L}}{W_0(\tilde{L}/e)}, \quad \tilde{w} = \log \tilde{z}. \quad (84.7)$$

For large y , $\tilde{L} > 0$ and W_0 is the positive real branch. Its defining identity gives the exact equation $\tilde{z}(\log \tilde{z} - 1) = \tilde{L}$. Stirling's formula gives

$$\mathcal{F}_K(\tilde{z}) - \log y = \frac{1}{2} \log(2\pi \tilde{z}) + O(\tilde{z}^{-1}), \quad \xi_K(y) = \tilde{z} - \frac{1}{2} + O(\tilde{w}^{-1}).$$

In particular, $\xi_K(y) = \tilde{z} + O(1)$, and eventually $\tilde{z} - 1 < \xi_K(y) < \tilde{z}$. Define the fully specified finite iteration

$$x_0 = \tilde{z}, \quad x_{j+1} = x_j - \frac{\log \Gamma(x_j + 1) - 4 + \log C_K(1/x_j) - \log y}{\psi(x_j + 1) - C'_K(1/x_j)/(x_j^2 C_K(1/x_j))}. \quad (84.8)$$

Proposition 84.2 (Newton iteration; Report 232). *For every fixed number m of iterations,*

$$0 \leq x_m - \xi_K(y) = O_{K,m} \left((\tilde{z}\tilde{w})^{-(2^m-1)} \right). \quad (84.9)$$

Proof. For all large $\tilde{z}$, the model is increasing and convex on $[\tilde{z} - 1, \tilde{z}]$, with $\mathcal{F}'_K \geq \tilde{w}/2$ and $0 < \mathcal{F}''_K \leq 2/\tilde{z}$ there. Convexity implies $\xi_K \leq x_{j+1} \leq x_j \leq \tilde{z}$ for a Newton step starting above the root. Taylor's theorem at x_j , with a point η_j between ξ_K and x_j , then gives

$$x_{j+1} - \xi_K = \frac{\mathcal{F}''_K(\eta_j)}{2\mathcal{F}'_K(x_j)}(x_j - \xi_K)^2 \leq \frac{2}{\tilde{z}\tilde{w}}(x_j - \xi_K)^2.$$

The initial error is at most 1, so induction even yields

$$0 \leq x_m - \xi_K(y) \leq \left(\frac{2}{\tilde{z}\tilde{w}} \right)^{2^m-1} \quad (84.10)$$

whenever the stated interval bounds hold. This proves (84.9). $\square$

Taking

$$m \geq \lceil \log_2(K+2) \rceil$$

makes the Newton error $O_K(\tilde{z}^{-K-1}/\log \tilde{z})$, no larger in asymptotic order than the counting-expansion uncertainty. Thus a finite number of standard special-function evaluations gives an all-fixed-order inverse centre. Combining (84.6) with the Newton estimate, for a suitable $E_K > 0$ one obtains

$$\left| x_m - E_K \frac{\tilde{z}^{-K-1}}{\log \tilde{z}} \right| \leq N(y) \leq \left| x_m + E_K \frac{\tilde{z}^{-K-1}}{\log \tilde{z}} \right| \quad (84.11)$$

for all sufficiently large y . If fewer iterations are used, add $O_{K,m}((\tilde{z} \log \tilde{z})^{-(2^m-1)})$ to the envelope radius.

84.4 An effective Newton certificate for the displayed quadratic models

Proposition 84.3 (Effective model bounds; Report 232). *For $C_2(t) = 1 + 8t + 48t^2$ or $C_2(t) = 1 + 4t + 28t^2$, all the interval conditions in the proof of [proposition 84.2](#) hold for every real $\tilde{z} \geq 32$. Consequently (84.10) is then an effective bound for the exact finite-model root. The same bounds cover the truncations C_0 and C_1 of these two polynomials.*

This certifies the finite models themselves, separately from the unevaluated counting remainder.

Proof. Write $C_2(1/x) = q(x) = 1 + a/x + b/x^2$ with $0 \leq a \leq 8$, $0 \leq b \leq 48$. For $x \geq 31$, the standard positive real Stirling and digamma bounds give

$$0 < \log \Gamma(x+1) - x \log x + x - \frac{1}{2} \log(2\pi x) < \frac{1}{12x}, \quad \psi(x+1) \geq \log x.$$

Also $0 < \psi'(x+1) = \sum_{j \geq 1} (x+j)^{-2} \leq 1/x$. The numerator of $(\log q)''$ is

$$q''q - (q')^2 = 2ax^{-3} + (a^2 + 6b)x^{-4} + 4abx^{-5} + 2b^2x^{-6} \geq 0.$$

It follows that on $[\tilde{z} - 1, \tilde{z}]$,

$$\begin{aligned}\mathcal{F}'_2(x) &\geq \log x - 8x^{-2} - 96x^{-3} \geq \tilde{w}/2, \\ 0 < \mathcal{F}''_2(x) &\leq x^{-1} + 16x^{-3} + 288x^{-4} \leq 2/\tilde{z}.\end{aligned}$$

The last inequalities follow by replacing x by $\tilde{z} - 1 \geq 31$ and using $\tilde{z}/(\tilde{z} - 1) \leq 32/31$; all remaining rational errors decrease with $\tilde{z}$. At $x = \tilde{z}$, $\mathcal{F}_2(\tilde{z}) - \log y > 0$. With $x_- = \tilde{z} - 1 \geq 31$, the opposite endpoint satisfies

$$\mathcal{F}_2(x_-) - \log y \leq -\frac{1}{2} \log \frac{x_-}{2\pi} + \frac{97}{12x_-} + \frac{48}{x_-^2} < 0.$$

The right side decreases with x_- and is negative at 31: indeed $31/(2\pi) > 4 > e$, and $97/(12 \cdot 31) + 48/31^2 < 1/2$. This proves the bracket and all claimed effective model bounds. No bound on the actual counting error has been inserted into this argument. $\square$

[Note of the batch-101 write: the intake checked the two endpoint estimates and the derivative bounds by hand. In the first displayed bound, $C'_2(1/x)/(x^2 C_2(1/x)) \leq (a + 2b/x)x^{-2} \leq 8x^{-2} + 96x^{-3}$; in the second, the three correction terms of $\mathcal{F}''_2$ are bounded by $2(a + 2b/x)x^{-3} + 2bx^{-4}$, since $q \geq 1$ and the last term is negative.]

84.5 Explicit Lambert corrections

For the second-order models define

$$\tilde{t} = \frac{\log(2\pi\tilde{z})}{2\tilde{w}}, \quad a_1 = c_1 + \frac{1}{12}, \quad a_2 = c_2 - \frac{c_1^2}{2}, \quad \tilde{B} = a_1 + \frac{\tilde{t}^2 - \tilde{t}}{2}.$$

[Note of the batch-101 write: a_1 and a_2 are Part I's logarithmic coefficients (9.4) for these counts (Report 232: b_1, b_2).]

Proposition 84.4 (Second-order inverse; Report 232).

$$\boxed{\begin{aligned}\xi_2(y) &= \tilde{z} - \tilde{t} - \frac{\tilde{B}}{\tilde{z}\tilde{w}} \\ &\quad - \frac{a_2 + a_1\tilde{t} + \tilde{t}^3/6 - \tilde{t}^2/4 + (\tilde{t} - 1/2)\tilde{B}/\tilde{w}}{\tilde{z}^2\tilde{w}} + O\left(\frac{\tilde{z}^{-3}}{\tilde{w}}\right).\end{aligned}} \quad (84.12)$$

The parameters are

$$(a_1, a_2) = \begin{cases} (97/12, 16), & \text{camel,} \\ (49/12, 20), & \text{knight.} \end{cases}$$

In particular the familiar common correction $-\tilde{t} = -1/2 - \log(2\pi)/(2\log \tilde{z})$ precedes the coloured-gap corrections. For the actual integer threshold, replace ξ_2 in (84.6) by the displayed expression without its remainder and enlarge the envelope constant. Two Newton steps from $\tilde{z}$ give another centre with the same $O(\tilde{z}^{-3}/\tilde{w})$ counting precision.

Derivation. Set $f(x) = x \log x - x$. Stirling and the explicit logarithm of C_2 give

$$\mathcal{F}_2(x) + 4 = f(x) + \frac{1}{2} \log(2\pi x) + a_1 x^{-1} + a_2 x^{-2} + O(x^{-3}).$$

Substitute $x = \tilde{z} + D$ and $D = d_0 + d_1/\tilde{z} + d_2/\tilde{z}^2$ into $\mathcal{F}_2(x) = \log y$, using $f(\tilde{z}) = \tilde{L}$. The coefficients of $1, \tilde{z}^{-1}, \tilde{z}^{-2}$ are respectively

$$\begin{aligned}\tilde{w}d_0 + \tilde{w}\tilde{t}, \\ \tilde{w}d_1 + (d_0^2 + d_0)/2 + a_1, \\ \tilde{w}d_2 + (d_0 + 1/2)d_1 - d_0^3/6 - d_0^2/4 - a_1d_0 + a_2.\end{aligned}$$

Their vanishing yields $d_0 = -\tilde{t}$, $d_1 = -\tilde{B}/\tilde{w}$, and the displayed d_2 . The substitution residual is $O(\tilde{z}^{-3})$, uniformly for the actual $\tilde{t} = 1/2 + \log(2\pi)/(2\tilde{w})$, whose value is bounded. Since $\mathcal{F}'_2 \asymp \tilde{w}$, the mean-value theorem turns that residual into $O(\tilde{z}^{-3}/\tilde{w})$ root error. $\square$

The same calculation provides an explicit recurrence at every fixed order. Let a_j be Part I's coefficients (9.4) for these counts, $a_j = \ell_j + s_j$ with $s_{2r-1} = \mathcal{B}_{2r}/(2r(2r-1))$, $s_{2r} = 0$ and $\mathcal{B}_{2r}$ a Bernoulli number (Report 232: $\beta_j = d_j + s_j$). Use the formal variable $\varsigma = 1/\tilde{z}$, set $D(\varsigma) = \sum_{j=0}^K d_j \varsigma^j$, and cancel successive coefficients through ς^K in

$$\tilde{w}(D + \tilde{t}) + \sum_{\rho=2}^{K+1} \frac{(-1)^\rho D^\rho \varsigma^{\rho-1}}{\rho(\rho-1)} + \frac{1}{2} \log(1 + \varsigma D) + \sum_{j=1}^K a_j \varsigma^j (1 + \varsigma D)^{-j}. \quad (84.13)$$

Terms above degree K are discarded. The coefficient at degree j is $\tilde{w}d_j$ plus an expression depending only on earlier d_i , so the recurrence is finite and unique. For $j \geq 1$, the resulting d_j are bounded $O_K(1/\tilde{w})$ as $\tilde{z} \rightarrow \infty$, whereas $d_0 = -\tilde{t} = O(1)$. The same residual argument proves

$$\xi_K(y) = \tilde{z} + \sum_{j=0}^K \frac{d_j(\tilde{w})}{\tilde{z}^j} + O_K\left(\frac{\tilde{z}^{-K-1}}{\tilde{w}}\right).$$

When $K = 0$, the sum over ρ and the sum over j in (84.13) are empty and its constant coefficient still correctly gives $d_0 = -\tilde{t}$. Only the established c_j through order K are needed. Expanding $\log C_2$ beyond second order does not establish the third coefficient of the counting sequence. [Note of the batch-101 write: this is Part I's recursion (9.13) in the shifted coordinates; there κ_θ carries the constant $\frac{1}{2} \log(2\pi)$, here $\tilde{t}$ does.]

84.6 Rounding and the distinction between a root and a staircase

Suppose a centre $\vartheta(y)$ has been proved to satisfy an envelope

$$\lceil \vartheta(y) - \varepsilon(y) \rceil \leq N(y) \leq \lceil \vartheta(y) + \varepsilon(y) \rceil, \quad \varepsilon(y) \longrightarrow 0.$$

If $\text{dist}(\vartheta(y), \mathbb{Z}) > \varepsilon(y)$, both ceilings coincide and $N(y) = \lceil \vartheta(y) \rceil$. If $2\varepsilon(y) < 1$, the two endpoints differ by at most one, so at most two adjacent threshold values remain. Close to an integer, an exact evaluation of $A(n)$ or an effective one-sided remainder bound is needed to resolve the ambiguity. In particular an assertion of unqualified eventual exact rounding of the centre is not supplied by a big-O root error. Nor is $N(y) = \xi_K(y) + o(1)$ valid uniformly for all large real y : the threshold is an integer-valued staircase. The inequalities above retain the essential integer convention.

At the special level $y = A(n)$, one does have $\xi_K(A(n)) = n + O_K(n^{-K-1}/\log n)$, by (84.3) and the mean-value theorem. This useful inverse-at-the-data statement should not be confused with a uniform noninteger interpolation of $N(y)$.

For comparison, the two counting expansions imply

$$\frac{A^{\text{camel}}(n)}{A^{\text{knight}}(n)} = 1 + \frac{4}{n} + \frac{4}{n^2} + O(n^{-3}),$$

so eventually $A^{\text{camel}}(n) > A^{\text{knight}}(n)$ and $A^{\text{camel}}(n) < A^{\text{knight}}(n+1)$. Thus, at every sufficiently large level y , $N_{\text{camel}}(y) \leq N_{\text{knight}}(y) \leq N_{\text{camel}}(y) + 1$. This also follows directly at integer indices and makes no smoothness assumption about either sequence.

Source boundary and reproducibility. The only sequence-specific input is the independently audited coloured-core expansion. Standard special-function facts used here are the defining Lambert identity and the positive-real Stirling/digamma expansions and bounds; see the DLMF [8, 7, 42]. Those sources do not provide the coloured-gap inverse theorem. The accompanying script verifies the inverse algebra by exact symbolic arithmetic, checks the finite-model Newton bound numerically at high precision, and independently enumerates small avoidance counts. Numerical tests are consistency checks, not a substitute for the analytic remainder or an effective sequence onset.

85 Literature comparison and scope of the result (Report 232)

85.1 Leading limits and earlier corrections

The leading e^{-4} limits for the two chess-piece permutation counts are classical. Kotěšovec’s sixth-edition compilation gives the general finite-leaper leading-limit discussion on printed pages 622–624, tables including A137774 and A189358 on page 636, probability plots on page 637, and successive-ratio plots on page 638 [34]. Those passages support the leading equivalences; they do not display the two correction polynomials proved here. The book’s separate treatment of unrestricted leaper placements on pages 685–686 concerns a different model from a permutation with exactly one point in every row and column.

Aspvall and Liang’s *Dinner Table Problem* [35] is a particularly important antecedent. Part I, printed pages 5–6, gives higher-order corrections for its original circular adjacency problem, derived from a recurrence and also accessible from its exact block formula. Higher-order Poisson corrections for permutation adjacency problems are therefore not a new general idea. Part II, printed pages 7–12, proves a Poisson limit under factorial-moment bounds; its concluding estimate on page 10 is additive $O(1/n)$. It does not by itself supply the arbitrary-order compact-complex remainder for the simultaneous swapped-gap model in (78.2). [Parts I and II here are those of Aspvall and Liang’s report, not of this one.]

Mendelsohn’s 1956 paper [30] develops asymptotic-series machinery for permutation restrictions, including a relative-position example. Its operator-recurrence framework in Section 5 requires a fixed-length constant-coefficient polynomial-in-shift recurrence; the subsequent coefficient calculation explicitly assumes existence of the complete asymptotic series. No such recurrence for the present counts is assumed or established here. The present proof supplies the existence and uniform remainder directly from finite cores and finite-range packings. [Note of the batch-101 write: Report 83 reads the same paper the same way; see [section 74](#).]

85.2 Which static-board results apply directly

Flajolet and Sedgewick’s Proposition V.11 [36], in the discussion on pages 369–373, gives a leading Poisson limit for fixed forbidden exceedances $\pi(i) - i$. The finite occupancy-mask transfer construction on pages 370–371 is methodological prior art and is credited as such. Their Note V.51 discusses succession gaps and broader forbidden gap sets. The source restrictions in the present problem concern *two* permutation entries at a time and their difference; they are not fixed forbidden positions in a static permutation board.

Similarly, Godsil and McKay [37] develop sophisticated sparse subgraph asymptotics, including small-cycle corrections, for Latin rectangles using fixed forbidden-position bipartite graphs and rook polynomials. The conceptual use of small subgraphs is classical. Their static-board formula cannot simply be applied by calling the present selected-event graph a forbidden-position board: its edges constrain pairs of images. The exact two-sided product $i_S(H)i_T(H)$ in (79.1) records that difference.

For the knight sequence, Bagno, Eisenberg, Reches and Sigron [38] identify the no-separator permutation objects and derive expectation results. Their Theorem 6.1 gives a generating

function for one separator direction, equivalently one marginal orientation. It does not state the joint all-orders avoidance expansion of both reciprocal constraints. General mod-Poisson approximation schemes such as [39] provide broader approximation methodology under suitable residue-control hypotheses. They do not remove the need to prove the fixed-parameter n^{-j} residue expansion in this particular coloured model.

85.3 A bounded comparison rather than a priority claim

The mathematical result established here is the two-sided reciprocal coloured finite-core expansion, its uniform factorial-scale summation bound, and the explicit mixed and cycle corrections for the two named sequences. Graph inclusion–exclusion, partition-lattice inversion, finite-state transfer matrices, analytic perturbation, and Poisson correction polynomials are standard tools. They are not claimed as inventions of this report.

The inspected sources do not give the theorem in the form proved here. That is a bounded source comparison, not a proof that no equivalent theorem exists elsewhere. In particular, finding only the leading equivalent on an OEIS entry does not establish novelty, and renaming a model or changing one gap would not establish a new theorem. The reason to treat the present family separately from a one-source/one-target forest calculation is the simultaneous coloured structure and its compatible cyclic components. External peer review remains appropriate for any publication claim. [Note of the batch-101 write: the one-source/one-target forest calculation is Parts I–IV of this report, whose marginal statements [theorem 78.1](#) reproduces ([remark 78.3](#)).]

86 Reproducibility and certificate design (Report 232)

86.1 Contents and exact arithmetic

The companion archive is a complete source package. Its principal files are

- `report.tex`, the single-file editable manuscript source;
- `certificates.py`, the standard-library exact graph, moment, catalogue, and packing checks;
- `inverse_checks.py`, exact inverse algebra and clearly labelled high-precision finite-model diagnostics;
- `build.py`, the deterministic full builder with guarded output;
- `guard_tests.py`, adversarial output-path and mutation checks;
- `fixtures/exact.json` and `fixtures/inverse.json`, the full expected outputs, regenerated and byte-compared during a build;
- `Report232.pdf`, the compiled report, and a digest manifest.

[Note of the batch-101 write: here the four programs are shipped in `code/` and the two fixtures in `data/`, each with the prefix `07-coloured-gaps-` (the fixtures with `fixtures-` after the prefix); `report.tex`, the PDF, the delivered README and the digest manifest are not shipped. The report README says how to rerun the programs.] The graph certificates use ascending arrays of rational coefficients, encoded as integers or strings such as $5/2$; there is no expression parser or arbitrary evaluation of fixture contents. Graph labels in the machine files use colours 0, 1 for A, B . Every graph row includes the canonical component list, automorphism order, and source and target unlabelled embedding polynomials. The safe polynomial threshold is stored at model level and stated in this report.

The checker reconstructs the catalogue and every polynomial from scratch, then verifies the full $\mathfrak{B}_1, \mathfrak{B}_2$, the independent permutation distributions, and the counterexample (83.11).

Stored fixtures are expected outputs, not computational inputs to the mathematical routines. All graph and packing arithmetic is exact. The inverse checker uses exact symbolic arithmetic for coefficient identities and separately uses 100-digit floating arithmetic for model roots. Its freely chosen higher-order method fixture exercises the formal inverse recurrence; it does not assert any uncomputed third or fourth coefficient of the counting sequence.

86.2 Running the build

The full build requires Python 3.11 or later, SymPy, mpmath, pdfLaTeX, and the ordinary LaTeX packages listed at the start of `report.tex`. No network access is used. From an extracted source package, run

```
python3 -B build.py --output /absolute/path/to/new-directory
```

The output directory must not exist, even as a dangling symlink. The builder rejects targets inside the source tree, targets whose parents cross a symlink, and existing targets. It uses exclusive creation after validation, so it never silently reuses a previous output directory. It snapshots the complete regular-file inventory and verifies the files' bytes after the build. Intermediate TeX files and regenerated fixtures are written in a separate temporary workspace. The output directory contains the PDF, the source ZIP, the flattened manuscript, the two regenerated fixtures, and an external build receipt.

All Python guard checks are explicit exceptions rather than `assert` statements, so optimization does not disable them. Every child Python invocation includes `-B`, and optimized parent runs also pass `-O`. The decimal integer-string limit is set to 640. No bytecode cache is written. The package uses fixed ZIP timestamps and member order, fixed TeX metadata, and no volatile build-time text in the PDF. The build receipt records the environment separately, so it does not change the mathematical artifacts. [Note of the batch-101 write: the build needs the delivered layout (`report.tex`, `README.md`, `SHA256SUMS` and `fixtures/` beside the scripts) and so runs only from a re-extraction of the archive in its arrival commit; the report README says so.]

86.3 What a successful replay establishes

A successful full build regenerates exact and inverse fixtures, compares them with the packaged fixtures, compiles the manuscript three times, checks for TeX errors and overfull boxes, and creates a deterministic archive. The normal and optimized builds must agree byte-for-byte on the PDF, ZIP, TeX, and fixture files. The release procedure tests this from two fresh extractions of the *actual produced ZIP*, rather than merely rerunning a working directory. The complete regular-file inventory of each extraction is hashed before and after, and those files must remain unchanged. The supplied guard tests challenge existing files and directories, a dangling symlink, a symlinked parent, a path inside the source tree, and tampered or missing source manifests.

The proof establishes the asymptotic theorem. These checks establish that the packaged finite calculations and build are replayable and that common boundary and normalization mistakes are caught. Neither a small- n comparison nor a high-precision model inverse is an effective certificate for the unknown onset of the sequence's asymptotic remainder. The package contains no third-party paper PDFs and no private review materials.

87 Further questions and research

[Added 5 October 2026, batch 101.] This section keeps Report 232's six questions, under its title "Further questions". They are the unproved claims of Report 232 that this write found: Report 232 asserts the existence of effective constants (question 1), of higher coefficients (question 2) and of extensions (questions 3–6) without proving them, and the write found no

demonstrably wrong claim. Its two steps argued at sketch level, the count of anchored shapes in lemma 80.4 and the rationality step of theorem 81.1, are completed in remarks 80.5 and 81.2; its slip in the Hölder step of lemma 80.3 is corrected in place with a note. Notes of the write after the list say what each continues.

- Q1. Effective constants.** The proof supplies finite constructions and eventual bounds. Extracting usable explicit constants and onsets for the complex-mark remainder would turn the threshold envelopes into practical certified finite-level inverse algorithms. The effective $\tilde{z} \geq 32$ Newton result concerns only the displayed smooth model.
- Q2. Higher coefficients.** The moment rule reduces order j to graphs with at most $2j$ selected edges. Efficient canonicalization, quotient caching, and sparse transfer operators could make orders three and above accessible. Their existence is proved here, but no numerical values for them are claimed.
- Q3. More colours and general gap sets.** The positive tail method and finite dictionaries suggest extensions to arbitrary fixed finite gap correspondences. Compatible triangles and denser cores change the deficit-to-edge relation and hence the marked-degree bound. Each such extension needs a stated structural inequality and its own proof.
- Q4. Growing offsets.** All constants here depend on fixed gaps. If r or s grows with n , dictionary sizes, spans, and analytic neighbourhoods change. The present compact-uniform theorem does not automatically control that regime.
- Q5. Alternative exact recurrences.** A finite-state recurrence for the full permutation counts, rather than the fixed-core packing polynomials, might yield efficient numerical counts or independent remainder estimates. No recurrence is inferred from the finite fixtures.
- Q6. Distributional refinements.** Uniformity on complex compact marking sets gives access to fixed mixed derivatives and correction measures. Quantitative norms for the whole joint distribution require stating and proving an appropriate inversion estimate, rather than reading a total-variation rate off a formal generating series.

Notes of the batch-101 write on these questions. Question 1 is Part VI's question 1 for these counts. Question 2: at order three remark 80.2 already says which cyclic cores enter (coloured cycles and other cores of deficit three); the intake's Richardson extrapolation of the knight b-file (OEIS data, $n \leq 30$) gives 27.2, 28.24, 28.13 for the second coefficient at orders 3–5, and of the camel b-file ($n \leq 24$) 49.1, 46.6, 50.0 at orders 2–4, consistent with 28 and 48; this is evidence for theorem 78.2, not a third coefficient. Question 3 continues Part I's question 6, Part II's question 4 and Part III's question 10, and with it question 50.5, all of which Part VII answers for reciprocal pairs only (the dated notes there say so): still open are two pairs sharing one gap (Part II's "first target"), the directed two-gap model (Part III), more than two colours, and moving gaps. Question 4 is Part I's question 5 and Part IV's corollary 46.1 for two colours; Part IV's local-density method is the natural starting point. Question 5 is the two-gap analogue of Part I's question 2. Question 6 is the two-mark analogue of Corollary 7.3 and corollary 38.2, which Part VII does not prove. Cyclic wraparound of positions or values, the other half of Part I's question 6, is not treated by Report 232. Nothing in Part VII is formalized.

Independent check of the write (dated note, 5 October 2026). An independent adversarial check of Parts VI and VII, made by the intake after the write (e9994117b), reread nine items: theorem 73.1 and proposition 73.2 with the write's proofs, the enclosures of table 27 with the checks printed after them, Remarks 78.3, 80.2, 80.5 and 81.2, the corrected Hölder step of lemma 80.3, and the second coefficients 48 and 28 of theorem 78.2. It found all nine items valid, no counterexample, and no gap in any proof chain. One sentence of the write

overstated what its numbers show: the checks after [table 27](#) called the E_{10} interval at $n = 1000$ an independent check of Part I's c_{11} and c_{12} . With width 9.35 that interval checks c_{11} and requires only a correction c_{12}/n of the printed sign and rough size. The sentence is corrected where it stands, with a dated note keeping the first wording. No other statement or number changed.

The check used neither the delivered programs nor the write's. For [theorem 73.1](#) it re-derived every inequality of the proof with $C = 2^{K+4}C_1$, including the case $K = 0$ (where $\ell_K = 0$) and general $r, s \geq 1$, for which only $c_0 = 1$ and [theorem 1.1](#) are used. For [proposition 73.2](#) it re-derived $\ell_2 = -\frac{5}{2}$, $\ell_3 = 4$ and the coefficient, sign and order of each of the five correction terms, and evaluated the display against the exact model root at 60 digits with $K = 10$. The error times $\mathfrak{X}^4 \log \mathfrak{X}$ is 13.29, ..., 10.75 for $\mathfrak{X} = 20, \dots, 800$, the intake's values, and 10.14 and 9.71 at $\mathfrak{X} = 3200$ and 12800, decreasing slowly toward the predicted constant $-\ell_4 = \frac{29}{4}$; the order $\mathfrak{X}^{-4}/\log \mathfrak{X}$ is thus attained, not merely an upper bound. For [Remark 80.2](#) it reread the case analysis and tested every pair $1 \leq r < s \leq 12$ by brute force (66 pairs, $s = 2r$ and $s = 3r$ among them): among all two-coloured graphs on four vertices that contain a cycle, the alternating square is the only one that embeds in both S_n and T_n , and $i_S = i_T = 4(n - r - s)$ by direct counting.

It recomputed the Bonferroni sums of [table 27](#) exactly, by a formula different from Report 83's: each moment is a sum over profiles of the exact number of vertex-disjoint subpath collections of that profile in the two residue paths, valid for every n . The full alternating sum reproduces brute-force permutation counts for $n \leq 9$ and the OEIS values of A189281 at $n = 10, 14, 20, 26$ and 30. The three pairs $[S_J, S_{J+1}]$ equal the shipped rationals exactly; with e bracketed by Taylor sums, every printed endpoint encloses the true interval, and the width 9.35 is right. Part III's truncation at order 40 lies inside all three enclosures, and Part I's at order 16 just outside at $n = 100$ and $n = 200$ and inside at $n = 1000$, as printed.

For [theorem 78.2](#) it computed $\mathfrak{B}_1$ and $\mathfrak{B}_2$ of both models exactly by a route independent of Report 232's Möbius inversion and sign propagation. For each n from 13 to 21 it enumerated every selected edge set with at most four edges in S_n and in T_n , canonicalized the coloured graphs by brute force over vertex labellings, fitted each embedding count by a polynomial (confirmed at all nine points) and expanded the moments exactly in $1/n$; the same finite graph sums equal brute-force permutation moments for $n = 6, \dots, 9$. It found 83 source-realizable types for the camel and 87 for the knight, the sizes of Report 232's catalogues. Every coefficient of (78.5), (78.6) and (78.7) is recovered, hence $\mathfrak{B}_2(-1, -1) = 48$ and 28, and the cyclic types contribute exactly $4u^2v^2$, as [Remark 80.2](#) says. Richardson extrapolation of the OEIS b-files (A189358 to $n = 24$, A137774 to $n = 30$, fetched 5 October 2026) is consistent with 48 and 28; it is support only, the exact computation being decisive. The marginal parts of $\mathfrak{B}_1$ and $\mathfrak{B}_2$ in [Remark 78.3](#) were recomputed from Part I's formulas; [Remarks 80.5](#) and [81.2](#) and the Hölder step were checked by reading. This was a careful reading with exact computations, not a formal verification.

88 A worked partition-lattice certificate (Report 232, Appendix A)

Consider two disjoint A edges in a single-length graph with gap d . There are $n - d$ total edges. Unordered distinct pairs number $\binom{n-d}{2}$. Incident pairs occur along each length- d path and total $\max(0, n - 2d)$. For $n \geq 2d$ the exact matching polynomial is therefore

$$N_{2A}(n) = \binom{n-d}{2} - (n-2d) = \frac{n^2}{2} - \frac{2d+3}{2}n + \frac{d^2+5d}{2}. \quad (88.1)$$

Its abstract graph has automorphism order $2^2 \cdot 2! = 8$. For the camel, the source uses $d = 1$ and target $d = 3$, giving

$$N_S = \frac{n^2 - 5n + 6}{2}, \quad N_T = \frac{n^2 - 9n + 24}{2}.$$

Its exact graph term is $8N_S N_T u^2 / n^4$ where these polynomials are valid. The partition-lattice algorithm obtains the same expressions by summing quotient homomorphisms with their Möbius factors, and does so for all graph types rather than requiring a separate geometric formula for each. For a mixed incident path the automorphism order is one, the source and target polynomial are both $4n - 4R - 2$, and its term is the first summand in (83.5). For a mixed disjoint pair the automorphism order is four, recovering the second summand. These small examples fix the normalization conventions of the machine-readable certificates.

89 Proof dependencies and limitations (Report 232, Appendix B)

The main theorem rests on four independently stated ingredients:

1. the exact two-sided injection identity, including cyclic graphs;
2. a uniform exponential-moment tail for the number of selected edges;
3. a positive geometric bound for high-deficit cores;
4. the factorial-scale packing expansion for the finite low-deficit dictionaries and its source–target normalization.

The degree bound is proved only after those ingredients are combined. It is what makes the finite four-edge computation conclusive for $\mathfrak{B}_2$. The inverse section then uses the resulting integer asymptotic expansion, standard gamma-function estimates, and monotonicity of an explicitly defined finite model. It never differentiates an unknown sequence remainder.

The reported computations establish exact rational identities and finite checks at stated ranges. They do not certify an effective counting-sequence onset, convergence of an infinite correction series, universal exact rounding of a smooth inverse, or worldwide historical priority. These limitations leave the proved compact-complex all-fixed-order expansion and the displayed mixed corrections intact.

A Proof dependencies and archive guide

The main theorem uses the following chain, all proved in the body of the article:

$$\begin{aligned} \text{selected-edge tilings} &\implies \text{stable polynomial moments} \\ &\implies \text{uniform profile expansion} \\ &\implies \text{finite coefficient formula.} \end{aligned}$$

The separate factorial-moment bound justifies discarding the tail and also proves the total-variation universality theorem. Stirling’s expansion and the definition of Lambert W are standard external analytic inputs for the inverse section. The recurrence conjectures, the rational-collapse conjecture, and the 2026 holonomicity report are not dependencies of the all-orders theorem.

The archive contains the L^AT_EX source and compiled PDF, the core exact coefficient/enumeration library, a validation program, the rational-certificate program, two optional exploratory programs, and their outputs. The README gives commands to reproduce the computations and compile the article. The symbolic collapse script asserts only its explicitly

bounded range of identities; its success is not presented as an infinite proof. [Added 1 October 2026, batch 73O2: in the shipped report the compiled PDF is a new build of this text, and the archive’s README is replaced by the report README; [section B](#) lists what is shipped.]

B Provenance and editorial notes

[Added 1 October 2026, batch 73O2.]

Source. The report was built from one manuscript, manuscript 60 of batch 73 of ProveIt’s incoming-reports intake: the archive `oeis_path_forests.zip` (main file `article.tex`, a 21-page PDF titled *Unconditional all-orders asymptotics for two OEIS permutation sequences*). It arrived in commit `aa43cc555` and was placed in commit `6e193dd4f`, which deleted the archive from `docs/incoming`; it survives in `aa43cc555`. The manuscript pins no repository revision. It cites the repository only by the path of the Fabius completion audit [9], which it uses as an editorial example and calls “not a dependency”. This report does not continue that audit, and nothing was filed in the Fabius tree. [Added 1 October 2026, batch 74: since batch 74 the report is built from two manuscripts; Part II and the paragraph “Part II (batch 74)” below describe the second.]

What the write changed.

- Every label gained the prefix `spf :`; no label was renamed otherwise and none was removed. The manuscript had 78 labels.
- Added: the editorial note after the status paragraph; Remark 7.2 on the offsets (1,1) (A000255 and A002464); Remark 9.1 on the repository’s inversion apparatus; bracketed notes on Padhi’s preprint, on integer thresholds and on the shipped files; three bibliography entries [10, 11, 12]; and this section. A separate reciprocal note, Remark 7.4, points to the local law of the companion report on balanced Smirnov words.
- Nothing was removed. Every non-claim of the manuscript is kept: the status paragraph, the Poincaré-only reading of Table 1, the limitation after [theorem 4.2](#), the fixed-offset limitation after the proof of [theorem 1.1](#), the conditional nature of Proposition 11.2, and the statement that the recurrences of A189281 and A110128 are not proved.

Status of the conjecture. Conjecture 11.1 (rational collapse) is a conjecture. It has been checked as an identity of rational functions for $h = 4, \dots, 16$ only, and the integrality of every $c_j^{(1)}(2, 2)$ is conditional on it (Proposition 11.2). [Added 3 October 2026, batch 85: this paragraph describes the report before batch 85. The conjecture is proved in Part III ([theorem 26.1](#)), and the integrality is unconditional ([corollary 26.3](#) and [theorem 28.1](#)); see the paragraph “Part III (batch 85)” below.]

Overloaded letters. The manuscript reuses a few letters; none is renamed here. C is a tiling count $C_{\mathcal{P}}(\mathbf{b})$ in [sections 2](#) and [3](#), the number of non-dimer components in [section 5](#), a constant $C_{r,s}$ in Lemma 6.1, and the series $C(x)$ in [section 9](#). K is a moment cutoff in Lemma 4.1 and Theorem 4.2 but the edge degree of the non-dimer part of a profile in [section 5](#). d is the defect of a profile, while $d_j(w)$ are the inverse coefficients of [section 9](#). $\mathcal{D}$ is a formal series in [section 3](#) and D the index correction in [section 9](#). In each case the section determines the meaning.

Checks by the intake. The intake reran every shipped program on a copy (all passed; the regenerated outputs equal the shipped ones apart from line endings, a final newline and timings), recounted both sequences for $n \leq 9$ by brute force, compared the two order-ten expansions

with the OEIS entries digit for digit, rederived d_1 in [theorem 9.2](#) by hand, and made the (1, 1) checks of [Remark 7.2](#). These checks do not replace an independent review of the proofs.

Part II (batch 74). [Added 1 October 2026, batch 74.] Part II is built from manuscript 03 of batch 74, the archive `ProveIt_Fixed_Gap_Asymptotics.zip` (14 files, a 25-page PDF), which arrived in commit `c664fc2f0` and was placed in commit `b669cff87`. Its pinned commit is `291fbe6bb` [14]. Its article, README, PDF and checksum ledger `SHA256SUMS.txt` (13/13 verified at placement) are not shipped; its other ten files are, byte-identical, with the prefix `02-fixed-gap-`: `code/fixed_gap.py` and `code/verify.py` in `code/`; the six files of its `data/` and `requirements.txt` in `data/`; and `sources.md` at the report root. Those files use the delivery names.

Where the batch-74 merge had to choose. Manuscript 60 stays the base, being the more general in u and the stronger in arithmetic (denominator theorem, order sixteen), in the whole-law statement and in its certificates. Manuscript 03’s re-derivations of Part I’s theorems are recorded as second routes in [table 5](#) and not reprinted; its own results form Part II, printed after Part I’s conclusion so that no existing number changes. Its symbols are renamed into Part I’s notation ([table 4](#)); the devices its new proofs need (the moment bound with $D_{n,k}$, the unit-disk Taylor bound, the defect-sensitive bound) are printed with them. Its coefficients through order twelve are printed only as a confirmation of [table 1](#).

What the batch-74 write changed. It added the part headings, the editorial note after the batch-7302 one, bracketed notes marked “[Added 1 October 2026, batch 74]” after the proof of [theorem 1.1](#), in questions 1 and 5 of [section 12](#) and in this appendix, Part II ([section 14](#) to [section 22](#)), two bibliography entries, and a table-of-contents line for the appendices. All its labels carry the prefix `fga:`. No existing label was renamed or removed, no existing number changed, and no delivered sentence of manuscript 60 was changed or removed. Manuscript 03’s text is restated in Part I’s notation, with some proofs condensed; none of its statements was strengthened.

Part III (batch 85). [Added 3 October 2026, batch 85.] Part III is built from manuscript 03 of batch 85 (a different manuscript from manuscript 03 of batch 74), the archive `OEIS_Rational_Collapse_and_Integrality.zip` (16 files, a 23-page PDF), which arrived in commit `317c1ce2e` and was placed in commit `713149ded`. Its pinned commit is `d68b65ea0` [19]. Its article, README, PDF and checksum ledger `MANIFEST.sha256` (15/15 verified at placement) are not shipped; its other twelve files are, byte-identical, with the prefix `03-rational-collapse-`: `code/verify.py`, `code/certify.py`, `code/inverse_demo.py` and the `Makefile` in `code/`; the six files of its `data/` and its `OEIS_proposed_note.md` in `data/`; and `SOURCES.md` at the report root. Those files use the delivery names.

Where the batch-85 merge had to choose. Manuscript 03 answers this report’s open conjecture, so it is an addition, printed as Part III after Part II so that no existing number changes. Its theorems that Parts I and II do not contain are printed in full with their proofs: the single-sum formula, the universal collapse, the A189281 product, integrality with the degree law, the kernel and auxiliary recurrence, the Borel transform, the coefficient tables beyond order 16 and for A189282–A189284, and its certificates. Its second proof of [theorem 1.1](#) at $\theta = 1$ is printed, because it runs through the new identity; its other re-derivations are recorded in [table 9](#). Its special cases that repeat Part II’s [propositions 17.2](#) and [18.1](#), which it does not cite, are printed as notes crediting Part II ([remarks 25.2](#) and [28.2](#)); its index reconstruction is printed as an instance of Part II’s [proposition 20.1](#). Its symbols are renamed into Part I’s

notation (table 8), its components are indexed by vertices rather than edges, and $p = r - 1$, $q = s - 1$ are written out. Its coefficient table is printed only from order 17, orders 11 to 16 being in table 1.

What the batch-85 write changed. It added two macros to the preamble, the batch-85 editorial note, bracketed notes marked “[Added 3 October 2026, batch 85]” after Conjecture 11.1, after Proposition 11.2, in questions 1 and 4 of section 12, after corollary 18.2, in the overlap paragraph and question 1 of section 22, and in the paragraph “Status of the conjecture” of this appendix; Part III (section 23 to section 35); these three paragraphs; and five bibliography entries. All its labels carry the prefix `spf:rc:`. No existing label was renamed or removed, no existing number changed, and no delivered sentence of either earlier manuscript was changed or removed: the conjecture keeps its heading and label, and the conditional proposition its statement. Manuscript 03’s text is restated in Part I’s notation, with some proofs condensed; none of its statements was strengthened, and every one of its non-claims is kept in section 23.

Parts IV and V (batch 97). [Added 4 October 2026, batch 97.] Part IV is built from manuscript 05 of batch 97, the archive `ProveIt_Moving_Gap_Permutations.zip` (21 files, a 26-page PDF), and Part V from manuscript 06, the archive `ProveIt_A189281_Borel_Completion.zip` (16 files, a 21-page PDF). Both arrived in commit `e88ed8bf6` and were placed in commit `5a69916e8`. Their pinned commits are `d1680cdfa` [20] and `79e7aab60` [23]. Their articles, READMEs, PDFs and checksum ledgers `SHA256SUMS.txt` (20/20 and 15/15 verified at placement) are not shipped, nor is manuscript 05’s `requirements-optional.txt`, a byte copy of `data/02-fixed-gap-requirements.txt`. Their other sixteen and twelve files are, byte-identical, with the prefixes `04-moving-gaps-` and `05-borel-completion-`; those files use the delivery names.

Where the batch-97 merge had to choose. The two manuscripts continue this report independently and share no theorem, so each is a Part of its own, in archive order, printed after Part III so that no existing number changes. Manuscript 05’s re-proofs of Part I’s profile identity and moment bound and of Part II’s one-path formula are kept as marked second routes, because its later proofs refer to them; manuscript 06’s inputs from Parts I and III are recalled as displays with their sources named, not re-proved. Manuscript 05’s correction polynomials, subpath counts and sensitivity polynomials are written in sans-serif type, to separate them from Part I’s B_J , S_j and H_h , and its inverse coordinates are those of Parts II and III (table 14). Manuscript 06’s E , $\mathcal{B}$ and b_m are Part III’s $\mathcal{E}$ and $\hat{C}$ and the new $\hat{c}_m$, and its doubly used letters are renamed (table 24). Following the intake’s standing rule on unproved and wrong claims, manuscript 05’s informal multiplicity remark is made precise (Remark 45.3), the remainder of its index reconstruction is completed (Remark 49.2), what in its proposition 43.1 rests on computer algebra is stated (Remark 43.2), and Penrose’s inequality is credited (Remark 40.4); manuscript 06’s two numerical slips are corrected in Remark 66.1, not in its delivered table or files; the questions of both manuscripts stand in their Parts’ sections “Further questions and research”. One consequence of Part IV for Part II’s question 2, Remark 45.4, is the write’s.

What the batch-97 write changed. It added to the preamble the package `mathrsfs`, a question environment with its `cleveref` names, and nine macros (`\mga`, `\mgb`, `\mgm`, `\mgl`, `\mgmf`, `\bcC`, `\bcA`, `\bcN`, `\bcI`); the batch-97 editorial note; bracketed notes marked “[Added 4 October 2026, batch 97]” in questions 2, 3, 4 and 5 of section 12, in Remark 16.8 and the caption of table 3, in question 2 of section 22, after the paragraph “What this does not imply” of section 30, and in questions 1, 2, 3, 4 and 8 of section 34; Parts IV and V (section 36 to section 70); these three paragraphs; and nine bibliography entries. All its labels carry the prefixes `spf:mg:` and `spf:bc:`. No existing label was renamed or removed, no existing

number changed, and no delivered sentence of an earlier manuscript was changed or removed. Manuscripts 05 and 06 are printed nearly in full, with renamed symbols, headings marked with the manuscript and bracketed notes of the write; their appendices are sections of their Parts. None of their statements was strengthened, and every one of their non-claims is kept in [sections 36](#) and [54](#).

Parts VI and VII (batch 101). [Added 5 October 2026, batch 101.] Part VI is built from Report 83 of the session bundle of Reports 1–243, the archive `Fixed_Displacement_Permutation_Asymptotics_Source.zip` (22 files, a 7-page PDF), and Part VII from Report 232, the archive `Report232.zip` (10 files, a 27-page PDF). Both arrived in commit `60f54ea06` and were placed in commit `f7c612c72`. Report 83 pins `1512ef835` [29]; Report 232 pins nothing. Their articles, READMEs, PDFs and checksum manifests (`manifest.json`, 21/21, and `SHA256SUMS`, 9/9, verified at placement) are not shipped, nor is Report 83’s `independent_review/replay.log`, a byte copy of its `check_coefficients.json`. Their other seventeen and six files are, byte-identical, with the prefixes `06-fixed-disp-` and `07-coloured-gaps-`; those files use the delivery names.

Where the batch-101 merge had to choose. Report 83 proves no theorem that Parts I and II lack, so on the precedent of Part II it is a short Part: its re-derivations are recorded in [table 25](#), and only its new material is printed, with the proof of its threshold bracket and the error bound of its digamma display completed by the write ([theorem 73.1](#) and [proposition 73.2](#)). Report 232 is new throughout apart from its single-gap marginals, so it is printed nearly in full, its appendices as sections of its Part, after Part VI so that no existing number changes. Its letters that clash are renamed ([table 30](#)): its correction polynomials are written in fraktur type, $\mathfrak{B}_j$, to separate them from Part I’s B_J and Part IV’s B_j ; its counts $A_{r,s}^{\text{cg}}(n)$; its triply used S , T and δ and doubly used A , B , h , L and P are split; and its inverse coordinates are Part I’s with a tilde. Its *deficit* is kept as a separate notion from Part I’s *defect*, with both readings printed ([section 77.3](#)). Following the intake’s standing rule, the write proves that the marginals are Part I’s ([Remark 78.3](#)) and that cycles first enter at order n^{-2} ([Remark 80.2](#)), completes Report 232’s two sketched steps ([Remarks 80.5](#) and [81.2](#)), corrects its slip in the Hölder step of [lemma 80.3](#) in place with a note, and keeps its six questions and Report 83’s four unproved items in the sections “Further questions and research” ([sections 76](#) and [87](#)). The two reports share no theorem; they share one reference, Mendelsohn’s paper, printed once.

What the batch-101 write changed. It added the batch-101 editorial note; bracketed notes marked “[Added 5 October 2026, batch 101]” at the end of [Remark 7.4](#) (the sibling report’s tail estimate, stale since 2 October 2026), at the end of [section 9](#), after the Bonferroni intervals of [section 10](#), in question 6 of [section 12](#), in question 4 of [section 22](#), in question 10 of [section 34](#), in the non-claims of [section 36](#) and after [question 50.5](#); Parts VI and VII ([section 71](#) to [section 89](#)); these three paragraphs; and fourteen bibliography entries. All its labels carry the prefixes `spf:fd:` and `spf:cg:`. No existing label was renamed or removed, no existing number changed, and no delivered sentence of an earlier manuscript was changed or removed. Report 232 is printed nearly in full with renamed symbols, headings marked with the report and bracketed notes of the write; none of the two reports’ statements was strengthened, and every one of their non-claims is kept in [sections 71](#) and [77](#).

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